# Libertinus and diacritics

## To do

Long term: add mark-to-mark positioning (stacking) for the common above-marks in all four styles: ring, dieresis, macron, acute, grave, circumflex, caron, tilde, breve. The designer's stacked precomposed glyphs (ȅ, ȍ, ã, …) could supply the mark-to-mark anchors, as the precomposed capitals do for stage 4.

Long term: audit the SFD sources for consistent component referencing, across the four styles, and perhaps redraw or redefine components for consistency.

- Copies that could be references: glyphs that copy another glyph's splines (a derived letter drawn as a copy of its base, a precomposed letter as a copy of base and mark), so that a correction to the base reaches every glyph built on it, and the derived and precomposed stages need not rediscover the base in the outline.
- Dots, in letters and marks. The precomposed dot-below letters reference the dot above, moved below the letter (`uni0307` in regular and semibold, the spacing `dotaccent` in italic and semibold italic), except ṃ (and regular ṇ), which reference the combining dot below `dotbelowcomb`, so the dot of a precomposed letter and of a composed one differ in size: regular 100 against 108, italic 98 against 110, semibold 124 against 109, semibold italic 120 against 122. The shapes differ too: four-point circles in regular and semibold; in italic, `uni0307` a circle but `dotaccent` and `dotbelowcomb` skewed ovals; in semibold italic `dotaccent` a four-point skewed oval and `dotbelowcomb` a six-point one. Regular ṇ references both dots and n twice, so its two dots overlap. The diaeresis below U+0324 is two `dotbelowcomb` in regular and italic, but the precomposed ụ̈ is built from a diaeresis (`uni0308` in regular, semibold and semibold italic, the spacing `dieresis` in italic), with smaller dots in regular and italic.
- Commas and horns, in letters and marks (2026-11-11). In semibold and semibold italic the combining commas are drawn at the lighter style's weight: U+0312–U+0315 have 0.98 to 1.05 times the ink of regular's (italic's), and semibold's comma below U+0326 0.91, where the designer's own semibold comma and quotes have 1.21 to 1.27. The horn U+031B is regular weight in semibold (1.00). The koronis U+0343 is a reference to `uni0313` in regular and italic, but a reference to the horn in semibold (which had no mark anchor) and a horn-sized outline of its own in semibold italic; the patch curates both anchors ("Greek koronis U+0343" under "Comma-style marks"). The spacing Greek and modifier signs are bolded only in part: U+02BC, U+02BB, psili U+1FBF and koronis U+1FBD about 1.11 in semibold, U+1FBF and U+1FBD 1.00 in semibold italic; reversed comma U+02BD, dasia U+1FFE and varia U+1FEF 1.00 (varia is regular's outline). The precomposed Greek letters reference neither combining mark: ὰ ἐ Ἀ ῥ̓ reference U+02BC (U+1FBD in semibold italic ὰ), and italic draws its own. After a base letter HarfBuzz shapes U+0343 as U+0313, its canonical equivalent, so a koronis on a letter shows the style's comma above: its weight matters more than the koronis glyph. Options: embolden the combining commas (as stage 3's `embolden()` does for the marks it copies) or derive them from the style's own comma; make U+0343 the style's U+0313.

Create the glyphs that are missing in the non-regular styles because the base groups now reach further than the designer's coverage (Ɛ R Ɣ ɣ Þ ɕ Ɱ ʊ ʏ exist only in regular, Ɉ f ſ not in every style, and Ɋ Ɪ Ɨ before them), as stage 2 already does for a few.

Revise how stage 8 moves $Y$ away from an added part (the hook, tail or stroke of a derived letter). Now the anchor moves by how much further the added part reaches than the base letter within 120 units of the mark's axis, and a tail at the bottom takes the candidate's $Y$, from the letter's lowest point; both use the extreme point, not where the marks actually come close, so most anchors get more clearance than they need and a few too little (ɢ above overlaps one mark). Alternatives measured on 2026-11-11 with every IPA mark: (A) the tightest mark on the derived letter gets the base letter's tightest gap, which removes over-clearance on 83 of 148 anchors but adds clearance on 43; (B) the same, but only toward the glyph, with a floor of about 20, so that only over-clearance and near-collisions change.

Write general instructions for using CGJ U+034F to block unwanted normalization. A below-mark typed after a precomposed letter that already has an above-mark is reordered and recomposed by normalization: ì + tilde below becomes ḭ + grave, which brings back the dot of i, and Ǎ + dot below becomes Ạ̌ + ring + acute (130 cases among the precomposed letters). That is a decomposition problem, to be avoided with a CGJ (or a dotless base) whether or not the font has mark-to-mark positioning; the sections "Lowercase i with tilde below and grave above, in italic and semibold italic" and following explore it, and "Floating tones" uses CGJ for a similar reason. The instructions should say when CGJ is needed, where it goes, and its cost (a precomposed glyph before CGJ has no below-anchor, so the below-mark falls back to a default position).

Recheck all curated anchors A–Z and a–z in all styles (2026-11-12). Curated anchors were made style by style and at different times, and some disagree between styles about where an anchor belongs: regular p has its below anchor $(120, -319)$ under the stem, but the other styles do not. For each letter, compare the four styles' curated anchors (and the scripted ones they overrule) against the anchor model and the test sheets, and make them agree, or note why they differ. Part of it is curation that the stem-axis rule leaves to a human ("Stem-axis rule" under

“Patch j regular, above-anchor”): in italic and semibold italic, the below anchors of i, ɪ and l and the above anchor of m, where a curl moves the optical centre off the stem (the curled foot of i ɪ l, right; the arches of m). None of the existing values (the designer’s i, ɪ and m, the curated l) is known to be reliable. Evidence from the precomposed glyphs, composed minus precomposed with the anchor on the stem: italic ị $-14$, ḷ $+4$ (median of the other dot-below letters $+11$); semibold italic ị $-33$, ḷ $-22$ (median $-19$, spread about $\pm 30$); italic ṁ $+7$, semibold italic $-21$. The ink-area centroid of the bottom zone (baseline to 100 above the stem’s straight part) came within 4 units of ị and ḷ in three of four cases. Test sheets `tex/stem_axis_test_sheet.pdf`, `tex/stem_angle_test_sheet.pdf`.

Audit the above anchors of the vowels against their precomposed glyphs, which are many and good evidence (survey of 2026-11-07). Where the patched anchor differs from the median mark centre of the precomposed glyphs by more than 15: italic o $+20$, O $+22$, E $-20$, Y $-17$; semibold y $-19$, Æ $-16$; semibold italic y $+29$ (regular’s anchor, copied), Y $-58$, a $+18$, A $-15$, Æ $-19$. Regular O is within 12, but its marks land 16–19 left of the centre line of O. Two cautions: italic and semibold italic Y and U were just curated so that shaped marks land on their axes, and their precomposed glyphs disagree by 14–58; and the semibold italic precomposed glyphs run 30–40 right of the anchor model for g and E too. The vowel base anchors have not changed since; with the symmetric marks on their axes (2026-11-09), the median offset of composed marks from the precomposed glyphs per letter confirms the lowercase figures (italic o $+19$, semibold y $-18$, semibold italic y $+30$, a $+22$), and adds i: marks on ɪ land 12–18 right of ì í î ï in every style (regular $+16$, italic $+18$, semibold $+18$, semibold italic $+12$). The capitals were not remeasured. O and o lean to the left, unlike most of the other humanist letters, so their optical centre at the top may lie right of the bbox midpoint: a reason to move their above-anchors right, to weigh in the audit.

## Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark’s below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

Every font has five vertical reference lines (with respective $Y$ coordinate values) – ascender line, capline, x-height or meanline, baseline, descender line – within which each glyph lives. These vertical reference lines are the gray vertical lines in Figures 1, 2, 3 and 4. Each glyph has a (tight) bounding box for real minimum and maximum $X$ and $Y$ coordinate values, as well as a so-called ideal bounding box where the glyph seems to live, but accounting for overshoots of bowls and serifs. The entire design goal is that the top and bottom of the ideal bounding box are two vertical reference lines, except for glyphs that are non-full ascenders or non-full descenders.

By clearance, I mean the (vertical) distance between a base and a mark, measured between the ideal bounding boxes of the base and the mark. For some designed fonts, the clearance is consistent for each vertical reference line, so that all glyphs that live at that reference line are separated from all marks with the same clearance; e.g. dots should not dance below the baseline for different letters that sit on the baseline, and acutes and graves sit at about the same height for different letters with similar top vertical reference line.

In roman styles of Libertinus, the clearance is encoded in the base anchor. The $Y$ coordinate of an above-anchor is the top of the base’s ideal bounding box plus the clearance of the nearest vertical reference line, and the $Y$ coordinate of a below-anchor is the bottom of the ideal bounding box minus that clearance. For most glyphs that edge of the ideal bounding box is the reference line itself. For a non-full ascender or non-full descender it is not, but the clearance is the same; see the section on vertical reference lines below. The $X$ coordinate of the anchor is at some horizontal optical center of the base glyph, but the meaning of the optical center depends on the glyph letterform, or whatever the designer thinks looks best. For a mostly symmetric letterform in the roman styles, a candidate for this $X$ coordinate is the midpoint of the bounding box of the glyph.

The anchor of a mark is at the horizontal center of the mark, if the mark is symmetric. An asymmetric mark, such as the acute or grave, has its anchor a few units to one side of the center of its outline. Vertically, the anchor of an above-mark is about 140 units above the bottom of the mark outline. That is near the top of a tall mark, such as the circumflex, and above the top of a short mark, such as the macron or dot. Because the above-anchor of a base at the x-height is 216 units above the x-height, the bottom of a tall mark is about 75 units above the base, and a short mark sits a little higher.

Other fonts use a different geometric model for anchors. In Noto, the base’s below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark’s below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark’s below-anchor is at the origin $(0, 0)$, and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark’s below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors. So in Noto the clearance is encoded in the mark anchor, and in Libertinus it is encoded in the base anchor.

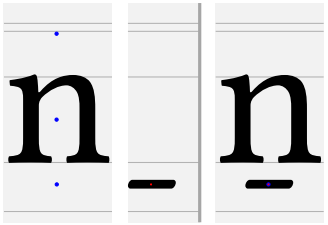


Figure 1: Libertinus n and macron with anchors.

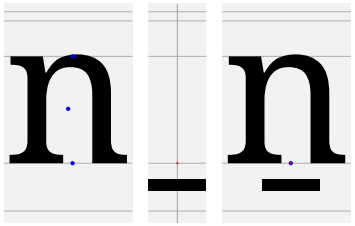


Figure 2: Noto n and macron with anchors.

### Italic anchor model

In italic, the Libertinus anchor model is similar to the roman styles, but it is applied along the italic axis. The italic lane of a base glyph is its advance-width box, slanted by the italic angle $\theta$ (12°). The midline of the italic lane is the line $X = W/2 + Y \tan\theta$, where $W$ is the advance width of the glyph. See the gray band and the dashed blue line in the left panel of Figure 3.

The $Y$ coordinate of an above- or below-anchor is set exactly as in the roman styles: the top or bottom of the glyph's ideal bounding box plus or minus the clearance of the nearest vertical reference line, measured vertically. So marks sit at the same height in roman and italic. The $X$ coordinate is the point of the midline at that height: $ax = W/2 + ay \tan\theta$ and $bx = W/2 + by \tan\theta$.

The height in this formula is the height of the anchor itself, that is, the edge of the ideal bounding box plus or minus the clearance, not the height of the reference line or of the ideal bounding box. So the $Y$ coordinate of the anchor must be determined before the $X$ coordinate can be calculated. In italic, the above-anchor is at $W/2 + 137$ for a letter at the x-height ($ay = 645$), at $W/2 + 181$ for a capital ($ay = 850$), and at $W/2 + 189$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 23$ for a letter on the baseline ($by = -110$) and at $W/2 - 68$ for a descender ($by = -319$). An error of 10 units in the $Y$ coordinate moves the $X$ coordinate by about 2 units.

The anchor of a mark lies on the mark's own slanted axis, through the center of its outline if the mark is symmetric. See the dashed red lines in the middle panel of Figure 3. That is, the mark anchor is at $X = C + Y \tan\theta$, where $Y$ is the height of the anchor, as in the roman styles, and $C$ is where the axis of the mark meets the baseline: the horizontal center of the mark outline after the slant is removed. An asymmetric mark keeps the same offset from its axis as in the roman styles. Superimposing the anchors lays the axis of the mark on the midline of the base, as in the right panel. The mark is displaced from the base along the italic axis, not vertically above or below it.

The midline is a geometric center, not an optical center, but it is a good candidate for the anchor $X$ coordinate of a mostly symmetric letterform. Asymmetric letterforms, and letters that overhang their advance width such as f and j, need curated anchors.

### Semibold italic anchor model

Semibold italic has its own anchor model, which is not the italic anchor model, and the patch keeps it. The outlines are slanted (the italic angle $\theta$ is 11.5°), but the anchors are, almost all, the anchors of regular, copied unchanged: 42 of the 44 above- and below-mark anchors, and 319 of the 384 above- and below-anchors of bases, have exactly the coordinates of regular.

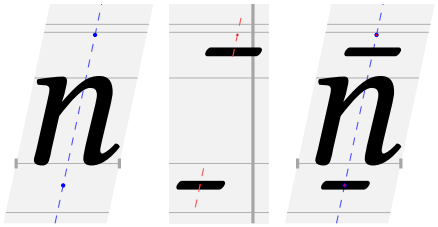


Figure 3: Libertinus italic n, macron and macron below, with anchors. The gray band is the italic lane and the dashed blue line is its midline.

So the anchors do not follow the slant. A slanted outline moves further right the higher it goes, but an anchor copied from regular stays where it was. An above-anchor therefore sits left of where the italic anchor model would put it, by roughly the amount the slant has moved the outline at that height. This is true of the base and of the mark alike: the base's above-anchor is left of the midline of its italic lane, and the mark's anchor is left of the mark's own slanted axis, by about the same amount. When the two anchors are superimposed, the two errors cancel, and the axis of the mark lands near the midline of the base, as it would in italic. See Figure 4, where the filled symbols are the designer's anchors and the hollow symbols are the positions given by the italic anchor model: the above-anchor of n and the anchor of the macron are both far left of the hollow symbols, but by nearly the same distance.

Each anchor in the font has its own shift $k$: how far it lies left of the position the italic anchor model gives it. For a base anchor, $k = W/2 + Y\tan\theta - X$. For a mark anchor, $k = C + Y\tan\theta - (X - d)$, where $C$ is where the mark's axis meets the baseline and $d$ is the mark's offset from its axis in regular ($d = 0$ for a symmetric mark; for the acute it is $-32$). In Figure 4, the above-anchor of n has $k = 170$ and the macron has $k = 141$. A mark lands $k_{\text{mark}} - k_{\text{base}}$ units right of the base's midline (left, if negative), so the macron lands $141 - 170 = -29$, that is, 29 units left of the midline of n.

The shifts differ from anchor to anchor, because the designer copied regular's $X$ coordinates, which carry no slant. So an anchor's shift is about $Y\tan\theta$ at its own height, plus a small amount that depends on how the slanted outline was drawn. A higher anchor has a larger shift: comma above U+0313, with its anchor at $Y = 605$, has $k = 125$; hook above U+0309, at $Y = 765$, has $k = 168$. Only the difference between a mark's shift and a base's shift moves the mark off the midline, so any single value works, provided every base anchor and every mark anchor uses it.

The semibold italic anchor model therefore fixes one value of $K$ for each anchor class and gives every anchor of that class the same shift: the base anchor is at $X = W/2 + Y\tan\theta - K$ and the mark anchor is at $X = C + Y\tan\theta + d - K$, with $\tan\theta = 0.2035$ and the same $Y$ coordinates as in the other styles.

- Above-anchors: $K = 150$, for every above-mark and every base's above-anchor. Above-anchors are high, where the slant has moved the outline far to the right, so their shift is large. 150 is the median shift of the designer's above-marks; most are between 132 and 168.
- Below-anchors: $K = -12$, for every below-mark and every base's below-anchor. Below-anchors are just under the baseline, where the slant has moved the outline a little to the left, so their shift is small and negative. $-12$ is the median shift of the designer's below-marks; most are between $-23$ and $-4$.

The values are taken from the marks rather than the bases so that most of the designer's mark anchors already conform. The designer's base anchors have median shifts of 139 above and $-1$ below on lowercase Latin and IPA letters, and 173 above and $-5$ below on capitals. An anchor conforms if its own shift $k$ is close to $K$. The italic anchor model is the case $K = 0$.

The candidates for a base are $ax = W/2 + ay\tan\theta - 150$ and $bx = W/2 + by\tan\theta + 12$. So the above-anchor is at $W/2 - 19$ for a letter at the x-height ($ay = 645$), at $W/2 + 23$ for a capital ($ay = 850$), and at $W/2 + 31$ for an ascender ($ay = 890$). The below-anchor is at $W/2 - 10$ for a letter on the baseline ($by = -110$) and at $W/2 - 53$ for a descender ($by = -319$). For semibold italic n, $W = 606$, so $ax = 284$ and $bx = 293$; the font has $ax = 264$ and $bx = 265$, the values of regular n, and the macron lands 29 units left of the midline.

The cancellation is only approximate, because the designer's anchors do not follow the slant at all, whereas the model's anchors do, apart from the constant $K$. A mark's shift is fixed by the height of its own anchor: for a mark, $k = Y\tan\theta$ plus a small amount, where $Y$ is the height of the mark anchor. But the designer's base anchors are near $W/2 - 9$ at every height, so they do not move right for a taller base, such as a capital or an ascender. The error of a mark on a base therefore grows with the difference in height of the two anchors, by 2 units for every 10. The median distance of a designer's base anchor from the candidate is about 20 units for a lowercase letter and 30 units for a capital.

The patch does not move this style to the italic anchor model. It keeps the designer's anchors where they conform to the semibold italic anchor model, and corrects only the anchors that do not, so that they conform to it: a base anchor is moved to the candidate above, and a mark anchor is moved to $X = C + Y \tan\theta - 150$ for an above-mark or $X = C + Y \tan\theta + 12$ for a below-mark. An anchor set by the italic anchor model would be wrong here: a base anchor without the shift would put every above-mark about 150 units too far right, because the marks carry the shift.

The following do not conform to the semibold italic anchor model and probably need to be patched, some of them manually.

- Capitals. The designer's above-anchors are about 25 units left of the candidate, since they are 205 units higher than the anchors of lowercase letters but no further right. The worst are B, D, I, N, O, R, W (40 to 60 units left) and L (159 units left; its anchor is over the stem, as in regular).
- The capital forms of the grave, acute, circumflex and breve (`grave.cap`, `acute.cap`, `circumflex.cap`, `breve.cap`), which the font substitutes over capitals. Their mark anchors were set with the slant ($k$ between $-13$ and 29), but the above-anchors of capitals were not, so these four marks land 160 to 190 units left of the middle of a capital. The other above-marks have no capital form in use and land 20 to 50 units left. The capital forms need mark anchors at $X = C + Y \tan\theta - 150$; `caron.cap` and `dieresis.cap` are not substituted.
- Above-anchors that were set with the slant, as in italic, so that an above-mark lands 130 to 190 units too far right: a, s, h and J.
- Other base anchors more than 40 units from the candidate. Above: c, g, Æ, Ɖ, ƕ, ƆȷƖ, Ƥ, ɲ, ɗ, ɦ, ʄ, Ħ, ɵ, ḥ. Below: A, C, D, J, a, d, f, j, q, r, t, Æ, ĸ, ƕ, ƆȷƖ, Ƥ, ɲ, ɾ, ɑ, ḍ, ɗ, ɤ, ɥ, ɯ, ɻ, Ħ, ɵ. Several of these are copies of the anchors of another glyph. The anchors of ɥ and ɥ are transposed, as in the other styles.
- Marks whose own shift $k$ is far from $K$ (150 above, $-12$ below), so that no base anchor can be right for them and for the other marks; each lands $k - K$ units right of where a conforming mark lands: dieresis U+0308 ($k = 111$, so it lands about 40 units left of the other above-marks), comma above U+0313 (125), inverted breve U+0311 (132), hook above U+0309 and double grave U+030F (168, about 18 units right), koronis U+0343 (371), right tack below U+0319 ($-30$) and tilde below U+0330 ($+10$). The double acute U+030B (156) conforms.
- Tilde U+0303, one of the two mark anchors that is not a copy of regular; its shift is 139. (These are the designer's values. In the patched font the dieresis and tilde are on the model by the mark-axis rule, and the hook above, inverted breve, U+0314, koronis, tilde below and right tack below by curated anchors; see "Mark anchor model".)
- The Hebrew ligatures U+05F0, U+05F1, U+05F2, which have above- and below-anchors in this lookup table in semibold italic only.
- Letters that overhang their advance width, such as f and j, which need curated anchors in every italic style.

The anchors of precomposed glyphs were not compared with these candidates.

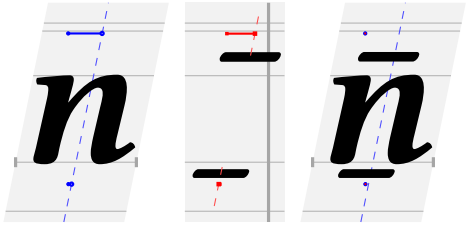


Figure 4: Libertinus semibold italic n, macron and macron below, with the designer's anchors (filled) and the positions given by the italic anchor model (hollow). The above-anchor of n is 170 units left of the midline ($k = 170$) and the anchor of the macron is 141 units left of its axis ($k = 141$), so the macron lands 29 units left of the midline of n. The model's own positions, with $K = 150$, would be 20 units right of the filled circle and 9 units left of the filled square.

**Mark anchor model**

The anchor of a mark follows the same model in every style. Horizontally, it lies on the mark's own axis, $X = C + Y \tan\theta - K$ ($\theta = 0$ and $K = 0$ in the roman styles, $K = 0$ in italic, $K = 150$ above and $-12$ below in semibold italic). An asymmetric mark, such as the grave, acute, double acute or double grave, keeps the offset from its axis that it has in regular. Vertically, the anchor is at a fixed distance from the mark's outline: the anchor of an above-mark is a certain height above the bottom of the outline, and the anchor of a below-mark is a certain depth below the top of the outline. These distances differ from mark to mark (a short mark such as the macron sits a little higher than a tall one), but since the base anchors have the same $Y$ coordinates in every style, a mark sits at the same height in every style only if its distance is the same as in regular.

So each mark can be checked in two ways: its $X$ offset from its axis, compared with regular, and the distance between its anchor and its outline, compared with regular. A third check is the designer's precomposed glyphs. They are not consistent between styles (the macron of ō was 18 units lower than the combining macron in regular, but 32 units higher in italic), and horizontally they also carry the offsets of the base anchors, so for $X$ they can confirm a finding but not settle one: the $X$ of a symmetric mark is set by the model (the mark-axis rule). Vertically they are better evidence: every x-height base has its above-anchor at 645, so where the offset of a composed mark from the precomposed glyphs is the same on every base, it belongs to the mark's anchor, and the anchor is moved by it ("Vertical moves from the precomposed glyphs"). In what follows, "lower" and "higher" and "left" and "right" describe where the mark lands on a base whose anchor conforms. A mark lands lower when its anchor is further above its outline, and left when its anchor is further right.

**Status.** The mark anchors have been reviewed style by style, against regular, the designer's precomposed glyphs and the mark zone figures (Figures 5–8). There is no scripted audit of them, as there is of base anchors (`anchor_audit.py`). They change in three ways:

- The capital forms (`grave.cap`, `acute.cap`, …) are set from the precomposed capitals (stage 4).
- A symmetric above mark whose $X$ is off its own axis is moved onto it (the mark-axis rule, stage 4, end of this section; no threshold since 2026-11-12).
- Everything else is changed one anchor at a time, as a curated anchor (stage 7), where the precomposed glyphs support the change: for $Y$, the median offset of the composed mark from the precomposed glyphs ("Vertical moves from the precomposed glyphs", below). A curated mark anchor is the final anchor and obeys the mark-axis rule itself; stage 4 skips it, and reports it if it does not.

Moving every mark anchor onto the model at once was tested and rejected (end of this section).

**Open decisions.**

- Semibold j: its dot is 14 units left of its stem (regular $-3$), but the dot rules apply only in the italic styles; ĵ and ǰ put the accent on the stem ($+2$).
- The axis of a mirror-symmetric mark is the centre of its ink bounds. A more precise one would be the line about which the outline best matches its mirror image, or its rotation by 180° for the tilde, zigzag and vertical tilde, whose terminals can push the bounds off centre.

**Decided: left as they are.**

- Italic and semibold italic, above marks on lowercase letters against the precomposed accents (2026-11-11). Over the 107 lowercase precomposed letters with one above accent, the composed mark lands a median 7 units (italic) and 8 (semibold italic) right of the precomposed accent, relative to the letter; with pure candidates as base anchors the medians are $+16$ and $+2$ (regular and semibold, bbox midpoint: $+9$ and $+6$, so the designer puts accents a little left of centre in roman too). This is not $K$, which is in both the base and the mark anchors and cancels when a mark attaches; not the slant (as a slant error the two italic styles would point opposite ways, by under 1°); and not the lane (the ink lane gives $+19$ and $+1$, with wider extremes). The offset varies by letter more than by mark (italic s $+35$, w $+25$, g $+23$, z $+23$, r $-29$; semibold italic y $+30$, c $+25$, z $+25$, a $+21$, g $-20$). Checked by eye in the grids: the precomposed accents are unusually far left on italic s and w (above all ẇ and ẘ), whose bowls support the composed marks further right, and the precomposed glyphs of semibold italic differ from italic's. The composed marks are as well centred as the precomposed accents, or better; no anchor is changed.

- Italic macron: its precomposed glyphs disagree ($-1$ on ā ē ḡ, $-32$ on ī ō ū ȳ), so it gets no vertical move. Across the styles the italic precomposed macrons are unusually loose (2026-11-12), but the combining macron sits where it should in its zone; nothing is changed.
- The capital hook above `hookabovecomb.cap` (2026-11-12): semibold italic's sits about 32 units higher in its capital zone than the other styles', but the designer is not consistent about how tight the hook should be; left as it is.
- Semibold double acute, diaeresis, hook above and circumflex, which sit 10 to 19 units lower than in regular: they are within 10 units of their precomposed glyphs vertically, which do not support a move (semibold's x-height is also 5 units higher than regular's).
- Semibold italic comma above U+0313 ($k = 125$) and double grave U+030F ($k = 168$): the precomposed glyphs do not support a move.
- Semibold italic tilde: the mark-axis rule moves it from 12 to 23 units right of its precomposed glyphs, the rule's one cost.
- Regular left half ring above U+0351: it shares the right half ring's anchor and lands 16 units right of centre (Gentium Plus puts it about 11 left).

**What the patch changes.**

- Stage 1, semibold only: the designer's second above class `top_punkt` (class 1) is merged into the above class. Its 12 marks (dot above, diaeresis, hook above, ring above, U+0312, U+0313, U+0314, U+031A, U+0351, U+0357, U+0363 and `hookabovecomb.cap`) move to class 0 with their anchors raised by 96 units, the difference between the two classes' base anchors. They sit where they did on a lowercase letter, but now attach to every base with an above anchor, not only the 24 with a `top_punkt` anchor.
- Stage 3, semibold and semibold italic: the combining marks the style lacks are added, and semibold's U+0358–U+0363, which were copies of the capital accents, are replaced. Their anchors are regular's; in semibold italic, regular's anchor with italic's outline moved right 12 units, which lands the added above marks 7 to 15 units left of the model, inside the spread of the designer's own ($k$ from 111 to 168).
- Stage 4, all styles, the capital forms, from the designer's precomposed capitals: on the axis for a symmetric mark, at the precomposed offset for an asymmetric one, with $K = 150$ in semibold italic. This fixes semibold italic, whose capital forms lacked $K$ and landed 120 to 160 units left, and italic, where only `grave.cap` had an anchor (in the wrong class). Italic's eight `.cap` anchors are curated (2026-11-07): each $X$ is fitted so that the accent lands, as a median over the capitals, where the designer drew it in the precomposed glyphs, as in regular (stage 4 had left the symmetric marks about 14 units left of them). The fit was made after italic O, U and Y took the median accent centre of their own precomposed glyphs as their above-anchor $X$ (493, 508, 441).
- Stage 4, all styles, the mark-axis rule (2026-11-09; threshold 0 and an axis measure per combining letter since 2026-11-12): $X$ only (end of this section).
- Stage 7, curated mark anchors (`data/fontanchors_human/`):
  - Vertical moves from the precomposed glyphs (2026-11-09): dot above in all four styles, macron in all but italic, semibold tilde, semibold italic ring, the double acute in all but semibold (below).
  - All styles, seven tall below marks (2026-11-08): tacks U+0318 and U+0319, half rings U+031C and U+0339, plus sign U+031F, ring U+0325, up arrow U+034E. They rose 82 to 94 units above their anchor, against 33 to 77 for the other below marks, so at the descender's below anchor ($Y = -319$) each overlapped 1 to 13 of the 26 to 31 Latin and IPA descenders, and under baseline letters their median gap was 7 to 26 units, against about 40 for marks such as the vertical line below. Each anchor is moved up, $X$ unchanged, so that the mark rises 62 above it, as U+0329 and U+0348 do. No below mark now overlaps a descender, and their median gap under baseline letters is 38 to 47. The precomposed Ạ̊ and ạ̊ confirm the ring (within 2 units in regular, italic and semibold italic). Lowering the descender reference to $-342$ instead also removed the overlaps, but left the crowding under baseline letters and moved every mark under descenders.

- All styles, up and down tacks below U+031D and U+031E (2026-11-11): both tops 62 above the anchor, so 48 below the baseline on an x-height letter, as the left and right tacks and the vertical line below; $X$ along the slant in the italic styles. The two are the same outline turned over. Other fonts level their boxes (the up tack's tip with the down tack's bar, within 0 to 11 units: Gentium Plus, Noto Serif, Junicode, DejaVu Serif, FreeSerif), and none levels the bars. Libertinus, like Linux Libertine O, put the down tack 37 units deeper than the up tack (gaps 33 and 70 on o; 45 and 58 in semibold italic), so the up tack was the tightest below mark, clearing descenders by 4 to 7 units, and the down tack among the loosest.
- All styles, tacks below U+0318 and U+0319: $X$ centred on the whole outline, along the slant in the italic styles, with $K = -12$ in semibold italic (regular earlier, the other styles 2026-11-11). The designer's anchors, the same as in Linux Libertine O, split the difference between the outline and the vertical stroke: the stroke landed 25 to 39 units off centre one way and the outline 19 to 36 the other. Fonts that centre the tacks centre the whole outline, which is also the middle of the horizontal stroke, and none centres the vertical stroke (Gentium Plus, Noto Serif, Junicode and DejaVu Serif, within about 10 units; 2026-11-11). The tacks have no precomposed glyphs, so the model decides. They now land with each style's other below marks.
- Semibold: grave and acute 30 units lower (713 to 683), as in regular; the grave keeps the designer's $X = -127$, 10 units (median) from the precomposed glyphs. The grave and acute tone marks U+0340 and U+0341, which stage 3 adds at 713, have the same anchors, as in regular (2026-11-11). Ring above lowered 27 to 715, the height of the ring in ů, and vertical line above lowered 31 to 688, as in the other styles (2026-11-01). Half rings above U+0351 and U+0357 (2026-11-08): after stage 1 they landed 20 to 32 units low; they now have regular's anchor-to-outline offsets, $(90, 623)$ and $(82, 613)$ (the $X$ of about $+100$ is right: semibold draws these marks right of the origin). Koronis U+0343 $(-92, 539)$ (2026-11-11): the designer's is a reference to the horn and had no mark anchor; placed as regular's U+0313, as in semibold italic (below). Inverted breve U+0311 $-217$ and tilde below U+0330 $-200$, on the model ($X = C + Y \tan\theta + d - K$; median distance from the precomposed glyphs 11 to 4 and 13 to 4; 2026-11-06).
- Italic: tilde below centred on its outline (8 units); double acute $(-55, 725)$, with $X$ moved 17 units from the designer's regular offset and $Y$ by the vertical move.
- Semibold italic: hook above U+0309 $-142$, inverted breve U+0311 $-223$ and U+0314 $-91$, on the model (median distance from the precomposed glyphs 26 to 8, 19 to 3, 19 to 4; 2026-11-06); tilde below centred on its outline (22 units off regular's offset); koronis U+0343 $(128, 532)$, from $(-96, 605)$ (2026-11-11): the designer drew it as a horn-sized outline right of the origin with the comma above's anchor, so $k$ was 371; it is now placed as regular's and italic's U+0313 (their U+0343 is the same glyph): the anchor 135 above the outline's bottom and 21 left of its ink centroid along the slant, with $K = 150$. The outline is left as it is (to-do list). After a base letter HarfBuzz replaces U+0343 by its canonical equivalent U+0313, so `uni0343` is shaped only on its own; macron below raised 52 units to $(-233, -58)$, 23 units below its outline as in the other styles (it sat 25 units under n, against 86 to 96 in the other styles; n̲ confirms; 2026-11-06). Curated grave and double acute that lacked $K$ (landing 142 and 190 units left on every conforming base) were removed; the designer's anchors conform. They had been fitted on a, whose above anchor carried the slant; stage 1 now deletes the above anchors of a, s, h and J and stage 6 gives them candidates (see the list above Figure 4).

**Vertical moves from the precomposed glyphs (2026-11-09).** For each above mark, $\Delta y$ is the median vertical offset of the composed mark (base + CGJ + mark) from the designer's precomposed glyphs, over the lowercase letters with an x-height base. Every such base has its above anchor at 645, and $\Delta y$ is the same on every base to within a few units, so it belongs to the mark's anchor. Where $|\Delta y| \geq 10$ the mark anchor is curated $\Delta y$ higher, so that the mark sits $\Delta y$ lower; in the italic styles its $X$ moves with the slant, by $\Delta y \tan\theta$, so that the mark keeps its $k$.

- Regular: macron $+18$, dot above $-45$, double acute $+11$.
- Italic: dot above $-11$, double acute $+11$.
- Semibold: tilde $+14$, macron $-12$, dot above $-21$.
- Semibold italic: macron $-32$, dot above $+11$, ring $+23$, double acute $+11$.

**Mark-zone consistency across styles (2026-11-12).** The base anchors have the same $Y$ in every style, so a mark sits at the same place among its style's marks (its mark zone) in every style only if its gap from its anchor, relative to the median gap of the style's marks, is the same. For most marks it is, within 8 units; test sheet `tex/mark_zone_consistency_sheet.pdf` shows the rest, each mark in its zone in the four styles, with the precomposed letters on every Latin and IPA base. Where the designer's precomposed letters are themselves inconsistent between styles, the composed marks were made consistent instead, with more clearance, and no longer match the tighter precomposed letters:

- Dot below U+0323 and diaeresis below U+0324: both at $+28$ in every style, the depth of the macron below and the minus sign below. The precomposed dots below are loose in regular (all 55 letters, 103 to 109 below the baseline on o) and tight in the other styles; ụ, the only precomposed diaeresis below, is unusually tight in every style. Regular's dots go up 12 and 10; italic's, semibold's and semibold italic's down 22 to 34.
- Dot above U+0307: $+30$ in every style (it was $+44$ in regular, $+13$ to $+20$ in the others, following each style's precomposed letters): regular down 14, the others up 10 to 17. This replaces the vertical moves of 2026-11-09 for the dot above.
- Semibold diaeresis U+0308 up 9 (to $+24$, as in the other styles), and `dieresis.cap` with it.
- Semibold italic ring U+030A up 16 (to $+3$, as in the other styles); å is not evidence (a and A are discarded). This replaces its vertical move of 2026-11-09.
- Semibold breve below U+032E down 6 (to $+1$, as in the other styles).

The capital dot `dotaccent.cap` follows the lowercase dot's old pattern (regular $+10$, semibold $-2$, relative to the capital forms) and is not changed.

**Mark-axis rule for symmetric marks (2026-11-09; below marks 2026-11-12).** Stage 4 (`font_patching_markaxis.py`) puts the anchor $X$ of a symmetric above mark on its own axis, $k = K$ (0; 150 in semibold italic), with $k$ measured on the mark's ink. Until 2026-11-12 it moved only anchors with $|k - K| \geq 10$; it now has no threshold, and moves every anchor whose $X$ rounds to a different value. The symmetric above marks are the mirror-symmetric ones (circumflex, tilde, macron, breve, dot above, dieresis, ring, caron, vertical line and double vertical line above, candrabindu, inverted breve, x above, vertical tilde, double overline, perispomeni, bridge above, not tilde, homothetic, almost equal to, fermata, zigzag) and the combining Latin letters U+0363–U+036F. Asymmetric marks keep their designed $k$: grave, acute and their tone-mark forms, double acute and double grave (a symmetric pair placed off centre), hook above, the comma-shaped marks, the half rings and the right arrowhead. The rule does not touch $Y$.

The axis is the centre of the ink bounds, with the slant removed, except for the combining letters that are not boxy (2026-11-12). For those, bounds and area centroid are more than about 4 units apart in regular (r 11, c 9, h 8, e 6, t 4), and the axis is the area centroid, which weights the stems over a thin arm, terminal or open side; it lies between the bounds and the stem for r ($-7$, $-18$, $-28$ relative to the old anchor in regular), as Gentium Plus, Noto Serif and Junicode place their combining r (75%, 40% and 30% of the way to the stem). Combining i U+0365 is measured on its stem (midline at the stem's top, carried at the slant), which its left top serif pulls the bounds 2.6 units off. Combining a, d, m, o, u, v, x are boxy enough for the bounds (bounds and centroid within 3). Since 2026-11-12 the rule also covers the mirror-symmetric below marks (`SYMMETRIC_BELOW`: plus and minus sign, dot, diaeresis and ring below, vertical line, bridge, inverted double arch, caron, circumflex, breve and inverted breve, tilde and macron below, double low line, inverted bridge, square, seagull, equals sign, double vertical line, left right arrow and upwards arrow below), with $K = -12$ in semibold italic, whose below marks are drawn with it as its base anchors are. The tacks keep their decided placement, and grave and acute below, the half rings, comma below and left angle below are asymmetric. It moves 19 or 20 anchors per style, most by 1 to 4 units; the largest are the inverted breve below in semibold and semibold italic (13, 12), the tilde below (8 in regular and semibold, 13 in semibold italic, whose curated anchor had left out $K$), the breve below in the italic styles (7), and the square and left right arrow below (5 to 7). Test sheet `tex/mark_below_axis_test_sheet.pdf`; the curated below anchors were moved with it (plus sign, ring, tilde, macron below, upwards arrow; 1 to 13 units). Curated anchors of symmetric marks were moved onto the axis with the change (X only; regular macron, italic dot above, semibold tilde, macron, dot above and inverted breve, semibold italic macron, dot above, ring and inverted breve; 1 to 9 units).

Matching each mark to the precomposed glyphs ($\Delta x = 0$ per mark) would be the wrong target: part of what a composed mark differs from the precomposed glyphs by is the base anchor's offset, common to every mark on that base, and fitting it into each mark would carry it to every base without a precomposed form (most IPA letters).

The rule moves 37 mark anchors. Among the common marks: italic tilde ($k$ 11, moved 11 units), semibold dieresis (12), semibold italic tilde (139, 11) and dieresis (112, 38); among the others, the vertical tilde (about 18, in regular, italic and semibold), the perispomeni (18, in regular and italic) and most of semibold italic's combining letters ($k$ 130 to 140). The semibold italic dieresis, which landed 26 units left of its precomposed glyphs where every other mark landed 5 to 16 right, now lands 12 right, like the others.

**Rejected: moving every mark anchor to the model (2026-11-08).** A stage that moved every mark anchor onto the model, $X = C + Y \tan\theta + d - K$, was tested and rejected: most moves were under 20 units, and as many marks got further from the precomposed glyphs as got closer. The mark-axis rule does not repeat it: it moves only symmetric marks, and only their $X$.

## Mark outline zones

Figures 5, 6, 7 and 8 overlay the outlines of all the above and below marks of the patched fonts on a solid o, at three times the scale of the anchor model figures. Each mark is placed so that its anchor is on the anchor of the o for its class (the blue dots), which is where the shaper puts it. The common marks are black outlines. The other marks are gray with a light fill, so the shading builds up where many marks overlap. The grid is in font units of the o, with lines every 100 units. In semibold italic the above marks lie well to the right of the blue dot, because both the base and the mark anchor carry the shift $K$.

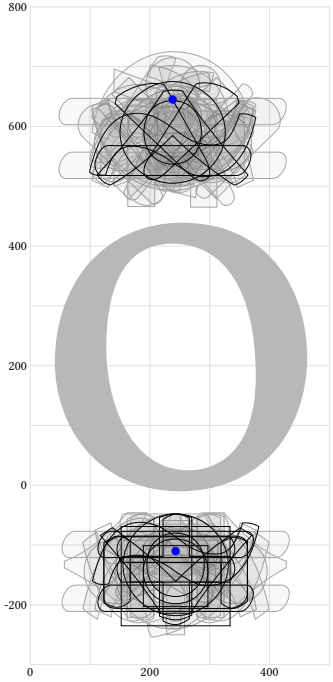



Figure 5: Mark zones, regular.

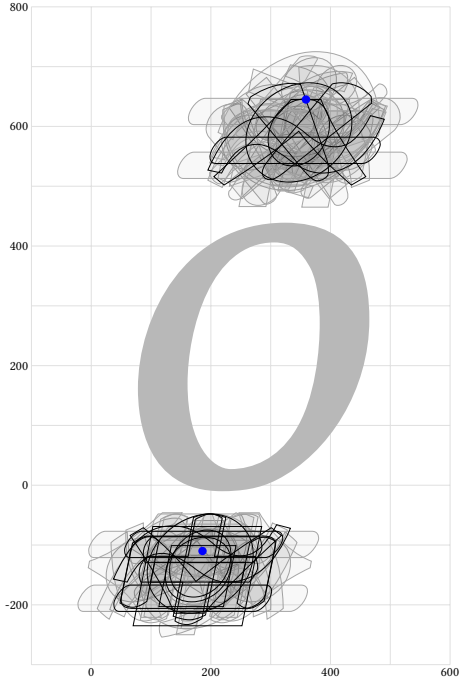



Figure 6: Mark zones, italic.

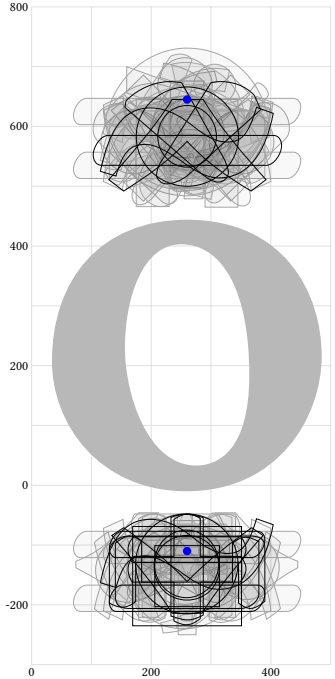



Figure 7: Mark zones, semibold.

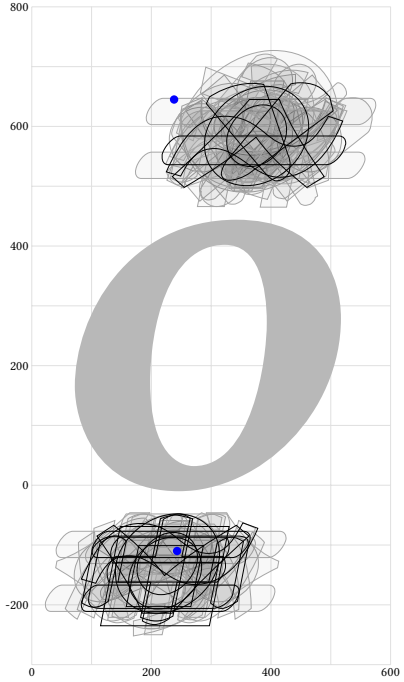



Figure 8: Mark zones, semibold italic.

## Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in the following tables (Tables 1–11), and the mark anchors of above- and below-marks (Tables 16–19). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for boldface styles.

### Table row headings

hex: Unicode character codepoint in 4-hex.

cat: Unicode character category. L = letter. Lu = uppercase (capital) letter, Ll = lowercase (small) letter, Lo = other letter. Used to determine expected ideal bounding box. M = mark. Mn = nonspacing mark.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show anchor positioning (via combining grapheme joiner) when the base has an anchor; otherwise HarfBuzz default positioning applies.

patch: Use the patched Libertinus font to print the character alone, with combining dot above, and with combining macron below to show anchor positioning of the patched font (via combining grapheme joiner). Where the original font has no above- or below-anchor, the patched font has the candidate (a?, b?), unless a human-curated anchor overrules it; existing anchors are not moved.

ideal: top and bottom ideal bounding box reference lines, with offset from $Y_{\text{max}}$ or $Y_{\text{min}}$ for non-full descender or non-full ascender. c = capline, x = x-height, a = ascender, b = baseline, d = descender.

a: $X, Y$ coordinates of the above-anchor.

a?: $X, Y$ coordinates of the above-anchor candidate.

a$\delta$: $X, Y$ coordinate differences between above-anchor and above-anchor candidate

b: $X, Y$ coordinates of the below-anchor.

b?: $X, Y$ coordinates of the below-anchor candidate.

b$\delta$: $X, Y$ coordinate differences between below-anchor and below-anchor candidate

In the tables of marks, each column is a mark and the rows are as follows.

reg, it, sb, si, patch: the mark on n and on H (via combining grapheme joiner), in the original and patched fonts.

m: $X, Y$ coordinates of the mark anchor. A class in parentheses is an anchor class other than above (0) or below (2): semibold puts the dot above, diaeresis, ring, hook above and comma above in class 1, whose base anchors are lower (n has 550 in class 1 and 646 in class 0). This is the designer's anchor class `top_punkt`; it places the marks at the same height as class 0, but only 24 bases have it. The patched semibold merges it into class 0.

m?: $X, Y$ coordinates of the mark-anchor candidate. $Y$ is the bottom of the mark outline plus the style's median distance of an above-mark anchor above it (regular 143, italic 142, semibold 157 in class 0 and 60 in class 1, semibold italic 147), or the top of the outline minus the median distance of a below-mark anchor below it (regular 46, italic 48, semibold 54, semibold italic 54). $X$ is on the mark's own axis at that height, $C + Y \tan\theta - K$.

m$\delta$: $X, Y$ coordinate differences between mark anchor and mark-anchor candidate. An asymmetric mark, such as the acute or grave, has an $X$ difference by design.

Table 1: Font metrics for A–Z 1/2

| hex | 0041 | 0042 | 0043 | 0044 | 0045 | 0046 | 0047 | 0048 | 0049 | 004A | 004B | 004C | 004D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu |
| reg | AȦẠ | BḂḄ | CĊÇ | DḊḌ | EĖẸ | FḞF | GĠG | HḢḤ | IİỊ | JJ̇J | KḴḲ | LĿḶ | MṀṂ |
| patch | AȦẠ | BḂḄ | CĊÇ | DḊḌ | EĖẸ | FḞF | GĠG | HḢḤ | IİỊ | JJ̇J | KḴḲ | LĿḶ | MṀṂ |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+60 | c, b | c, b | c, b |
| a | 354, 850 | 267, 850 | 340, 850 | 306, 850 | 281, 850 | 284, 850 | 365, 850 | 361, 850 | 150, 850 | 182, 850 | 336, 850 | 151, 850 | 431, 850 |
| a? | 346, 850 | 280, 850 | 324, 850 | 336, 850 | 271, 850 | 244, 850 | 352, 850 | 362, 850 | 148, 850 | 120, 850 | 323, 850 | 260, 850 | 418, 850 |
| a$\delta$ | 8, 0 | -14, 0 | 16, 0 | -30, 0 | 10, 0 | 40, 0 | 14, 0 | -1, 0 | 2, 0 | 62, 0 | 13, 0 | -110, 0 | 13, 0 |
| b | 308, -110 | 268, -110 | 363, -110 | 309, -110 | 287, -110 | 209, -110 | 367, -110 | 362, -110 | 147, -110 | 109, -319 | 342, -110 | 262, -110 | 393, -110 |
| b? | 346, -110 | 280, -110 | 324, -110 | 336, -110 | 271, -110 | 244, -110 | 352, -110 | 362, -110 | 148, -110 | 120, -257 | 323, -110 | 260, -110 | 418, -110 |
| b$\delta$ | -38, 0 | -12, 0 | 39, 0 | -27, 0 | 16, 0 | -36, 0 | 16, 0 | 0, 0 | -1, 0 | -11, -62 | 19, 0 | 2, 0 | -25, 0 |
| it | *AȦẠ* | *BḂḄ* | *CĊÇ* | *DḊḌ* | *EĖẸ* | *FḞF* | *GĠG* | *HḢḤ* | *IİỊ* | *JJ̇J* | *KḴḲ* | *LĿḶ* | *MṀṂ* |
| patch | *AȦẠ* | *BḂḄ* | *CĊÇ* | *DḊḌ* | *EĖẸ* | *FḞF* | *GĠG* | *HḢḤ* | *IİỊ* | *JJ̇J* | *KḴḲ* | *LĿḶ* | *MṀṂ* |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c+32, d+32 | c, b | c, b | c, b |
| a | 505, 850 | 447, 850 | 500, 850 | 396, 850 | 413, 850 | 406, 850 | 513, 850 | 481, 850 | 316, 850 | 392, 850 | 466, 850 | 389, 850 | 557, 850 |
| a? | 514, 850 | 459, 850 | 489, 850 | 514, 850 | 444, 850 | 409, 850 | 513, 850 | 517, 850 | 321, 850 | 388, 850 | 499, 850 | 440, 850 | 583, 850 |
| a$\delta$ | -9, 0 | -12, 0 | 11, 0 | -118, 0 | -31, 0 | -3, 0 | 0, 0 | -36, 0 | -5, 0 | 4, 0 | -33, 0 | -51, 0 | -26, 0 |
| b | 308, -108 | 238, -108 | 333, -110 | 279, -107 | 269, -107 | 131, -111 | 345, -110 | 320, -107 | 127, -109 | 109, -319 | 292, -110 | 252, -110 | 363, -110 |
| b? | 310, -110 | 255, -110 | 285, -110 | 310, -110 | 240, -110 | 205, -110 | 309, -110 | 313, -110 | 117, -110 | 146, -285 | 295, -110 | 236, -110 | 379, -110 |
| b$\delta$ | -2, 2 | -17, 2 | 48, 0 | -31, 3 | 29, 3 | -74, -1 | 36, 0 | 7, 3 | 10, 1 | -37, -34 | -3, 0 | 16, 0 | -16, 0 |
| sb | **AȦẠ** | **BḂḄ** | **CĊÇ** | **DḊḌ** | **EĖẸ** | **FḞF** | **GĠG** | **HḢḤ** | **IİỊ** | **JJ̇J** | **KḴḲ** | **LĿḶ** | **MṀṂ** |
| patch | **AȦẠ** | **BḂḄ** | **CĊÇ** | **DḊḌ** | **EĖẸ** | **FḞF** | **GĠG** | **HḢḤ** | **IİỊ** | **JJ̇J** | **KḴḲ** | **LĿḶ** | **MṀṂ** |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, d+50 | c, b | c, b | c, b |
| a | | | | | | | | | | 203, 806 | | | |
| a? | 350, 850 | 300, 850 | 332, 850 | 352, 850 | 283, 850 | 268, 850 | 368, 850 | 362, 850 | 160, 850 | 131, 850 | 340, 850 | 267, 850 | 428, 850 |
| a$\delta$ | | | | | | | | | | 72, -44 | | | |
| b | | | | | | | | 362, -110 | | 109, -319 | | | |
| b? | 350, -110 | 300, -110 | 332, -110 | 352, -110 | 283, -110 | 268, -110 | 368, -110 | 362, -110 | 160, -110 | 131, -267 | 340, -110 | 267, -110 | 428, -110 |
| b$\delta$ | | | | | | | | 0, 0 | | -22, -52 | | | |
| si | ***AȦẠ*** | ***BḂḄ*** | ***CĊÇ*** | ***DḊḌ*** | ***EĖẸ*** | ***FḞF*** | ***GĠG*** | ***HḢḤ*** | ***IİỊ*** | ***JJ̇J*** | ***KḴḲ*** | ***LĿḶ*** | ***MṀṂ*** |
| patch | ***AȦẠ*** | ***BḂḄ*** | ***CĊÇ*** | ***DḊḌ*** | ***EĖẸ*** | ***FḞF*** | ***GĠG*** | ***HḢḤ*** | ***IİỊ*** | ***JJ̇J*** | ***KḴḲ*** | ***LĿḶ*** | ***MṀṂ*** |
| ideal | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c+34, d+68 | c, b | c, b | c, b |
| a | 354, 850 | 267, 850 | 340, 850 | 306, 850 | 273, 850 | 276, 850 | 365, 850 | 361, 850 | 150, 850 | 432, 850 | 336, 850 | 149, 850 | 431, 850 |
| a? | 382, 850 | 319, 850 | 341, 850 | 367, 850 | 292, 850 | 269, 850 | 371, 850 | 396, 850 | 194, 850 | 249, 850 | 369, 850 | 310, 850 | 441, 850 |
| a$\delta$ | -28, 0 | -52, 0 | -1, 0 | -61, 0 | -19, 0 | 7, 0 | -6, 0 | -35, 0 | -44, 0 | 183, 0 | -33, 0 | -161, 0 | -10, 0 |
| b | 308, -110 | 268, -110 | 363, -110 | 279, -110 | 279, -110 | 201, -110 | 367, -110 | 365, -110 | 147, -110 | 109, -319 | 342, -110 | 260, -110 | 393, -110 |
| b? | 349, -110 | 286, -110 | 308, -110 | 334, -110 | 259, -110 | 236, -110 | 338, -110 | 363, -110 | 161, -110 | 187, -249 | 336, -110 | 277, -110 | 408, -110 |
| b$\delta$ | -41, 0 | -18, 0 | 55, 0 | -55, 0 | 20, 0 | -35, 0 | 29, 0 | 2, 0 | -14, 0 | -78, -70 | 6, 0 | -17, 0 | -15, 0 |

Table 2: Font metrics for A–Z 2/2

| hex | 004E | 004F | 0050 | 0051 | 0052 | 0053 | 0054 | 0055 | 0056 | 0057 | 0058 | 0059 | 005A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu | Lu |
| reg | NN̈N̲ | OO̤O̲ | PP̈P̲ | QQ̤Q̲ | RR̈R̲ | SS̈S̲ | TT̈T̲ | UṲU̲ | VV̈V̲ | WẄW̲ | XẌX̲ | YŸY̲ | ZŻZ̲ |
| patch | NN̈N̲ | OO̤O̲ | PP̈P̲ | QQ̤Q̲ | RR̈R̲ | SS̈S̲ | TT̈T̲ | UṲU̲ | VV̈V̲ | WẄW̲ | XẌX̲ | YŸY̲ | ZŻZ̲ |
| ideal | c, b | c, b | c, b | c, d+23 | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 346, 850 | 335, 850 | 277, 850 | 348, 850 | 269, 850 | 232, 850 | 307, 850 | 346, 850 | 336, 850 | 450, 850 | 332, 850 | 295, 850 | 320, 850 |
| a? | 353, 850 | 352, 850 | 260, 850 | 359, 850 | 301, 850 | 236, 850 | 297, 850 | 331, 850 | 326, 850 | 474, 850 | 323, 850 | 288, 850 | 310, 850 |
| a$\delta$ | -7, 0 | -16, 0 | 18, 0 | -11, 0 | -32, 0 | -4, 0 | 10, 0 | 15, 0 | 10, 0 | -24, 0 | 9, 0 | 8, 0 | 10, 0 |
| b | 348, -110 | 350, -110 | 240, -110 |  | 258, -110 | 234, -110 | 300, -110 | 328, -110 | 318, -110 | 479, -110 | 307, -110 | 295, -110 | 287, -110 |
| b? | 353, -110 | 352, -110 | 260, -110 | 359, -294 | 301, -110 | 236, -110 | 297, -110 | 331, -110 | 326, -110 | 474, -110 | 323, -110 | 288, -110 | 310, -110 |
| b$\delta$ | -5, 0 | -2, 0 | -20, 0 |  | -43, 0 | -2, 0 | 3, 0 | -3, 0 | -8, 0 | 5, 0 | -16, 0 | 8, 0 | -22, 0 |
| it | *NN̈N̲* | *OO̤O̲* | *PP̈P̲* | *QQ̤Q̲* | *RR̈R̲* | *SS̈S̲* | *TT̈T̲* | *UṲU̲* | *VV̈V̲* | *WẄW̲* | *XẌX̲* | *YŸY̲* | *ZŻZ̲* |
| patch | *NN̈N̲* | *OO̤O̲* | *PP̈P̲* | *QQ̤Q̲* | *RR̈R̲* | *SS̈S̲* | *TT̈T̲* | *UṲU̲* | *VV̈V̲* | *WẄW̲* | *XẌX̲* | *YŸY̲* | *ZŻZ̲* |
| ideal | c, b | c, b | c, b | c, d+23 | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 476, 850 | 515, 850 | 426, 850 | 498, 850 | 419, 850 | 402, 850 | 457, 850 | 486, 850 | 481, 850 | 610, 850 | 482, 850 | 435, 850 | 450, 850 |
| a? | 514, 850 | 515, 850 | 430, 850 | 515, 850 | 458, 850 | 408, 850 | 453, 850 | 498, 850 | 479, 850 | 635, 850 | 495, 850 | 457, 850 | 470, 850 |
| a$\delta$ | -38, 0 | 0, 0 | -4, 0 | -17, 0 | -39, 0 | -6, 0 | 4, 0 | -12, 0 | 2, 0 | -25, 0 | -13, 0 | -22, 0 | -20, 0 |
| b | 328, -109 | 310, -113 | 140, -110 |  | 248, -111 | 204, -111 | 277, -109 | 308, -109 | 263, -110 | 439, -107 | 287, -107 | 270, -109 | 267, -107 |
| b? | 310, -110 | 311, -110 | 226, -110 | 272, -294 | 254, -110 | 204, -110 | 249, -110 | 294, -110 | 275, -110 | 431, -110 | 291, -110 | 253, -110 | 266, -110 |
| b$\delta$ | 18, 1 | -1, -3 | -86, 0 |  | -6, -1 | 0, -1 | 28, 1 | 14, 1 | -12, 0 | 8, 3 | -4, 3 | 17, 1 | 1, 3 |
| sb | **NNN̲** | **OOO̲** | **PPP̲** | **QQ̤Q̲** | **RR̈R̲** | **SS̈S̲** | **TT̈T̲** | **UUU̲** | **VV̈V̲** | **WẄW̲** | **XẌX̲** | **YŸY̲** | **ZŻZ̲** |
| patch | **NN̈N̲** | **OO̤O̲** | **PP̈P̲** | **QQ̤Q̲** | **RR̈R̲** | **SS̈S̲** | **TT̈T̲** | **UṲU̲** | **VV̈V̲** | **WẄW̲** | **XẌX̲** | **YŸY̲** | **ZŻZ̲** |
| ideal | c, b | c, b | c, b | c, d | c, b | c, b | c+28, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a |  | 363, 802 |  | 363, 802 | 304, 808 | 252, 802 | 298, 802 |  | 331, 805 | 493, 805 | 364, 804 | 316, 805 | 321, 803 |
| a? | 354, 850 | 366, 850 | 278, 850 | 366, 850 | 320, 850 | 238, 850 | 299, 850 | 339, 850 | 318, 850 | 489, 850 | 354, 850 | 308, 850 | 309, 850 |
| a$\delta$ |  | -2, -48 |  | -2, -48 | -16, -42 | 14, -48 | -1, -48 |  | 12, -45 | 4, -45 | 10, -46 | 8, -45 | 12, -47 |
| b |  |  |  |  | 292, -111 | 234, -111 | 301, -107 |  | 303, -108 | 470, -107 | 318, -107 | 316, -109 | 308, -107 |
| b? | 354, -110 | 366, -110 | 278, -110 | 366, -319 | 320, -110 | 238, -110 | 299, -110 | 339, -110 | 318, -110 | 489, -110 | 354, -110 | 308, -110 | 309, -110 |
| b$\delta$ |  |  |  |  | -28, -1 | -4, -1 | 2, 3 |  | -16, 2 | -19, 3 | -36, 3 | 8, 1 | -1, 3 |
| si | ***NN̈N̲*** | ***OO̤O̲*** | ***PP̈P̲*** | ***QQ̤Q̲*** | ***RR̈R̲*** | ***SS̈S̲*** | ***TT̈T̲*** | ***UṲU̲*** | ***VV̈V̲*** | ***WẄW̲*** | ***XẌX̲*** | ***YŸY̲*** | ***ZŻZ̲*** |
| patch | ***NN̈N̲*** | ***OO̤O̲*** | ***PP̈P̲*** | ***QQ̤Q̲*** | ***RR̈R̲*** | ***SS̈S̲*** | ***TT̈T̲*** | ***UṲU̲*** | ***VV̈V̲*** | ***WẄW̲*** | ***XẌX̲*** | ***YŸY̲*** | ***ZŻZ̲*** |
| ideal | c, b | c, b | c, b | c, d | c, b | c, b | c+28, b | c, b | c, b | c, b | c, b | c, b | c, b |
| a | 346, 850 | 335, 850 | 316, 850 | 348, 850 | 269, 850 | 232, 850 | 307, 850 | 346, 850 | 336, 850 | 450, 850 | 332, 850 | 295, 850 | 320, 850 |
| a? | 394, 850 | 390, 850 | 284, 850 | 390, 850 | 315, 850 | 250, 850 | 309, 850 | 364, 850 | 328, 850 | 501, 850 | 362, 850 | 319, 850 | 322, 850 |
| a$\delta$ | -48, 0 | -55, 0 | 32, 0 | -42, 0 | -46, 0 | -18, 0 | -2, 0 | -18, 0 | 8, 0 | -51, 0 | -30, 0 | -24, 0 | -2, 0 |
| b | 348, -110 | 350, -110 | 240, -110 |  | 258, -110 | 234, -110 | 300, -110 | 328, -110 | 318, -110 | 479, -110 | 307, -110 | 295, -110 | 287, -110 |
| b? | 361, -110 | 357, -110 | 251, -110 | 314, -319 | 282, -110 | 217, -110 | 276, -110 | 331, -110 | 295, -110 | 468, -110 | 329, -110 | 286, -110 | 289, -110 |
| b$\delta$ | -13, 0 | -7, 0 | -11, 0 |  | -24, 0 | 17, 0 | 24, 0 | -3, 0 | 23, 0 | 11, 0 | -22, 0 | 9, 0 | -2, 0 |

Table 3: Font metrics for a–z 1/2

| hex | 0061 | 0062 | 0063 | 0064 | 0065 | 0066 | 0067 | 0068 | 0069 | 006A | 006B | 006C | 006D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | aȧạ | bḃḅ | cċc̣ | dḋḍ | eėẹ | fḟf̣ | gġg̣ | hḣḥ | iiį | jjj̣ | kk̇ḳ | l̇l̇ḷ | ṁṁṃ |
| patch | aȧạ | bḃḅ | cċc̣ | dḋḍ | eėẹ | fḟf̣ | gġg̣ | hḣḥ | iiį | jjj̣ | kk̇ḳ | l̇l̇ḷ | ṁṁṃ |
| ideal | x, b | a, b | x, b | a, b | x, b | a, b | x+33, d | a, b | a-51, b | a-51, d+9 | a, b | a, b | x, b |
| a | 209, 645 | | 242, 645 | 218, 645 | 243, 645 | | 231, 645 | 299, 645 | 137, 795 | | 306, 784 | 135, 885 | 392, 645 |
| a? | 246, 645 | 232, 885 | 218, 645 | 270, 885 | 222, 645 | 208, 885 | 256, 676 | 270, 885 | 142, 834 | 76, 834 | 261, 885 | 134, 885 | 402, 645 |
| a$\delta$ | -36, 0 | | 24, 0 | -52, -240 | 21, 0 | | -26, -31 | 28, -240 | -5, -39 | | 45, -101 | 1, 0 | -10, 0 |
| b | 196, -110 | 234, -110 | 236, -110 | 221, -110 | 234, -110 | 132, -110 | 229, -319 | 272, -110 | 136, -110 | 109, -319 | 290, -214 | 135, -110 | 408, -110 |
| b? | 246, -110 | 232, -110 | 218, -110 | 270, -110 | 222, -110 | 208, -110 | 256, -319 | 270, -110 | 142, -110 | 76, -308 | 261, -110 | 134, -110 | 402, -110 |
| b$\delta$ | -50, 0 | 2, 0 | 18, 0 | -50, 0 | 12, 0 | -76, 0 | -28, 0 | 2, 0 | -6, 0 | 34, -11 | 29, -104 | 1, 0 | 6, 0 |
| it | *aȧạ* | *bḃḅ* | *cċc̣* | *dḋḍ* | *eėẹ* | *fḟf̣* | *gġg̣* | *hḣḥ* | *iiį* | *jjj̣* | *kk̇ḳ* | *l̇l̇ḷ* | *ṁṁṃ* |
| patch | *aȧạ* | *bḃḅ* | *cċc̣* | *dḋḍ* | *eėẹ* | *fḟf̣* | *gġg̣* | *hḣḥ* | *iiį* | *jjj̣* | *kk̇ḳ* | *l̇l̇ḷ* | *ṁṁṃ* |
| ideal | x, b | a, b | x, b | a, b | x, b | a, d | x, d | a, b | a-80, b | a-80, d+36 | a, b | a, b | x, b |
| a | 341, 647 | | 343, 645 | 218, 645 | 332, 645 | | 360, 645 | 409, 645 | | | 405, 646 | 292, 890 | 522, 645 |
| a? | 380, 645 | 428, 890 | 332, 645 | 434, 890 | 338, 645 | 346, 890 | 376, 645 | 449, 890 | 310, 810 | 302, 810 | 432, 890 | 322, 890 | 529, 645 |
| a$\delta$ | -39, 2 | | 11, 0 | -216, -245 | -6, 0 | | -16, 0 | -40, -245 | | | -27, -244 | -30, 0 | -7, 0 |
| b | 174, -108 | 214, -108 | 163, -113 | 221, -107 | 158, -108 | 222, -111 | 197, -319 | 229, -111 | 136, -107 | 0, -319 | 223, -107 | 115, -107 | 382, -105 |
| b? | 220, -110 | 216, -110 | 171, -110 | 221, -110 | 177, -110 | 89, -319 | 171, -319 | 236, -110 | 115, -110 | 70, -281 | 220, -110 | 110, -110 | 368, -110 |
| b$\delta$ | -46, 2 | -2, 2 | -8, -3 | 0, 3 | -19, 2 | 133, 208 | 26, 0 | -7, -1 | 21, 3 | -70, -38 | 3, 3 | 5, 3 | 14, 5 |
| sb | **aȧạ** | **bḃḅ** | **cċc̣** | **dḋḍ** | **eėẹ** | **fḟf̣** | **gġg̣** | **hḣḥ** | **iiį** | **jjj̣** | **kk̇ḳ** | **l̇l̇ḷ** | **ṁṁṃ** |
| patch | **aȧạ** | **bḃḅ** | **cċc̣** | **dḋḍ** | **eėẹ** | **fḟf̣** | **gġg̣** | **hḣḥ** | **iiį** | **jjj̣** | **kk̇ḳ** | **l̇l̇ḷ** | **ṁṁṃ** |
| ideal | x, b | a, b-13 | x, b | a, b | x, b | a, b | x+33, d | a, b | a-48, b | a-48, d | a, b | a, b | x, b |
| a | | | | | | | 231, 645 | | 137, 795 | | | 142, 885 | 396, 645 |
| a? | 255, 645 | 236, 885 | 224, 645 | 274, 885 | 232, 645 | 226, 885 | 264, 676 | 275, 885 | 156, 837 | 78, 837 | 262, 885 | 148, 885 | 408, 645 |
| a$\delta$ | | | | | | | -32, -31 | | -18, -42 | | | -6, 0 | -12, 0 |
| b | | | | | | | 229, -319 | 277, -110 | 136, -107 | 109, -319 | | 135, -110 | 396, -105 |
| b? | 255, -110 | 236, -121 | 224, -110 | 274, -110 | 232, -110 | 226, -110 | 264, -319 | 275, -110 | 156, -110 | 78, -319 | 262, -110 | 148, -110 | 408, -110 |
| b$\delta$ | | | | | | | -34, 0 | 2, 0 | -20, 3 | 31, 0 | | -14, 0 | -12, 5 |
| si | ***aȧạ*** | ***bḃḅ*** | ***cċc̣*** | ***dḋḍ*** | ***eėẹ*** | ***fḟf̣*** | ***gġg̣*** | ***hḣḥ*** | ***iiį*** | ***jjj̣*** | ***kk̇ḳ*** | ***l̇l̇ḷ*** | ***ṁṁṃ*** |
| patch | ***aȧạ*** | ***bḃḅ*** | ***cċc̣*** | ***dḋḍ*** | ***eėẹ*** | ***fḟf̣*** | ***gġg̣*** | ***hḣḥ*** | ***iiį*** | ***jjj̣*** | ***kk̇ḳ*** | ***l̇l̇ḷ*** | ***ṁṁṃ*** |
| ideal | x, b | a, b | x, b | a, b | x, b | a, d | x, d | a, b | a-67, b | a-67, d+25 | a, b | a, b | x, b |
| a | 379, 645 | | 242, 645 | 218, 645 | 243, 645 | | 291, 645 | 469, 645 | 137, 795 | | 306, 784 | 174, 890 | 392, 645 |
| a? | 251, 645 | 290, 890 | 203, 645 | 309, 890 | 218, 645 | 216, 890 | 220, 645 | 331, 890 | 180, 823 | 192, 823 | 316, 890 | 194, 890 | 407, 645 |
| a$\delta$ | 128, 0 | | 39, 0 | -91, -245 | 25, 0 | | 71, 0 | 138, -245 | -43, -28 | | -10, -106 | -20, 0 | -15, 0 |
| b | 196, -110 | 234, -110 | 236, -110 | 221, -110 | 234, -110 | 132, -110 | 199, -319 | 272, -110 | 136, -110 | 67, -319 | 290, -214 | 135, -110 | 400, -110 |
| b? | 260, -110 | 248, -110 | 212, -110 | 268, -110 | 227, -110 | 132, -319 | 186, -319 | 290, -110 | 153, -110 | 127, -292 | 275, -110 | 153, -110 | 415, -110 |
| b$\delta$ | -64, 0 | -14, 0 | 24, 0 | -47, 0 | 7, 0 | 0, 209 | 13, 0 | -18, 0 | -17, 0 | -60, -27 | 15, -104 | -18, 0 | -15, 0 |

Table 4: Font metrics for a–z 2/2

| hex | 006E | 006F | 0070 | 0071 | 0072 | 0073 | 0074 | 0075 | 0076 | 0077 | 0078 | 0079 | 007A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | nn̰ | oo̰ | pp̰ | qq̰ | rr̰ | ss̰ | tt̰ | uṵ | vv̰ | ww̰ | xx̰ | yy̰ | zz̰ |
| patch | nn̰ | oo̰ | pp̰ | qq̰ | rr̰ | ss̰ | tt̰ | uṵ | vv̰ | ww̰ | xx̰ | yy̰ | zz̰ |
| ideal | x, b | x, b | x, d | x, d | x, b | x, b | a-116, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 264, 646 | 238, 645 | 271, 646 | 238, 645 | 191, 646 | 206, 645 |  | 255, 645 | 256, 646 | 345, 646 | 256, 646 | 262, 645 | 220, 646 |
| a? | 276, 645 | 252, 645 | 242, 645 | 268, 645 | 190, 645 | 199, 645 | 163, 769 | 271, 645 | 249, 645 | 373, 645 | 245, 645 | 260, 645 | 220, 645 |
| a$\delta$ | -12, 1 | -14, 0 | 28, 1 | -30, 0 | 1, 1 | 7, 0 |  | -16, 0 | 7, 1 | -28, 1 | 11, 1 | 2, 0 | 0, 1 |
| b | 265, -110 | 243, -110 | 179, -319 | 348, -319 | 135, -110 | 196, -110 | 163, -110 | 239, -110 | 241, -110 | 352, -110 | 224, -110 | 242, -319 | 207, -110 |
| b? | 276, -110 | 252, -110 | 242, -319 | 268, -319 | 190, -110 | 199, -110 | 163, -110 | 271, -110 | 249, -110 | 373, -110 | 245, -110 | 260, -319 | 220, -110 |
| b$\delta$ | -10, 0 | -9, 0 | -64, 0 | 80, 0 | -55, 0 | -3, 0 | 0, 0 | -32, 0 | -8, 0 | -21, 0 | -21, 0 | -18, 0 | -14, 0 |
| it | *nn̰* | *oo̰* | *pp̰* | *qq̰* | *rr̰* | *ss̰* | *tt̰* | *uṵ* | *vv̰* | *ww̰* | *xx̰* | *yy̰* | *zz̰* |
| patch | *nn̰* | *oo̰* | *pp̰* | *qq̰* | *rr̰* | *ss̰* | *tt̰* | *uṵ* | *vv̰* | *ww̰* | *xx̰* | *yy̰* | *zz̰* |
| ideal | x, b | x, b | x+62, d | x, d | x, b | x, b | x+132, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 394, 646 | 359, 645 | 411, 646 | 369, 645 | 312, 646 | 327, 645 |  | 386, 646 |  | 348, 646 | 366, 646 | 367, 644 | 360, 646 |
| a? | 396, 645 | 361, 645 | 394, 705 | 383, 645 | 316, 645 | 314, 645 | 318, 775 | 398, 645 | 373, 645 | 481, 645 | 375, 645 | 389, 645 | 355, 645 |
| a$\delta$ | -2, 1 | -2, 0 | 17, -59 | -14, 0 | -4, 1 | 13, 0 |  | -12, 1 |  | -133, 1 | -9, 1 | -22, -1 | 5, 1 |
| b | 235, -111 | 186, -107 | 248, -110 | 169, -110 | 113, -110 | 169, -111 | 133, -107 | 226, -111 |  | 355, -110 | 224, -110 | 368, -107 | 187, -111 |
| b? | 236, -110 | 200, -110 | 177, -319 | 178, -319 | 155, -110 | 153, -110 | 130, -110 | 237, -110 | 213, -110 | 321, -110 | 214, -110 | 184, -319 | 195, -110 |
| b$\delta$ | -1, -1 | -14, 3 | 71, 209 | -9, 209 | -42, 0 | 16, -1 | 3, 3 | -11, -1 |  | 34, 0 | 10, 0 | 184, 212 | -8, -1 |
| sb | **nn̰** | **oo̰** | **pp̰** | **qq̰** | **rr̰** | **ss̰** | **tt̰** | **uṵ** | **vv̰** | **ww̰** | **xx̰** | **yy̰** | **zz̰** |
| patch | **nn̰** | **oo̰** | **pp̰** | **qq̰** | **rr̰** | **ss̰** | **tt̰** | **uṵ** | **vv̰** | **ww̰** | **xx̰** | **yy̰** | **zz̰** |
| ideal | x, b | x, b | x, d | x, d | x, b | x, b | a-103, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 264, 646 |  |  |  |  |  |  | 264, 646 |  |  |  |  |  |
| a? | 290, 645 | 260, 645 | 254, 645 | 273, 645 | 212, 645 | 201, 645 | 168, 782 | 276, 645 | 262, 645 | 384, 645 | 260, 645 | 264, 645 | 222, 645 |
| a$\delta$ | -26, 1 |  |  |  |  |  |  | -12, 1 |  |  |  |  |  |
| b | 265, -111 |  |  |  |  |  | 163, -107 | 248, -111 |  |  |  |  |  |
| b? | 290, -110 | 260, -110 | 254, -319 | 273, -319 | 212, -110 | 201, -110 | 168, -110 | 276, -110 | 262, -110 | 384, -110 | 260, -110 | 264, -319 | 222, -110 |
| b$\delta$ | -24, -1 |  |  |  |  |  | -5, 3 | -28, -1 |  |  |  |  |  |
| si | ***nn̰*** | ***oo̰*** | ***pp̰*** | ***qq̰*** | ***rr̰*** | ***ss̰*** | ***tt̰*** | ***uṵ*** | ***vv̰*** | ***ww̰*** | ***xx̰*** | ***yy̰*** | ***zz̰*** |
| patch | ***nn̰*** | ***oo̰*** | ***pp̰*** | ***qq̰*** | ***rr̰*** | ***ss̰*** | ***tt̰*** | ***uṵ*** | ***vv̰*** | ***ww̰*** | ***xx̰*** | ***yy̰*** | ***zz̰*** |
| ideal | x, b | x, b | x+36, d | x, d | x, b | x, b | a-125, b | x, b | x, b | x, b | x, b | x, d | x, b |
| a | 264, 646 | 238, 645 | 271, 646 | 238, 645 | 191, 646 | 326, 645 |  | 255, 645 | 256, 646 | 345, 646 |  | 262, 645 | 220, 646 |
| a? | 284, 645 | 235, 645 | 257, 679 | 267, 645 | 203, 645 | 190, 645 | 189, 765 | 262, 645 | 220, 645 | 332, 645 | 240, 645 | 236, 645 | 210, 645 |
| a$\delta$ | -20, 1 | 3, 0 | 14, -33 | -29, 0 | -12, 1 | 136, 0 |  | -7, 0 | 36, 1 | 13, 1 |  | 26, 0 | 10, 1 |
| b | 265, -110 | 243, -110 | 179, -319 | 348, -319 | 135, -110 | 196, -110 | 133, -107 | 239, -110 | 241, -110 | 352, -110 |  | 242, -319 | 207, -110 |
| b? | 293, -110 | 243, -110 | 216, -319 | 233, -319 | 212, -110 | 198, -110 | 173, -110 | 270, -110 | 228, -110 | 341, -110 | 248, -110 | 202, -319 | 218, -110 |
| b$\delta$ | -28, 0 | 0, 0 | -37, 0 | 115, 0 | -77, 0 | -2, 0 | -40, 3 | -31, 0 | 13, 0 | 11, 0 |  | 40, 0 | -11, 0 |

Table 5: Font metrics for Latin and IPA 1/11

| hex | 00C6 | 00D0 | 00D8 | 00DE | 00DF | 00E6 | 00F0 | 00F8 | 00FE | 0110 | 0111 | 0126 | 0127 | 0131 | 013F | 0140 | 0141 | 0142 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Lu | Ll | Ll | Ll | Ll | Ll | Lu | Ll | Lu | Ll | Ll | Lu | Ll | Lu | Ll |
| reg | Æ Æ̇ Æ̱ | Ð Ð̇ Ð̱ | Ø Ø̇ Ø̱ | Þ Þ̇ Þ̱ | ß ß̇ ß̱ | æ æ̇ æ̱ | ð ð̇ ð̱ | ø ø̇ ø̱ | þ þ̇ þ̱ | Ð Ð̇ Ð̱ | đ đ̇ đ̱ | Ħ Ħ̇ Ħ̱ | ħ ḣ ħ̱ | ı i̇ ı̱ | Ŀ L̇ Ḻ | ŀ l̇ ḻ | Ł L̇ Ḻ | ł l̇ ḻ |
| patch | Æ Æ̇ Æ̱ | Ð Ð̇ Ð̱ | Ø Ø̇ Ø̱ | Þ Þ̇ Þ̱ | ß ß̇ ß̱ | æ æ̇ æ̱ | ð ð̇ ð̱ | ø ø̇ ø̱ | þ þ̇ þ̱ | Ð Ð̇ Ð̱ | đ đ̇ đ̱ | Ħ Ħ̇ Ħ̱ | ħ ḣ ħ̱ | ı i̇ ı̱ | Ŀ L̇ Ḻ | ŀ l̇ ḻ | Ł L̇ Ḻ | ł l̇ ḻ |
| ideal | c, b | c, b | c, b-13 | c, b | a, b | x, b | a, b | x+25, b-28 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | c, b | a, b |
| a | 480, 850 | 341, 850 | 351, 850 | 281, 850 | | 350, 645 | | | 278, 646 | | | | | 132, 645 | | | | |
| a? | 426, 850 | 333, 850 | 352, 850 | 250, 850 | 259, 885 | 344, 645 | 240, 885 | 250, 668 | 236, 885 | 336, 850 | 270, 885 | 362, 850 | 270, 885 | 142, 645 | 260, 850 | 163, 885 | 256, 850 | 133, 885 |
| a$\delta$ | 54, 0 | 8, 0 | 0, 0 | 32, 0 | | 6, 0 | | | 42, -239 | | | | | -10, 0 | | | | |
| b | 549, -107 | | 350, -110 | | | 327, -109 | | 243, -110 | | | | | | 139, -108 | | | | |
| b? | 426, -110 | 333, -110 | 352, -121 | 250, -110 | 259, -110 | 344, -110 | 240, -110 | 250, -136 | 236, -319 | 336, -110 | 270, -110 | 362, -110 | 270, -110 | 142, -110 | 260, -110 | 163, -110 | 256, -110 | 133, -110 |
| b$\delta$ | 122, 3 | | -2, 11 | | | -16, 1 | | -8, 26 | | | | | | -3, 2 | | | | |
| it | *Æ Æ̇ Æ̱* | *Ð Ð̇ Ð̱* | *Ø Ø̇ Ø̱* | *Þ Þ̇ Þ̱* | *ß ß̇ ß̱* | *æ æ̇ æ̱* | *ð ð̇ ð̱* | *ø ø̇ ø̱* | *þ þ̇ þ̱* | *Ð Ð̇ Ð̱* | *đ đ̇ đ̱* | *Ħ Ħ̇ Ħ̱* | *ħ ḣ ħ̱* | *ı i̇ ı̱* | *Ŀ L̇ Ḻ* | *ŀ l̇ ḻ* | *Ł L̇ Ḻ* | *ł l̇ ḻ* |
| patch | *Æ Æ̇ Æ̱* | *Ð Ð̇ Ð̱* | *Ø Ø̇ Ø̱* | *Þ Þ̇ Þ̱* | *ß ß̇ ß̱* | *æ æ̇ æ̱* | *ð ð̇ ð̱* | *ø ø̇ ø̱* | *þ þ̇ þ̱* | *Ð Ð̇ Ð̱* | *đ đ̇ đ̱* | *Ħ Ħ̇ Ħ̱* | *ħ ḣ ħ̱* | *ı i̇ ı̱* | *Ŀ L̇ Ḻ* | *ŀ l̇ ḻ* | *Ł L̇ Ḻ* | *ł l̇ ḻ* |
| ideal | c, b | c, b | c+25, b-43 | c, b | a, d | x, b | a+6, b | x+48, b-48 | a, d-7 | c, b | a, b | c, b | a, b | x, b | c, b | a, b | c, b | a, b |
| a | 615, 850 | | | | | 480, 645 | | | 278, 646 | | | | | 267, 645 | | | | |
| a? | 595, 850 | 514, 850 | 515, 850 | 431, 850 | 461, 890 | 481, 645 | 429, 896 | 370, 691 | 439, 890 | 514, 850 | 434, 890 | 517, 850 | 449, 890 | 275, 645 | 440, 850 | 349, 890 | 440, 850 | 322, 890 |
| a$\delta$ | 20, 0 | | | | | -1, 0 | | | -161, -244 | | | | | -8, 0 | | | | |
| b | 549, -107 | | 310, -113 | | | 327, -109 | | 186, -107 | | | | | | 139, -108 | | | | |
| b? | 391, -110 | 310, -110 | 302, -151 | 227, -110 | 204, -319 | 320, -110 | 216, -110 | 190, -156 | 181, -324 | 310, -110 | 221, -110 | 313, -110 | 236, -110 | 115, -110 | 236, -110 | 136, -110 | 236, -110 | 110, -110 |
| b$\delta$ | 158, 3 | | 8, 38 | | | 7, 1 | | -4, 49 | | | | | | 24, 2 | | | | |
| sb | **Æ Æ̇ Æ̱** | **Ð Ð̇ Ð̱** | **Ø Ø̇ Ø̱** | **Þ Þ̇ Þ̱** | **ß ß̇ ß̱** | **æ æ̇ æ̱** | **ð ð̇ ð̱** | **ø ø̇ ø̱** | **þ þ̇ þ̱** | **Ð Ð̇ Ð̱** | **đ đ̇ đ̱** | **Ħ Ħ̇ Ħ̱** | **ħ ḣ ħ̱** | **ı i̇ ı̱** | **Ŀ L̇ Ḻ** | **ŀ l̇ ḻ** | **Ł L̇ Ḻ** | **ł l̇ ḻ** |
| patch | **Æ Æ̇ Æ̱** | **Ð Ð̇ Ð̱** | **Ø Ø̇ Ø̱** | **Þ Þ̇ Þ̱** | **ß ß̇ ß̱** | **æ æ̇ æ̱** | **ð ð̇ ð̱** | **ø ø̇ ø̱** | **þ þ̇ þ̱** | **Ð Ð̇ Ð̱** | **đ đ̇ đ̱** | **Ħ Ħ̇ Ħ̱** | **ħ ḣ ħ̱** | **ı i̇ ı̱** | **Ŀ L̇ Ḻ** | **ŀ l̇ ḻ** | **Ł L̇ Ḻ** | **ł l̇ ḻ** |
| ideal | c, b | c, b | c+25, b-45 | c, b | a, b | x, b | a+19, b | x+41, b-34 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | c, b | a, b |
| a | 510, 850 | | | | | 355, 645 | | | | | | | | 151, 645 | | | | |
| a? | 446, 850 | 349, 850 | 366, 850 | 287, 850 | 284, 885 | 362, 645 | 256, 904 | 260, 684 | 260, 885 | 349, 850 | 274, 885 | 362, 850 | 275, 885 | 156, 645 | 267, 850 | 212, 885 | 258, 850 | 142, 885 |
| a$\delta$ | 64, 0 | | | | | -8, 0 | | | | | | | | -4, 0 | | | | |
| b | | | | | | | | | | | | | | 135, -108 | | | | |
| b? | 446, -110 | 349, -110 | 366, -153 | 287, -110 | 284, -110 | 362, -110 | 256, -110 | 260, -142 | 260, -319 | 349, -110 | 274, -110 | 362, -110 | 275, -110 | 156, -110 | 267, -110 | 212, -110 | 258, -110 | 142, -110 |
| b$\delta$ | | | | | | | | | | | | | | -20, 2 | | | | |
| si | ***Æ Æ̇ Æ̱*** | ***Ð Ð̇ Ð̱*** | ***Ø Ø̇ Ø̱*** | ***Þ Þ̇ Þ̱*** | ***ß ß̇ ß̱*** | ***æ æ̇ æ̱*** | ***ð ð̇ ð̱*** | ***ø ø̇ ø̱*** | ***þ þ̇ þ̱*** | ***Ð Ð̇ Ð̱*** | ***đ đ̇ đ̱*** | ***Ħ Ħ̇ Ħ̱*** | ***ħ ḣ ħ̱*** | ***ı i̇ ı̱*** | ***Ŀ L̇ Ḻ*** | ***ŀ l̇ ḻ*** | ***Ł L̇ Ḻ*** | ***ł l̇ ḻ*** |
| patch | ***Æ Æ̇ Æ̱*** | ***Ð Ð̇ Ð̱*** | ***Ø Ø̇ Ø̱*** | ***Þ Þ̇ Þ̱*** | ***ß ß̇ ß̱*** | ***æ æ̇ æ̱*** | ***ð ð̇ ð̱*** | ***ø ø̇ ø̱*** | ***þ þ̇ þ̱*** | ***Ð Ð̇ Ð̱*** | ***đ đ̇ đ̱*** | ***Ħ Ħ̇ Ħ̱*** | ***ħ ḣ ħ̱*** | ***ı i̇ ı̱*** | ***Ŀ L̇ Ḻ*** | ***ŀ l̇ ḻ*** | ***Ł L̇ Ḻ*** | ***ł l̇ ḻ*** |
| ideal | c, b | c, b | c+25, b-40 | c, b | a, d | x, b | a+19, b | x, b-30 | a, d | c, b | a, b | c, b | a, b | x, b | c, b | a, b | c, b | a, b |
| a | 500, 850 | 341, 850 | 351, 850 | 281, 850 | | 350, 645 | | | 278, 646 | | | | | 132, 645 | | | | |
| a? | 459, 850 | 367, 850 | 390, 850 | 303, 850 | 325, 890 | 339, 645 | 291, 909 | 235, 645 | 300, 890 | 367, 850 | 309, 890 | 396, 850 | 331, 890 | 144, 645 | 310, 850 | 234, 890 | 300, 850 | 190, 890 |
| a$\delta$ | 41, 0 | -26, 0 | -39, 0 | -22, 0 | | 11, 0 | | | -22, -244 | | | | | -12, 0 | | | | |
| b | 549, -107 | | 350, -110 | | | 327, -109 | | 243, -110 | | | | | | 139, -108 | | | | |
| b? | 426, -110 | 334, -110 | 349, -148 | 270, -110 | 241, -319 | 347, -110 | 246, -110 | 237, -138 | 216, -319 | 334, -110 | 268, -110 | 363, -110 | 290, -110 | 153, -110 | 277, -110 | 192, -110 | 267, -110 | 148, -110 |
| b$\delta$ | 123, 3 | | 1, 38 | | | -20, 1 | | 6, 28 | | | | | | -14, 2 | | | | |

Table 6: Font metrics for Latin and IPA 2/11

| hex | 014A | 014B | 0152 | 0153 | 0166 | 0167 | 017F | 0180 | 0181 | 0182 | 0183 | 0184 | 0186 | 0187 | 0188 | 0189 | 018A | 018B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Ll | Lu | Ll | Lu | Ll | Ll | Ll | Lu | Lu | Ll | Lu | Lu | Lu | Ll | Lu | Lu | Lu |
| reg | ŋŋŋ | ŋŋŋ | ŒŒŒ | œœœ | ŦŦŦ | ŧŧŧ | ſſſ | ƀƀƀ | ƁƂƁ | ƂƂƂ | ƃƃƃ | ƄƅƄ | ƆƆƆ | ƇƈƇ | ƈƈƈ | ƉƊƉ | ƊƋƊ | ƋƌƋ |
| patch | ŋŋŋ | ŋŋŋ | ŒŒŒ | œœœ | ŦŦŦ | ŧŧŧ | ſſſ | ƀƀƀ | ƁƂƁ | ƂƂƂ | ƃƃƃ | ƄƅƄ | ƆƆƆ | ƇƈƇ | ƈƈƈ | ƉƊƉ | ƊƋƊ | ƋƌƋ |
| ideal | c, d+60 | x, d+9 | c, b | x, b | c, b | a-116, b | a, b | a, b | c, b | c, b | a, b | c, b | c, b | c+98, b | x+70, b | c, b | c, b | c, b |
| a | 346, 804 | 264, 646 | 460, 850 | 405, 645 | | | | | 327, 805 | | | 267, 805 | | | | | 306, 805 | |
| a? | 316, 850 | 240, 645 | 436, 850 | 388, 645 | 297, 850 | 160, 769 | 190, 885 | 231, 885 | 303, 850 | 274, 850 | 228, 885 | 278, 850 | 322, 850 | 369, 850 | 264, 713 | 333, 850 | 368, 850 | 292, 850 |
| a$\delta$ | 30, -46 | 24, 1 | 24, 0 | 18, 0 | | | | | 24, -45 | | | -10, -45 | | | | | -62, -45 | |
| b | 353, -109 | 265, -111 | | | | | | | 328, -108 | | 234, -109 | 268, -108 | | | | | 369, -107 | |
| b? | 316, -257 | 240, -308 | 436, -110 | 388, -110 | 297, -110 | 160, -110 | 190, -110 | 231, -110 | 303, -110 | 274, -110 | 228, -110 | 278, -110 | 322, -110 | 369, -110 | 264, -110 | 333, -110 | 368, -110 | 292, -110 |
| b$\delta$ | 37, 148 | 25, 197 | | | | | | | 25, 2 | | 6, 1 | -10, 2 | | | | | 0, 3 | |
| it | *ŋŋŋ* | *ŋŋŋ* | *ŒŒŒ* | *œœœ* | *ŦŦŦ* | *ŧŧŧ* | *ſſſ* | *ƀƀƀ* | *ƁƂƁ* | *ƂƂƂ* | *ƃƃƃ* | *ƄƅƄ* | *ƆƆƆ* | *ƇƈƇ* | *ƈƈƈ* | *ƉƊƉ* | *ƊƋƊ* | *ƋƌƋ* |
| patch | *ŋŋŋ* | *ŋŋŋ* | *ŒŒŒ* | *œœœ* | *ŦŦŦ* | *ŧŧŧ* | *ſſſ* | *ƀƀƀ* | *ƁƂƁ* | *ƂƂƂ* | *ƃƃƃ* | *ƄƅƄ* | *ƆƆƆ* | *ƇƈƇ* | *ƈƈƈ* | *ƉƊƉ* | *ƊƋƊ* | *ƋƌƋ* |
| ideal | c, d+32 | x, d | c, b | x, b | c, b | x+132, b | a, d | a, b | c, b | c, b | a, b | c, b | c+14, b | c+98, b | x+70, b | c, b | c, b | c, b |
| a | 346, 804 | 264, 646 | 575, 850 | 480, 645 | | | | | 327, 805 | | | 267, 805 | | | | | 306, 805 | |
| a? | 495, 850 | 382, 645 | 587, 850 | 490, 645 | 453, 850 | 318, 775 | 339, 890 | 428, 890 | 488, 850 | 465, 850 | 413, 890 | 455, 850 | 503, 850 | 503, 850 | 398, 713 | 534, 850 | 553, 850 | 450, 850 |
| a$\delta$ | -149, -46 | -118, 1 | -12, 0 | -10, 0 | | | | | -161, -45 | | | -188, -45 | | | | | -247, -45 | |
| b | 364, -109 | 265, -111 | | | | | | | 328, -108 | | 230, -109 | 268, -108 | | | | | 369, -107 | |
| b? | 253, -285 | 177, -319 | 383, -110 | 329, -110 | 249, -110 | 130, -110 | 82, -319 | 216, -110 | 284, -110 | 261, -110 | 201, -110 | 251, -110 | 299, -110 | 299, -110 | 223, -110 | 330, -110 | 349, -110 | 246, -110 |
| b$\delta$ | 111, 176 | 88, 208 | | | | | | | 44, 2 | | 29, 1 | 17, 2 | | | | | 20, 3 | |
| sb | **ŋŋŋ** | **ŋŋŋ** | **ŒŒŒ** | **œœœ** | **ŦŦŦ** | **ŧŧŧ** | **ſſſ** | **ƀƀƀ** | **ƁƂƁ** | **ƂƂƂ** | **ƃƃƃ** | **ƄƅƄ** | **ƆƆƆ** | **ƇƈƇ** | **ƈƈƈ** | **ƉƊƉ** | **ƊƋƊ** | **ƋƌƋ** |
| patch | **ŋŋŋ** | **ŋŋŋ** | **ŒŒŒ** | **œœœ** | **ŦŦŦ** | **ŧŧŧ** | **ſſſ** | **ƀƀƀ** | **ƁƂƁ** | **ƂƂƂ** | **ƃƃƃ** | **ƄƅƄ** | **ƆƆƆ** | **ƇƈƇ** | **ƈƈƈ** | **ƉƊƉ** | **ƊƋƊ** | **ƋƌƋ** |
| ideal | c, d+50 | x, d+36 | c, b | x, b | c+28, b | a-103, b | a, b | a, b-13 | c, b | c, b | a-15, b-16 | c, b | c, b | c+98, b | x+65, b | c, b | c, b | c, b |
| a | | | 450, 850 | 430, 645 | | | | | | | | | 275, 833 | | | | | |
| a? | 328, 850 | 248, 645 | 432, 850 | 402, 645 | 299, 850 | 168, 782 | 217, 885 | 236, 885 | 316, 850 | 268, 850 | 223, 870 | 274, 850 | 314, 850 | 368, 850 | 264, 708 | 349, 850 | 381, 850 | 292, 850 |
| a$\delta$ | | | 18, 0 | 28, 0 | | | | | | | | | -38, -17 | | | | | |
| b | | | | | | | | | | | | | | | | | | |
| b? | 328, -267 | 248, -281 | 432, -110 | 402, -110 | 299, -110 | 168, -110 | 217, -110 | 236, -121 | 316, -110 | 268, -110 | 223, -124 | 274, -110 | 314, -110 | 368, -110 | 264, -110 | 349, -110 | 381, -110 | 292, -110 |
| b$\delta$ | | | | | | | | | | | | | | | | | | |
| si | ***ŋŋŋ*** | ***ŋŋŋ*** | ***ŒŒŒ*** | ***œœœ*** | ***ŦŦŦ*** | ***ŧŧŧ*** | ***ſſſ*** | ***ƀƀƀ*** | ***ƁƂƁ*** | ***ƂƂƂ*** | ***ƃƃƃ*** | ***ƄƅƄ*** | ***ƆƆƆ*** | ***ƇƈƇ*** | ***ƈƈƈ*** | ***ƉƊƉ*** | ***ƊƋƊ*** | ***ƋƌƋ*** |
| patch | ***ŋŋŋ*** | ***ŋŋŋ*** | ***ŒŒŒ*** | ***œœœ*** | ***ŦŦŦ*** | ***ŧŧŧ*** | ***ſſſ*** | ***ƀƀƀ*** | ***ƁƂƁ*** | ***ƂƂƂ*** | ***ƃƃƃ*** | ***ƄƅƄ*** | ***ƆƆƆ*** | ***ƇƈƇ*** | ***ƈƈƈ*** | ***ƉƊƉ*** | ***ƊƋƊ*** | ***ƋƌƋ*** |
| ideal | c, d+49 | x, d+16 | c, b | x, b | c+28, b | a-125, b | a+13, d-9 | a, b | c, b | c, b | a-15, b | c, b | c, b | c+98, b | x+65, b | c, b | c, b | c, b |
| a | 359, 804 | 264, 646 | 480, 850 | 415, 645 | | | | | 327, 805 | | | 267, 805 | | | | | 306, 805 | |
| a? | 390, 850 | 253, 645 | 450, 850 | 384, 645 | 325, 850 | 175, 765 | 209, 903 | 290, 890 | 359, 850 | 303, 850 | 269, 875 | 306, 850 | 357, 850 | 359, 850 | 253, 708 | 388, 850 | 424, 850 | 304, 850 |
| a$\delta$ | -31, -46 | 11, 1 | 30, 0 | 31, 0 | | | | | -32, -45 | | | -39, -45 | | | | | -118, -45 | |
| b | 373, -109 | 265, -111 | | | | | | | 328, -108 | | 234, -109 | 268, -108 | | | | | 369, -107 | |
| b? | 325, -268 | 222, -301 | 417, -110 | 393, -110 | 292, -110 | 159, -110 | 121, -326 | 248, -110 | 326, -110 | 270, -110 | 231, -110 | 273, -110 | 324, -110 | 326, -110 | 249, -110 | 355, -110 | 391, -110 | 271, -110 |
| b$\delta$ | 48, 159 | 43, 190 | | | | | | | 2, 2 | | 3, 1 | -5, 2 | | | | | -22, 3 | |

Table 7: Font metrics for Latin and IPA 3/11

| hex | 018C | 018E | 018F | 0190 | 0191 | 0192 | 0193 | 0194 | 0196 | 0197 | 0198 | 0199 | 019A | 019C | 019D | 019F | 01A2 | 01A4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Lu | Lu | Lu | Lu | Ll | Lu | Lu | Lu | Lu | Lu | Ll | Ll | Lu | Lu | Lu | Lu | Lu |
| reg | ƍƌƍ | ƎƎƎ | ƏƏƏ | ƐƐƐ | ƑƑƑ | *ƒƒƒ* | ƓƓƓ | ƔƔƔ | ƖƖƖ | ƗƗƗ | ƘƘƘ | ƙƙƙ | ƚƚƚ | ƜƜƜ | ƝƝƝ | ƟƟƟ | ƢƢƢ | ƤƤƤ |
| patch | ƍȧƍ | ƎȧƎ | ƏȧƏ | ƐȧƐ | ƑƒƑ | *ƒƒ̇ƒ* | ƓƓ̇Ɠ | ƔƔ̇Ɣ | ƖƖ̇Ɩ | ƗƗ̇Ɨ | ƘƘ̇Ƙ | ƙƙ̇ƙ | ƚƚ̇ƚ | ƜƜ̇Ɯ | ƝƝ̇Ɲ | ƟƟ̇Ɵ | ƢƢ̇Ƣ | ƤƤ̇Ƥ |
| ideal | a, b | c, b | c, b | c, b | c, d+60 | a, d | c+98, b | c, d | a, b | c, b | c, b | a, b | a, b | c, b | c, d+45 | c, b | c, d+52 | c, b |
| a | 214, 650 | | | | | | | 331, 805 | | | | | | 431, 801 | | | 359, 804 | 400, 850 |
| a? | 275, 885 | 280, 850 | 326, 850 | 256, 850 | 183, 850 | 201, 885 | 372, 850 | 308, 850 | 152, 885 | 148, 850 | 339, 850 | 262, 885 | 132, 885 | 434, 850 | 324, 850 | 352, 850 | 469, 850 | 318, 850 |
| a$\delta$ | -61, -235 | | | | | | | 22, -45 | | | | | | -2, -49 | | | -110, -46 | 82, 0 |
| b | 217, -107 | | | | | | | | | | | 263, -107 | | 450, -108 | | 350, -110 | 354, -113 | 363, -110 |
| b? | 275, -110 | 280, -110 | 326, -110 | 256, -110 | 183, -257 | 201, -319 | 372, -110 | 308, -319 | 152, -110 | 148, -110 | 339, -110 | 262, -110 | 132, -110 | 434, -110 | 324, -272 | 352, -110 | 469, -265 | 318, -110 |
| b$\delta$ | -58, 3 | | | | | | | | | | | 1, 3 | | 16, 2 | | -2, 0 | -115, 152 | 45, 0 |
| it | *ƍƌƍ* | *ƎƎƎ* | *ƏƏƏ* | *ƐƐƐ* | *ƑƑƑ* | *ƒƒƒ* | *ƓƓƓ* | *ƔƔƔ* | *ƖƖƖ* | *ƗƗƗ* | *ƘƘƘ* | *ƙƙƙ* | *ƚƚƚ* | *ƜƜƜ* | *ƝƝƝ* | *ƟƟƟ* | *ƢƢƢ* | *ƤƤƤ* |
| patch | *ƍȧƍ* | *ƎȧƎ* | *ƏȧƏ* | *ƐȧƐ* | *ƑƒƑ* | *ƒƒ̇ƒ* | *ƓƓ̇Ɠ* | *ƔƔ̇Ɣ* | *ƖƖ̇Ɩ* | *ƗƗ̇Ɨ* | *ƘƘ̇Ƙ* | *ƙƙ̇ƙ* | *ƚƚ̇ƚ* | *ƜƜ̇Ɯ* | *ƝƝ̇Ɲ* | *ƟƟ̇Ɵ* | *ƢƢ̇Ƣ* | *ƤƤ̇Ƥ* |
| ideal | a, b | c, b | c, b | c, b | c, d+60 | a, d+60 | c+98, b | c, d+20 | a, b | c, b | c, b | a, b | a, b | c, b | c, d+45 | c, b | c, d+89 | c, b |
| a | 209, 650 | | | | | | | 331, 805 | | | | | | 431, 801 | | | 355, 804 | |
| a? | 423, 890 | 465, 850 | 515, 850 | 439, 850 | 419, 850 | 414, 890 | 541, 850 | 462, 850 | 322, 890 | 321, 850 | 499, 850 | 429, 890 | 342, 890 | 609, 850 | 528, 850 | 515, 850 | 622, 850 | 473, 850 |
| a$\delta$ | -214, -240 | | | | | | | -131, -45 | | | | | | -178, -49 | | | -267, -46 | |
| b | 212, -107 | | | | | | | | | | | 263, -107 | | 450, -108 | | | 350, -113 | |
| b? | 211, -110 | 261, -110 | 311, -110 | 235, -110 | 184, -257 | 170, -257 | 337, -110 | 218, -297 | 110, -110 | 117, -110 | 295, -110 | 216, -110 | 130, -110 | 405, -110 | 290, -272 | 311, -110 | 393, -228 | 269, -110 |
| b$\delta$ | 1, 3 | | | | | | | | | | | 47, 3 | | 45, 2 | | | -43, 115 | |
| sb | **ƍƌƍ** | **ƎƎƎ** | **ƏƏƏ** | **ƐƐƐ** | **ƑƑƑ** | ***ƒƒƒ*** | **ƓƓƓ** | **ƔƔƔ** | **ƖƖƖ** | **ƗƗƗ** | **ƘƘƘ** | **ƙƙƙ** | **ƚƚƚ** | **ƜƜƜ** | **ƝƝƝ** | **ƟƟƟ** | **ƢƢƢ** | **ƤƤƤ** |
| patch | **ƍȧƍ** | **ƎȧƎ** | **ƏȧƏ** | **ƐȧƐ** | **ƑƒƑ** | ***ƒƒ̇ƒ*** | **ƓƓ̇Ɠ** | **ƔƔ̇Ɣ** | **ƖƖ̇Ɩ** | **ƗƗ̇Ɨ** | **ƘƘ̇Ƙ** | **ƙƙ̇ƙ** | **ƚƚ̇ƚ** | **ƜƜ̇Ɯ** | **ƝƝ̇Ɲ** | **ƟƟ̇Ɵ** | **ƢƢ̇Ƣ** | **ƤƤ̇Ƥ** |
| ideal | a-15, b | c, b | c, b | c, b | c, d+36 | a, d+62 | c+98, b | c, d+22 | a, b | c, b | c, b | a, b | a, b | c, b | c, d+47 | c, b | c, d+90 | c, b |
| a | | | | | | | | | | | | | | | | | | |
| a? | 261, 870 | 302, 850 | 353, 850 | 269, 850 | 236, 850 | 221, 885 | 374, 850 | 294, 850 | 170, 885 | 158, 850 | 338, 850 | 259, 885 | 138, 885 | 444, 850 | 336, 850 | 366, 850 | 454, 850 | 296, 850 |
| a$\delta$ | | | | | | | | | | | | | | | | | | |
| b | | | | | | | | | | | | | | | | | | |
| b? | 261, -110 | 302, -110 | 353, -110 | 269, -110 | 236, -281 | 221, -255 | 374, -110 | 294, -295 | 170, -110 | 158, -110 | 338, -110 | 259, -110 | 138, -110 | 444, -110 | 336, -270 | 366, -110 | 454, -227 | 296, -110 |
| b$\delta$ | | | | | | | | | | | | | | | | | | |
| si | ***ƍƌƍ*** | ***ƎƎƎ*** | ***ƏƏƏ*** | ***ƐƐƐ*** | ***ƑƑƑ*** | ***ƒƒƒ*** | ***ƓƓƓ*** | ***ƔƔƔ*** | ***ƖƖƖ*** | ***ƗƗƗ*** | ***ƘƘƘ*** | ***ƙƙƙ*** | ***ƚƚƚ*** | ***ƜƜƜ*** | ***ƝƝƝ*** | ***ƟƟƟ*** | ***ƢƢƢ*** | ***ƤƤƤ*** |
| patch | ***ƍȧƍ*** | ***ƎȧƎ*** | ***ƏȧƏ*** | ***ƐȧƐ*** | ***ƑƒƑ*** | ***ƒƒ̇ƒ*** | ***ƓƓ̇Ɠ*** | ***ƔƔ̇Ɣ*** | ***ƖƖ̇Ɩ*** | ***ƗƗ̇Ɨ*** | ***ƘƘ̇Ƙ*** | ***ƙƙ̇ƙ*** | ***ƚƚ̇ƚ*** | ***ƜƜ̇Ɯ*** | ***ƝƝ̇Ɲ*** | ***ƟƟ̇Ɵ*** | ***ƢƢ̇Ƣ*** | ***ƤƤ̇Ƥ*** |
| ideal | a-15, b | c, b | c, b | c, b | c, d+36 | a, d+62 | c+98, b | c, d+22 | a, b | c, b | c, b | a, b | a, b | c, b | c, d+47 | c, b | c, d+90 | c, b |
| a | 214, 650 | | | | | | | 331, 805 | | | | | | 431, 801 | | | 359, 804 | 400, 850 |
| a? | 275, 875 | 320, 850 | 369, 850 | 293, 850 | 288, 850 | 269, 890 | 395, 850 | 316, 850 | 194, 890 | 181, 850 | 353, 850 | 283, 890 | 174, 890 | 463, 850 | 383, 850 | 387, 850 | 464, 850 | 315, 850 |
| a$\delta$ | -61, -225 | | | | | | | 15, -45 | | | | | | -32, -49 | | | -105, -46 | 85, 0 |
| b | 217, -107 | | | | | | | | | | | 263, -107 | | 450, -108 | | | 354, -113 | 363, -110 |
| b? | 237, -110 | 287, -110 | 336, -110 | 260, -110 | 220, -281 | 198, -255 | 362, -110 | 245, -295 | 153, -110 | 148, -110 | 320, -110 | 241, -110 | 132, -110 | 430, -110 | 317, -270 | 354, -110 | 407, -227 | 282, -110 |
| b$\delta$ | -20, 3 | | | | | | | | | | | 22, 3 | | 20, 2 | | | -53, 114 | 81, 0 |

Table 8: Font metrics for Latin and IPA 4/11

| hex | 01A5 | 01A9 | 01AB | 01AC | 01AD | 01AE | 01B3 | 01B4 | 01B5 | 01B6 | 01B7 | 01E4 | 01E5 | 0221 | 0234 | 0235 | 0236 | 0237 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Lu | Ll | Lu | Ll | Lu | Lu | Ll | Lu | Ll | Lu | Lu | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ƥƥ̲ | ΣΣΣ̲ | ʇʇ̲ | TTT̲ | ffff̲ | ΤΤΤ̲ | ΥΥΥ̲ | yẏy̲ | ZŻZ̲ | zżz̲ | 333̲ | GĠG̲ | gġg̲ | ɗɗɗ̲ | llll̲ | nṅn̲ | ttt̲ | jȷj̲ |
| patch | ƥƥ̲ | ΣΣΣ̲ | ʇʇ̲ | TṪT̲ | ffff̲ | ΤΤΤ̲ | ΥΥΥ̲ | yẏy̲ | ZŻZ̲ | zżz̲ | 333̲ | GĠG̲ | gġg̲ | ɗɗɗ̲ | llll̲ | nṅn̲ | ttt̲ | jȷj̲ |
| ideal | a, d | c, b | a-116, d+87 | c, b | a, b | c, d+56 | c+14, b | x+128, d | c, b | x, b | c+22, d+21 | c, b | x+33, d | a, d+107 | a, b-108 | x, b-93 | a-115, b-93 | x, d+9 |
| a | | | | | | | | 270, 644 | | | | | | | | 261, 646 | | 140, 645 |
| a? | 242, 885 | 277, 850 | 163, 769 | 308, 850 | 186, 885 | 297, 850 | 321, 850 | 316, 771 | 310, 850 | 220, 645 | 303, 850 | 366, 850 | 256, 676 | 326, 885 | 175, 885 | 333, 645 | 196, 770 | 74, 645 |
| aδ | | | | | | | | -46, -127 | | | | | | | | -72, 1 | | 66, 0 |
| b | | | | | 163, -107 | | | | | | 272, -319 | | | | | 262, -111 | 163, -109 | |
| b? | 242, -319 | 277, -110 | 163, -230 | 308, -110 | 186, -110 | 297, -261 | 321, -110 | 316, -319 | 310, -110 | 220, -110 | 303, -296 | 366, -110 | 256, -319 | 326, -210 | 175, -216 | 333, -201 | 196, -201 | 74, -308 |
| bδ | | | | | -24, 3 | | | | | | -31, -23 | | | | | -71, 90 | -33, 92 | |
| it | *ƥƥ̲* | *ΣΣΣ̲* | *ʇʇ̲* | *TTT̲* | *ffff̲* | *ΤΤΤ̲* | *ΥΥΥ̲* | *yẏy̲* | *ZŻZ̲* | *zżz̲* | *333̲* | *GĠG̲* | *gġg̲* | *ɗɗɗ̲* | *llll̲* | *nṅn̲* | *ttt̲* | *jȷj̲* |
| patch | *ƥƥ̲* | *ΣΣΣ̲* | *ʇʇ̲* | *TṪT̲* | *ffff̲* | *ΤΤΤ̲* | *ΥΥΥ̲* | *yẏy̲* | *ZŻZ̲* | *zżz̲* | *333̲* | *GĠG̲* | *gġg̲* | *ɗɗɗ̲* | *llll̲* | *nṅn̲* | *ttt̲* | *jȷj̲* |
| ideal | a, d | c, b | a-114, d+87 | c+24, b | a, b | c, d+43 | c+14, b | x+111, d | c, b | x, b | c+22, d+21 | c, b | x, d | a, d+112 | a, b-106 | x, b-89 | a-114, b-96 | x, d+36 |
| a | | | | | | | | 270, 644 | | | | | | | | 261, 646 | | 140, 645 |
| a? | 429, 890 | 477, 850 | 316, 776 | 488, 850 | 341, 890 | 453, 850 | 501, 850 | 471, 754 | 470, 850 | 343, 645 | 485, 850 | 524, 850 | 376, 645 | 507, 890 | 389, 890 | 460, 645 | 352, 776 | 267, 645 |
| aδ | | | | | | | | -201, -110 | | | | | | | | -199, 1 | | -127, 0 |
| b | | | | | 163, -107 | | | | | | | | | | | 262, -111 | 163, -109 | |
| b? | 172, -319 | 273, -110 | 102, -230 | 284, -110 | 129, -110 | 214, -274 | 297, -110 | 243, -319 | 266, -110 | 182, -110 | 241, -296 | 320, -110 | 171, -319 | 274, -205 | 155, -214 | 281, -197 | 144, -204 | 70, -281 |
| bδ | | | | | 34, 3 | | | | | | | | | | | -19, 86 | 19, 95 | |
| sb | **ƥƥ̲** | **ΣΣΣ̲** | **ʇʇ̲** | **TTT̲** | **ffff̲** | **ΤΤΤ̲** | **ΥΥΥ̲** | **yẏy̲** | **ZŻZ̲** | **zżz̲** | **333̲** | **GĠG̲** | **gġg̲** | **ɗɗɗ̲** | **llll̲** | **nṅn̲** | **ttt̲** | **jȷj̲** |
| patch | **ƥƥ̲** | **ΣΣΣ̲** | **ʇʇ̲** | **TṪT̲** | **ffff̲** | **ΤΤΤ̲** | **ΥΥΥ̲** | **yẏy̲** | **ZŻZ̲** | **zżz̲** | **333̲** | **GĠG̲** | **gġg̲** | **ɗɗɗ̲** | **llll̲** | **nṅn̲** | **ttt̲** | **jȷj̲** |
| ideal | a, d | c, b | a-103, d+88 | c+30, b | a, b | c+30, d+36 | c, b | x+125, d | c, b | x, b | c+29, d+24 | c, b | x+33, d | a, d+101 | a, b-115 | x, b-99 | a-103, b-106 | x, d |
| a | | | | | | | | | | | | | | | | | | 163, 645 |
| a? | 236, 885 | 293, 850 | 168, 782 | 320, 850 | 186, 885 | 306, 850 | 333, 850 | 328, 768 | 309, 850 | 222, 645 | 316, 850 | 368, 850 | 258, 676 | 338, 885 | 212, 885 | 345, 645 | 206, 782 | 78, 645 |
| aδ | | | | | | | | | | | | | | | | | | 85, 0 |
| b | | | | | | | | | | | 272, -319 | | | | | | | |
| b? | 236, -319 | 293, -110 | 168, -229 | 320, -110 | 186, -110 | 306, -281 | 333, -110 | 328, -319 | 309, -110 | 222, -110 | 316, -293 | 368, -110 | 258, -319 | 338, -216 | 212, -223 | 345, -207 | 206, -214 | 78, -319 |
| bδ | | | | | | | | | | | -44, -26 | | | | | | | |
| si | ***ƥƥ̲*** | ***ΣΣΣ̲*** | ***ʇʇ̲*** | ***TTT̲*** | ***ffff̲*** | ***ΤΤΤ̲*** | ***ΥΥΥ̲*** | ***yẏy̲*** | ***ZŻZ̲*** | ***zżz̲*** | ***333̲*** | ***GĠG̲*** | ***gġg̲*** | ***ɗɗɗ̲*** | ***llll̲*** | ***nṅn̲*** | ***ttt̲*** | ***jȷj̲*** |
| patch | ***ƥƥ̲*** | ***ΣΣΣ̲*** | ***ʇʇ̲*** | ***TṪT̲*** | ***ffff̲*** | ***ΤΤΤ̲*** | ***ΥΥΥ̲*** | ***yẏy̲*** | ***ZŻZ̲*** | ***zżz̲*** | ***333̲*** | ***GĠG̲*** | ***gġg̲*** | ***ɗɗɗ̲*** | ***llll̲*** | ***nṅn̲*** | ***ttt̲*** | ***jȷj̲*** |
| ideal | a, d | c, b | a-103, d+89 | c+30, b | a, b | c+30, d+36 | c, b | x+125, d+20 | c+25, b | x, b | c+28, d+24 | c, b | x, d | a, d+101 | a, b-115 | x, b-99 | a-103, b-106 | x, d+25 |
| a | | | | | | | | 270, 644 | | | | | | | | 261, 646 | | 140, 645 |
| a? | 271, 890 | 319, 850 | 161, 787 | 330, 850 | 183, 890 | 314, 850 | 368, 850 | 317, 768 | 332, 850 | 187, 645 | 329, 850 | 388, 850 | 220, 645 | 362, 890 | 233, 890 | 324, 645 | 209, 787 | 121, 645 |
| aδ | | | | | | | | -47, -124 | | | | | | | | -63, 1 | | 19, 0 |
| b | | | | | 163, -107 | | | | | | 236, -319 | | | | | 262, -111 | 163, -109 | |
| b? | 187, -319 | 286, -110 | 117, -228 | 297, -110 | 142, -110 | 246, -281 | 335, -110 | 262, -297 | 299, -110 | 195, -110 | 258, -293 | 355, -110 | 186, -319 | 299, -216 | 169, -223 | 313, -207 | 167, -214 | 93, -292 |
| bδ | | | | | 21, 3 | | | | | | -22, -26 | | | | | -51, 96 | -4, 105 | |

Table 9: Font metrics for Latin and IPA 5/11

| hex | 023A | 023B | 023C | 023D | 023E | 0243 | 0246 | 0247 | 0248 | 0249 | 024A | 024C | 024D | 024E | 024F | 0250 | 0251 | 0252 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Ll | Lu | Lu | Lu | Lu | Ll | Lu | Ll | Lu | Lu | Ll | Lu | Ll | Ll | Ll | Ll |
| reg | Ⱥ | Ƀ | ȼ | Ƚ | Ɇ | Ƀ | Ɇ | ɇ | Ɉ | ɉ | Ɋ | Ɍ | ɍ | Ɏ | ɏ | ɐ | ɑ | ɒ |
| patch | Ⱥ | Ƀ | ȼ | Ƚ | Ɇ | Ƀ | Ɇ | ɇ | Ɉ | ɉ | Ɋ | Ɍ | ɍ | Ɏ | ɏ | ɐ | ɑ | ɒ |
| ideal | c+53, b-110 | c+53, b-110 | x+25, b-28 | c, b | c+109, b-57 | c, b | c, b-13 | x+25, b-28 | c, d+60 | a-51, d+9 | c, d+40 | c, b | x, b | c, b | x, d | x, b | x, b | x, b |
| a | | | | | | | | | | | 355, 804 | | | | | 228, 645 | 228, 645 | 264, 645 |
| a? | 346, 850 | 324, 850 | 216, 668 | 248, 850 | 297, 850 | 255, 850 | 250, 850 | 216, 668 | 186, 850 | 116, 834 | 380, 850 | 301, 850 | 188, 645 | 288, 850 | 260, 645 | 216, 645 | 272, 645 | 247, 645 |
| aδ | | | | | | | | | | | -26, -46 | | | | | 12, 0 | -44, 0 | 17, 0 |
| b | | | | | | | | | | | 350, -113 | | | | | 229, -114 | 212, -114 | 259, -114 |
| b? | 346, -218 | 324, -218 | 216, -136 | 248, -110 | 297, -165 | 255, -110 | 250, -121 | 216, -136 | 186, -257 | 116, -308 | 380, -277 | 301, -110 | 188, -110 | 288, -110 | 260, -319 | 216, -110 | 272, -110 | 247, -110 |
| bδ | | | | | | | | | | | -30, 164 | | | | | 12, -4 | -60, -4 | 12, -4 |
| it | *Ⱥ* | *Ƀ* | *ȼ* | *Ƚ* | *Ɇ* | *Ƀ* | ▯ | *ɇ* | ▯ | *ɉ* | ▯ | ▯ | *ɍ* | ▯ | ▯ | *ɐ* | *ɑ* | *ɒ* |
| patch | *Ⱥ* | *Ƀ* | *ȼ* | *Ƚ* | *Ɇ* | *Ƀ* | ▯ | *ɇ* | ▯ | *ɉ* | ▯ | ▯ | *ɍ* | ▯ | ▯ | *ɐ* | *ɑ* | *ɒ* |
| ideal | c+107, b-56 | c+109, b-57 | x+25, b-29 | c, b | c+109, b-57 | c, b | | x+49, b-35 | | a-80, d+36 | | | a+118, d+60 | | | x, b | x, b | x, b |
| a | | | | | | | | 332, 645 | | | | | 312, 646 | | | | | |
| a? | 528, 850 | 503, 850 | 352, 668 | 440, 850 | 453, 850 | 477, 850 | | 348, 692 | | 315, 810 | | | 515, 1008 | | | 365, 645 | 380, 645 | 391, 645 |
| aδ | | | | | | | | -16, -47 | | | | | -203, -362 | | | | | |
| b | | | | | | | | 158, -108 | | | | | 113, -110 | | | 229, -114 | 212, -114 | 243, -114 |
| b? | 313, -164 | 287, -165 | 181, -137 | 236, -110 | 237, -165 | 273, -110 | | 170, -143 | | 83, -281 | | | 246, -257 | | | 205, -110 | 220, -110 | 231, -110 |
| bδ | | | | | | | | -12, 35 | | | | | -133, 147 | | | 24, -4 | -8, -4 | 12, -4 |
| sb | **Ⱥ** | **Ƀ** | **ȼ** | **Ƚ** | **Ɇ** | **Ƀ** | ▯ | **ɇ** | **Ɉ** | **ɉ** | ▯ | ▯ | **ɍ** | ▯ | ▯ | **ɐ** | **ɑ** | **ɒ** |
| patch | **Ⱥ** | **Ƀ** | **ȼ** | **Ƚ** | **Ɇ** | **Ƀ** | ▯ | **ɇ** | **Ɉ** | **ɉ** | ▯ | ▯ | **ɍ** | ▯ | ▯ | **ɐ** | **ɑ** | **ɒ** |
| ideal | c+53, b-110 | c+53, b-110 | x+27, b-21 | c, b | c+109, b-57 | c, b | | x+43, b-34 | c, d+50 | a-48, d | | | x, b | | | x, b-14 | x, b | x, b |
| a | | | | | | | | | 203, 806 | | | | | | | | | |
| a? | 350, 850 | 332, 850 | 200, 670 | 263, 850 | 299, 850 | 294, 850 | | 232, 686 | 131, 850 | 112, 837 | | | 210, 645 | | | 207, 645 | 276, 645 | 251, 645 |
| aδ | | | | | | | | | 72, -44 | | | | | | | | | |
| b | | | | | | | | | | | | | | | | 228, -114 | | |
| b? | 350, -218 | 332, -218 | 200, -129 | 263, -110 | 299, -165 | 294, -110 | | 232, -142 | 131, -267 | 112, -319 | | | 210, -110 | | | 207, -122 | 276, -110 | 251, -110 |
| bδ | | | | | | | | | | | | | | | | 21, 8 | | |
| si | ***Ⱥ*** | ***Ƀ*** | ***ȼ*** | ***Ƚ*** | ***Ɇ*** | ***Ƀ*** | ▯ | ***ɇ*** | ▯ | ***ɉ*** | ▯ | ▯ | ***ɍ*** | ▯ | ▯ | ***ɐ*** | ***ɑ*** | ***ɒ*** |
| patch | ***Ⱥ*** | ***Ƀ*** | ***ȼ*** | ***Ƚ*** | ***Ɇ*** | ***Ƀ*** | ▯ | ***ɇ*** | ▯ | ***ɉ*** | ▯ | ▯ | ***ɍ*** | ▯ | ▯ | ***ɐ*** | ***ɑ*** | ***ɒ*** |
| ideal | c+53, b-110 | c+53, b-110 | x+27, b-21 | c, b | c+109, b-57 | c, b | | x+41, b-32 | | a-67, d+25 | | | x, b | | | x, b | x, b | x, b |
| a | | | | | | | | 243, 645 | | | | | 191, 646 | | | | 228, 645 | 264, 645 |
| a? | 382, 850 | 357, 850 | 204, 670 | 300, 850 | 325, 850 | 319, 850 | | 226, 684 | | 192, 823 | | | 203, 645 | | | 221, 645 | 247, 645 | 247, 645 |
| aδ | | | | | | | | 17, -39 | | | | | -12, 1 | | | | -19, 0 | 17, 0 |
| b | | | | | | | | 234, -110 | | 109, -319 | | | 135, -110 | | | 229, -114 | 212, -114 | 259, -114 |
| b? | 327, -218 | 302, -218 | 203, -129 | 267, -110 | 281, -165 | 286, -110 | | 221, -140 | | 127, -292 | | | 212, -110 | | | 230, -110 | 256, -110 | 256, -110 |
| bδ | | | | | | | | 13, 30 | | -18, -27 | | | -77, 0 | | | -1, -4 | -44, -4 | 3, -4 |

Table 10: Font metrics for Latin and IPA 6/11

| hex | 0253 | 0254 | 0255 | 0256 | 0257 | 0258 | 0259 | 025A | 025B | 025C | 025D | 025E | 025F | 0260 | 0261 | 0262 | 0263 | 0264 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɓ | ɔ | ɕ | ɖ | ɗ | ɘ | ə | ɚ | ɛ | ɜ | ɝ | ɞ | ɟ | ɠ | ɡ | ɢ | ɣ | ɤ |
| patch | ɓ | ɔ | ɕ | ɖ | ɗ | ɘ | ə | ɚ | ɛ | ɜ | ɝ | ɞ | ɟ | ɠ | ɡ | ɢ | ɣ | ɤ |
| ideal | a, b | x, b | x, b-106 | a, d+44 | a, b | x, b | x, b | x+45, b | x, b | x, b | a-90, b | x, b | x, d+26 | a, d | x, d | x, b | x, d+30 | x, b |
| a | 480, 630 | 216, 645 | 209, 645 | 218, 645 | 218, 645 | 204, 645 | 195, 645 | 229, 645 | 209, 645 | 194, 645 | 209, 645 | 239, 645 | 169, 645 | 209, 645 | 239, 645 | 265, 645 | 259, 645 | 248, 645 |
| a? | 265, 885 | 216, 645 | 222, 645 | 314, 885 | 322, 885 | 220, 645 | 232, 645 | 306, 688 | 202, 645 | 196, 645 | 234, 795 | 242, 645 | 124, 645 | 320, 885 | 228, 645 | 259, 645 | 258, 645 | 221, 645 |
| a$\delta$ | 215, -255 | 0, 0 | -13, 0 | -96, -240 | -104, -240 | -16, 0 | -37, 0 | -78, -43 | 8, 0 | -2, 0 | -26, -150 | -2, 0 | 44, 0 | -112, -240 | 11, 0 | 6, 0 | 1, 0 | 27, 0 |
| b | 212, -114 | 177, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | 186, -114 | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | 196, -110 | 236, -110 | 226, -320 | 248, -110 | 251, -277 | 216, -114 |
| b? | 265, -110 | 216, -110 | 222, -214 | 314, -273 | 322, -110 | 220, -110 | 232, -110 | 306, -110 | 202, -110 | 196, -110 | 234, -110 | 242, -110 | 124, -291 | 320, -319 | 228, -319 | 259, -110 | 258, -287 | 221, -110 |
| b$\delta$ | -53, -4 | -38, -4 | -10, 100 | -102, 159 | -110, -4 | -8, -4 | -46, -4 | -94, -4 | 10, -4 | -12, -4 | -22, -4 | 14, -4 | 72, 181 | -84, 209 | -2, -1 | -11, 0 | -7, 10 | -5, -4 |
| it | *ɓ* | *ɔ* | *ɕ* | *ɖ* | *ɗ* | *ɘ* | *ə* | *ɚ* | *ɛ* | *ɜ* | *ɝ* | *ɞ* | *ɟ* | *ɠ* | *ɡ* | *ɢ* | *ɣ* | *ɤ* |
| patch | *ɓ* | *ɔ* | *ɕ* | *ɖ* | *ɗ* | *ɘ* | *ə* | *ɚ* | *ɛ* | *ɜ* | *ɝ* | *ɞ* | *ɟ* | *ɠ* | *ɡ* | *ɢ* | *ɣ* | *ɤ* |
| ideal | a+6, b | x, b | x, b-100 | a+9, d+44 | a, b | x, b | x, b | x+44, b | x, b | x, b | x+132, b | x, b-25 | x, d | a+6, d+31 | x, d+31 | x, b | x, d+45 | x, b+2 |
| a | | 322, 654 | | 218, 645 | 218, 645 | 204, 645 | | | 322, 654 | | | | | | | 265, 645 | | |
| a? | 431, 896 | 332, 645 | 352, 645 | 465, 899 | 464, 890 | 356, 645 | 356, 645 | 418, 687 | 338, 645 | 331, 645 | 413, 775 | 375, 645 | 305, 645 | 431, 896 | 378, 645 | 388, 645 | 387, 645 | 338, 645 |
| a$\delta$ | | -10, 9 | | -247, -254 | -246, -245 | -152, 0 | | | -16, 9 | | | | | | | -123, 0 | | |
| b | 212, -114 | 172, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | | | | 248, -114 | | 177, -114 |
| b? | 218, -110 | 171, -110 | 171, -208 | 215, -273 | 251, -110 | 195, -110 | 195, -110 | 248, -110 | 178, -110 | 170, -110 | 225, -110 | 209, -133 | 100, -319 | 180, -286 | 180, -286 | 227, -110 | 192, -272 | 178, -106 |
| b$\delta$ | -6, -4 | 1, -4 | 41, 94 | -3, 159 | -39, -4 | 17, -4 | 17, -4 | -36, -4 | 34, -4 | 14, -4 | -13, -4 | 47, 19 | | | | 21, -4 | | -1, -8 |
| sb | **ɓ** | **ɔ** | **ɕ** | **ɖ** | **ɗ** | **ɘ** | **ə** | **ɚ** | **ɛ** | **ɜ** | **ɝ** | **ɞ** | **ɟ** | **ɠ** | **ɡ** | **ɢ** | **ɣ** | **ɤ** |
| patch | **ɓ** | **ɔ** | **ɕ** | **ɖ** | **ɗ** | **ɘ** | **ə** | **ɚ** | **ɛ** | **ɜ** | **ɝ** | **ɞ** | **ɟ** | **ɠ** | **ɡ** | **ɢ** | **ɣ** | **ɤ** |
| ideal | a, b | x, b-14 | x, b-101 | a, d+46 | a, b | x, b | x, b | x+50, b | x, b | x, b | a-128, b | x, b | x, d | a+6, d+33 | x, d | x, b | x, d+32 | x, b |
| a | | 199, 645 | | | | | | | 229, 645 | | | | | | | | | |
| a? | 268, 885 | 199, 645 | 232, 645 | 322, 885 | 324, 885 | 230, 645 | 228, 645 | 319, 693 | 212, 645 | 216, 645 | 276, 757 | 249, 645 | 144, 645 | 336, 891 | 235, 645 | 266, 645 | 272, 645 | 202, 645 |
| a$\delta$ | | 0, 0 | | | | | | | 16, 0 | | | | | | | | | |
| b | | 184, -114 | | | | | 212, -114 | | | | | | | | | | | |
| b? | 268, -110 | 199, -122 | 232, -209 | 322, -271 | 324, -110 | 230, -110 | 228, -110 | 319, -110 | 212, -110 | 216, -110 | 276, -110 | 249, -110 | 144, -319 | 336, -284 | 235, -319 | 266, -110 | 272, -285 | 202, -110 |
| b$\delta$ | | -15, 8 | | | | | -16, -4 | | | | | | | | | | | |
| si | ***ɓ*** | ***ɔ*** | ***ɕ*** | ***ɖ*** | ***ɗ*** | ***ɘ*** | ***ə*** | ***ɚ*** | ***ɛ*** | ***ɜ*** | ***ɝ*** | ***ɞ*** | ***ɟ*** | ***ɠ*** | ***ɡ*** | ***ɢ*** | ***ɣ*** | ***ɤ*** |
| patch | ***ɓ*** | ***ɔ*** | ***ɕ*** | ***ɖ*** | ***ɗ*** | ***ɘ*** | ***ə*** | ***ɚ*** | ***ɛ*** | ***ɜ*** | ***ɝ*** | ***ɞ*** | ***ɟ*** | ***ɠ*** | ***ɡ*** | ***ɢ*** | ***ɣ*** | ***ɤ*** |
| ideal | a, b | x, b | x, b-101 | a, d+46 | a, b | x, b | x, b | x+50, b | x, b | x, b | a-128, b | x, b | x, d | a, d+33 | x, d | x, b | x, d+32 | x, b |
| a | | 160, 645 | | 218, 645 | 218, 645 | 204, 645 | | | 209, 645 | 194, 645 | | | | | | 265, 645 | | |
| a? | 284, 890 | 199, 645 | 204, 645 | 302, 890 | 318, 890 | 207, 645 | 218, 645 | 299, 693 | 194, 645 | 194, 645 | 261, 762 | 232, 645 | 141, 645 | 309, 890 | 238, 645 | 244, 645 | 255, 645 | 185, 645 |
| a$\delta$ | | -39, 0 | | -84, -245 | -100, -245 | -3, 0 | | | 15, 0 | 0, 0 | | | | | | 21, 0 | | |
| b | 212, -114 | 177, -114 | 212, -114 | 212, -114 | 212, -114 | 212, -114 | | 212, -114 | 212, -114 | 184, -114 | 212, -114 | 256, -114 | | | | 248, -110 | | 177, -114 |
| b? | 243, -110 | 207, -110 | 192, -209 | 227, -271 | 276, -110 | 216, -110 | 227, -110 | 298, -110 | 203, -110 | 203, -110 | 246, -110 | 241, -110 | 107, -319 | 232, -284 | 204, -319 | 252, -110 | 228, -285 | 194, -110 |
| b$\delta$ | -31, -4 | -30, -4 | 20, 95 | -15, 157 | -64, -4 | -4, -4 | | -86, -4 | 9, -4 | -19, -4 | -34, -4 | 15, -4 | | | | -4, 0 | | -17, -4 |

Table 11: Font metrics for Latin and IPA 7/11

| hex | 0265 | 0266 | 0267 | 0268 | 0269 | 026A | 026B | 026C | 026D | 026E | 026F | 0270 | 0271 | 0272 | 0273 | 0274 | 0275 | 0276 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɥɥ̧ | ɦɦ̧ | ɧɧ̧ | ɨɨ̧ | ɩɩ̧ | ɪɪ̧ | ɬɬ̧ | ɬɬ̧ | ɭɭ̧ | ʞʞ̧ | ɯɯ̧ | ɰɰ̧ | ɱɱ̧ | ɲɲ̧ | ɳɳ̧ | ɴɴ̧ | ɵɵ̧ | ɶɶ̧ |
| patch | ɥɥ̧ | ɦɦ̧ | ɧɧ̧ | ɨɨ̧ | ɩɩ̧ | ɪɪ̧ | ɬɬ̧ | ɬɬ̧ | ɭɭ̧ | ʞʞ̧ | ɯɯ̧ | ɰɰ̧ | ɱɱ̧ | ɲɲ̧ | ɳɳ̧ | ɴɴ̧ | ɵɵ̧ | ɶɶ̧ |
| ideal | x, d | a, b | a, d+11 | a-51, b | x, b | x, b | a, b | a, b | a, d+44 | a, d | x, b-13 | x, d | x, d+11 | x, d+11 | x, d+11 | x, b | x, b | x, b |
| a | 268, 645 | 299, 565 | 299, 565 | 155, 795 | 123, 645 | 133, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 407, 645 | 209, 645 | 383, 645 | 264, 646 | 263, 645 | 209, 645 | 209, 645 | 209, 645 |
| a? | 272, 645 | 262, 885 | 228, 885 | 150, 834 | 170, 645 | 136, 645 | 144, 885 | 182, 885 | 182, 885 | 250, 885 | 390, 645 | 379, 645 | 366, 645 | 234, 645 | 296, 645 | 278, 645 | 252, 645 | 340, 645 |
| a$\delta$ | -4, 0 | 36, -320 | 72, -320 | 6, -39 | -47, 0 | -3, 0 | 65, -240 | 27, -240 | 27, -240 | -42, -240 | 16, 0 | -170, 0 | 18, 0 | 30, 1 | -32, 0 | -70, 0 | -43, 0 | -132, 0 |
| b | 219, -129 | 240, -114 | 240, -114 | 134, -114 | 153, -114 | 139, -114 | 146, -114 | 218, -120 | 196, -110 | 196, -110 | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | 196, -110 | 196, -110 |
| b? | 272, -319 | 262, -110 | 228, -306 | 150, -110 | 170, -110 | 136, -110 | 144, -110 | 182, -110 | 182, -273 | 250, -319 | 390, -121 | 379, -319 | 366, -306 | 234, -306 | 296, -306 | 278, -110 | 252, -110 | 340, -110 |
| b$\delta$ | -52, 190 | -22, -4 | 12, 192 | -16, -4 | -17, -4 | 3, -4 | 2, -4 | 36, -10 | 14, 163 | -54, 209 | 20, 7 | -85, 205 | 18, 201 | 30, 195 | -34, 194 | -32, -4 | -56, 0 | -144, 0 |
| it | *ɥɥ̧* | *ɦɦ̧* | *ɧɧ̧* | *ɨɨ̧* | *ɩɩ̧* | *ɪɪ̧* | *ɬɬ̧* | *ɬɬ̧* | *ɭɭ̧* | *ʞʞ̧* | *ɯɯ̧* | *ɰɰ̧* | *ɱɱ̧* | *ɲɲ̧* | *ɳɳ̧* | *ɴɴ̧* | *ɵɵ̧* | *ɶɶ̧* |
| patch | *ɥɥ̧* | *ɦɦ̧* | *ɧɧ̧* | *ɨɨ̧* | *ɩɩ̧* | *ɪɪ̧* | *ɬɬ̧* | *ɬɬ̧* | *ɭɭ̧* | *ʞʞ̧* | *ɯɯ̧* | *ɰɰ̧* | *ɱɱ̧* | *ɲɲ̧* | *ɳɳ̧* | *ɴɴ̧* | *ɵɵ̧* | *ɶɶ̧* |
| ideal | x, d | a, b | a, d | a-80, b | x, b | x, b | a, b | a, b | a, d+30 | a, d+8 | x, b-15 | x, d | x, d | x, d | x, d | x, b | x, b | x, b |
| a | 258, 645 | 299, 565 | | | 123, 645 | 128, 645 | | | | | 407, 645 | | 383, 645 | 264, 646 | 263, 645 | | | |
| a? | 396, 645 | 449, 890 | 449, 890 | 316, 810 | 266, 645 | 281, 645 | 333, 890 | 368, 890 | 341, 890 | 447, 890 | 517, 645 | 515, 645 | 515, 645 | 396, 645 | 367, 645 | 418, 645 | 379, 645 | 457, 645 |
| a$\delta$ | -138, 0 | -150, -325 | | | -143, 0 | -153, 0 | | | | | -110, 0 | | -132, 0 | -132, 1 | -104, 0 | | | |
| b | 219, -129 | 240, -114 | | 134, -114 | 153, -114 | 139, -114 | 146, -114 | 218, -120 | | | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | | |
| b? | 191, -319 | 236, -110 | 192, -319 | 121, -110 | 106, -110 | 120, -110 | 120, -110 | 156, -110 | 91, -287 | 192, -309 | 354, -123 | 310, -319 | 310, -319 | 191, -319 | 162, -319 | 258, -110 | 218, -110 | 296, -110 |
| b$\delta$ | 28, 190 | 4, -4 | | 13, -4 | 47, -4 | 19, -4 | 26, -4 | 62, -10 | | | 56, 9 | -16, 205 | 73, 214 | 74, 208 | 100, 207 | -11, -4 | | |
| sb | **ɥɥ̧** | **ɦɦ̧** | **ɧɧ̧** | **ɨɨ̧** | **ɩɩ̧** | **ɪɪ̧** | **ɬɬ̧** | **ɬɬ̧** | **ɭɭ̧** | **ʞʞ̧** | **ɯɯ̧** | **ɰɰ̧** | **ɱɱ̧** | **ɲɲ̧** | **ɳɳ̧** | **ɴɴ̧** | **ɵɵ̧** | **ɶɶ̧** |
| patch | **ɥɥ̧** | **ɦɦ̧** | **ɧɧ̧** | **ɨɨ̧** | **ɩɩ̧** | **ɪɪ̧** | **ɬɬ̧** | **ɬɬ̧** | **ɭɭ̧** | **ʞʞ̧** | **ɯɯ̧** | **ɰɰ̧** | **ɱɱ̧** | **ɲɲ̧** | **ɳɳ̧** | **ɴɴ̧** | **ɵɵ̧** | **ɶɶ̧** |
| ideal | x, d | a, b | a, d | a-67, b | x, b | x, b | a, b | a, b | a, d | a, d | x-2, b-17 | x, d | x, d | x, d | x, d | x, b | x, b | x, b |
| a | | | | | | 146, 645 | | | | | 377, 645 | | | | | | | |
| a? | 278, 645 | 274, 885 | 243, 885 | 152, 818 | 164, 645 | 152, 645 | 156, 885 | 190, 885 | 194, 885 | 274, 885 | 354, 641 | 388, 645 | 363, 645 | 246, 645 | 306, 645 | 290, 645 | 260, 645 | 334, 645 |
| a$\delta$ | | | | | | -6, 0 | | | | | 23, 4 | | | | | | | |
| b | | | | 134, -114 | | 156, -114 | 146, -114 | 218, -120 | | | 380, -114 | | | | | | 229, -114 | |
| b? | 278, -319 | 274, -110 | 243, -319 | 152, -110 | 164, -110 | 152, -110 | 156, -110 | 190, -110 | 194, -319 | 274, -319 | 354, -125 | 388, -319 | 363, -319 | 246, -319 | 306, -319 | 290, -110 | 260, -110 | 334, -110 |
| b$\delta$ | | | | -18, -4 | | 4, -4 | -10, -4 | 28, -10 | | | 26, 11 | | | | | | -30, -4 | |
| si | ***ɥɥ̧*** | ***ɦɦ̧*** | ***ɧɧ̧*** | ***ɨɨ̧*** | ***ɩɩ̧*** | ***ɪɪ̧*** | ***ɬɬ̧*** | ***ɬɬ̧*** | ***ɭɭ̧*** | ***ʞʞ̧*** | ***ɯɯ̧*** | ***ɰɰ̧*** | ***ɱɱ̧*** | ***ɲɲ̧*** | ***ɳɳ̧*** | ***ɴɴ̧*** | ***ɵɵ̧*** | ***ɶɶ̧*** |
| patch | ***ɥɥ̧*** | ***ɦɦ̧*** | ***ɧɧ̧*** | ***ɨɨ̧*** | ***ɩɩ̧*** | ***ɪɪ̧*** | ***ɬɬ̧*** | ***ɬɬ̧*** | ***ɭɭ̧*** | ***ʞʞ̧*** | ***ɯɯ̧*** | ***ɰɰ̧*** | ***ɱɱ̧*** | ***ɲɲ̧*** | ***ɳɳ̧*** | ***ɴɴ̧*** | ***ɵɵ̧*** | ***ɶɶ̧*** |
| ideal | x, d | a, b | a, d | a-67, b | x, b | x, b | a, b | a, b | a, d | a, d | x-3, b | x, d | x, d | x, d | x, d | x, b | x, b | x-1, b |
| a | 258, 645 | 299, 565 | 299, 565 | 137, 795 | 123, 645 | 128, 645 | | | | | 407, 645 | | 383, 645 | 264, 646 | 263, 645 | | | |
| a? | 258, 645 | 302, 890 | 301, 890 | 180, 823 | 138, 645 | 137, 645 | 187, 890 | 223, 890 | 187, 890 | 301, 890 | 372, 640 | 373, 645 | 373, 645 | 254, 645 | 254, 645 | 275, 645 | 229, 645 | 313, 642 |
| a$\delta$ | 0, 0 | -3, -325 | -2, -325 | -43, -28 | -15, 0 | -9, 0 | | | | | 35, 5 | | 10, 0 | 10, 1 | 9, 0 | | | |
| b | 219, -129 | 240, -114 | 240, -114 | 136, -110 | 153, -114 | 139, -114 | 146, -114 | 218, -120 | | | 410, -114 | 294, -114 | 383, -105 | 265, -111 | 262, -112 | 247, -114 | | |
| b? | 224, -319 | 260, -110 | 217, -319 | 153, -110 | 147, -110 | 145, -110 | 145, -110 | 181, -110 | 103, -319 | 217, -319 | 382, -110 | 339, -319 | 339, -319 | 220, -319 | 220, -319 | 283, -110 | 238, -110 | 322, -110 |
| b$\delta$ | -5, 190 | -20, -4 | 23, 205 | -17, 0 | 6, -4 | -6, -4 | 1, -4 | 37, -10 | | | 28, -4 | -45, 205 | 44, 214 | 45, 208 | 42, 207 | -36, -4 | | |

Table 12: Font metrics for Latin and IPA 8/11

| hex | 0277 | 0278 | 0279 | 027A | 027B | 027C | 027D | 027E | 027F | 0280 | 0281 | 0282 | 0283 | 0284 | 0285 | 0286 | 0287 | 0288 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll |
| reg | ɷ | ɸ | ɹ | ɺ | ɾ | ɽ | ɻ | ɹ | ɰ | ʀ | ʁ | ʂ | ʃ | ʄ | ɯ | ʃ | ɕ | ʈ |
| patch | ɷ | ɸ | ɹ | ɺ | ɾ | ɽ | ɻ | ɹ | ɰ | ʀ | ʁ | ʂ | ʃ | ʄ | ɯ | ʃ | ɕ | ʈ |
| ideal | x, b | a, d | x, b-13 | a-51, b-13 | x, d+58 | x, d | x, d | x, b | x, b | x, b | x, b | x, d+56 | a, d+18 | a, d+18 | x+121, d+18 | a, d+7 | x, d+79 | a-114, d+11 |
| a | 209, 645 | 209, 645 | 209, 645 | 209, 645 | | 222, 703 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 |
| a? | 357, 645 | 286, 885 | 178, 645 | 180, 834 | 226, 645 | 186, 645 | 186, 645 | 174, 645 | 170, 645 | 238, 645 | 238, 645 | 199, 645 | 140, 885 | 140, 885 | 170, 764 | 150, 885 | 150, 645 | 158, 771 |
| aδ | -148, 0 | -76, -240 | 32, 0 | 30, -189 | | 36, 58 | 22, 0 | 34, 0 | 38, 0 | -30, 0 | -30, 0 | 10, 0 | 68, -240 | 68, -240 | 38, -119 | 60, -240 | 59, 0 | 52, -126 |
| b | 341, -114 | 196, -110 | 190, -114 | 203, -114 | 196, -110 | 127, -305 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 |
| b? | 357, -110 | 286, -319 | 178, -121 | 180, -121 | 226, -259 | 186, -319 | 186, -319 | 174, -110 | 170, -110 | 238, -110 | 238, -110 | 199, -261 | 140, -299 | 140, -299 | 170, -299 | 150, -310 | 150, -238 | 158, -306 |
| bδ | -16, -4 | -90, 209 | 12, 7 | 24, 7 | -30, 149 | -60, 14 | 10, 209 | 22, 0 | 26, 0 | -42, 0 | -42, 0 | -3, 151 | 56, 189 | 56, 189 | 26, 189 | 46, 200 | 46, 128 | 38, 196 |
| it | *ɷ* | *ɸ* | *ɹ* | *ɺ* | *ɾ* | *ɽ* | *ɻ* | *ɹ* | *ɰ* | *ʀ* | *ʁ* | *ʂ* | *ʃ* | *ʄ* | *ɯ* | *ʃ* | *ɕ* | *ʈ* |
| patch | *ɷ* | *ɸ* | *ɹ* | *ɺ* | *ɾ* | *ɽ* | *ɻ* | *ɹ* | *ɰ* | *ʀ* | *ʁ* | *ʂ* | *ʃ* | *ʄ* | *ɯ* | *ʃ* | *ɕ* | *ʈ* |
| ideal | x, b-18 | a, d+32 | x, b | a-71, b-19 | x, d+58 | x, d+34 | x, d+38 | x, b | x, b | x, b | x, b | x, d+52 | a+10, d+18 | a+10, d+18 | a+10, d+18 | a+10, d+34 | x, d+78 | a-114, d+6 |
| a | | | | | | 222, 703 | | | | | | | | | | | | |
| a? | 494, 645 | 481, 890 | 315, 645 | 362, 819 | 352, 645 | 325, 645 | 320, 645 | 311, 645 | 307, 645 | 358, 645 | 349, 645 | 314, 645 | 343, 900 | 343, 900 | 343, 900 | 343, 900 | 288, 645 | 314, 776 |
| aδ | | | | | | -103, 58 | | | | | | | | | | | | |
| b | 341, -114 | | 190, -114 | 203, -114 | | 127, -305 | | | | | | | | | | | | |
| b? | 330, -126 | 231, -285 | 155, -110 | 161, -127 | 160, -259 | 128, -283 | 124, -279 | 151, -110 | 147, -110 | 197, -110 | 189, -110 | 121, -265 | 88, -299 | 88, -299 | 88, -299 | 92, -283 | 100, -239 | 83, -311 |
| bδ | 11, 12 | | 35, -4 | 42, 13 | | -1, -22 | | | | | | | | | | | | |
| sb | **ɷ** | **ɸ** | **ɹ** | **ɺ** | **ɾ** | **ɽ** | **ɻ** | **ɹ** | **ɰ** | **ʀ** | **ʁ** | **ʂ** | **ʃ** | **ʄ** | **ɯ** | **ʃ** | **ɕ** | **ʈ** |
| patch | **ɷ** | **ɸ** | **ɹ** | **ɺ** | **ɾ** | **ɽ** | **ɻ** | **ɹ** | **ɰ** | **ʀ** | **ʁ** | **ʂ** | **ʃ** | **ʄ** | **ɯ** | **ʃ** | **ɕ** | **ʈ** |
| ideal | x, b | a, d+34 | x-2, b-17 | a-69, b | x, d+60 | x, d+36 | x, d | x, b | x, b | x, b | x, b | x, d+53 | a, d+21 | a, d+21 | a, d+21 | a, d | x, d+67 | a-103, d |
| a | | | | | | | | | | | | | | | | | | |
| a? | 369, 645 | 297, 885 | 144, 641 | 205, 816 | 238, 645 | 206, 645 | 198, 645 | 186, 645 | 182, 645 | 236, 645 | 220, 645 | 201, 645 | 152, 885 | 152, 885 | 152, 885 | 162, 885 | 157, 645 | 182, 782 |
| aδ | | | | | | | | | | | | | | | | | | |
| b | | | 190, -114 | | | | | | | | | | | | | | | |
| b? | 369, -110 | 297, -283 | 144, -125 | 205, -110 | 238, -257 | 206, -281 | 198, -319 | 186, -110 | 182, -110 | 236, -110 | 220, -110 | 201, -264 | 152, -296 | 152, -296 | 152, -296 | 162, -319 | 157, -250 | 182, -319 |
| bδ | | | 46, 11 | | | | | | | | | | | | | | | |
| si | ***ɷ*** | ***ɸ*** | ***ɹ*** | ***ɺ*** | ***ɾ*** | ***ɽ*** | ***ɻ*** | ***ɹ*** | ***ɰ*** | ***ʀ*** | ***ʁ*** | ***ʂ*** | ***ʃ*** | ***ʄ*** | ***ɯ*** | ***ʃ*** | ***ɕ*** | ***ʈ*** |
| patch | ***ɷ*** | ***ɸ*** | ***ɹ*** | ***ɺ*** | ***ɾ*** | ***ɽ*** | ***ɻ*** | ***ɹ*** | ***ɰ*** | ***ʀ*** | ***ʁ*** | ***ʂ*** | ***ʃ*** | ***ʄ*** | ***ɯ*** | ***ʃ*** | ***ɕ*** | ***ʈ*** |
| ideal | x, b | a, d+34 | x-3, b | a-69, b | x, d+60 | x, d+36 | x, d | x, b | x, b | x-1, b | x, b | x, d+52 | a, d+21 | a, d+21 | a, d+21 | a, d | x, d+67 | a-103, d |
| a | | | | | | 222, 703 | | | | | | | | | | | | |
| a? | 350, 645 | 328, 890 | 171, 640 | 217, 821 | 194, 645 | 181, 645 | 177, 645 | 167, 645 | 167, 645 | 214, 642 | 214, 645 | 178, 645 | 195, 890 | 195, 890 | 195, 890 | 189, 890 | 149, 645 | 172, 787 |
| aδ | | | | | | 41, 58 | | | | | | | | | | | | |
| b | 341, -114 | | 190, -114 | 203, -114 | | 127, -305 | | | | | | | | | | | | |
| b? | 359, -110 | 251, -283 | 180, -110 | 190, -110 | 172, -257 | 155, -281 | 143, -319 | 176, -110 | 176, -110 | 223, -110 | 223, -110 | 155, -265 | 116, -296 | 116, -296 | 116, -296 | 105, -319 | 129, -250 | 109, -319 |
| bδ | -18, -4 | | 10, -4 | 13, -4 | | -28, -24 | | | | | | | | | | | | |

Table 13: Font metrics for Latin and IPA 9/11

| hex | 0289 | 028A | 028B | 028C | 028D | 028E | 028F | 0290 | 0291 | 0292 | 0293 | 0294 | 0295 | 0296 | 0297 | 0299 | 029B | 029C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Lu | Lu | Lu | Ll | Ll | Ll | Ll |
| reg | ʉ̵ | ʊ̵ | ʋ̵ | ʌ̵ | ʍ̵ | ʎ̵ | ʏ̵ | ʐ̵ | ʑ̵ | ʒ̵ | ʓ̵ | ʔ̵ | ʕ̵ | ʖ̵ | ɕ̵ | ʙ̵ | ɠ̵ | ʜ̵ |
| patch | ʉ̵ | ʊ̵ | ʋ̵ | ʌ̵ | ʍ̵ | ʎ̵ | ʏ̵ | ʐ̵ | ʑ̵ | ʒ̵ | ʓ̵ | ʔ̵ | ʕ̵ | ʖ̵ | ɕ̵ | ʙ̵ | ɠ̵ | ʜ̵ |
| ideal | x, b | x, b | x, b | x, b | x, b | a-37, b | x, b | x, d+47 | x+19, d+110 | x, d | x, d | c, b | c, b | c, b | x, d+11 | x, b | a-121, b | x, b |
| a | 209, 645 | 209, 645 | 259, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 218, 646 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 |
| a? | 271, 645 | 254, 645 | 235, 645 | 244, 645 | 367, 645 | 248, 848 | 236, 645 | 262, 645 | 218, 662 | 216, 645 | 220, 645 | 211, 850 | 221, 850 | 208, 850 | 221, 645 | 232, 645 | 294, 764 | 293, 645 |
| a$\delta$ | -62, 0 | -44, 0 | 24, 0 | -35, 0 | -158, 0 | -39, -203 | -27, 0 | -54, 0 | -8, -17 | 2, 1 | -11, 0 | -2, -205 | -12, -205 | 1, -205 | -12, 0 | -24, 0 | -85, -119 | -84, 0 |
| b | 196, -110 | 258, -114 | 251, -114 | 196, -110 | 196, -110 | 196, -110 | 239, -107 | 196, -110 | 196, -110 | 185, -319 | 196, -110 | 202, -107 | 230, -107 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 305, -110 |
| b? | 271, -110 | 254, -110 | 235, -110 | 244, -110 | 367, -110 | 248, -110 | 236, -110 | 262, -270 | 218, -207 | 216, -319 | 220, -319 | 211, -110 | 221, -110 | 208, -110 | 221, -306 | 232, -110 | 294, -110 | 293, -110 |
| b$\delta$ | -75, 0 | 4, -4 | 16, -4 | -48, 0 | -171, 0 | -52, 0 | 3, 3 | -66, 160 | -22, 97 | -32, 0 | -24, 209 | -9, 3 | 9, 3 | -12, 0 | -25, 196 | -36, 0 | -98, 0 | 12, 0 |
| it | *ʉ̵* | *ʊ̵* | *ʋ̵* | *ʌ̵* | *ʍ̵* | *ʎ̵* | *ʏ̵* | *ʐ̵* | *ʑ̵* | *ʒ̵* | *ʓ̵* | *ʔ̵* | *ʕ̵* | *ʖ̵* | *ɕ̵* | *ʙ̵* | *ɠ̵* | *ʜ̵* |
| patch | *ʉ̵* | *ʊ̵* | *ʋ̵* | *ʌ̵* | *ʍ̵* | *ʎ̵* | *ʏ̵* | *ʐ̵* | *ʑ̵* | *ʒ̵* | *ʓ̵* | *ʔ̵* | *ʕ̵* | *ʖ̵* | *ɕ̵* | *ʙ̵* | *ɠ̵* | *ʜ̵* |
| ideal | x, b | x, b-13 | x, b-16 | x, b | x, b | a-52, b | x, b | x, d+41 | x, d+110 | x, d | x, d-80 | c+27, b | c+38, b | c, b-23 | x, d | x, b | a-94, b | x, b |
| a | | | 259, 645 | | | | | | | 328, 646 | | | | | | | | |
| a? | 398, 645 | 392, 645 | 382, 645 | 383, 645 | 506, 645 | 433, 838 | 372, 645 | 343, 645 | 352, 645 | 350, 645 | 355, 645 | 373, 850 | 373, 850 | 373, 850 | 411, 645 | 352, 645 | 410, 796 | 416, 645 |
| a$\delta$ | | | -123, 0 | | | | | | | -22, 1 | | | | | | | | |
| b | | 258, -114 | 251, -114 | | | | 239, -107 | | | 130, -319 | | 202, -107 | 230, -107 | | | | | 273, -110 |
| b? | 238, -110 | 229, -121 | 218, -124 | 222, -110 | 346, -110 | 231, -110 | 212, -110 | 147, -276 | 171, -207 | 145, -319 | 133, -397 | 169, -110 | 169, -110 | 164, -131 | 206, -319 | 192, -110 | 217, -110 | 256, -110 |
| b$\delta$ | | 29, 7 | 33, 10 | | | | 27, 3 | | | -15, 0 | | 33, 3 | 61, 3 | | | | | 17, 0 |
| sb | **ʉ̵** | **ʊ̵** | **ʋ̵** | **ʌ̵** | **ʍ̵** | **ʎ̵** | **ʏ̵** | **ʐ̵** | **ʑ̵** | **ʒ̵** | **ʓ̵** | **ʔ̵** | **ʕ̵** | **ʖ̵** | **ɕ̵** | **ʙ̵** | **ɠ̵** | **ʜ̵** |
| patch | **ʉ̵** | **ʊ̵** | **ʋ̵** | **ʌ̵** | **ʍ̵** | **ʎ̵** | **ʏ̵** | **ʐ̵** | **ʑ̵** | **ʒ̵** | **ʓ̵** | **ʔ̵** | **ʕ̵** | **ʖ̵** | **ɕ̵** | **ʙ̵** | **ɠ̵** | **ʜ̵** |
| ideal | x, b | x, b | x, b | x, b | x, b | a-36, b | x, b | x, d+41 | x, d+107 | x, d | x, d-96 | c, b | c, b | c, b | x, d | x, b | a-121, b | x, b |
| a | | 266, 645 | | | | | | | | 234, 645 | | | | | | | | |
| a? | 274, 645 | 267, 645 | 256, 645 | 231, 645 | 356, 645 | 243, 849 | 244, 645 | 254, 645 | 230, 645 | 227, 645 | 236, 645 | 206, 850 | 206, 850 | 206, 850 | 273, 645 | 246, 645 | 290, 764 | 316, 645 |
| a$\delta$ | | -1, 0 | | | | | | | | 7, 0 | | | | | | | | |
| b | | | | | | | | | | 193, -319 | | | | | | | | 351, -110 |
| b? | 274, -110 | 267, -110 | 256, -110 | 231, -110 | 356, -110 | 243, -110 | 244, -110 | 254, -276 | 230, -210 | 227, -319 | 236, -413 | 206, -110 | 206, -110 | 206, -110 | 273, -319 | 246, -110 | 290, -110 | 316, -110 |
| b$\delta$ | | | | | | | | | | -34, 0 | | | | | | | | 35, 0 |
| si | ***ʉ̵*** | ***ʊ̵*** | ***ʋ̵*** | ***ʌ̵*** | ***ʍ̵*** | ***ʎ̵*** | ***ʏ̵*** | ***ʐ̵*** | ***ʑ̵*** | ***ʒ̵*** | ***ʓ̵*** | ***ʔ̵*** | ***ʕ̵*** | ***ʖ̵*** | ***ɕ̵*** | ***ʙ̵*** | ***ɠ̵*** | ***ʜ̵*** |
| patch | ***ʉ̵*** | ***ʊ̵*** | ***ʋ̵*** | ***ʌ̵*** | ***ʍ̵*** | ***ʎ̵*** | ***ʏ̵*** | ***ʐ̵*** | ***ʑ̵*** | ***ʒ̵*** | ***ʓ̵*** | ***ʔ̵*** | ***ʕ̵*** | ***ʖ̵*** | ***ɕ̵*** | ***ʙ̵*** | ***ɠ̵*** | ***ʜ̵*** |
| ideal | x, b | x-1, b | x, b | x, b | x, b | a-56, b | x, b | x, d+38 | x, d+107 | x, d | x, d-96 | c, b | c, b | c, b | x, d+15 | x, b | a-94, b | x+28, b |
| a | | | 259, 645 | | | | | | | 218, 646 | | | | | | | | |
| a? | 255, 645 | 248, 642 | 241, 645 | 239, 645 | 363, 645 | 286, 834 | 228, 645 | 202, 645 | 208, 645 | 206, 645 | 211, 645 | 227, 850 | 227, 850 | 227, 850 | 268, 645 | 213, 645 | 278, 796 | 342, 671 |
| a$\delta$ | | | 18, 0 | | | | | | | 12, 1 | | | | | | | | |
| b | | 258, -114 | 251, -114 | | | | 239, -107 | | | 141, -319 | | 202, -107 | 230, -107 | | | | | 327, -110 |
| b? | 264, -110 | 257, -110 | 249, -110 | 247, -110 | 371, -110 | 256, -110 | 237, -110 | 176, -279 | 196, -210 | 172, -319 | 157, -413 | 194, -110 | 194, -110 | 194, -110 | 237, -302 | 221, -110 | 256, -110 | 345, -110 |
| b$\delta$ | | 1, -4 | 2, -4 | | | | 2, 3 | | | -31, 0 | | 8, 3 | 36, 3 | | | | | -18, 0 |

Table 14: Font metrics for Latin and IPA 10/11

| hex | 029D | 029F | 02A0 | 02A1 | 02A2 | 02AE | 02AF | 03B2 | 03B8 | 03C7 | 1E9A | 1E9C | 1E9D | 1EFA | 1EFF | 2132 | 2C60 | 2C61 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Ll | Lu | Ll | Lu | Lu | Ll |
| reg | ɟʝɟ | ʟɻʟ | ɠɢɠ | ʔʔʔ | ʕʕʕ | ɥɣɥ | ɥɣɥ | βββ | θθθ | χχ̇χ | aʾaʾaʾ | fff | fff | ⅬⅬ̈Ⅼ | yẏy | ⅎⅎ | ɬɬɬ | ɨɨ |
| patch | ɟʝɟ | ʟɻʟ | ɠɢɠ | ʔʔʔ | ʕʕʕ | ɥɣɥ | ɥɣɥ | ββ̇β | θθ̇θ | χχ̇χ | aʾaʾaʾ | fff | fff | ⅬⅬ̈Ⅼ | yẏy | ⅎⅎ | ɬɬɬ | ɨɨ |
| ideal | a-51, d+55 | x, b | a, d | a-42, b | a-42, b | x, d | x, d | a, d | a, b | x, d | a-47, b | a, b | a, b | c, b | x, d | c, b | c, b | a, b |
| a | 209, 645 | 209, 645 | 238, 645 | 209, 645 | 209, 645 | 278, -214 | 278, -214 | | | | | | | 276, 804 | 270, 644 | 201, -203 | | |
| a? | 126, 834 | 207, 645 | 320, 885 | 211, 843 | 221, 843 | 288, 645 | 328, 645 | 262, 885 | 245, 885 | 254, 645 | 272, 838 | 182, 885 | 190, 885 | 388, 850 | 256, 645 | 241, 850 | 260, 850 | 132, 885 |
| a$\delta$ | 83, -189 | 2, 0 | -82, -240 | -2, -198 | -12, -198 | -10, -859 | -50, -859 | | | | | | | -112, -46 | 14, -1 | -40, -1053 | | |
| b | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 318, 542 | 318, 542 | 275, -114 | | | | | | 387, -110 | | 276, 757 | | |
| b? | 126, -262 | 207, -110 | 320, -319 | 211, -110 | 221, -110 | 288, -319 | 328, -319 | 262, -319 | 245, -110 | 254, -319 | 272, -110 | 182, -110 | 190, -110 | 388, -110 | 256, -319 | 241, -110 | 260, -110 | 132, -110 |
| b$\delta$ | 70, 152 | -11, 0 | -124, 209 | -15, 0 | -25, 0 | 30, 861 | -10, 861 | 14, 205 | | | | | | -1, 0 | | 35, 867 | | |
| it | *ɟʝɟ* | *ʟɻʟ* | *ɠɢɠ* | *ʔʔʔ* | *ʕʕʕ* | *ɥɣɥ* | *ɥɣɥ* | *βββ* | *θθθ* | *χχ̇χ* | *aʾaʾaʾ* | *fff* | *fff* | | | | *ɬɬɬ* | *ɨɨ* |
| patch | *ɟʝɟ* | *ʟɻʟ* | *ɠɢɠ* | *ʔʔʔ* | *ʕʕʕ* | *ɥɣɥ* | *ɥɣɥ* | *ββ̇β* | *θθ̇θ* | *χχ̇χ* | *aʾaʾaʾ* | *fff* | *fff* | | | | *ɬɬɬ* | *ɨɨ* |
| ideal | a-80, d+63 | x, b | a-19, d | a-26, b | a-15, b | x, d+12 | x, d-9 | a, d | a, b | x, d | a-47, b | a, d | a, d | | | | c, b | a, b |
| a | | | 238, 645 | | | 278, -214 | 278, -214 | | | | | | | | | | | |
| a? | 303, 810 | 355, 645 | 460, 871 | 376, 864 | 378, 875 | 422, 645 | 433, 645 | 425, 890 | 413, 890 | 372, 645 | 489, 843 | 339, 890 | 339, 890 | | | | 444, 850 | 330, 890 |
| a$\delta$ | | | -222, -226 | | | -144, -859 | -155, -859 | | | | | | | | | | | |
| b | | | | | | 318, 542 | 318, 542 | 275, -114 | | | | | | | | | | |
| b? | 77, -254 | 194, -110 | 207, -319 | 169, -110 | 169, -110 | 220, -305 | 227, -326 | 168, -319 | 200, -110 | 167, -319 | 286, -110 | 82, -319 | 82, -319 | | | | 240, -110 | 117, -110 |
| b$\delta$ | | | | | | 98, 847 | 91, 868 | 107, 205 | | | | | | | | | | |
| sb | **ɟʝɟ** | **ʟɻʟ** | **ɠɢɠ** | **ʔʔʔ** | **ʕʕʕ** | **ɥɣɥ** | **ɥɣɥ** | **βββ** | **θθθ** | **χχ̇χ** | **aʾaʾaʾ** | | | | | | **ɬɬɬ** | **ɨɨ** |
| patch | **ɟʝɟ** | **ʟɻʟ** | **ɠɢɠ** | **ʔʔʔ** | **ʕʕʕ** | **ɥɣɥ** | **ɥɣɥ** | **ββ̇β** | **θθ̇θ** | **χχ̇χ** | **aʾaʾaʾ** | | | | | | **ɬɬɬ** | **ɨɨ** |
| ideal | a-64, d+65 | x, b | a, d | a-40, b | a-40, b | x, d | x, d | a, d | a-17, b | x, d+42 | a-45, b | | | | | | c, b | a, b |
| a | | | | | | | | | | | | | | | | | | |
| a? | 132, 821 | 219, 645 | 337, 885 | 206, 845 | 206, 845 | 300, 645 | 340, 645 | 290, 885 | 249, 868 | 279, 645 | 288, 840 | | | | | | 266, 850 | 144, 885 |
| a$\delta$ | | | | | | | | | | | | | | | | | | |
| b | | | | | | | | | | | | | | | | | | |
| b? | 132, -252 | 219, -110 | 337, -319 | 206, -110 | 206, -110 | 300, -319 | 340, -319 | 290, -319 | 249, -110 | 279, -275 | 288, -110 | | | | | | 266, -110 | 144, -110 |
| b$\delta$ | | | | | | | | | | | | | | | | | | |
| si | ***ɟʝɟ*** | ***ʟɻʟ*** | ***ɠɢɠ*** | ***ʔʔʔ*** | ***ʕʕʕ*** | ***ɥɣɥ*** | ***ɥɣɥ*** | ***βββ*** | ***θθθ*** | ***χχ̇χ*** | ***aʾaʾaʾ*** | | | | | | ***ɬɬɬ*** | ***ɨɨ*** |
| patch | ***ɟʝɟ*** | ***ʟɻʟ*** | ***ɠɢɠ*** | ***ʔʔʔ*** | ***ʕʕʕ*** | ***ɥɣɥ*** | ***ɥɣɥ*** | ***ββ̇β*** | ***θθ̇θ*** | ***χχ̇χ*** | ***aʾaʾaʾ*** | | | | | | ***ɬɬɬ*** | ***ɨɨ*** |
| ideal | a-78, d+65 | x, b | a, d | a-40, b | a-40, b | x, d | x, d | a, d | a-17, b | x, d+42 | a-46, b | | | | | | c, b | a, b |
| a | | | 238, 645 | | | 278, -214 | 278, -214 | | | | | | | | | | | |
| a? | 158, 812 | 211, 645 | 318, 890 | 227, 850 | 227, 850 | 279, 645 | 289, 645 | 302, 890 | 277, 873 | 256, 645 | 371, 844 | | | | | | 300, 850 | 184, 890 |
| a$\delta$ | | | -80, -245 | | | -1, -859 | -11, -859 | | | | | | | | | | | |
| b | | | | | | 318, 542 | 318, 542 | 275, -114 | | | | | | | | | | |
| b? | 103, -252 | 219, -110 | 234, -319 | 194, -110 | 194, -110 | 245, -319 | 255, -319 | 218, -319 | 239, -110 | 231, -275 | 339, -110 | | | | | | 267, -110 | 142, -110 |
| b$\delta$ | | | | | | 73, 861 | 63, 861 | 57, 205 | | | | | | | | | | |

Table 15: Font metrics for Latin and IPA 11/11

| hex | 2C62 | 2C63 | 2C64 | 2C65 | 2C66 | 2C67 | 2C68 | 2C69 | 2C6A | 2C6B | 2C6C | 2C6E | A765 | A793 | AB52 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Lu | Lu | Lu | Ll | Ll | Lu | Ll | Lu | Ll | Lu | Ll | Lu | Ll | Ll | Ll |
| reg | ⱢꞍ | Ᵽꞑ | ⱤꞒ | ⱥɑ | ⱦꞇ | Ⱨꞕ | ⱨꞔ | ⱪꞖ | Ⱪꞗ | ⱬꞙ | Ⱬꞓ | ⱭꞚ | Ⱶꞛ | ꜧꞕ | ꬲꬳ |
| patch | ⱢꞍ | Ᵽꞑ | ⱤꞒ | ⱥɑ | ⱦꞇ | Ⱨꞕ | ⱨꞔ | ⱪꞖ | Ⱪꞗ | ⱬꞙ | Ⱬꞓ | ⱭꞚ | Ⱶꞛ | ꜧꞕ | ꬲꬳ |
| ideal | c, b | c, b | c, d+60 | x+47, b-62 | a-116, b-62 | c, d+91 | a, d+104 | c, d+91 | a, d+104 | c+22, d+91 | x+22, d+104 | c, d+60 | a, d | x, b | x, b |
| a | | | | | | | 299, 645 | 336, 798 | 275, 646 | 320, 803 | 218, 646 | 431, 850 | | 327, 645 | 376, 645 |
| a? | 235, 850 | 260, 850 | 301, 850 | 246, 690 | 146, 769 | 345, 850 | 262, 885 | 326, 850 | 259, 885 | 322, 850 | 206, 665 | 399, 850 | 240, 885 | 260, 645 | 332, 645 |
| a$\delta$ | | | | | | | 36, -240 | 10, -52 | 16, -239 | -2, -47 | 12, -19 | 32, 0 | | 67, 0 | 44, 0 |
| b | | | | | | | 259, -111 | 342, -109 | 263, -107 | 287, -107 | 204, -111 | 393, -110 | | 321, -110 | 360, -110 |
| b? | 235, -110 | 260, -110 | 301, -257 | 246, -170 | 146, -170 | 345, -226 | 262, -213 | 326, -226 | 259, -213 | 322, -226 | 206, -213 | 399, -257 | 240, -319 | 260, -110 | 332, -110 |
| b$\delta$ | | | | | | | -4, 102 | 16, 117 | 4, 106 | -34, 119 | -2, 102 | -6, 147 | | 61, 0 | 28, 0 |
| it | *ⱢꞍ* | *Ᵽꞑ* | *ⱤꞒ* | *ⱥɑ* | *ⱦꞇ* | *Ⱨꞕ* | *ⱨꞔ* | *ⱪꞖ* | *Ⱪꞗ* | *ⱬꞙ* | *Ⱬꞓ* | *ⱭꞚ* | *Ⱶꞛ* | *ꜧꞕ* | *ꬲꬳ* |
| patch | *ⱢꞍ* | *Ᵽꞑ* | *ⱤꞒ* | *ⱥɑ* | *ⱦꞇ* | *Ⱨꞕ* | *ⱨꞔ* | *ⱪꞖ* | *Ⱪꞗ* | *ⱬꞙ* | *Ⱬꞓ* | *ⱭꞚ* | *Ⱶꞛ* | *ꜧꞕ* | *ꬲꬳ* |
| ideal | c, b | c, b | c, d+12 | x+25, b-28 | x+132, b-28 | c, b-115 | a, b-115 | c, b-115 | a, b-115 | c+22, b-99 | x+22, b-99 | | | | |
| a | | | | | | | 299, 645 | 336, 798 | 275, 646 | 320, 803 | 218, 646 | | | | |
| a? | 444, 850 | 430, 850 | 465, 850 | 385, 668 | 364, 775 | 528, 850 | 448, 890 | 516, 850 | 453, 890 | 490, 850 | 347, 665 | | | | |
| a$\delta$ | | | | | | | -149, -245 | -180, -52 | -178, -244 | -170, -47 | -129, -19 | | | | |
| b | | | | | | | 259, -111 | 342, -109 | 263, -107 | 287, -107 | 204, -111 | | | | |
| b? | 240, -110 | 226, -110 | 220, -305 | 214, -136 | 170, -136 | 300, -223 | 211, -223 | 288, -223 | 216, -223 | 265, -207 | 162, -207 | | | | |
| b$\delta$ | | | | | | | 48, 112 | 54, 114 | 47, 116 | 22, 100 | 42, 96 | | | | |
| sb | **ⱢꞍ** | **Ᵽꞑ** | **ⱤꞒ** | **ⱥɑ** | **ⱦꞇ** | **Ⱨꞕ** | **ⱨꞔ** | **ⱪꞖ** | **Ⱪꞗ** | **ⱬꞙ** | **Ⱬꞓ** | **ⱭꞚ** | **Ⱶꞛ** | **ꜧꞕ** | **ꬲꬳ** |
| patch | **ⱢꞍ** | **Ᵽꞑ** | **ⱤꞒ** | **ⱥɑ** | **ⱦꞇ** | **Ⱨꞕ** | **ⱨꞔ** | **ⱪꞖ** | **Ⱪꞗ** | **ⱬꞙ** | **Ⱬꞓ** | **ⱭꞚ** | **Ⱶꞛ** | **ꜧꞕ** | **ꬲꬳ** |
| ideal | c, b | c, b | c, d | x+42, b-62 | a-103, b-62 | c, b-114 | a, b-114 | c, b-113 | a, b-114 | c+25, b-98 | x+24, b-98 | | | | |
| a | | | | | | | | | | | | | | | |
| a? | 238, 850 | 276, 850 | 292, 850 | 255, 685 | 146, 782 | 345, 850 | 264, 885 | 345, 850 | 273, 885 | 327, 850 | 212, 667 | | | | |
| a$\delta$ | | | | | | | | | | | | | | | |
| b | | | | | | | | | | | | | | | |
| b? | 238, -110 | 276, -110 | 292, -319 | 255, -170 | 146, -170 | 345, -222 | 264, -222 | 345, -221 | 273, -222 | 327, -206 | 212, -206 | | | | |
| b$\delta$ | | | | | | | | | | | | | | | |
| si | ***ⱢꞍ*** | ***Ᵽꞑ*** | ***ⱤꞒ*** | ***ⱥɑ*** | ***ⱦꞇ*** | ***Ⱨꞕ*** | ***ⱨꞔ*** | ***ⱪꞖ*** | ***Ⱪꞗ*** | ***ⱬꞙ*** | ***Ⱬꞓ*** | ***ⱭꞚ*** | ***Ⱶꞛ*** | ***ꜧꞕ*** | ***ꬲꬳ*** |
| patch | ***ⱢꞍ*** | ***Ᵽꞑ*** | ***ⱤꞒ*** | ***ⱥɑ*** | ***ⱦꞇ*** | ***Ⱨꞕ*** | ***ⱨꞔ*** | ***ⱪꞖ*** | ***Ⱪꞗ*** | ***ⱬꞙ*** | ***Ⱬꞓ*** | ***ⱭꞚ*** | ***Ⱶꞛ*** | ***ꜧꞕ*** | ***ꬲꬳ*** |
| ideal | c, b | c, b | c, d+15 | x+42, b-62 | a-125, b-62 | c, b-114 | a, b-114 | c, b-113 | a, b-114 | c+25, b-98 | x, b-98 | | | | |
| a | | | | | | | 299, 645 | 336, 798 | 275, 646 | 320, 803 | 218, 646 | | | | |
| a? | 300, 850 | 296, 850 | 307, 850 | 234, 685 | 187, 765 | 370, 850 | 290, 890 | 358, 850 | 295, 890 | 332, 850 | 187, 645 | | | | |
| a$\delta$ | | | | | | | 9, -245 | -22, -52 | -20, -244 | -12, -47 | 31, 1 | | | | |
| b | | | | | | | 259, -111 | 342, -109 | 263, -107 | 287, -107 | 204, -111 | | | | |
| b? | 267, -110 | 263, -110 | 235, -302 | 222, -170 | 159, -170 | 314, -222 | 225, -222 | 302, -221 | 230, -222 | 279, -206 | 176, -206 | | | | |
| b$\delta$ | | | | | | | 34, 111 | 40, 112 | 33, 115 | 8, 99 | 28, 95 | | | | |

Table 16: Font metrics for above marks 1/2

| hex | 0300 | 0301 | 0302 | 0303 | 0304 | 0306 | 0307 | 0308 | 0309 | 030A | 030B | 030C | 030D | 030F | 0310 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | ǹH̃ | ńH́ | n̂H̃ | ñH̃ | n̄H̃ | n̆H̃ | ṅH̃ | n̈H̃ | n̉H̃ | n̊H̃ | ő H̃ | ňH̃ | n̍H̃ | n̏H̃ | n̑H̃ |
| patch | ǹH̃ | ńH́ | n̂H̃ | ñH̃ | n̄H̃ | n̆H̃ | ṅH̃ | n̈H̃ | n̉H̃ | n̊H̃ | ő H̃ | ňH̃ | n̍H̃ | n̏H̃ | n̑H̃ |
| m | -142, 683 | -191, 682 | -139, 697 | -149, 697 | -209, 644 | -161, 681 | -180, 667 | -210, 645 | -160, 765 | -184, 708 | -219, 714 | -205, 676 | -155, 688 | -140, 758 | -251, 675 |
| m? | -159, 690 | -160, 691 | -140, 697 | -149, 711 | -212, 678 | -161, 672 | -180, 667 | -210, 669 | -161, 755 | -182, 711 | -170, 700 | -204, 676 | -154, 691 | -168, 724 | -251, 672 |
| m$\delta$ | 17, -7 | -31, -9 | 1, 0 | 0, -14 | 3, -34 | 0, 9 | 0, 0 | 0, -24 | 1, 10 | -2, -3 | -49, 14 | -1, 0 | -1, -3 | 28, 34 | 0, 3 |
| it | *ǹH* | *ńH* | *n̂H* | *ñH̃* | *n̄H̃* | *n̆H* | *ṅH̃* | *n̈H̃* | *n̉H̃* | *n̊H̃* | *ő H̃* | *ňH̃* | *n̍H̃* | *n̏H̃* | *n̑H̃* |
| patch | *ǹH̃* | *ńH́* | *n̂H̃* | *ñH̃* | *n̄H̃* | *n̆H̃* | *ṅH̃* | *n̈H̃* | *n̉H̃* | *n̊H̃* | *ő H̃* | *ňH̃* | *n̍H̃* | *n̏H̃* | *n̑H̃* |
| m | -3, 683 | -52, 682 | 3, 697 | -17, 697 | -78, 645 | -22, 681 | -40, 689 | -79, 645 | -4, 765 | -40, 708 | -74, 714 | -67, 676 | -15, 688 | 14, 758 | -114, 675 |
| m? | -18, 689 | -18, 690 | 3, 696 | -3, 716 | -71, 680 | -24, 671 | -35, 698 | -74, 669 | -8, 754 | -37, 712 | -26, 699 | -66, 675 | -13, 690 | -19, 723 | -114, 671 |
| m$\delta$ | 15, -6 | -34, -8 | 0, 1 | -14, -19 | -7, -35 | 2, 10 | -5, -9 | -5, -24 | 4, 11 | -3, -4 | -48, 15 | -1, 1 | -2, -2 | 33, 35 | 0, 4 |
| sb | **ǹH** | **ńH** | **n̂H** | **ñH** | **n̄H** | **n̆H** | **ṅH** | **n̈H** | **n̉H** | **n̊H** | **ő H** | **ňH** | **n̍H** | **n̏H** | **n̑H** |
| patch | **ǹH̃** | **ńH́** | **n̂H̃** | **ñH̃** | **n̄H̃** | **n̆H̃** | **ṅH̃** | **n̈H̃** | **n̉H̃** | **n̊H̃** | **ő H̃** | **ňH̃** | **n̍H̃** | **n̏H̃** | **n̑H̃** |
| m | -127, 713 | -180, 713 | -127, 700 | -133, 679 | -205, 625 | -149, 684 | -169, 603 (1) | -198, 557 (1) | -162, 682 (1) | -170, 646 (1) | -197, 733 | -193, 676 | -137, 719 | -127, 758 | -239, 675 |
| m? | -144, 702 | -144, 699 | -128, 704 | -136, 719 | -200, 660 | -149, 680 | -168, 603 | -186, 580 | -148, 672 | -170, 630 | -158, 714 | -192, 686 | -137, 696 | -152, 737 | -235, 680 |
| m$\delta$ | 17, 11 | -36, 14 | 1, -4 | 3, -40 | -5, -35 | 0, 4 | -1, 0 | -12, -23 | -14, 10 | 0, 16 | -39, 19 | -1, -10 | 0, 23 | 25, 21 | -4, -5 |
| si | ***ǹH̃*** | ***ńH́*** | ***n̂H̃*** | ***ñH̃*** | ***n̄H̃*** | ***n̆H̃*** | ***ṅH̃*** | ***n̈H̃*** | ***n̉H̃*** | ***n̊H̃*** | ***ő H̃*** | ***ňH̃*** | ***n̍H̃*** | ***n̏H̃*** | ***n̑H̃*** |
| patch | ***ǹH̃*** | ***ńH́*** | ***n̂H̃*** | ***ñH̃*** | ***n̄H̃*** | ***n̆H̃*** | ***ṅH̃*** | ***n̈H̃*** | ***n̉H̃*** | ***n̊H̃*** | ***ő H̃*** | ***ňH̃*** | ***n̍H̃*** | ***n̏H̃*** | ***n̑H̃*** |
| m | -142, 683 | -191, 682 | -139, 697 | -134, 700 | -209, 645 | -161, 681 | -180, 667 | -210, 645 | -160, 765 | -184, 708 | -219, 714 | -205, 676 | -155, 688 | -140, 758 | -251, 675 |
| m? | -154, 694 | -152, 694 | -136, 697 | -141, 717 | -217, 651 | -162, 673 | -177, 692 | -243, 671 | -145, 759 | -174, 717 | -166, 704 | -205, 676 | -147, 686 | -156, 728 | -248, 673 |
| m$\delta$ | 12, -11 | -39, -12 | -3, 0 | 7, -17 | 8, -6 | 1, 8 | -3, -25 | 33, -26 | -15, 6 | -10, -9 | -53, 10 | 0, 0 | -8, 2 | 16, 30 | -3, 2 |

Table 17: Font metrics for above marks 2/2

| hex | 0311 | 0312 | 033D | 033E | 033F | 030E | 0346 | 034A | 034B | 034C | 0350 | 0351 | 0352 | 0357 | 035B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̑H̑ | ṅH̑ | ǹH̑ | ńH̑ | n̄H̄ | n̋H̋ | n̑H̑ | n̑H̑ | n̊H̊ | n̑H̑ | ǹH̑ | ṅH̑ | n̂Ĥ | ṅH̑ | ǹH̑ |
| patch | n̑H̑ | ṅH̑ | ǹH̑ | ńH̑ | n̄H̄ | n̋H̋ | n̑H̑ | n̑H̑ | n̊H̊ | n̄H̄ | ǹH̑ | ṅH̑ | n̂Ĥ | ṅH̑ | ǹH̑ |
| m | -205, 645 | -110, 605 | -265, 605 | -265, 605 | -235, 611 | -245, 719 | -251, 621 | -215, 645 | -205, 645 | -235, 645 | -201, 645 | -171, 645 | -251, 645 | -171, 645 | -164, 645 |
| m? | -208, 641 | -102, 611 | -260, 582 | -247, 584 | -240, 622 | -244, 682 | -255, 624 | -217, 637 | -217, 611 | -238, 647 | -183, 643 | -156, 652 | -251, 652 | -141, 662 | -174, 632 |
| m$\delta$ | 3, 4 | -8, -6 | -5, 23 | -18, 21 | 5, -11 | -1, 37 | 4, -3 | 2, 8 | 12, 34 | 3, -2 | -18, 2 | -15, -7 | 0, -7 | -30, -17 | 10, 13 |
| it | *n̑H̑* | *ṅH̑* | *ǹH̑* | *ńH̑* | *n̄H̄* | *n̋H̋* | *n̑H̑* | *n̑H̑* | *n̊H̊* | *n̄H̄* | *ǹH̑* | *ṅH̑* | *n̂Ĥ* | *ṅH̑* | *ǹH̑* |
| patch | *n̑H̑* | *ṅH̑* | *ǹH̑* | *ńH̑* | *n̄H̄* | *n̋H̋* | *n̑H̑* | *n̑H̑* | *n̊H̊* | *n̄H̄* | *ǹH̑* | *ṅH̑* | *n̂Ĥ* | *ṅH̑* | *ǹH̑* |
| m | -74, 645 | 13, 605 | -142, 605 | -142, 605 | -111, 611 | -99, 719 | -125, 621 | -84, 645 | -74, 645 | -104, 645 | -70, 645 | -40, 645 | -120, 645 | -40, 645 | -33, 645 |
| m? | -76, 640 | 24, 610 | -142, 581 | -128, 583 | -113, 621 | -105, 681 | -128, 623 | -87, 636 | -92, 610 | -107, 646 | -53, 642 | -23, 651 | -118, 651 | -6, 661 | -46, 631 |
| m$\delta$ | 2, 5 | -11, -5 | 0, 24 | -14, 22 | 2, -10 | 6, 38 | 3, -2 | 3, 9 | 18, 35 | 3, -1 | -17, 3 | -17, -6 | -2, -6 | -34, -16 | 13, 14 |
| sb | **n̑H̑** | **ṅH** | **n□H□** | **n□H□** | **n□H□** | **ǹH̑** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **ṅH‘** | **n□H□** | **ṅH’** | **ǹH** |
| patch | **n̑H̑** | **ṅH̑** | **ǹH̑** | **ńH̑** | **n̄H̄** | **n̋H̋** | **n̑H̑** | **n̑H̑** | **n̊H̊** | **n̄H̄** | **ǹH̑** | **ṅH̑** | **n̂Ĥ** | **ṅH̑** | **ǹH̑** |
| m | -205, 645 | -97, 504 (1) | | | | -232, 719 | | | | | | 100, 547 (1) | | 100, 549 (1) | -180, 639 |
| m? | -220, 650 | -102, 528 | | | | -232, 696 | | | | | | 106, 547 | | 112, 547 | -184, 681 |
| m$\delta$ | 15, -5 | 5, -24 | | | | 0, 23 | | | | | | -6, 0 | | -12, 2 | 4, -42 |
| si | ***n̑H̑*** | ***ṅH̑*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***ǹH̑*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** |
| patch | ***n̑H̑*** | ***ṅH̑*** | ***ǹH̑*** | ***ńH̑*** | ***n̄H̄*** | ***n̋H̋*** | ***n̑H̑*** | ***n̑H̑*** | ***n̊H̊*** | ***n̄H̄*** | ***ǹH̑*** | ***ṅH̑*** | ***n̂Ĥ*** | ***ṅH̑*** | ***ǹH̑*** |
| m | -205, 645 | -110, 605 | | | | -245, 719 | | | | | | | | | |
| m? | -226, 644 | -116, 614 | | | | -242, 686 | | | | | | | | | |
| m$\delta$ | 21, 1 | 6, -9 | | | | -3, 33 | | | | | | | | | |

Table 18: Font metrics for below marks 1/2

| hex | 0316 | 0317 | 0318 | 0319 | 031C | 0320 | 0323 | 0324 | 0325 | 0326 | 0329 | 032A | 032B | 032C | 032D | 032E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̖Ḥ | n̗Ḥ | n̘Ḥ | n̙Ḥ | n̚Ḥ | n̛Ḥ | n̜Ḥ | n̝Ḥ | n̞Ḥ | n̟Ḥ | n̠Ḥ | n̡Ḥ | n̢Ḥ | ṇḤ | n̤Ḥ | n̥Ḥ |
| patch | n̖Ḥ | n̗Ḥ | n̘Ḥ | n̙Ḥ | n̚Ḥ | n̛Ḥ | n̜Ḥ | n̝Ḥ | n̞Ḥ | n̟Ḥ | n̠Ḥ | n̡Ḥ | n̢Ḥ | ṇḤ | n̤Ḥ | n̥Ḥ |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -148 | -216, -113 | -218, -116 | -218, -116 | -255, -82 | -257, -122 | -253, -113 | -251, -108 | -204, -116 | -195, -133 | -251, -110 |
| m? | -248, -124 | -224, -114 | -249, -102 | -249, -101 | -262, -98 | -252, -170 | -216, -152 | -217, -152 | -217, -78 | -245, -87 | -256, -106 | -255, -134 | -252, -119 | -204, -123 | -195, -128 | -252, -109 |
| m$\delta$ | 29, -9 | 14, -18 | 25, -46 | -36, -46 | 46, -48 | -1, 22 | 0, 39 | -1, 36 | -1, -38 | -10, 5 | -1, -16 | 2, 21 | 1, 11 | 0, 7 | 0, -5 | 1, -1 |
| it | *n̖Ḥ* | *n̗Ḥ* | *n̘Ḥ* | *n̙Ḥ* | *n̚Ḥ* | *n̛Ḥ* | *n̜Ḥ* | *n̝Ḥ* | *n̞Ḥ* | *n̟Ḥ* | *n̠Ḥ* | *n̡Ḥ* | *n̢Ḥ* | *ṇḤ* | *n̤Ḥ* | *n̥Ḥ* |
| patch | *n̖Ḥ* | *n̗Ḥ* | *n̘Ḥ* | *n̙Ḥ* | *n̚Ḥ* | *n̛Ḥ* | *n̜Ḥ* | *n̝Ḥ* | *n̞Ḥ* | *n̟Ḥ* | *n̠Ḥ* | *n̡Ḥ* | *n̢Ḥ* | *ṇḤ* | *n̤Ḥ* | *n̥Ḥ* |
| m | -246, -133 | -237, -132 | -254, -148 | -315, -147 | -246, -146 | -283, -148 | -239, -113 | -242, -116 | -242, -116 | -272, -82 | -282, -122 | -276, -113 | -273, -108 | -228, -116 | -222, -133 | -275, -110 |
| m? | -274, -126 | -248, -116 | -270, -104 | -270, -103 | -281, -100 | -287, -172 | -238, -111 | -239, -111 | -233, -82 | -263, -89 | -278, -108 | -283, -136 | -277, -121 | -229, -125 | -221, -130 | -269, -111 |
| m$\delta$ | 28, -7 | 11, -16 | 16, -44 | -45, -44 | 35, -46 | 4, 24 | -1, -2 | -3, -5 | -9, -34 | -9, 7 | -4, -14 | 7, 23 | 4, 13 | 1, 9 | -1, -3 | -6, 1 |
| sb | **n̖Ḥ** | **n̗Ḥ** | **n̘Ḥ** | **n̙Ḥ** | **n̚Ḥ** | **n̛Ḥ** | **n̜Ḥ** | **n̝Ḥ** | **n̞Ḥ** | **n̟Ḥ** | **n̠Ḥ** | **n̡Ḥ** | **n̢Ḥ** | **ṇḤ** | **n̤Ḥ** | **n̥Ḥ** |
| patch | **n̖Ḥ** | **n̗Ḥ** | **n̘Ḥ** | **n̙Ḥ** | **n̚Ḥ** | **n̛Ḥ** | **n̜Ḥ** | **n̝Ḥ** | **n̞Ḥ** | **n̟Ḥ** | **n̠Ḥ** | **n̡Ḥ** | **n̢Ḥ** | **ṇḤ** | **n̤Ḥ** | **n̥Ḥ** |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -149 | -216, -113 | -218, -116 | -218, -116 | -255, -73 | -257, -124 | -253, -113 | -251, -108 | -208, -116 | -195, -133 | -239, -110 |
| m? | -248, -132 | -224, -122 | -249, -110 | -249, -110 | -262, -106 | -251, -179 | -216, -118 | -218, -113 | -216, -88 | -244, -86 | -256, -116 | -255, -142 | -252, -127 | -204, -131 | -195, -136 | -240, -105 |
| m$\delta$ | 29, -1 | 14, -10 | 25, -38 | -36, -37 | 46, -40 | -2, 30 | 0, 5 | 0, -3 | -2, -28 | -11, 13 | -1, -8 | 2, 29 | 1, 19 | -4, 15 | 0, 3 | 1, -5 |
| si | ***n̖Ḥ*** | ***n̗Ḥ*** | ***n̘Ḥ*** | ***n̙Ḥ*** | ***n̚Ḥ*** | ***n̛Ḥ*** | ***n̜Ḥ*** | ***n̝Ḥ*** | ***n̞Ḥ*** | ***n̟Ḥ*** | ***n̠Ḥ*** | ***n̡Ḥ*** | ***n̢Ḥ*** | ***ṇḤ*** | ***n̤Ḥ*** | ***n̥Ḥ*** |
| patch | ***n̖Ḥ*** | ***n̗Ḥ*** | ***n̘Ḥ*** | ***n̙Ḥ*** | ***n̚Ḥ*** | ***n̛Ḥ*** | ***n̜Ḥ*** | ***n̝Ḥ*** | ***n̞Ḥ*** | ***n̟Ḥ*** | ***n̠Ḥ*** | ***n̡Ḥ*** | ***n̢Ḥ*** | ***ṇḤ*** | ***n̤Ḥ*** | ***n̥Ḥ*** |
| m | -219, -133 | -210, -132 | -224, -148 | -285, -147 | -216, -146 | -253, -148 | -216, -113 | -218, -116 | -218, -116 | -255, -82 | -257, -122 | -253, -113 | -251, -108 | -204, -116 | -195, -133 | -251, -110 |
| m? | -251, -132 | -225, -122 | -246, -110 | -259, -110 | -260, -106 | -263, -179 | -214, -117 | -216, -113 | -211, -88 | -237, -86 | -256, -116 | -260, -142 | -253, -127 | -206, -127 | -199, -136 | -244, -113 |
| m$\delta$ | 32, -1 | 15, -10 | 22, -38 | -26, -37 | 44, -40 | 10, 31 | -2, 4 | -2, -3 | -7, -28 | -18, 4 | -1, -6 | 7, 29 | 2, 19 | 2, 11 | 4, 3 | -7, 3 |

Table 19: Font metrics for below marks 2/2

| hex | 032F | 031D | 031E | 031F | 0330 | 0331 | 0333 | 0339 | 033A | 033B | 033C | 0347 | 0348 | 0349 | 034D | 034E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| cat | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn | Mn |
| reg | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ |
| patch | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ | n̯H̯ |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -145 | -213, -70 | -244, -110 | -244, -110 | -216, -146 | -253, -113 | -246, -113 | -251, -108 | -244, -110 | -197, -110 | -192, -110 | -222, -110 | -227, -110 |
| m? | -251, -125 | -249, -112 | -249, -112 | -252, -99 | -205, -68 | -240, -133 | -240, -123 | -178, -98 | -255, -134 | -252, -118 | -252, -119 | -240, -123 | -195, -94 | -189, -111 | -228, -92 | -228, -74 |
| m$\delta$ | 1, 9 | 0, -31 | 2, 6 | 3, -46 | -8, -2 | -4, 23 | -4, 13 | -38, -48 | 2, 21 | 6, 5 | 1, 11 | -4, 13 | -2, -16 | -3, 1 | 6, -18 | 1, -36 |
| it | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* |
| patch | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* | *n̯H̯* |
| m | -274, -116 | -278, -143 | -269, -106 | -278, -145 | -227, -70 | -266, -110 | -266, -110 | -246, -146 | -276, -113 | -269, -113 | -273, -108 | -266, -110 | -219, -110 | -214, -110 | -244, -110 | -249, -110 |
| m? | -276, -127 | -271, -114 | -272, -114 | -271, -101 | -220, -74 | -268, -135 | -266, -125 | -198, -100 | -283, -136 | -277, -120 | -275, -121 | -266, -125 | -214, -96 | -213, -113 | -247, -94 | -243, -76 |
| m$\delta$ | 2, 11 | -7, -29 | 3, 8 | -7, -44 | -7, 4 | 2, 25 | 0, 15 | -48, -46 | 7, 23 | 8, 7 | 2, 13 | 0, 15 | -5, -14 | -1, 3 | 3, -16 | -6, -34 |
| sb | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** | **n□H□** |
| patch | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** | **n̯H̯** |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -146 | -213, -70 | -244, -58 | | | | | | | | | | |
| m? | -263, -127 | -249, -121 | -249, -120 | -251, -108 | -192, -70 | -240, -89 | | | | | | | | | | |
| m$\delta$ | 13, 11 | 0, -22 | 2, 14 | 2, -38 | -21, 0 | -4, 31 | | | | | | | | | | |
| si | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** | ***n□H□*** |
| patch | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** | ***n̯H̯*** |
| m | -250, -116 | -249, -143 | -247, -106 | -249, -145 | -213, -70 | -244, -110 | | | | | | | | | | |
| m? | -265, -131 | -252, -132 | -247, -108 | -249, -108 | -184, -77 | -234, -89 | | | | | | | | | | |
| m$\delta$ | 15, 15 | 3, -11 | 0, 2 | 0, -37 | -29, 7 | -10, -21 | | | | | | | | | | |

## Vertical references, anchor $Y$ coordinates and reference, and clearance

A font has at least five vertical reference lines where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these $Y$ coordinates are in the first five rows of Table 20. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as $yra$, $yrc$, $yrx$, $yr0$, $yrd$ because they are $Y$ values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects. These values (and the ranges that set the bands) are in the SFD, in a line that starts with `BlueValues`, e.g. in Libertinus serif regular,

```
BlueValues 47 [-12 0 429 442 460 472 600 610 645 658 688 698]
OtherBlues 11 [-238 -227]
```

means $yra = 698, [688, 698]$; $yrc = 645, [645, 658]$; $yrx = 429, [429, 442]$; $yr0 = 0, [-12, 0]$; and $yrd = -232, [-238, -227]$.

In the anchor model for Libertinus, the $Y$ coordinate of an above-anchor ($ay$) or a below-anchor ($by$) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows a and b in Tables 1–4 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor $Y$ coordinate reference values by ascender aspect, so I will denote them $ayra$, $ayrc$, $ayrx$, $byr0$, $byrd$, These are the middle five rows of Table 20. For example, all glyphs that sit on the baseline have below-anchor $Y$ coordinate $by = -110$, and thus $byr0 = -110$. All capitals A–Z have $ay = 850$, and thus $ayrc = 850$. Any glyph that descends to the descender value $yrd = -232$ or descender range $[-238, -227]$ should have $by = byrd = -319$.

The clearance is defined as the difference between the anchor reference value and the vertical aspect $Y$ value. I will denote these clearances as $ay\delta a$, $ay\delta c$, $ay\delta x$, $by\delta 0$, $by\delta d$. So, $ay\delta a = ayra - yra$ etc. For example, capital letters A–Z have a capital height $yrc = 645$, and all capital letters A–Z have an above-anchor $Y$ coordinate at $ayrc = 850$. So the above-anchor clearance for capitals is $ay\delta c = ayrc - yec = 850 - 645 = 205$. Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor $Y$ coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the $Y$ coordinate of the above- or below-anchor ($ay$ or $by$) of a glyph that is not a full ascender or descender.

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender $a$, capital $c$, or x-height $x$. Then $ay = Y_{\text{max}} + ay\delta[acx] - \epsilon$, where $Y_{\text{max}}$ is the maximum $Y$ coordinate of bounding box of the glyph, and $\epsilon$ is the correction for the vertical aspect range, usually for an overshoot.

Capitals are the exception: their top reference line is always the capline, and their above-anchor always has the capital reference $Y$ ($ay = 850$), even where the outline rises above the capline. In humanist letterforms capitals do not have ascenders: an ascender is a stroke of the pen, not merely a high $Y_{\text{max}}$. What rises above the capline in Libertinus is away from the horizontal center of the glyph, where the mark sits: the top-left stroke of J (italic and semibold italic, 34 units above the capline), the serifs of T (semibold and semibold italic, 28), the slash of Ø (25), the top of Ʒ (up to 29), and the hooks of Ƈ and Ɠ (98). A mark at the capital reference $Y$ does not collide with these. The one capital with a genuine ascender is the capital iota ɩ (U+0196), whose stem reaches the ascender line; it takes the ascender reference ($ay = 885$; italic styles 890). The glottal stop ʔ (U+0294) and pharyngeal ʕ (U+0295), counted here as capitals, rise above the capline only in italic (672 and 683); there they have curated anchors at $ay = 850 + (Y_{\text{max}} - 645) - 10$, with the bowl overshoot correction $\epsilon = 10$ instead of the default 2, which gives 867 and 878.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or desender $d$. Then $by = Y_{\text{min}} + by\delta[0d] + \epsilon$, where $Y_{\text{min}}$ in the minimum $Y$ coordinate of the bounding box of the glyph, and $\epsilon$ is the correction for the vertical aspect range, usually for an overshoot.

The capital anchor $Y$ reference and clearance of semibold are the patched values. The x-height bands marked * are widened above the designer's BlueValues (442, 447, 447) on 2026-11-07: several x-height letters of those styles rise 1–11 units above the band (ɐ ɔ ɡ ɵ ɽ ʂ ʐ ʑ in italic, ɢ ʑ in semibold, ʐ ʑ in semibold italic), which gave their candidates a $Y$ of 657–666, above the base letters s, r, z, o at 645. Snapped to 645, no above-mark of the IPA set overlaps them (the nearest is 19 away); semibold italic ʜ (462) and p (470) stay outside.

The designer set semibold's capitals near 805 (clearance 160) and compensated only in the capital alternates of the marks, so a mark without a capital alternate crashed into the capital: in 225 of the 780 combinations of A–Z with 30 above marks, the bottom of the mark was below the top of the letter. The patch moves the capitals to 850 and adds 45 to the mark anchors of the capital alternates, which stay where they were. That leaves 4 such combinations, all on T, whose serif tips rise to 673 away from the marks; under the marks the gap is 21–26. A few capitals have the same problem in the other styles (Ɲ Ƅ Ƈ Ɋ Ħ Ɪ Ƙ Ƶ at 798–805), so since 2026-11-08 stage 5 moves every capital with an above-anchor in 795–835 to 850 in every style. In italic and semibold italic the designer's $X$ of Ɲ Ƅ Ƈ Ƙ Ƶ was regular's, without the slant (italic Ƈ 355 against the candidate 622), so they are deleted in stage 1 and get the candidate; Ħ, outside the Latin/IPA scope, keeps regular's $X$ there. These capitals are also in `@capitals`, so the twelve marks with a capital alternate take it on them.

Table 20: Vertical aspects, anchor $Y$ references, and clearances of Libertinus

| | v. aspect | regular | italic | semibold | semibold italic |
|---|---|---|---|---|---|
| $yra$ | ascender | $698, [688, 698]$ | $698, [688, 698]$ | $698, [690, 698]$ | $698, [696, 705]$ |
| $yrc$ | capital | $645, [645, 658]$ | $645, [645, 658]$ | $645, [645, 662]$ | $645, [645, 662]$ |
| $yrx$ | x-height | $429, [429, 442]$ | $429, [429, 450]^*$ | $434, [433, 457]^*$ | $434, [434, 458]^*$ |
| $yr0$ | baseline | $0, [-12, 0]$ | $0, [-12, 1]$ | $0, [-12, 1]$, | $0, [-20, 0]$ |
| $yrd$ | descender | $-232, [-238, -227]$ | $-232, [-238, -227]$ | $-232, [-238, -212]$ | $-232, [-239, -219]$ |
| $ayra$ | ascender | 885 | 890 | 885 | 890 |
| $ayrc$ | capital | 850 | 850 | 850 | 850 |
| $ayrx$ | x-height | 645 | 645 | 645 | 645 |
| $byr0$ | baseline | $-110$ | $-110$ | $-110$ | $-110$ |
| $byrd$ | descender | $-319$ | $-319$ | $-319$ | $-319$ |
| $ay\delta a$ | ascender | 187 | 192 | 187 | 192 |
| $ay\delta c$ | capital | 205 | 205 | 205 | 205 |
| $ay\delta x$ | x-height | 216 | 216 | 211 | 211 |
| $by\delta 0$ | baseline | 110 | 110 | 110 | 110 |
| $by\delta d$ | descender | 87 | 87 | 87 | 87 |

Of the other designer anchors more than 5 from a reference (stage 5 lists them), most are legitimate: stacked marks on precomposed glyphs, superscripts at 805–815, small capitals at their own height, 685. Fixed on 2026-11-08: above and below swapped on ɥ ɥ̨ (regular, italic, semibold italic) and ƚ (regular), now deleted in stage 1 and given the candidates; and į i̇ ɨ, whose above-anchors were 9–59 units below i's, so that a mark keeping the dot crowded it, now curated at i's anchor.

For the vertical aspect, the first single value is the optical value of the vertical aspect. In practice, this is the y-coordinate of the central control point of a (vertical) stem that meets that aspect. In the table, the range for the vertical aspects are the BlueValues set by the designer in the SFD and seen as guidelines in a FontForge glyph outline window view.

I need to keep track of serif and bowl corrections relative to the (single) vertical aspect value. In Libertinus serif regular, a glyph that meets an aspect with a stem needs a 2 unit correction. For example, the letter x meets the baseline and x-height aspects, with stems, but $y_{\text{min}} = 0 - 2 = -2$ and $y_{\text{max}} = 429 + 2 = 431$. For bowls, the corrections tend to be different for each aspect. For Libertinus serif regular, a glyph that is a bowl full-descender (usually) has a 6 unit overshoot, i.e. $y_{\text{min}} = -232 - 6 = -238$. A bowl on the baseline (usually) has a 10 unit overshoot, i.e. $y_{\text{min}} = 0 - 10 = -10$. A bowl at the x-height aspect (usually) has a 10 unit overshoot, i.e. $y_{\text{max}} = 429 + 10 = 439$. A bowl at the capital aspect (usually) has a 13 unit overshoot, i.e. $y_{\text{max}} = 645 + 13 = 658$. For full ascenders, there is no correction in practice, regardless of the shape of the glyph's ascender, because $y_{\text{max}} = 698$ for all such glyphs.

That is, these stem and bowl corrections are 2 and 6, 10, 10, 13.

For the other font styles (italic, semibold, semibold italic), these corrections are remarkably consistent for all aspects, even though e.g. italic small letters do not really have flat terminal stems. But I think that the correction principle is valid for glyphs that are not within the threshold of an expected vertical aspect. For a non-full descender, if $y_{\text{min}}$ is closest to the descender aspect, and the glyph looks like a bowl at this lowest point, then the correction is 6; otherwise I assume that the glyph must look like a flat stem and the correction is 2. But for a non-full descender that is closer to the baseline, the bowl correction is 10. Full descender bowls tend to be flatter because the lower threhold of the descender really is a hard lower bound.

# Anchor survey, with patches

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show that some anchors are not set to good values. The base coverage gets worse for italic and semibold.

I will systematically survey above- and below anchors in the four styles, commenting on any features and patching any problems. Refer to Tables 1–4 for anchor values. Missing anchors are readily apparaent as blank entries. Large anchor-midpoint offset values — $a\delta$ and $b\delta$ — indicate possible problems, but these can often be explained by glyph symmetry.

## Copied anchors

Many designer anchors are copies: within a style, clusters of letters share one above- or below-anchor exactly (a's $(209, 645)$ and $(196, -110)$ on some 40 IPA letters each. Some copies are harmless coincidences of similar letterforms; most are not.

`run_anchor_copy_report.py` lists the clusters (`tex/input/copied_anchors.tex`, no longer printed here). On 2026-11-07 every member of the regular clusters, and its counterparts in the other styles, was reviewed per character: each anchor is now the candidate, inherited from its base letter, or curated by a stated rule, the same way in all four styles.

## A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D, $ax = 306$. Consider the precomposed characters Ď (U+010E) and Ḋ (U+1E0A). The designer drew their glyphs with the mark centered at $X = 312$ and 296 respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F, $ax = 284$. For Ḟ (U+1E1E), the dot is centered at $X = 254$. So, the designer is deliberately choosing to keep the mark cloder to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

#### Patch J regular, below-anchor

J's below-anchor is too low. For J, $by = -319$, but J is not a full descender; its bbox $Y_{\text{min}} = -172$, which is well above $yrd = -232$. Yes, the designer set $by = byrd = -319$ for J in all fout font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -172 - 87 + 5 = -254$, where $\epsilon = 5$ accounts for the overshoot of J's tail.

#### Patch Q regular, below-anchor

Q has bbox $Y_{\text{min}} = -209$. Again, according to the heuristic algorithm, the below-anchor anchor should be at $by = Y_{\text{min}} - by\delta d + \epsilon = -209 - 87 + 5 = -291$, where $\epsilon = 5$ accounts for the overshoot of Q's tail. For $bx$, observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at $bx = 350$, the same as that for O.

## a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have $ay$ regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h, $ay$ is at the x-height anchor reference, and therefore

wide marks collide with the ascender. For k, $ay$ is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.
The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

**Patch k regular, below-anchor**

For k, $by - 214$. But it should be at $byr0 = -110$, like all other letters that sit on the baseline.
Here is this positioning of k in a patched version of the font (with o for comparison only).
Original: k̠ k̠o ok̠
Patched: k̠ k̠o ok̠

**Patch b, d, f, h regular, above-anchor**

For b, d, f, and h, set $ay = ayra = 885$.
For $ax$ for b, d, f, and h, there are at least two strategies. First, set $ax$ at the bowl apex. So, for f, set $ax = 279$ at the apex of the bowl. But for b, d, and h, you could set $ax$ centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

**Patch j regular, above-anchor**

j needs an above anchor for the rare case of an above-mark that is placed over the dot. The ccmp feature swaps i and j for their dotless forms before every above-mark in its list, so only a mark outside the list attaches to i or j itself; among the IPA marks that is only the double overline U+033F. Such a mark must clear the dot. The designer's $(137, 795)$ on i (and the earlier curated j at $(126, 795)$) put the double overline only 13–16 above the dot in regular and semibold, against 69–74 above o. The patch (2026-11-07) curates i and j in every style on the stem's axis, at the candidate's $Y$, the dot top plus the ascender clearance (regular 834, italic 810, semibold 837, semibold italic 823), so the double overline sits 55–60 above the dot. The candidate's $X$ is right for i but not for j, whose hook pulls the bounding-box midpoint left of the stem (by 84 in semibold).

**Stem-axis rule (2026-11-12).** Letters with a central stem and no curl have their anchors on the axis of the stem (`font_patching_stemaxis.py`, stage 7b, after the curated anchors: it sets $X$ over the designer's and the curated anchors alike, and reports each curated $X$ it changes; $Y$ is not touched). The group is the basest such letters: i, ɪ, j, ɉ, l, m (its middle stem), I, T, ɪ, above and below (j and ɉ above only, their below anchor being under the hook). The letters derived from them (ɨ ɪ ɬ ꟊ T ɱ …) move with their base letter in stage 8; ɾ and ɹ are not in the group, because their hooked top puts the optical centre 40 to 55 units off the stem.

The stem's midline is the mean of its two straight edges, read from the control points (a pair of line segments on opposite sides of one ink interval, overlapping in height; where one side is a curve, as in the middle stem of semibold italic m, the straight edge moved by the median half-width). It is taken at the top of the straight part for the above anchor and at its bottom for the below anchor, and carried to the anchor's height at the style's declared slant (12°, 11.5°), less $K$ in semibold italic.

The stems are not drawn at the declared slant: the lowercase stems are steeper (italic 12.1° to 13.5°, semibold italic 12.8° to 13.4°), the capitals I and T shallower (11°, 10°). Carrying the midline at the stem's own angle was tested and rejected (test sheet `tex/stem_angle_test_sheet.pdf`): it moves the above anchors of ɪ, ɉ and l 4 to 9 units right in italic and 9 to 12 in semibold italic, and the evidence points the other way: the dot of i and j, and the accents of the precomposed ì í î ĩ ī ĭ ï ỉ î ȉ ǰ, sit left of the line at the declared slant, not right of it. Against the precomposed ḿ ṃ İ Į Ī Ṫ Ṭ Ṯ neither slant was consistently closer.

In the italic styles, curls change the optical centre, and the rule is adapted:

- Dotted i and j: the above anchor is over the centre of the dot, 11 to 13 units left of the stem's axis (only U+033F attaches there, right over the dot).
- Dotless ɪ and ɉ: the above anchor is on the axis of the dot of i and j, so that an accent takes the dot's place. Their curved entry stroke at the top left pulls the optical centre left of the stem, and the precomposed accents agree: composed minus precomposed (median), on the stem's axis and on the dot's: italic ɪ $+20$, $+8$, ɉ $+28$, $+15$; semibold italic ɪ $+14$, $+1$, ɉ $+10$, $-14$.

- The below anchors of i, ɪ and l (curled foot) and the above anchor of m (arches) are left out, to be curated (“Open decisions”).

In regular and semibold the rule applies to the whole group, above and below. The accents of the precomposed accented i are 16 to 20 units left of ɪ’s stem there too, but the designer puts accents about 9 left of centre on lowercase letters generally, and the rule looked better on the test sheets.

Moves, in units: regular and semibold m $+11$ to $+12$, T below $+8$ (regular), others 0 to 2; italic ɪ above $-10$, ɟ $-13$, I above $-5$ and below $-12$, T below $-16$, m below $-9$, i and j above (dot) $-12$, $-13$; semibold italic ɪ above $-11$, ɟ $-21$, T above $-12$ and below $-26$, i and j above (dot) $-11$; the derived and precomposed letters move with them.

**Patch n, p, r, v, w, x, z regular, above-anchor**

In my patched version of the font, I regularize $ay$ to $ayrx = 645$ for all letters that sit at the x-height, including n, p, r, v, w, x, z.

## Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

## Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

## Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, cicrumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A cirmumflex Â Ầ Ấ Ậ Ẩ
O circumflex Ô Ồ Ố Ộ Ổ
AE ligature Æ Ǽ Æ Ǽ Ǽ
Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ
Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ Ṩ Ṯ Ṽ

## Regular, above-anchor, capitals, $X$ coordinate

Figure 9 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad $X$ coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor $X$ coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor $X$ coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 9 for an illustration of increasingly bad (spoofed) $X$ coordinates of N’s above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

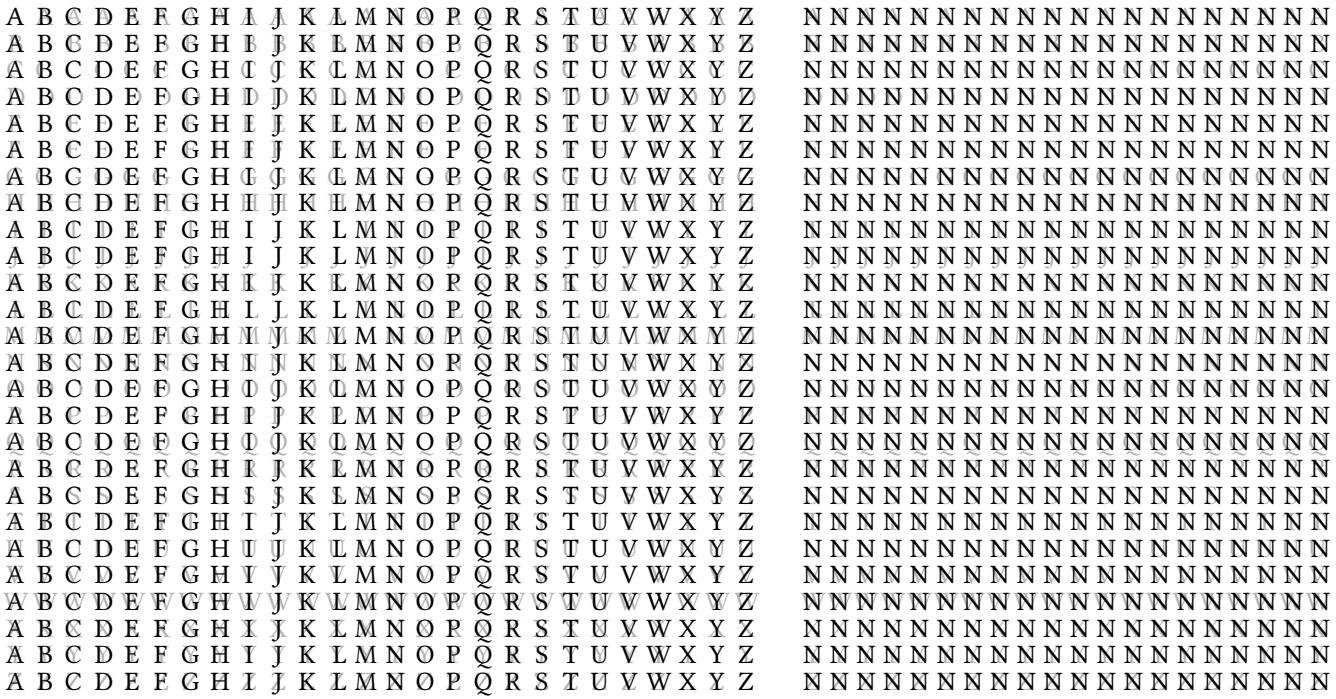


Figure 9: Above-anchor $X$ coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at the their above-anchor. The right panel shows how hard it is to discern bad $X$ coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the $X$ coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed) $X$ coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

## Comma-style marks

Libertinus has a some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

### Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` ď *ď* **ď** ***ď***

`\char"013D` Ľ *Ľ* **Ľ** ***Ľ***

`\char"013E` ľ *ľ* **ľ** ***ľ***

`\char"0148` ň *ň* **ň** ***ň***

`\char"0165` ť *ť* **ť** ***ť***

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{l\char"030C}` ʟ̬

`\textsc{\char"013E}` ʟ̬

### Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

| | |
|---|---|
| `\char"00C7` Ç *Ç* **Ç** ***Ç*** | `\char"00E7` ç *ç* **ç** ***ç*** |
| `\char"1E08` Ḉ *Ḉ* **Ḉ** ***Ḉ*** | `\char"1E08` ḉ *ḉ* **ḉ** ***ḉ*** |
| `\char"1E10` Ḑ *Ḑ* **Ḑ** ***Ḑ*** | `\char"1E11` ḑ *ḑ* **ḑ** ***ḑ*** |
| `\char"0228` Ȩ *Ȩ* **Ȩ** ***Ȩ*** | `\char"0229` ȩ *ȩ* **ȩ** ***ȩ*** |
| `\char"1E1C` Ḝ *Ḝ* **Ḝ** ***Ḝ*** | `\char"1E1D` ḝ *ḝ* **ḝ** ***ḝ*** |
| `\char"0122` Ģ *Ģ* **Ģ** ***Ģ*** | `\char"0123` ģ *ģ* **ģ** ***ģ*** |
| `\char"1E28` Ḩ *Ḩ* **Ḩ** ***Ḩ*** | `\char"1E29` ḩ *ḩ* **ḩ** ***ḩ*** |
| `\char"0136` Ķ *Ķ* **Ķ** ***Ķ*** | `\char"0137` ķ *ķ* **ķ** ***ķ*** |
| `\char"013B` Ļ *Ļ* **Ļ** ***Ļ*** | `\char"013C` ļ *ļ* **ļ** ***ļ*** |
| `\char"0145` Ņ *Ņ* **Ņ** ***Ņ*** | `\char"0146` ņ *ņ* **ņ** ***ņ*** |
| `\char"0156` Ŗ *Ŗ* **Ŗ** ***Ŗ*** | `\char"0157` ŗ *ŗ* **ŗ** ***ŗ*** |
| `\char"015E` Ş *Ş* **Ş** ***Ş*** | `\char"015F` ş *ş* **ş** ***ş*** |
| `\char"0162` Ţ *Ţ* **Ţ** ***Ţ*** | `\char"0163` ţ *ţ* **ţ** ***ţ*** |

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

`c\char"0327` ç *ç* **ç** ***ç***
`d\char"0327` ḑ *ḑ* **ḑ** ***ḑ***
`e\char"0327` ȩ *ȩ* **ȩ** ***ȩ***
`g\char"0327` ģ *ģ* **ģ** ***ģ***
`h\char"0327` ḩ *ḩ* **ḩ** ***ḩ***
`k\char"0327` ķ *ķ* **ķ** ***ķ***
`l\char"0327` ļ *ļ* **ļ** ***ļ***
`n\char"0327` ņ *ņ* **ņ** ***ņ***
`r\char"0327` ŗ *ŗ* **ŗ** ***ŗ***
`s\char"0327` ş *ş* **ş** ***ş***
`t\char"0327` ţ *ţ* **ţ** ***ţ***

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`. This behavior is tolerable for the other bases – c, d, e, h, s, t.

#### Greek koronis U+0343

*Purpose.* In polytonic (ancient) Greek the koronis marks crasis, two words run together with their vowels contracted: κἀγώ for καὶ ἐγώ. It has exactly the shape of the smooth breathing (psili), and Unicode writes both with one character, U+0313 COMBINING COMMA ABOVE. U+0343 COMBINING GREEK KORONIS was encoded for compatibility with older standards: its canonical decomposition is the single character U+0313, so any normalization (NFC or NFD) replaces it with U+0313, and no precomposed letter decomposes to it (ἀ is α + U+0313). The spacing forms U+1FBD GREEK KORONIS and U+1FBF GREEK PSILI are both compatibility equivalents of space + U+0313. Modern monotonic Greek uses neither. In every style and weight the koronis should look like the style's smooth breathing. Text should use U+0313 or the precomposed letters; nobody needs to type U+0343.

*XeLaTeX and HarfBuzz.* HarfBuzz applies the singleton decomposition: after a base letter (or a no-break space) U+0343 is shaped with the glyph of U+0313, and α + U+0343 composes to ἀ. The glyph `uni0343` is shaped only when the mark begins a word, as after a space, which XeTeX passes to HarfBuzz as glue, not as a character. Below, the first row is a + U+0313, the second a + U+0343 (the same glyphs), the third α + U+0343 (the precomposed ἀ), and the last U+0343 alone after a space (the glyph `uni0343`); each in the original fonts, then the patched fonts.

`a\char"0313` ả *ả* **ả** ***ả*** ả *ả* **ả** ***ả***
`a\char"0343` ả *ả* **ả** ***ả*** ả *ả* **ả** ***ả***
`\char"03B1\char"0343` ἀ *ἀ* **ἀ** ***ἀ*** ἀ *ἀ* **ἀ** ***ἀ***
`\ \char"0343` ᾽ ᾽ ᾽ ᾽ ᾽ ᾽

*Libertinus.* In regular and italic, `uni0343` is a reference to `uni0313`, with the same anchor: correct. In semibold it is a reference to the horn `uni031B`, moved 72 up and 72 right, and it had no mark anchor, so it attached to nothing. In semibold italic it is an outline of its own, the horn's size (188 by 178, against the comma's 134 by 180), drawn right of the origin with the comma above's anchor, so $k$ was 371 against $K = 150$. The patch leaves both outlines and curates their anchors as regular's and italic's U+0313: 135 above the outline's bottom and 21 left of its ink centroid (along the slant, with $K$, in semibold italic): $(-92, 539)$ in semibold, $(128, 532)$ in semibold italic (2026-11-11). The glyph a reader actually sees on a letter is U+0313, which in semibold and semibold italic is drawn at the lighter style's weight (and in the original semibold attached only to the 24 bases with a `top_punkt` anchor; stage 1 merges that class). The precomposed Greek letters reference neither mark: ἀ ἐ Ἀ ῥ̓ are built from the modifier apostrophe U+02BC (U+1FBD in semibold italic ἀ), and italic draws its own. The outlines are an item of the to-do list (commas and horns).

## Æ: a letter the midpoint cannot centre

Why Æ has curated anchors in every style rather than the anchor model's candidate (see "Where candidates fail" in `ANCHOR-HEURISTICS.md`).

Over Æ, accents do not go at the midpoint of the glyph. The left leg of the A stops short of the top, so the top edge runs from the apex of the A to the end of the arm of the E, and its centre lies well to the right. Other fonts agree: Charis SIL, Gentium Plus, EB Garamond, Junicode, DejaVu Serif and Andika put the above-anchor at the macron of Ǣ, within 5 units, and below they use the common stroke or the same $X$. In Libertinus the designer's above-anchors missed the accents of Ǣ and Ǽ by 55–88 and 16–68 units, so the curated above-anchors lie between the two (555, 674, 555, 545 for regular, italic, semibold, semibold italic). In the italic styles the macron and the acute cannot both be met, and each lands about 30 units from its precomposed position. Below, semibold's anchor is curated under the common stroke, at 510. The designer's 549 of the other three styles stands: nothing collides there.

## IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

IJ `\char"0132` U+0132 capital ligature IJ

ij `\char"0133` U+0133 small ligature ij

Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,

```
\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}
```

ĲĳĲ́ `{\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute

ĳ́ `{\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute

ĳ́ `{\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute

In an English LaTeX document, `ij` and `IJ` are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

## Lowercase i with tilde below and grave above, in italic and semibold italic

First, start with i with tilde below (precomposed).

ḭ *ḭ* **ḭ** ***ḭ*** `\char"1E2D` i with tilde below (precomposed)

You cannot simply put a grave on it because it collides with the dot.

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"1E2D\char"0300` i with tilde below (precomposed) with grave

Start over with i with grave (precomposed).

ì *ì* **ì** ***ì*** `\char"00EC` i with grave (precomposed)

Try putting a grave below.

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\char"0330` i with grave (precomposed) with tilde below

This fails, because normalization breaks down the input "i with grave (precomposed) with tilde below" into "i with grave with tilde below", then into canonical order "i with tilde below with grave", the recomposes it into "i with tilde below (precomposed) with grave", which reintroduces i's dot.

Next, try to block normalization with the combining grapheme joiner (CGJ).

ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀*** `\char"00EC\cgj\char"0330` i with grave (precomposed) with combining grapheme joiner with tilde below

This looks bad because i with grave (precomposed) does not have a below-anchor and fallback shaping looks bad.

Finally, try starting with dotless i, which has OK above- and below-anchors.

ì̗ *ì̗* **ì̗** ***ì̗*** `\char"0131\char"0330\char"0300` dotless i with tilde below with grave

Normalization does not affect the dotless i, so it looks the same with the CGJ.

ì̗ *ì̗* **ì̗** ***ì̗*** `\char"0131\cgj\char"0330\char"0300` dotless i with CGJ with tilde below with grave

All together now, for comparison.

ḭ ì ḭ ì ḭ ḭ ì ḭ ì ḭ \char"1E2D \char"00EC \char"0131\char"0330\char"0300 i with tilde below (precomposed); i with grave (precomposed); dotless i with tilde below with grave

The same strategy works with other above marks.

ḭ ȉ ḭ ȉ \char"0131\char"0330\char"030F dotless i with tilde below with double grave

ḭ ́ ḭ ́ \char"0131\char"0330\char"0301 dotless i with tilde below with acute

ḭ ̋ ḭ ̋ \char"0131\char"0330\char"030B dotless i with tilde below with double acute

## Lowercase a with double grave, followed by floating double grave, in italic

*Superseded by "Floating tones" below:* write `\floattone{a}{\char"030F}`.

In italic, we want to produce a (precomposed) vowel with a double grave, followed by floating double grave. Ideally, the two double graves should be aligned. We cannot use the combining double grave as the floater, because it has nothing to combine to, and we cannot control its height. So we use a spacing character, specifically the double prime character (‶ `\char"2036`), which looks like a double grave, in italic.

*ä*‶ `\textit{ä\char"2036}}`

Then we lower it by a few points.

*ä*‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

The distance to move the glyph can be calculated from its SFD splineset, or by trial and error.

Note that only a, e, and i have precomposed glyphs with double acute.

*ä*‶ `\textit{ä\raisebox{-0.05em}{\char"2036}}`

*ë*‶ `\textit{ë\raisebox{-0.05em}{\char"2036}}`

*ë*‶ `\textit{ë\raisebox{-0.045em}{\char"2036}}`

*ɛ*‶ `\textit{\char"025B\raisebox{-0.05em}{\char"2036}}`

## Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` a̰. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ḛ ḛ Ḭ ḭ Ṵ ṵ *Ḛ ḛ Ḭ ḭ Ṵ ṵ* **Ḛ ḛ Ḭ ḭ Ṵ ṵ** ***Ḛ ḛ Ḭ ḭ Ṵ ṵ***

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330 = \char"1E2D` ḭ = ḭ is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301 = \char"1E2D\char"0301` ḭ ́ ḭ ́ = ḭ ́ ḭ ́

`\char"0131\char"0330\char"0301` ı̰ ́ ı̰ ́

Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

`i\char"0301\char"0330` í ̰ í ̰ `i\char"0301` i ́ i ́ `i\char"0330` ḭ ́ ḭ ́

\char"00EC\char"0330 ḭ̀ *ḭ̀* **ḭ̀** ***ḭ̀***

\char"1E2D\char"0301 ḭ́ *ḭ́* **ḭ́** ***ḭ́***

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, \char"034F, or \cgj if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

1 Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ* **Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ** ***Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ***

2 À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ* **À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ** ***À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ***

3 Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ***

4 Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ *Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ* **Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ** ***Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ***

5 Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ *Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ* **Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ** ***Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ***

6 Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ* **Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ** ***Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ***

7 Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ***

7c Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ***

8 Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ***

9 Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ***

10 Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ***

11 Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ɛ Ɔ ą̏ ę̏ į̏ ǫ̏ ų̏ ɛ ɔ *Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ɛ Ɔ ą̏ ę̏ į̏ ǫ̏ ų̏ ɛ ɔ* **Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ɛ Ɔ ą̏ ę̏ į̏ ǫ̏ ų̏ ɛ ɔ** ***Ą̏ Ę̏ Į̏ Ǫ̏ Ų̏ Ɛ Ɔ ą̏ ę̏ į̏ ǫ̏ ų̏ ɛ ɔ***

Patched

A E I O U Ɛ Ɔ a e i o u ɛ ɔ *A E I O U Ɛ Ɔ a e i o u ɛ ɔ* **A E I O U Ɛ Ɔ a e i o u ɛ ɔ** ***A E I O U Ɛ Ɔ a e i o u ɛ ɔ***

Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ *Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ* **Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ** ***Á É Í Ó Ú Ɛ Ɔ á é í ó ú ɛ ɔ***

À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ *À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ* **À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ** ***À È Ì Ò Ù Ɛ Ɔ à è ì ò ù ɛ ɔ***

Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ *Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ* **Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ** ***Ă Ĕ Ĭ Ŏ Ŭ Ɛ Ɔ ă ĕ ĭ ŏ ŭ ɛ ɔ***

Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ *Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ* **Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ** ***Ắ Ế Ḯ Ố Ű Ɛ Ɔ ắ ế ḯ ố ű ɛ ɔ***

Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ *Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ* **Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ** ***Ằ Ề Ỉ Ồ Ừ Ɛ Ɔ ằ ề ỉ ồ ừ ɛ ɔ***

Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ* **Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ** ***Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ***

cgj Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ *Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ* **Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ** ***Ą Ę Į Ǫ Ų Ɛ Ɔ ą ę į ǫ ų ɛ ɔ***

Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ***

cgj Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ***

Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ *Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ* **Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ** ***Ą́ Ę́ Į́ Ǫ́ Ų́ Ɛ Ɔ ą́ ę́ į́ ǫ́ ų́ ɛ ɔ***

Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ***

cgj Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ *Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ* **Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ** ***Ą̀ Ę̀ Į̀ Ǫ̀ Ų̀ Ɛ Ɔ ą̀ ę̀ į̀ ǫ̀ ų̀ ɛ ɔ***

Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ***

cgj Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ *Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ* **Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ** ***Ą̆ Ę̆ Į̆ Ǫ̆ Ų̆ Ɛ Ɔ ą̆ ę̆ į̆ ǫ̆ ų̆ ɛ ɔ***

Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ *Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ* **Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ** ***Ą̋ Ę̋ Į̋ Ǫ̋ Ų̋ Ɛ Ɔ ą̋ ę̋ į̋ ǫ̋ ų̋ ɛ ɔ***

cgj Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏ *Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏*
Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏ **Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏**
cgj Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏ *Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏* **Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏** ***Ạ̏ Ệ̏ Ị̏ Ọ̏ Ụ̏ Ệ̏ Ỏ̏ ạ̏ ệ̏ ị̏ ọ̏ ụ̏ ệ̏ ỏ̏***
Patched not grave
À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏ *À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏* **À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏** ***À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏***
Patched grave
À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏ *À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏* **À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏** ***À ȂÈ Ȉ Ò Ù Ȅ Ỏ̏ à ȃè ȉ ò ù ȅ ỏ̏***

## Floating tones

A floating tone is written as a tone mark with no vowel of its own: after a vowel that bears the same tone (ȅ̏), or standing alone as a morpheme in a gloss. By Unicode convention, a mark that stands alone is applied to U+00A0 no-break space (the Unicode Standard prefers NBSP over U+0020 space, which allows a line break between the space and its mark). The patched fonts give U+00A0 an above and a below anchor at the lowercase references (645 above, −110 below), with X on each style's anchor model, so a mark on NBSP sits at exactly the height it has on a lowercase vowel, in all four styles.

### How to write a floating tone in XeLaTeX

1. Load the patched fonts with `fontspec`, e.g.

```
\setmainfont{LibertinusSerif-Regular-patch.otf}[
    Path           = ../fonts/,
    ItalicFont     = LibertinusSerif-Italic-patch.otf,
    BoldFont       = LibertinusSerif-Semibold-patch.otf,
    BoldItalicFont = LibertinusSerif-SemiboldItalic-patch.otf]
```

2. Define three commands in the preamble (this report does):

```
\newcommand{\cgj}{\char"034F }                % combining grapheme joiner
\newcommand{\nbsp}{\char"00A0 }               % no-break space, base for a floating mark
\newcommand{\floattone}[2]{#1\cgj#2\nbsp#2}   % vowel + mark, then the same mark floating
```

3. Write a vowel followed by a floating tone of the same specification as `\floattone{`*vowel*`}{`*mark*`}`, e.g. `\floattone{e}{\char"030F}` gives ȅ̏. The mark may also be typed as the literal combining character between the braces. The vowel may carry other marks (`\floattone{a\char"0330}{\char"0300}` gives ạ̀), and may be any lowercase tone-bearing unit, such as a syllabic nasal (`\floattone{ŋ}{\char"0300}` gives ŋ̀).

4. Write a floating tone on its own as `\nbsp` followed by the mark: `-\nbsp\char"0300` gives -̀.

5. Punctuation may follow directly: `\floattone{e}{\char"0300}.` gives è̀. NBSP is a quarter em wide (250 units; 274 in semibold italic), so there is that much space before the punctuation.

Two things to avoid. Do not write the base as ~ or `\nobreakspace`: in TeX these are glue, not a character, so HarfBuzz sees no base and the mark is set on its own at its default position. Do not leave out the CGJ when the floating tone repeats the vowel's mark: without it, HarfBuzz substitutes the precomposed glyph where one exists (ȅ), and a precomposed glyph may

carry its mark at a different height from the anchor (the dot above U+0307 is 45 units higher in the precomposed glyphs of regular); with it, both marks are placed by anchors at the same height. `\floattone` inserts the CGJ. When typing `\nbsp` by hand before a literal combining mark, write `\nbsp{}`, since XeTeX reads a combining mark as a letter of the command name; `\floattone` has no such problem.

The anchors are lowercase ones, so a floating mark after a capital does not take its `.cap` form or the capital height. In fonts without NBSP anchors (original Libertinus, Noto Serif) the mark still appears, at its default position; Gentium Plus and Charis SIL have NBSP anchors of their own.

**Comparison**

Patched fonts, with `\floattone`: `-k\floattone{e}{\char"030F}`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Patched fonts, without CGJ: `-ke\char"030F\nbsp\char"030F`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Original Libertinus, with `\floattone`

-kȅ̏ *-kȅ̏* **-kȅ̏** ***-kȅ̏***

Gentium Plus, with `\floattone`

-kȅ̏

Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

**Showcase**

Each item is `\floattone{`*vowel*`}{`*mark*`}` for the vowels a e i o u ɛ ɔ ə ɪ ʊ ɨ ʉ ʌ æ and the syllabic nasals m n ŋ, in the patched fonts (regular, italic, semibold, semibold italic), at `\Large` font size. The floating mark matches the height of the vowel's mark throughout, since every x-height letter and NBSP share the same above reference (645) and below reference (−110).

U+0301 acute

á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́

*á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́*

**á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́**

***á́ é́ í́ ó́ ú́ ɛ́ ɔ́ ə́ ɪ́ ʊ́ ɨ́ ʉ́ ʌ́ ǽ ḿ ń ŋ́***

U+0300 grave

à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀

*à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀*

**à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀**

***à̀ è̀ ì̀ ò̀ ù̀ ɛ̀ ɔ̀ ə̀ ɪ̀ ʊ̀ ɨ̀ ʉ̀ ʌ̀ æ̀ m̀ ǹ ŋ̀***

U+030B double acute

a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋

*a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋*

**a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋**

***a̋ e̋ i̋ ő ű ɛ̋ ɔ̋ ə̋ ɪ̋ ʊ̋ ɨ̋ ʉ̋ ʌ̋ æ̋ m̋ n̋ ŋ̋***

U+030F double grave

ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏

*ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏*

**ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏**

***ȁ ȅ ȉ ȍ ȕ ɛ̏ ɔ̏ ə̏ ɪ̏ ʊ̏ ɨ̏ ʉ̏ ʌ̏ æ̏ m̏ n̏ ŋ̏***

U+0304 macron

ā¯ ē¯ ī¯ ō¯ ū¯ ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄

*ā¯ ē¯ ī¯ ō¯ ū¯ ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄*

**ā¯ ē¯ ī¯ ō¯ ū¯ ɛ̄ ɔ̄ ə̄ ɪ̄ ʊ̄ ɨ̄ ʉ̄ ʌ̄ ǣ m̄ n̄ ŋ̄**

***ā̄*** ***ē̄*** ***ī̄*** ***ō̄*** ***ū̄*** ***ɛ̄̄*** ***ɔ̄*** ***ə̄*** ***ɪ̄*** ***ʊ̄*** ***ɨ̄*** ***ʉ̄*** ***ʌ̄*** ***ǣ***  ***m̄*** ***n̄*** ***ŋ̄***

U+0302 circumflex

âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ əˆ ɪˆ ʊ̂ ɨˆ ʉˆ ʌˆ æˆ  m̂ n̂ ŋ̂

*âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ ə̂ ɪˆ ʊ̂ ɨˆ ʉˆ ʌˆ æˆ  m̂ n̂ ŋ̂*

**âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ əˆ ɪˆ ʊ̂ ɨˆ ʉˆ ʌˆ æˆ  m̂ n̂ ŋ̂**

***âˆ êˆ îˆ ôˆ ûˆ ɛ̂ ɔ̂ ə̂ ɪˆ ʊ̂ ɨˆ ʉˆ ʌˆ æˆ  m̂ n̂ ŋ̂***

U+030C caron

ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛˇ ǰˇ əˇ ǐˇ ʊˇ ɨˇ ʉˇ ʌˇ æˇ  m̌ ňˇ ŋˇ

*ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛˇ ǰˇ əˇ ǐˇ ʊˇ ɨˇ ʉˇ ʌˇ æˇ  m̌ ňˇ ŋˇ*

**ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛˇ ǰˇ əˇ ǐˇ ʊˇ ɨˇ ʉˇ ʌˇ æˇ  m̌ ňˇ ŋˇ**

***ǎˇ ěˇ ǐˇ ǒˇ ǔˇ ɛˇ ǰˇ əˇ ǐˇ ʊˇ ɨˇ ʉˇ ʌˇ æˇ  m̌ ňˇ ŋˇ***

## Squat reversed esh (Sinological apical vowels)

U+0285 LATIN SMALL LETTER SQUAT REVERSED ESH ʅ is the apical vowel of Sinological transcription after retroflex sibilants (Mandarin *shī* [ʂʅ$^{55}$], *zhī* [tʂʅ$^{55}$], *rì* [ʐʅ$^{51}$]). Its partner ɿ U+027F follows dental sibilants (*sī* [sɿ$^{55}$], *cí* [tsʰɿ$^{35}$]). ʅ is ɿ with a retroflex hook, the hook of ɳ ʈ ʂ ʐ ɭ: its top is ɿ's fishhook, at the x-height.

The designer's ʅ was 550 high in regular, 119 above the x-height, and in italic, semibold and semibold italic it is ∫ mirrored, rising to the ascender (semibold exactly so: every point $x \mapsto 305 - x$, hence the same bounding box as ∫). In a survey of 22 fonts (2026-11-09) every font that draws the letter rather than mirroring ∫ tops it at the x-height, level with its own ɿ: Gentium Plus, Charis SIL, Andika, Noto Serif, STIX Two, Junicode, EB Garamond, DejaVu Serif, Tinos, Roboto. The ascender form occurs only as a mirrored ∫ (Gentium 1.x, which Gentium Plus replaced with the squat form; Lato) and in Libertinus and Langscitinus, which has Libertinus's glyph unchanged.

Stage 2 of the patch redraws ʅ in all four styles by control points: ɿ's fishhook and stem, with ɳ's retroflex hook in place of the foot serif. The hook reaches as deep as ɳ's (−221 in regular, −229 in the other styles), short of the descender (−234), like the other retroflex hooks. The left sidebearing is ɿ's, the right ɳ's at the hook. The above-anchor is ɿ's (stage 8, which treats ʅ as ɿ with a bottom tail); the below-anchor is the anchor model's candidate on the new outline (stage 6), at the descender reference. The designer's anchors, placed for the old outline, are deleted in stage 1, including regular's ogonek and cedilla anchors.

| | | | | |
|---|---|---|---|---|
| Original Libertinus | ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$ | aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$* | *aʅa* **ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$** | **aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$*** | ***aʅa*** |
| Langscitinus | ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$ | aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$* | *aʅa* **ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$** | **aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$*** | ***aʅa*** |
| Patched | ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$ | aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$* | *aʅa* **ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$** | **aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$*** | ***aʅa*** |
| Gentium Plus | ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$ | aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$* | *aʅa* **ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$** | **aʅa *ʂʅ$^{55}$ tʂʰʅ$^{55}$ ʐʅ$^{51}$ sɿ$^{55}$*** | ***aʅa*** |

With tone and other marks, patched: [ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ḷ] *[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ḷ]* **[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ḷ]** ***[ʂ̄ ʂ́ ʂ̌ ʂ̀] [ʂ̲] [ḷ]***

## Reversed r with fishhook

U+027F LATIN SMALL LETTER REVERSED R WITH FISHHOOK ɿ is the apical vowel after dental sibilants (see the previous section). Fonts draw it in two ways: on the baseline, as ɾ mirrored (Libertinus, Langscitinus, Gentium 1.x, Junicode, STIX Two, DejaVu Serif, EB Garamond), or with a straight descender (Gentium Plus, Charis SIL, Andika, Noto, Tinos, Roboto). SIL changed it from the baseline form in Gentium 1.x to the descending form in Gentium Plus, and the current SIL fonts offer no way back: they have one glyph for U+027F, no character variant or stylistic set for it, and no PUA alternate. Both forms are established, and the patch keeps Libertinus's baseline ɿ as U+027F.

For users who want the descending form, the patched fonts add it at a Private Use Area code point, U+E27F (glyph `uni027F.desc`), in all four styles. It is ɿ's fishhook and stem continued down to q's descender serif, with ɿ's advance width. The above-anchor is ɿ's (stage 8); the below-anchor is curated, under the stem at the descender reference (−319), as q's.

| | | | | |
|---|---|---|---|---|
| U+027F, patched | sɿ$^{55}$ tsʰɿ$^{35}$ [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲] *sɿ$^{55}$ tsʰɿ$^{35}$ [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]* **sɿ$^{55}$ tsʰɿ$^{35}$ [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]** ***sɿ$^{55}$ tsʰɿ$^{35}$ [sɪ̄ sɪ́ sɪ̌ sɪ̀] [sɪ̲]*** |
| U+E27F, patched | sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲] *sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]* **sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]** ***sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]*** |
| Gentium Plus | sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲] *sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]* **sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]** ***sɻ$^{55}$ tsʰɻ$^{35}$ [sɻ̄ sɻ́ sɻ̌ sɻ̀] [sɻ̲]*** |

### How to use the descending form

- In XeLaTeX or LuaLaTeX with the patched fonts, write `\char"E27F` or define a command, as this report does:

  ```
  \newcommand{\rfishdesc}{\char"E27F }   % reversed r with fishhook and descender
  ```

  `s\rfishdesc` gives sɻ. Before a literal combining mark write `\rfishdesc{}`, since XeTeX reads a combining mark as a letter of the command name (`s\rfishdesc{}` gives sɻ́).

- In other applications, enter the code point E27F with the application's Unicode input (e.g. type `e27f` then Alt+X in Word, or Ctrl+Shift+U e27f in GTK applications).

A Private Use code point has no meaning outside the font: in any other font U+E27F shows as a missing glyph, and searching, sorting, spell-checking and copied text see U+E27F, not U+027F. Use it for display where the descending form is wanted, and keep U+027F wherever the text must be interchanged or searched.

## f with dot below

tiɣə́ ṛfa

*tiɣə́ ṛfa*

**tiɣə́ ṛfa**

***tiɣə́ ṛfa***

ṛ = `\char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

f̣ = `f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̣ g̣ j̣ p̣ q̣ x̣

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̣ g̣ j̣ p̣ q̣ x̣*

**ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̣ g̣ j̣ p̣ q̣ x̣**

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ  ç f̣ g̣ j̣ p̣ q̣ x̣***

The left group of letters – a b d e h i k l m n o r s t u v w y z – have a precomposed character. For the right group of letters – c f g j p q x – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take a bigger dot. In semibold, the size difference is reversed: `uni0307` has larger diameter 124, and `dotbelowcomb` has small diameter 108. Actually, semibold `dotbelowcomb` is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, `dotaccent` is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by $(\pm 6, \pm 7.5)$ from the center — and `dotbelowcomb` is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at $Y = -319$, so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of `dotbelowcomb`. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonbly faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

For italic, the descender's U-pocket is defined by the control points at $(14, -238)$ and $(33, -205)$, giving a U-center around $x \approx 24$. Using the (symmetric) italic `dotbelowcomb` anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is $X \approx 30$.

For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of $X = -33.5$. The semibold italic `dotbelowcomb` itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is $X \approx 62$.

Here are examples with the patched fonts, with f's below-anchor in italic at $(30, -319)$ and semibold italic at $(62, -319)$. This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

| Original | Patched italics (f only) |
|---|---|
| tiɣə ṛfa | tiɣə ṛfa |
| *tiɣə ṛfa* | *tiɣə ṛfa* |
| **tiɣə ṛfa** | **tiɣə ṛfa** |
| ***tiɣə ṛfa*** | ***tiɣə ṛfa*** |

Original (repeated from above)

ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱*

**ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱**

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱***

Patched italics (f only)

*ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱*

***ạ ḅ ḍ ẹ ḥ ị ḳ ḷ ṃ ṇ ọ ṛ ṣ ṭ ụ ṿ ẉ ỵ ẓ ç f g̱ j̱ p̱ q̱ x̱***

## Ashkun

`a\char"0304\char"0303` ã *ã* **ã** ***ã***

`\char"0101\char"0303` ã *ã* **ã** ***ã***

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Patched

`u\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`u\cgj\char"0304\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\cgj\char"0303` ũ *ũ* **ũ** ***ũ***

`\char"016B\char"0303` ũ *ũ* **ũ** ***ũ***

Move box without patching font

`\char"016B\kern-4.132pt\raisebox{0.88pt}{\char"02DC}` ũ *ũ* **ũ** ***ũ***

## Tacks

a̰ḛḭo̰ṵa̰ḛḭo̰ṵ
a̰ḛḭo̰ṵa̰ḛḭo̰ṵ

## Upsilon and grave

Original, then patched, versions of upsilon and combining grave, after setting upsilon's above-anchor.

ὺ ύ̀ **ὺ** **ύ̀**

ὺ ύ̀ **ὺ** **ύ̀**

New upsilon above-anchor: regular $(253, 645)$; italic $(358, 645)$; semibold $(267, 645)$; semibold italic $(369, 645)$

Revised combining grave above-anchor: semibold italic $(0, 683)$. Moved from $(-142, 683)$, which is far left of the optical center of the mark, which has the effect of attaching the mark too far right.

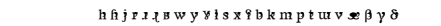
Spoof to move modifier grave (which has the exact same outline as combining grave) to an anchored position, if upsilon had an above-anchor at $(358, 645)$ (keeping combining grave's above-anchor as is). Note upsilon is U+028A and modifier grave is U+02CB.

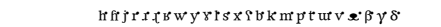
```
\textit{\char"028A\kern-4.66pt\raisebox{-0.38pt}{\char"02CB}}
```

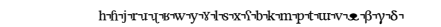
*ὺ*

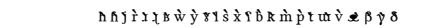
## Patch superscript consonant anchors

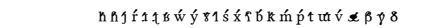
Libertinus serif regular

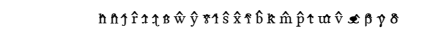




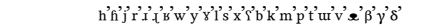


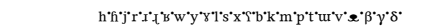


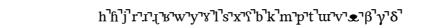


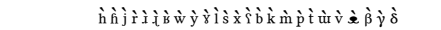


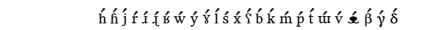
Libertinus serif regular, patched

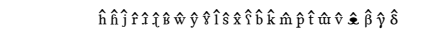






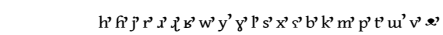


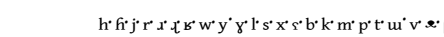


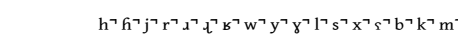


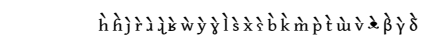


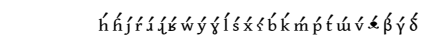
Gentium Plus

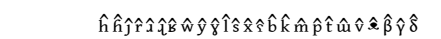
















A B C D E F G H I J K L M N O P Q R S T U V W X Y Z a b c d e f g h i j k l m n o p q r s t u v w x y z Æ æ Œ œ Ø ø Đ ð Þ þ ß Ɽ Ð đ Ħ ħ ı ȷ Ł ł Ƀ ƀ
À Ḃ Ċ Ḋ È Ḟ Ġ Ḣ Ì J̀ Ǩ Ĺ Ṁ Ǹ Ò Ṗ Q̀ Ṙ Ṡ Ṫ Ù V̀ Ẁ Ẋ Ỳ Ż à ḃ ċ ḋ è ḟ ġ ḣ ì j̀ ǩ ĺ ṁ ǹ ò ṗ q̀ ṙ ṡ ṫ ù v̀ ẁ ẋ ỳ ż Æ̀ æ̀ Œ̀ œ̀ Ø̀ ø̀ Đ̀ ð̀ Þ̀ þ̀ ß̀ Ɽ̀ Đ̀ đ̀ Ħ̀ ħ̀ ì j̀ Ł̀ ł̀ Ƀ̀ ƀ̀

## Dot removal

To do. Check GSUB to see if dot removal for i and j is complete.

## Precomposed characters

To do. Check GSUB if an available precomposed character is encoded by a doublet of the base and a combining mark, or a triplet of the base and the two combining marks, in either mark order, then the precomposed character should be be produced by the font's GSUB table.

# Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

## Above marks – GPOS anchor 0

x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿ x̀ x́ x̂ x̃ x̄ x̆ ẋ ẍ x̉ x̊ x̋ x̌ x̍ x̎ x̏ x̐ x̑ x̒ x̓ x̔ x̕ x̚ x̛ x̽ x̾ x̿

## Below marks – GPOS anchor 2

x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳ x̖ x̗ x̘ x̙ x̚ x̛ x̜ x̝ x̞ x̟ x̠ x̡ x̢ x̣ x̤ x̥ x̦ x̧ x̨ x̩ x̪ x̫ x̬ x̭ x̮ x̯ x̰ x̱ x̲ x̳

## Left angle – GPOS anchor 3

x̚ x̛ x̜ x̝

## Below-right marks – GPOS anchor 4

x̹ x̺ x̻
x̹ x̺ x̻
x̹ x̺ x̻
x̹ x̺ x̻

## Cedilla – GPOS anchor 5

x̧ x̨ x̧ x̨

## Overlay marks – GPOS anchor 6

o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴
o̶ o̷ o̸ o̵ o̴

## Double marks

x͡x x͡x x͞x x͟x x̃x x͡x x̲x
x͡x x͡x x͞x x͟x x̃x x͡x x̲x
x̆x x̋x x̏x x̂x x̆x x̊x ẍx
x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x x⃞x

## Greek marks

t̀t́t̃t̄t̆ṫẗṭ
t̀́t̃ṭ
t̂t̂t̀t́t̃t̄t̆ṫẗṭ
t̀t́t̃t̄t̆ṫẗṭ
t̀́t̃ṭ
t̂t̂t̀t́t̃t̄t̆ṫẗṭ
**t̀t́t̃t̄t̆ṫẗṭ**
**t̀́t̃t**
**t̂t̂t̀t́t̃t̄t̆ṫẗṭ**
**t̀t́t̃t̄t̆ṫẗṭ**
**t̀́t̃t**
**t̂t̂t̀t́t̃t̄t̆ṫẗṭ**

## Adding missing marks

Semibold and semibold italic lack many of the combining marks of regular and italic. The patch adds them, with mark anchors, in the range U+0332–U+036F. No mark is drawn from scratch: each is an outline that the designer drew in another style. A semibold mark is the regular outline (an above-mark is moved right by 12 units, with its anchor, as the designer's semibold above-marks are). A semibold italic mark is the italic outline moved right by 12 units, with the anchor of regular, as in the semibold italic anchor model. In semibold, U+0358–U+0363 were copies of the capital accents, and they are replaced. The patch also adds `dotaccent.cap` to italic and semibold italic: the dot of the precomposed İ, at the height it has in the precomposed dotted capitals. It adds `tildecomb.cap` to every style (2026-11-08): the tilde of the precomposed capitals, which is `tildecomb` itself in regular, italic and semibold, and in semibold italic the taller tilde of Ã (also of Ñ Õ Ĩ Ẽ Ỹ Ṽ and ã ñ õ), which differs from `tildecomb` there.

The designer does not thicken below-marks between weights, but thickens above-marks by about 12 percent (mean stroke thickness, measured as twice the area of the outline divided by its perimeter: tilde 40 to 45, caron 36 to 42). So the added above-marks are made bolder. Where a mark is, or contains, another mark, the bold style's own mark is used. The rest of the outline is pushed outward by 7.5 percent of its stroke thickness on each side, about 2 to 3 units. These algorithmically emboldened marks are shown in the table on the next page, on a few bases that have above-anchors in all four patched fonts, with the designer's tilde and caron for comparison of weight. The regular and italic columns show the outlines that were copied.

Also added, but not shown in the table: the low line and double low line (U+0332, U+0333), the tilde, stroke and solidus overlays (U+0334–U+0337), the below-marks U+0339–U+033C, U+0347–U+0349, U+034D, U+034E, U+0353–U+0356, U+0359, U+035A, the double overline (U+033F), the dot above right (U+0358), the double marks (U+035C–U+0362) and the combining letters (U+0363–U+036F). These are copied without emboldening, since the designer's lines, below-marks and modifier letters are not thicker in the bold styles; the dot above right is the bold style's own dot. The double marks have no anchors, as in regular. In semibold, the overlays and the dot above right have no anchors, because the semibold lookup table has no anchor class for them. The half rings above (U+0351, U+0357) already exist in semibold, drawn to the right of the origin, so they sit beside the base; they are left as they are. Not added: the custom-width low lines and overlines of the next section, U+0344, U+0345 and U+0483.

| | | regular | semibold | italic | semibold italic | |
|---|---|---|---|---|---|---|
| U+0303 | tilde (designer's, for comparison) | ãñũÕ | **ãñũÕ** | *ãñũÕ* | ***ãñũÕ*** | |
| U+030C | caron (designer's, for comparison) | ǎňǔǑ | **ǎňǔǑ** | *ǎňǔǑ* | ***ǎňǔǑ*** | |
| U+033D | x above | x̌x̌ǔx̌ | **x̌x̌ǔx̌** | *x̌x̌ǔx̌* | ***x̌x̌ǔx̌*** | emboldened |
| U+033E | vertical tilde | a̾n̾u̾O̾ | **a̾n̾u̾O̾** | *a̾n̾u̾O̾* | ***a̾n̾u̾O̾*** | emboldened |
| U+034A | not tilde above | x̃ñ̃ũ̃Õ̃ | **x̃ñ̃ũ̃Õ̃** | *x̃ñ̃ũ̃Õ̃* | ***x̃ñ̃ũ̃Õ̃*** | emboldened |
| U+034B | homothetic above | x̃̃ñ̃̃ũ̃̃Õ̃̃ | **x̃̃ñ̃̃ũ̃̃Õ̃̃** | *x̃̃ñ̃̃ũ̃̃Õ̃̃* | ***x̃̃ñ̃̃ũ̃̃Õ̃̃*** | own tilde; dots emboldened |
| U+034C | almost equal to above | x̃̃ñ̃̃ũ̃̃Õ̃̃ | **x̃̃ñ̃̃ũ̃̃Õ̃̃** | *x̃̃ñ̃̃ũ̃̃Õ̃̃* | ***x̃̃ñ̃̃ũ̃̃Õ̃̃*** | emboldened |
| U+0350 | right arrowhead above | x̚n̚ǔx̚ | **x̚n̚ǔx̚** | *x̚n̚ǔx̚* | ***x̚n̚ǔx̚*** | emboldened |
| U+0351 | left half ring above | x́ńǔx́ | **x́ńǔx́** | *x́ńǔx́* | ***x́ńǔx́*** | emboldened (semibold italic only) |
| U+0352 | fermata | x̑n̑ǔx̑ | **x̑n̑ǔx̑** | *x̑n̑ǔx̑* | ***x̑n̑ǔx̑*** | own inverted breve; dot emboldened |
| U+0357 | right half ring above | a̾n̾u̾O̾ | **a̾n̾u̾O̾** | *a̾n̾u̾O̾* | ***a̾n̾u̾O̾*** | emboldened (semibold italic only) |
| U+035B | zigzag above | x́ńǔx́ | **x́ńǔx́** | *x́ńǔx́* | ***x́ńǔx́*** | emboldened |
| U+0340 | grave tone mark | àǹùÒ | **àǹùÒ** | *àǹùÒ* | ***àǹùÒ*** | own grave |
| U+0341 | acute tone mark | áńúÓ | **áńúÓ** | *áńúÓ* | ***áńúÓ*** | own acute |
| U+0346 | bridge above | x̄n̄ūŌ | **x̄n̄ūŌ** | *x̄n̄ūŌ* | ***x̄n̄ūŌ*** | own bridge below, raised |

## Overline and low line

U+0305 combining overline and U+0332 combining low line are applied by GSUB lookups that substitute a custom-width line mark according to width classes. GSUB lookup 51 effectively classifies the glyphs by width, and GSUB lookups 52–82 provide the appropriate custom-width mark for the base width class. Semibold and semibold italic do not have any combining low lines; the patch adds U+0332 and U+0333 with below-anchors, but not the custom-width variants. Custom-width line marks do not have anchors. They are positioned by HarfBuzz fallback shaping. The overlines are above the ascender line, so they are above all bases. The low lines are below the baseline, but above the descender line, and so they cut though any descenders. If a new glyph is added, and you want to support custom-width line marks for it, then remember to classify the new glyph in GSUB lookup 51.



## Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

### Horn

The Vietnamese alphabet has two horn vowels – Ơơ Ưư– and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

ỜớỜớỞởỠỡỢợỨứỪừỬửỮữỰự

*Ờớ Ừừ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự*

**Ờớ Ừừ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự**

***Ờớ Ừừ Ởở Ỡỡ Ợợ Ứứ Ừừ Ửử Ữữ Ựự***

There is a Unicode character for the combining horn mark: ̛ `\char"031B` U+031B, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

### Comma above right

There is a Unicode character for the combining comma above right mark: ̕ `\char"0315` U+0315.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *k̕wkt* [k̕$^{wkx}$t] ‘to scrape’ (Nuxalk)

(2) *p’ak* [p’$^{kx}$ak] ‘he/she cut it’ (Upper Chehalis)

In (1), the superscript consonant cluster is $^{wkx}$ and it is glottalized. In (2), the superscript consonant $^{x}$ is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

### Dot above right

There is a Unicode character for the combining dot above right mark: ̙ `\char"0358` U+0358 dot above right

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.

Libertinus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k˒ʷᵏˣ˒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣ˓t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k˒ʷᵏˣʼt]

Gentium Plus

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t]` [k˒ʷᵏˣ˒t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣ˓t]

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t]` [k˒ʷᵏˣʼt]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ʰ˒ɦ˒ʲ˒ʳ˒ɹ˒ɻ˒ʁ˒ʷ˒ʸ˒ʏ˒ˡ˒ˢ˒ˣ˒ʕ˒ᵇ˒ᵏ˒ᵐ˒ᵖ˒ᵗ˒ᵚ˒ᵛ˒ᵜ˒ᵝ˒ᵞ˒ᵟ˒

ʰ˓ɦ˓ʲ˓ʳ˓ɹ˓ɻ˓ʁ˓ʷ˓ʸ˓ʏ˓ˡ˓ˢ˓ˣ˓ʕ˓ᵇ˓ᵏ˓ᵐ˓ᵖ˓ᵗ˓ᵚ˓ᵛ˓ᵜ˓ᵝ˓ᵞ˓ᵟ˓

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣ˓t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ʰ˒ɦ˒ʲ˒ʳ˒ɹ˒ɻ˒ʁ˒ʷ˒ʸ˒ʏ˒ˡ˒ˢ˒ˣ˒ʕ˒ᵇ˒ᵏ˒ᵐ˒ᵖ˒ᵗ˒ᵚ˒ᵛ˒ᵜ˒ᵝ˒ᵞ˒ᵟ˒

ʰ˓ɦ˓ʲ˓ʳ˓ɹ˓ɻ˓ʁ˓ʷ˓ʸ˓ʏ˓ˡ˓ˢ˓ˣ˓ʕ˓ᵇ˓ᵏ˓ᵐ˓ᵖ˓ᵗ˓ᵚ˓ᵛ˓ᵜ˓ᵝ˓ᵞ˓ᵟ˓

`[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t]` [k˒ʷᵏˣ˓t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at $Y = 630$. So their anchor will be at $Y = 630 - 62 + 80 = 648$. The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at $X = \text{xMax} + 26 + 12$. These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.

# Bases

Here are some subgroups of bases.

## Latin

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ
*ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ*
**ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ**
***ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyzÆæŒœØøÐðÞþßſĐđĦħıȷŁłƁƀ***

## IPA

ƅↃƇƉƊɑɋƎƏƐƑƒƓɣƕlǂKⱮɲƟɭʒƨ||ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɞɵɟɠɡɢɣɤɥɦɧɨɪɪɬɭɮɯɰɱɲɳɴɵœɷɸɹɻɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ
*ƅↃƇƉƊɑɋƎƏƐƑƒƓɣƕlǂKⱮɲƟɭʒƨ||ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɞɵɟɠɡɢɣɤɥɦɧɨɪɪɬɭɮɯɰɱɲɳɴɵœɷɸɹɻɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ*
**ƅↃƇƉƊɑɋƎƏƐƑƒƓɣƕlǂKⱮɲƟɭʒƨ||ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɞɵɟɠɡɢɣɤɥɦɧɨɪɪɬɭɮɯɰɱɲɳɴɵœɷɸɹɻɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ**
***ƅↃƇƉƊɑɋƎƏƐƑƒƓɣƕlǂKⱮɲƟɭʒƨ||ǂ!ɐɑɒɔɕɖɗɘəɚɛɜɞɵɟɠɡɢɣɤɥɦɧɨɪɪɬɭɮɯɰɱɲɳɴɵœɷɸɹɻɽɾʁʂʃʅʆʇʉʊʌʍʎʏʐʑʒʓʔʕʖǀ***

## Superscript consonants

ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ
*ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ*
**ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ**
***ʰʱʲʳʴʵʷʸˡˢˣˤᵇᵏᵐᵖᵗᵡᵛ◌ᵝᵞᵟ***

## Greek

ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω
*ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω*
**ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω**
***ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩαβγδεζηθικλμνξοπρστυφχψω***

## Cyrillic

АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ
*АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ*
**АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ**
***АБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяЁёЄєЇїЎўӘәӨөҺһІ***

## Precomposed capital vowels

ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ

*ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ*

**ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ**

***ÀÁÂÃÄÅĀĂĄǍǞǠǺȀȂẠẢẤẦẨẪẬẮẰẲẴẶÈÉÊËĒĔĖĘĚȄȆẸẺẾỀỂỄỆÌÍÎÏĨĪĬĮİǏȈȊỈỊÒÓÔÕÖŌŎŐǑǪǬȌȎỌỎỐỒỔỖỘỚỜỞỠỢÙÚÛÜŨŪŬŮŰŲǓǕǗǙǛỤỦỨỪỬỮỰ***

## Precomposed small vowels

àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűųǔǖǘǚǜụủứừửữự

*àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűųǔǖǘǚǜụủứừửữự*

**àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűųǔǖǘǚǜụủứừửữự**

***àáâãäåāăąǎǟǡǻȁȃạảấầẩẫậắằẳẵặèéêëēĕėęěȅȇẹẻếềểễệìíîïĩīĭįǐȉȋỉịòóôõöōŏőǒǫǭȍȏọỏốồổỗộớờởỡợùúûüũūŭůűųǔǖǘǚǜụủứừửữự***

## Precomposed capital consonants

ḂḆÇĆĈČĊÇḈĎĐḊḌḎḐḒḞĜĞĠĢḠǦĤĦḤḦḪḨĴĶḰḴǨḲĹĻĽĿŁḶḸḼḺḾṀṂŃŅŇṄṆṊƝǾṔṖṕŖŘŖŖŖŚŜŞŠṠṢṦṨṤȘŢŤṪṰṬṮȚƬƮŨṼỴỤỸẀẂẄỲẀẊẌÝŶŸẎỲỴỸỾŸŹŻŽẒẐẔƵ

*ḂḆÇĆĈČĊÇḈĎĐḊḌḎḐḒḞĜĞĠĢḠǦĤĦḤḦḪḨĴĶḰḴǨḲĹĻĽĿŁḶḸḼḺḾṀṂŃŅŇṄṆṊƝǾṔṖṕŖŘŖŖŖŚŜŞŠṠṢṦṨṤȘŢŤṪṰṬṮȚƬƮŨṼỴỤỸẀẂẄỲẀẊẌÝŶŸẎỲỴỸỾŸŹŻŽẒẐẔƵ*

**ḂḆÇĆĈČĊÇḈĎĐḊḌḎḐḒḞĜĞĠĢḠǦĤĦḤḦḪḨĴĶḰḴǨḲĹĻĽĿŁḶḸḼḺḾṀṂŃŅŇṄṆṊƝǾṔṖṕŖŘŖŖŖŚŜŞŠṠṢṦṨṤȘŢŤṪṰṬṮȚƬƮŨṼỴỤỸẀẂẄỲẀẊẌÝŶŸẎỲỴỸỾŸŹŻŽẒẐẔƵ**

***ḂḆÇĆĈČĊÇḈĎĐḊḌḎḐḒḞĜĞĠĢḠǦĤĦḤḦḪḨĴĶḰḴǨḲĹĻĽĿŁḶḸḼḺḾṀṂŃŅŇṄṆṊƝǾṔṖṕŖŘŖŖŖŚŜŞŠṠṢṦṨṤȘŢŤṪṰṬṮȚƬƮŨṼỴỤỸẀẂẄỲẀẊẌÝŶŸẎỲỴỸỾŸŹŻŽẒẐẔƵ***

## Precomposed small consonants

ƀƅƃɓƈƈƈƈƈƈƈƈƌƍƌƌƌƌƌƌƒĝğġģǧǥǧǵĥħȟɦḥḫẖḩĵǰķĸǩǩƙĺļľŀłɬḷḽḻɬḿṁṃṅņňŉŋṇṋṉɲȵǫɋƥƿƥƾɍɾřŗȓȑŗŗŕśŝşšṡṣṥṧṩẛʂţťŧṫṭṯṱƭʉṽỽẁẅẃẘẍẋýÿŷỳẏỹƴźżžẑẓẕȝƨƽƿ

*ƀƅƃɓƈƈƈƈƈƈƈƈƌƍƌƌƌƌƌƌƒĝğġģǧǥǧǵĥħȟɦḥḫẖḩĵǰķĸǩǩƙĺļľŀłɬḷḽḻɬḿṁṃṅņňŉŋṇṋṉɲȵǫɋƥƿƥƾɍɾřŗȓȑŗŗŕśŝşšṡṣṥṧṩẛʂţťŧṫṭṯṱƭʉṽỽẁẅẃẘẍẋýÿŷỳẏỹƴźżžẑẓẕȝƨƽƿ*

**ƀƅƃɓƈƈƈƈƈƈƈƈƌƍƌƌƌƌƌƌƒĝğġģǧǥǧǵĥħȟɦḥḫẖḩĵǰķĸǩǩƙĺļľŀłɬḷḽḻɬḿṁṃṅņňŉŋṇṋṉɲȵǫɋƥƿƥƾɍɾřŗȓȑŗŗŕśŝşšṡṣṥṧṩẛʂţťŧṫṭṯṱƭʉṽỽẁẅẃẘẍẋýÿŷỳẏỹƴźżžẑẓẕȝƨƽƿ**

***ƀƅƃɓƈƈƈƈƈƈƈƈƌƍƌƌƌƌƌƌƒĝğġģǧǥǧǵĥħȟɦḥḫẖḩĵǰķĸǩǩƙĺļľŀłɬḷḽḻɬḿṁṃṅņňŉŋṇṋṉɲȵǫɋƥƿƥƾɍɾřŗȓȑŗŗŕśŝşšṡṣṥṧṩẛʂţťŧṫṭṯṱƭʉṽỽẁẅẃẘẍẋýÿŷỳẏỹƴźżžẑẓẕȝƨƽƿ***

## No marks: rare, digraph etc.

Ƹ̵ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj
*Ƹ̵ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj*
**Ƹ̵ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj**
***Ƹ̵ ƹ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj***

## Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ
*A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ*
**A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ**
***A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Ð Œ Þ IJ ß Ŋ***
AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ
*AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ*
**AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ**
***AÀÁÂÃÄÅĂĄ CÇĆČ DĎ EÈÉÊËĘĚ GĞ IÌÍÎÏI LĹŁĽ NÑŃŇ OÒÓÔÖŐÕØ RŔŘ SŚŞŠŞ TŢŦŤŢ UÙÚÛÜŰŮ YÝŸ ZŹŻŽ ÆÁÆ***

## Precomposed characters of anchor-relevant bases and marks

àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ
*àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ*
**àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ**
***àáâãāåạặầắẳẵấẫèéêẽēęệếềễẽìíîĩīịòóôõōọộốồỗùúûũūụỳýŷỹȳẏḍḷṇṛṣṭẓ***

## Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A blue color means a precomposed glyph.

A red color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: o͇ (U+0333 double macron below), o͚ (U+033A inverted bridge below), and o͛ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

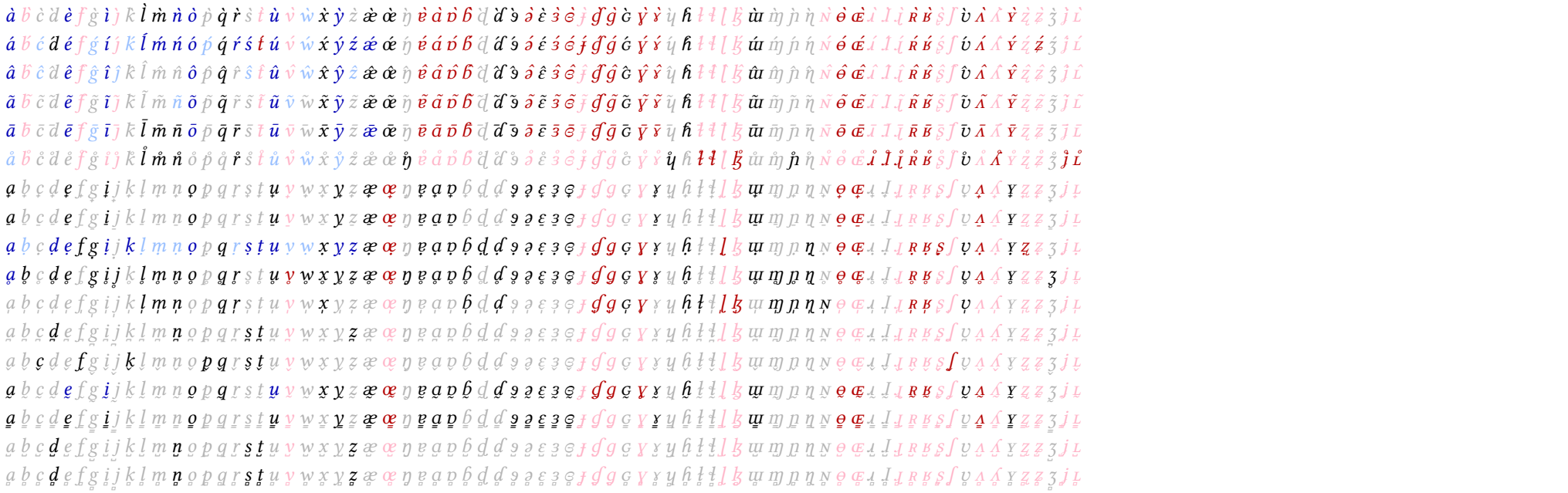
## Regular, Bases requiring anchors for common marks, Common combining marks



## Regular patched, Bases requiring anchors for common marks, Common combining marks



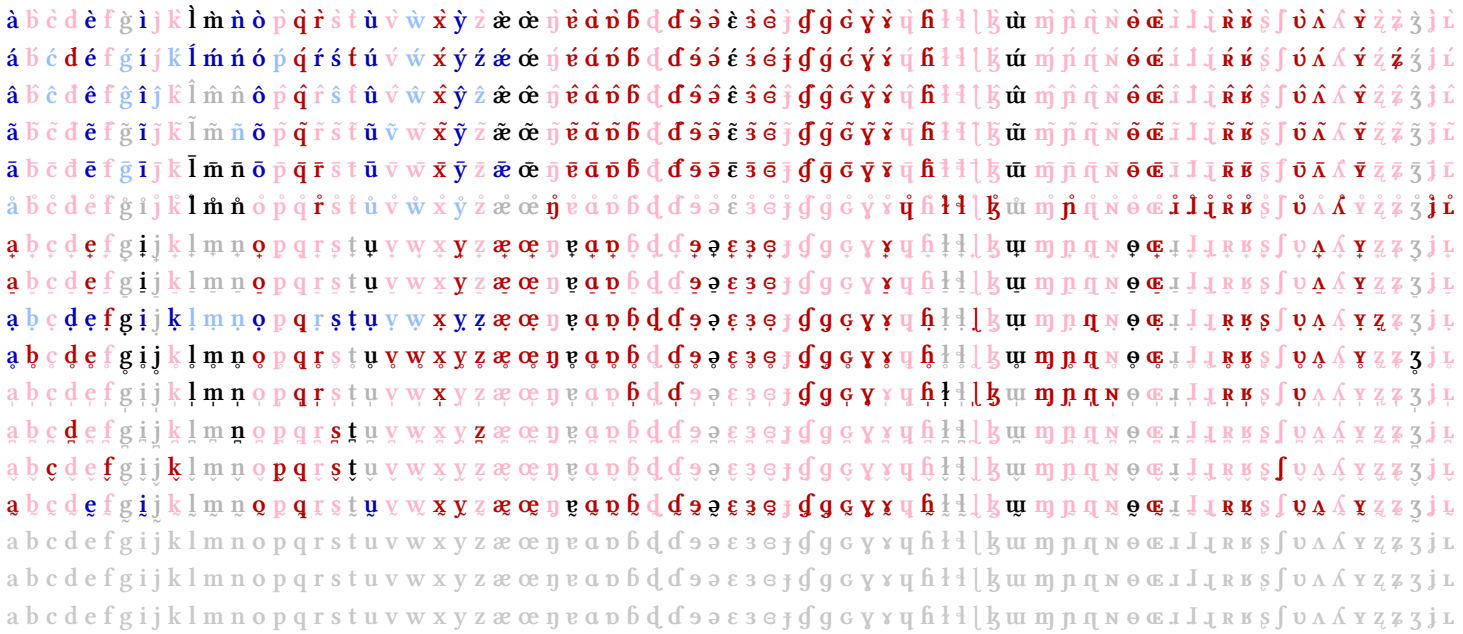
## Italic, Bases requiring anchors for common marks, Common combining marks



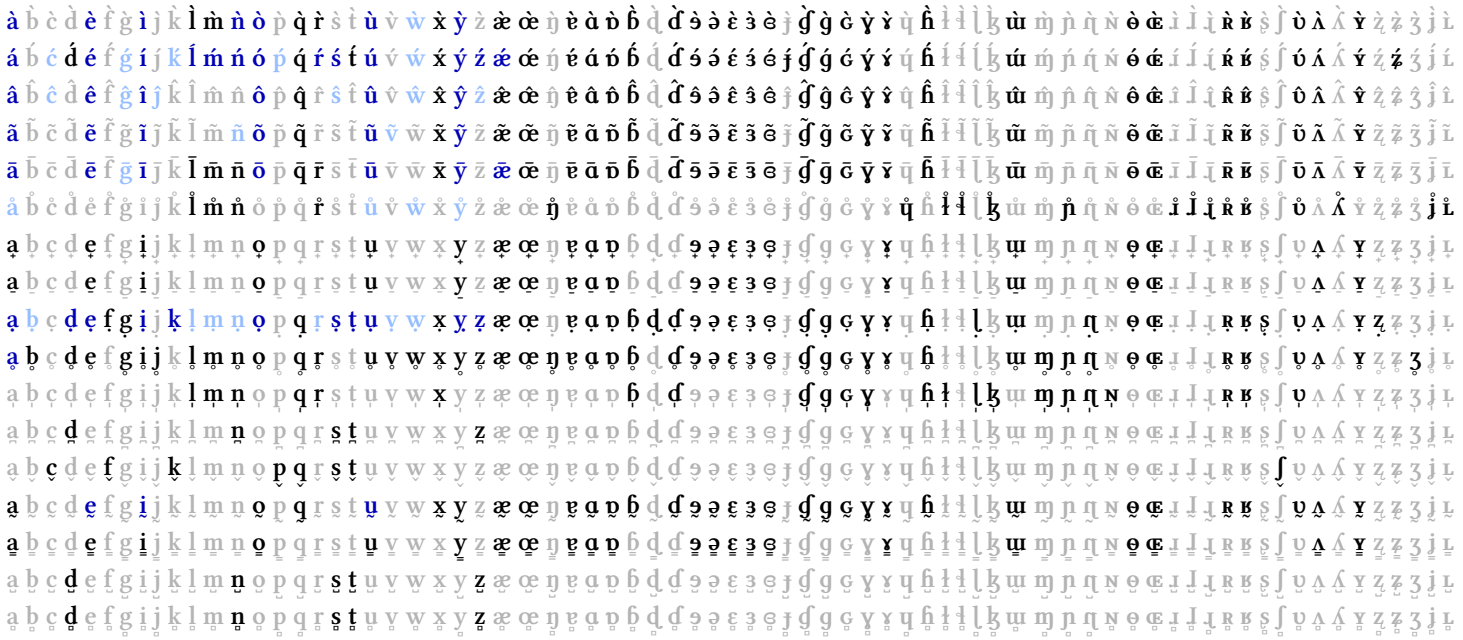
## Italic patched, Bases requiring anchors for common marks, Common combining marks



## Semibold, Bases requiring anchors for common marks, Common combining marks



## Semibold patched, Bases requiring anchors for common marks, Common combining marks



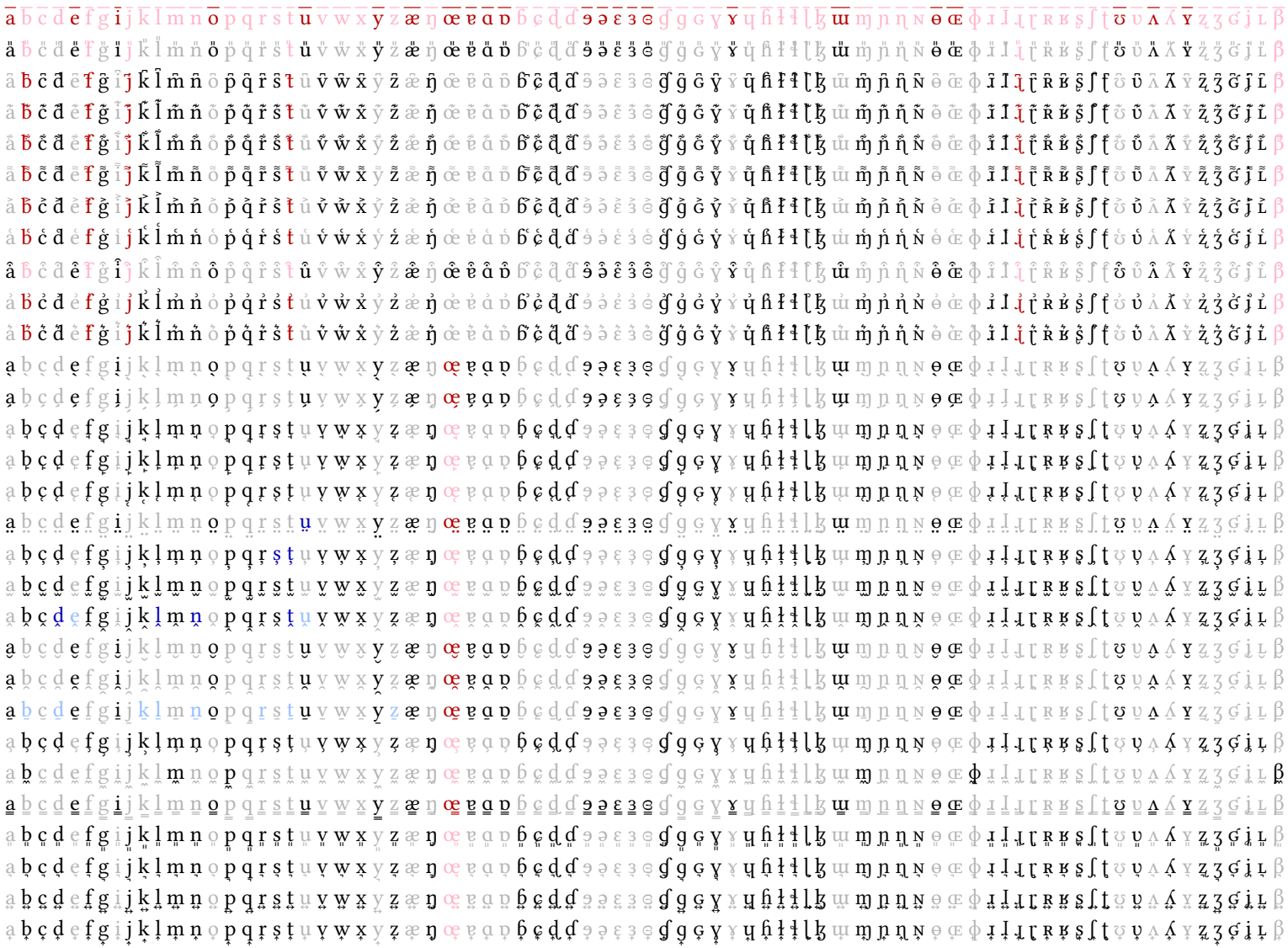
## Semibold italic, Bases requiring anchors for common marks, Common combining marks



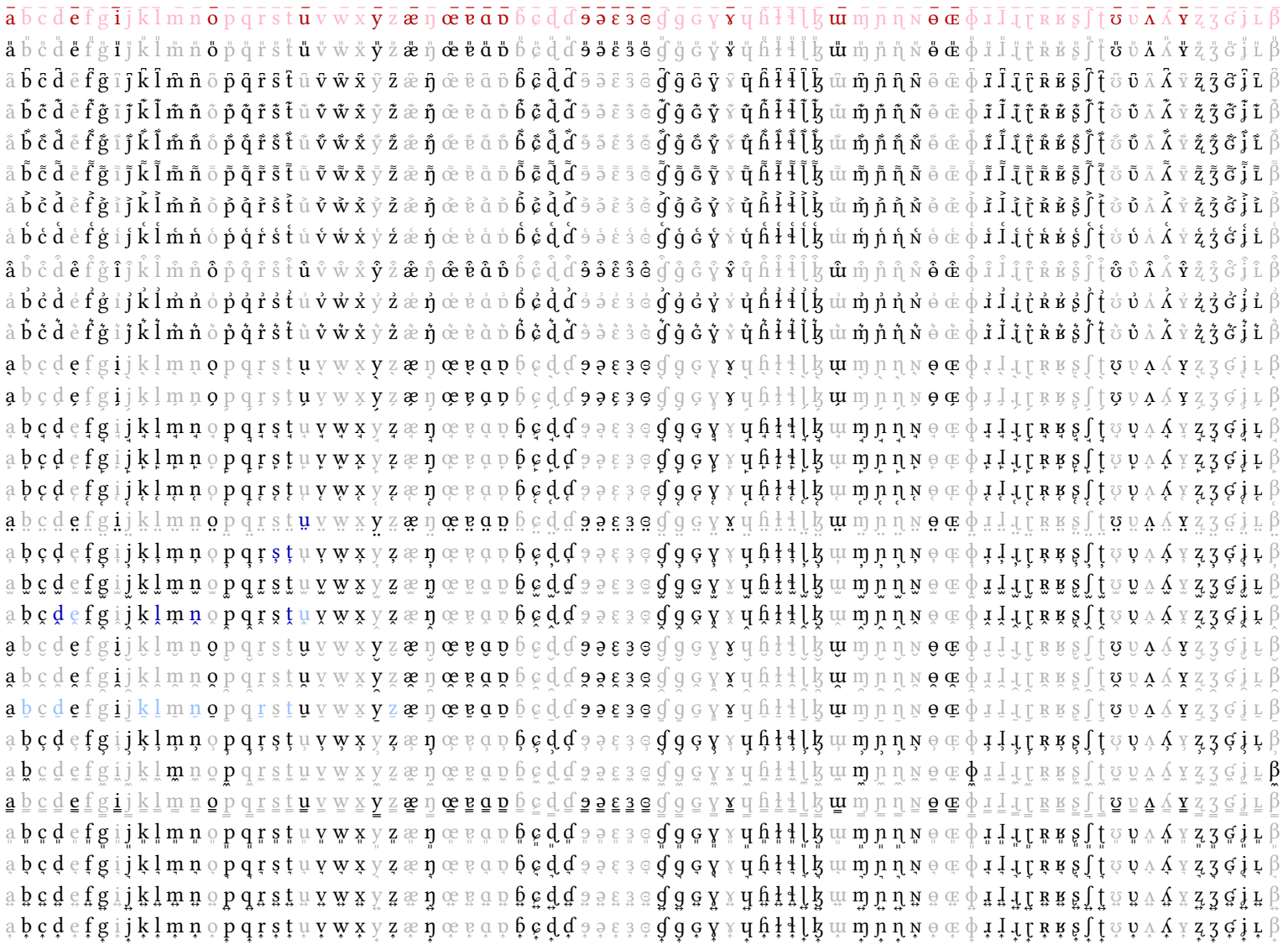
## Semibold italic patched, Bases requiring anchors for common marks, Common combining marks



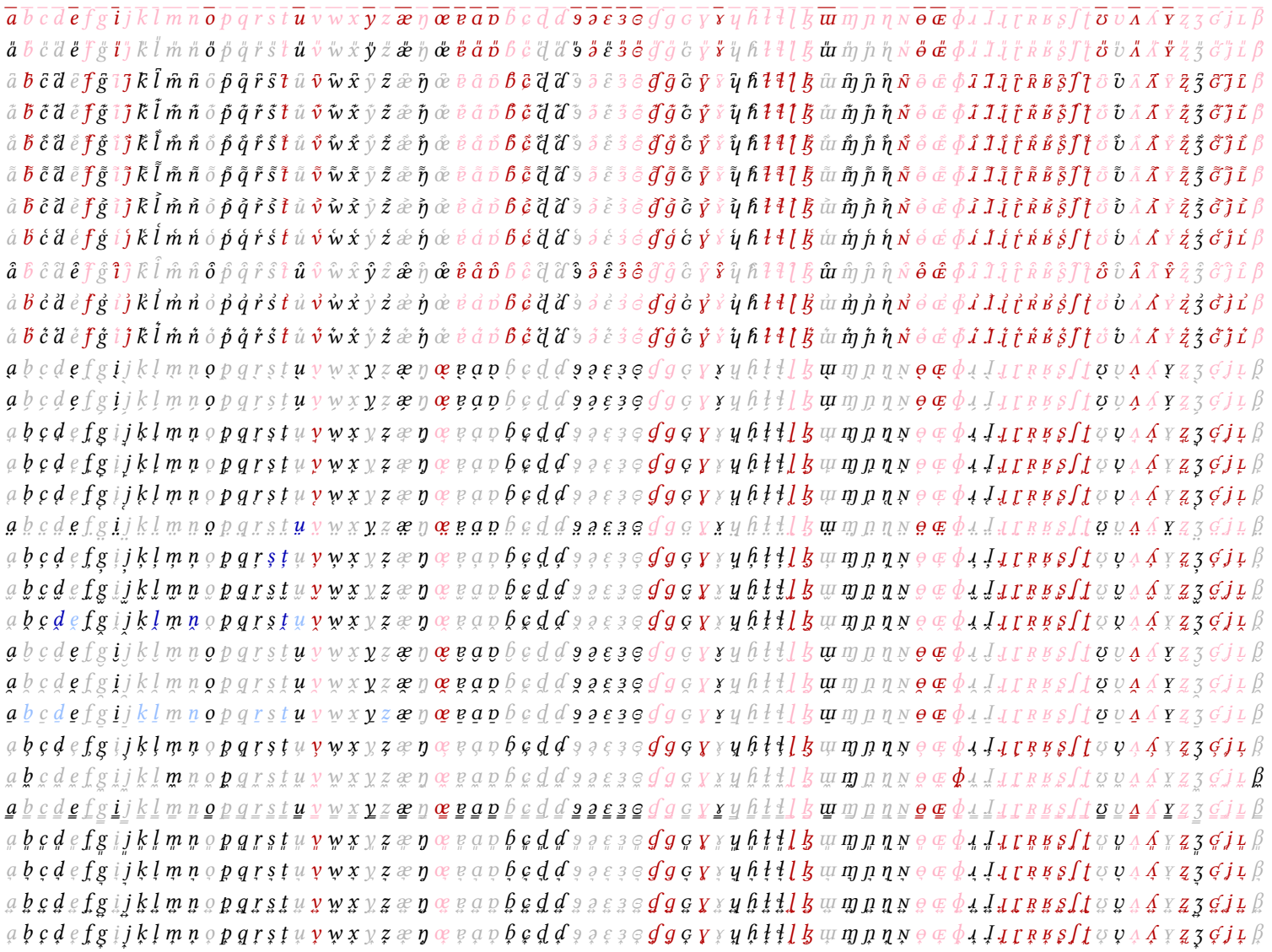
## Regular, Bases requiring anchors for rare marks, Rare combining marks



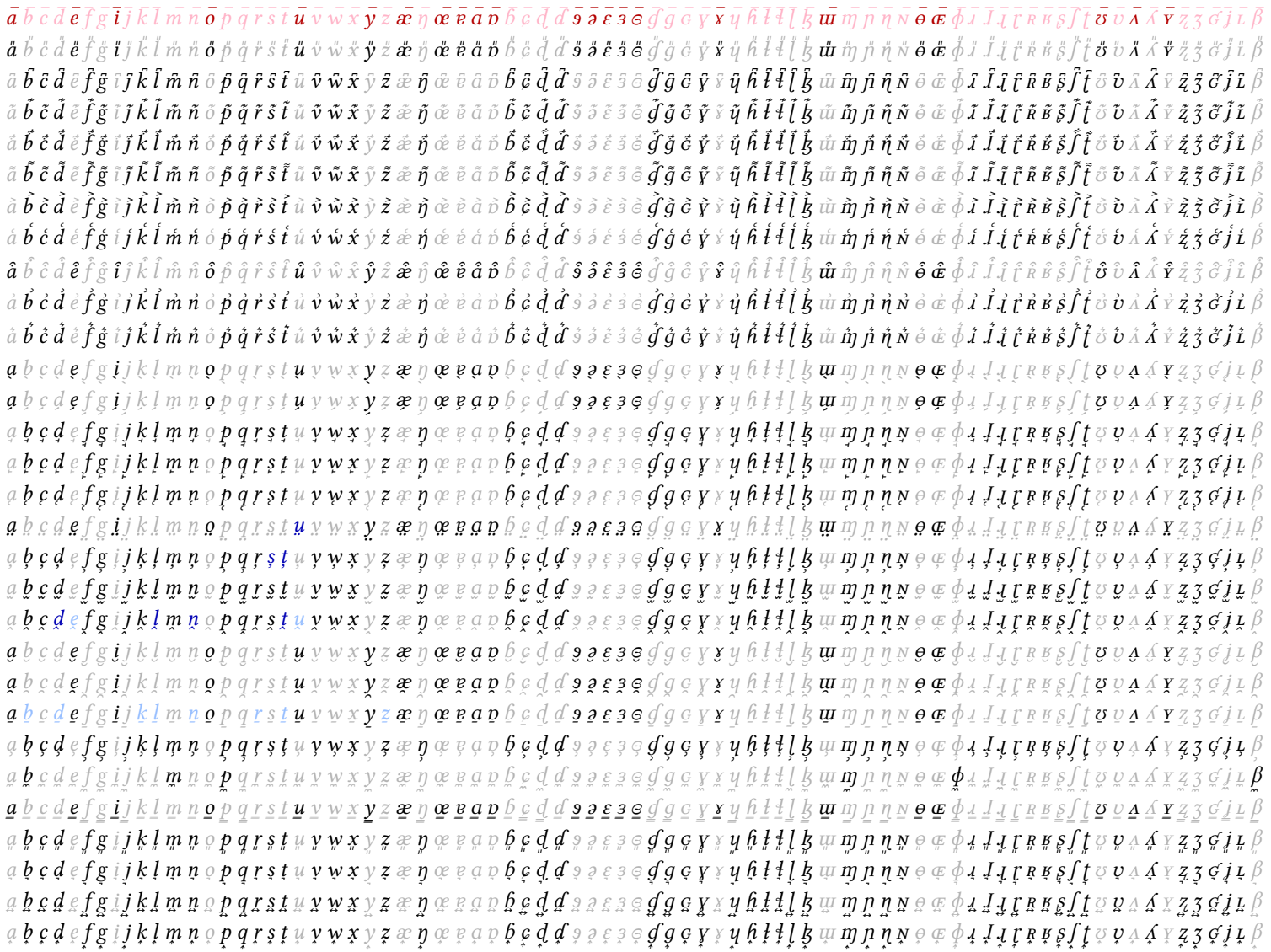
## Regular patched, Bases requiring anchors for rare marks, Rare combining marks



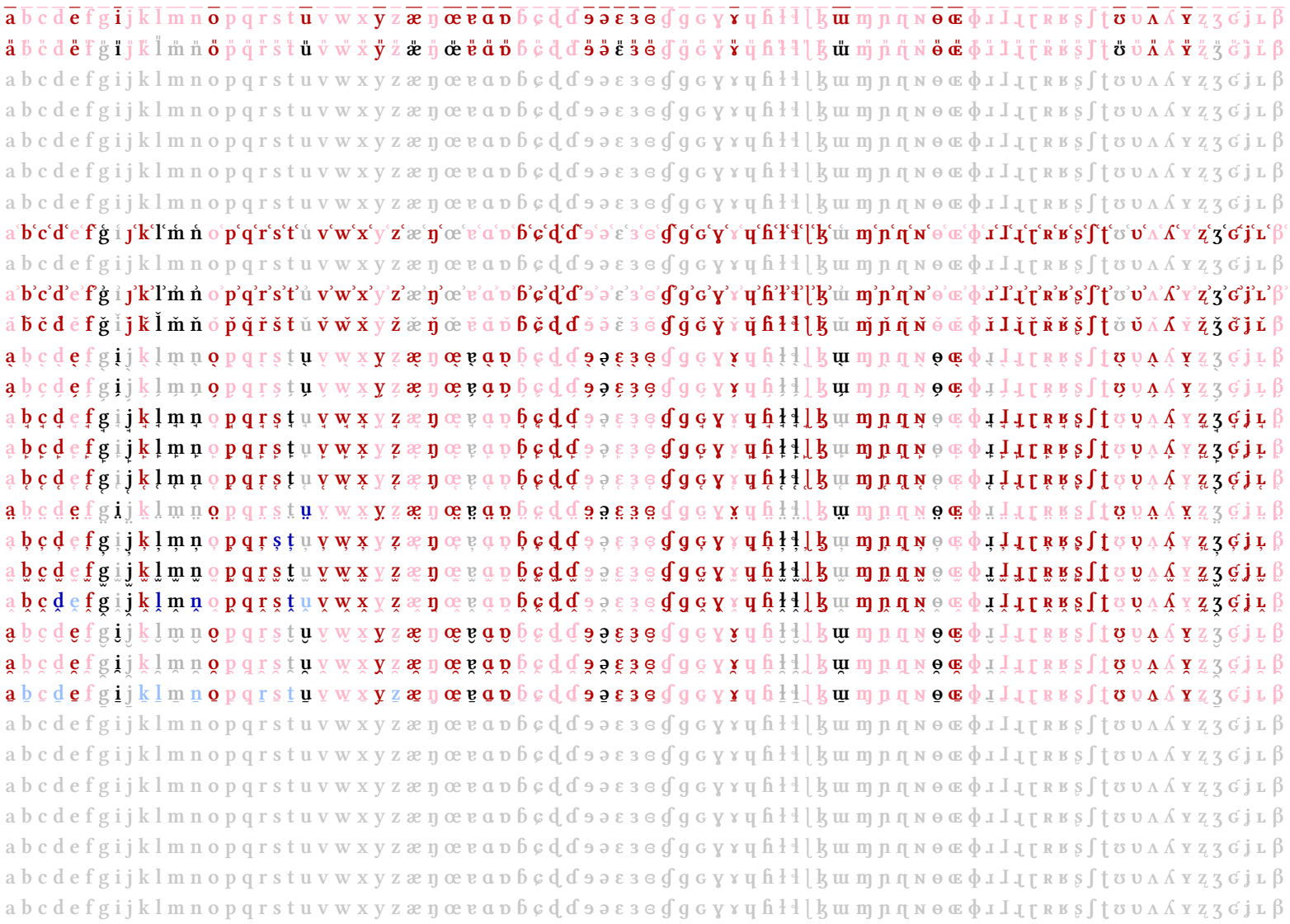
## Italic, Bases requiring anchors for rare marks, Rare combining marks



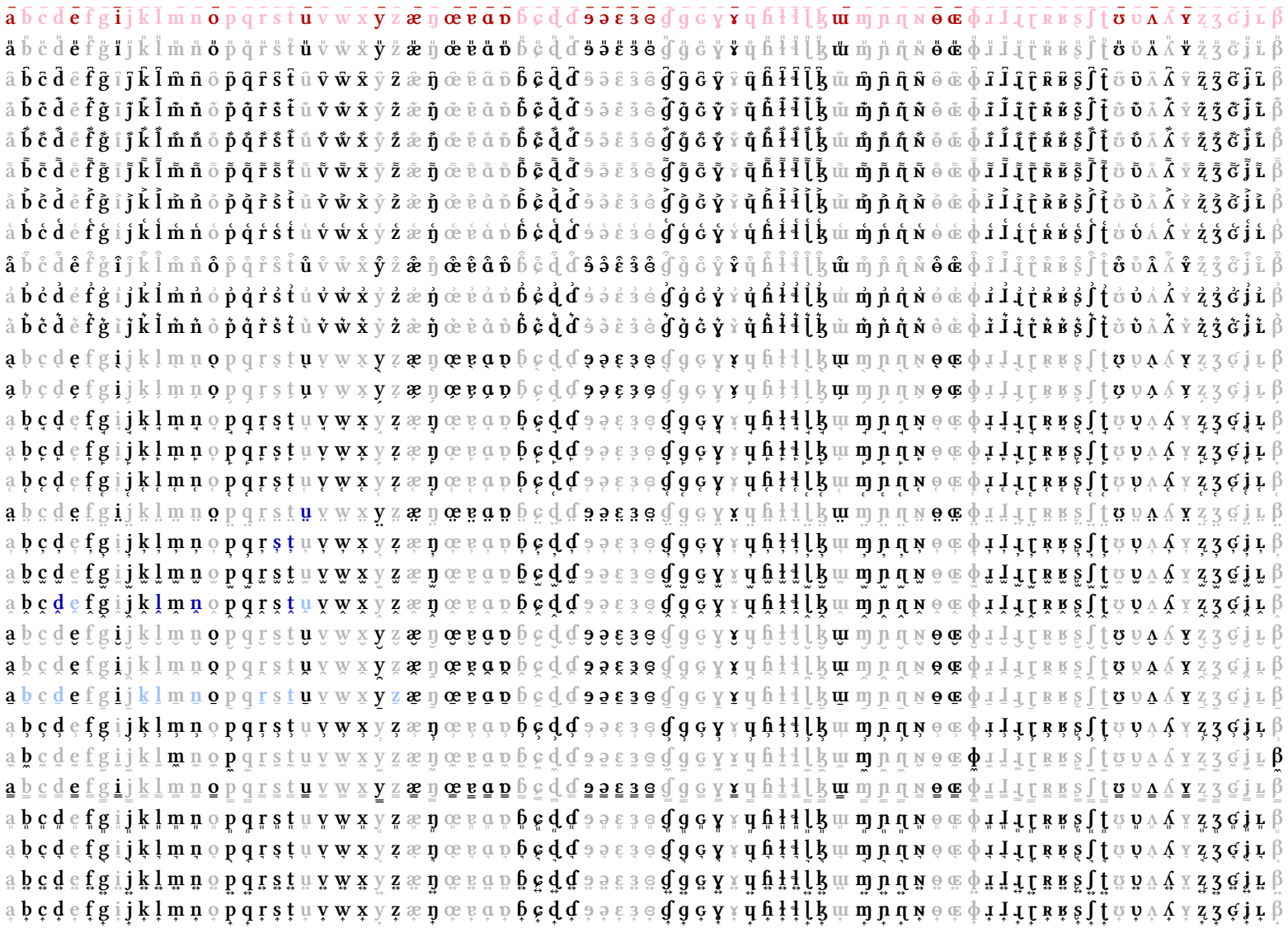
## Italic patched, Bases requiring anchors for rare marks, Rare combining marks



## Semibold, Bases requiring anchors for rare marks, Rare combining marks



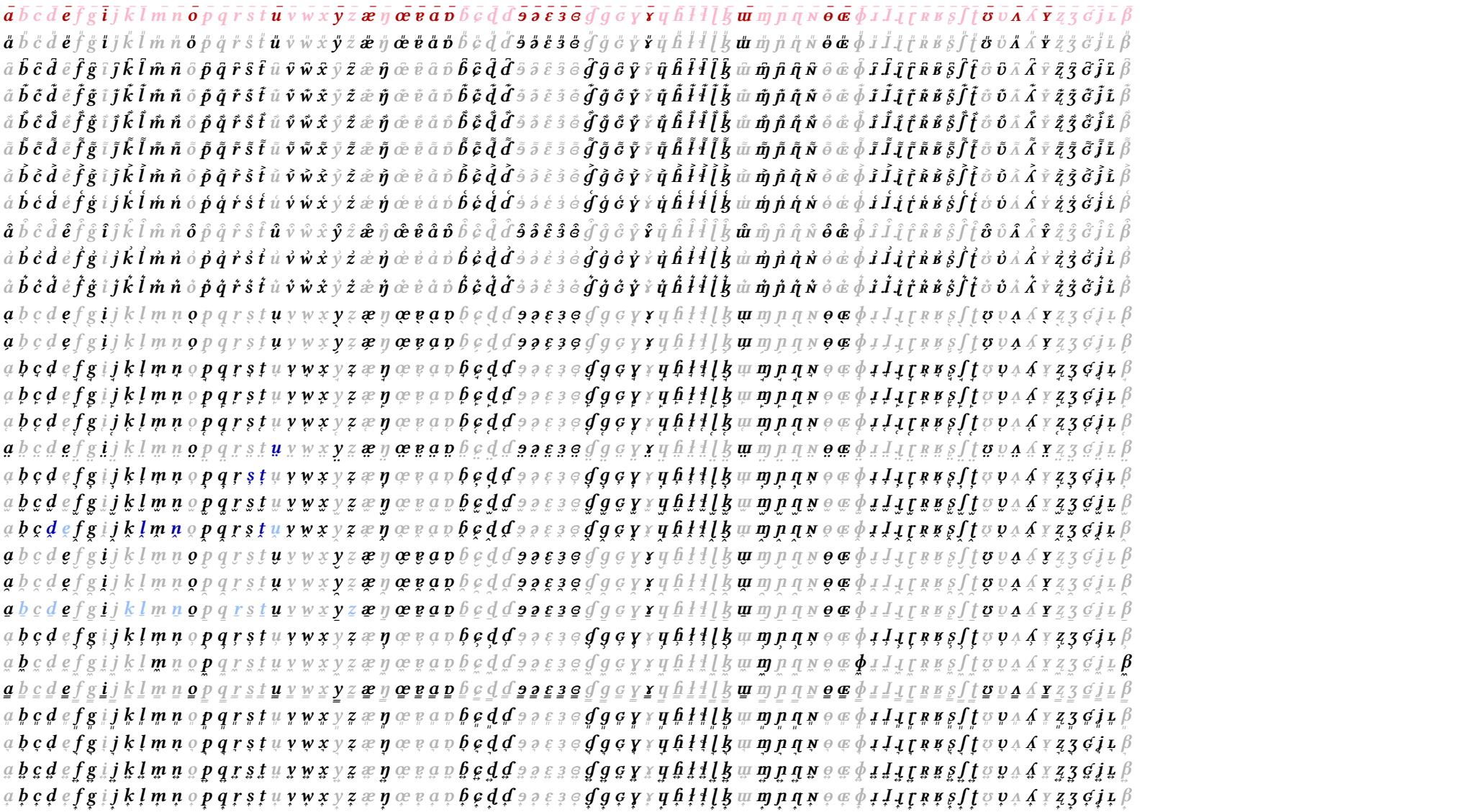
## Semibold patched, Bases requiring anchors for rare marks, Rare combining marks



## Semibold italic, Bases requiring anchors for rare marks, Rare combining marks



## Semibold italic patched, Bases requiring anchors for rare marks, Rare combining marks



## Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks. They leave out the letters whose anchors stage 8 or stage 9 sets, the derived and precomposed letters, which have their own grids after their base letters (sections “Derived letters” and “Precomposed characters”).

Blue indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

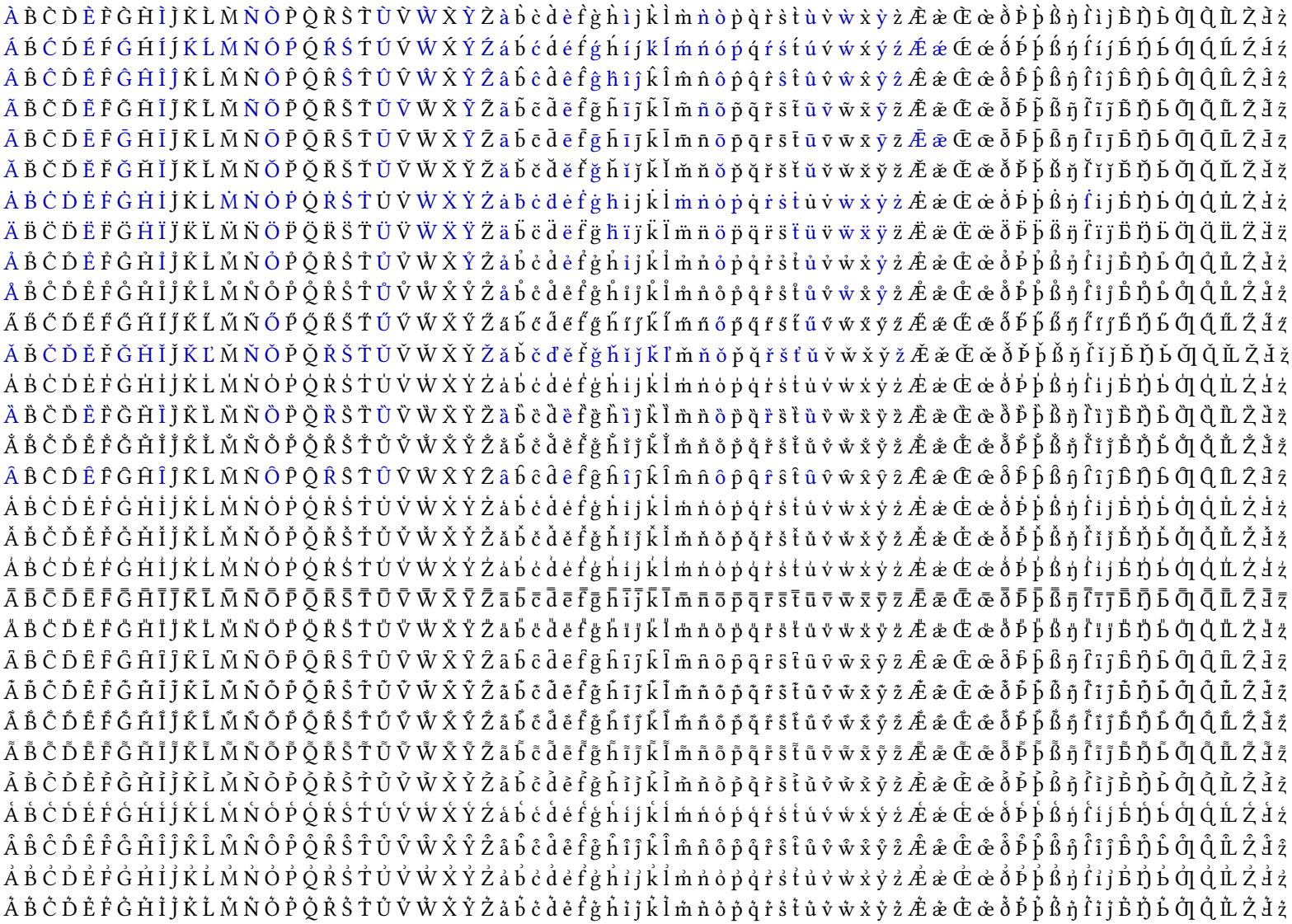
Red indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

Light gray indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.

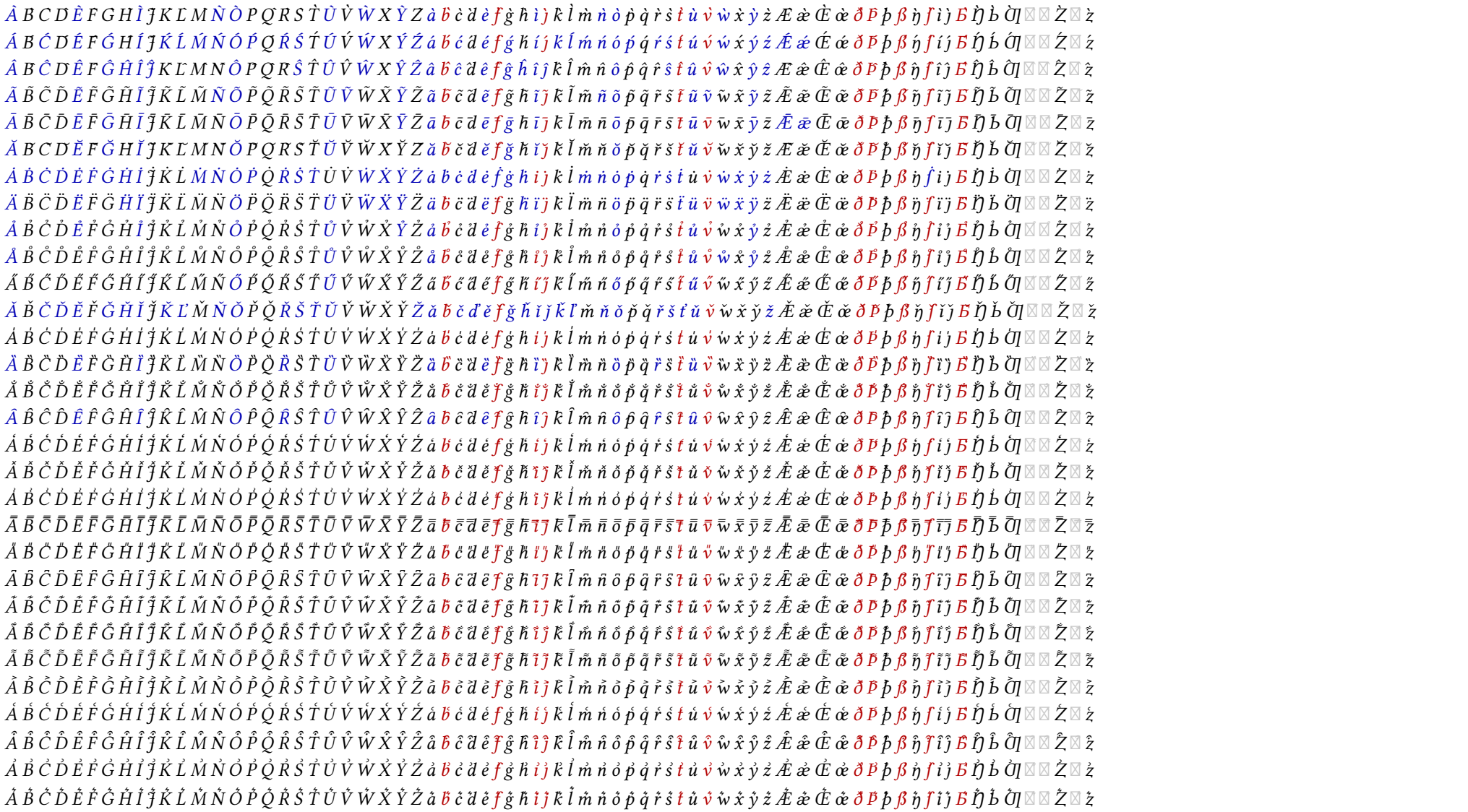
## Regular, Latin, Above



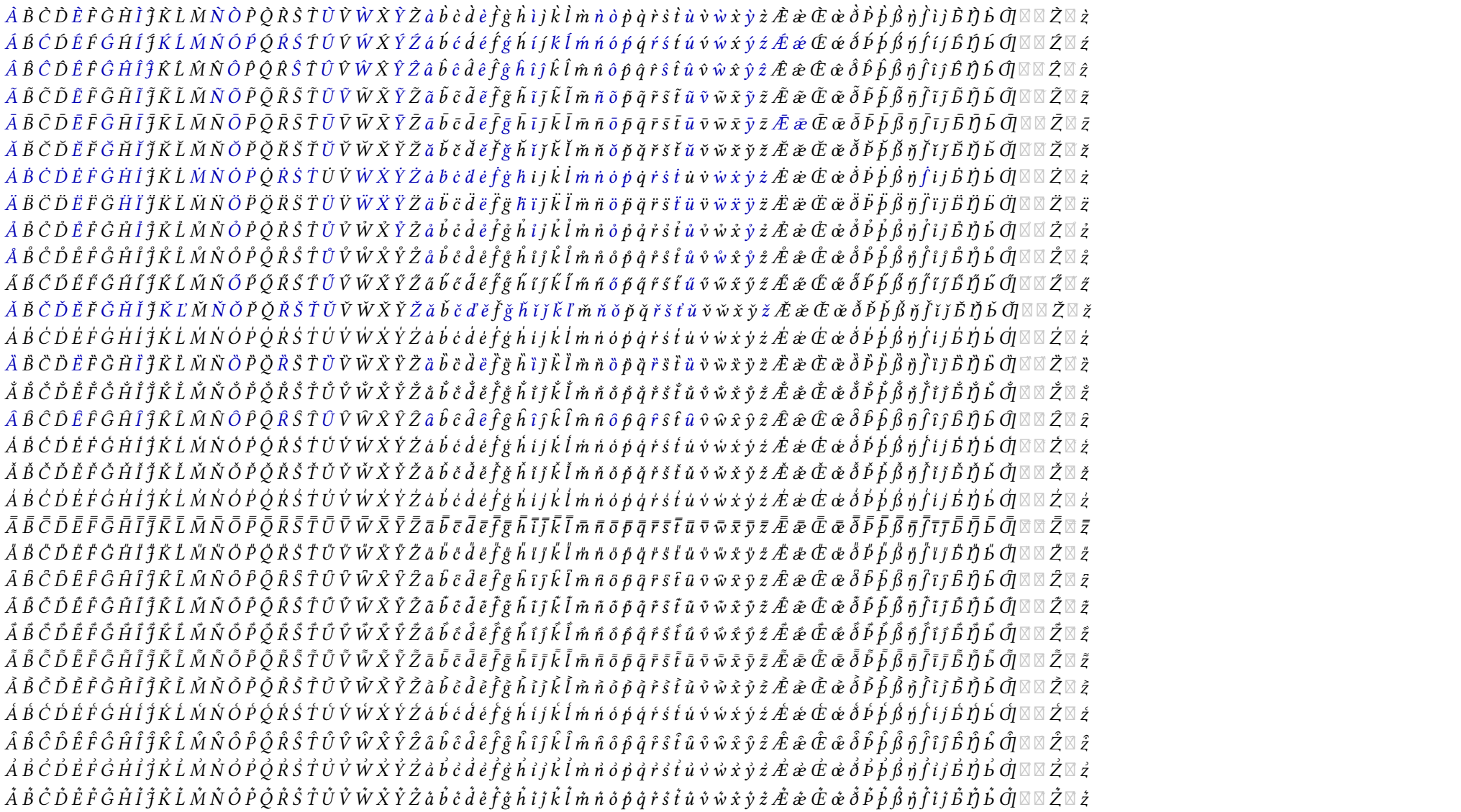
## Regular patched, Latin, Above



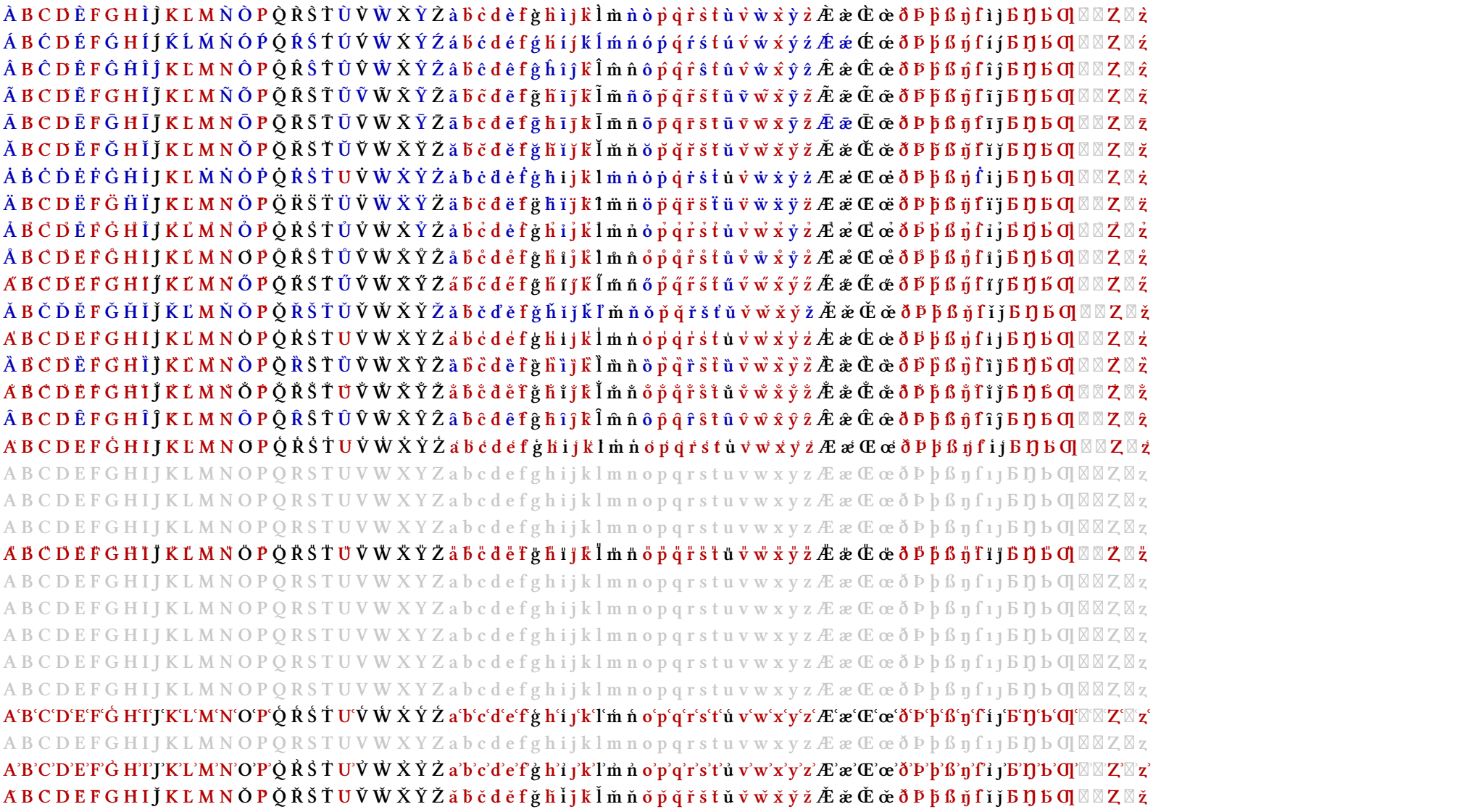
## Italic, Latin, Above



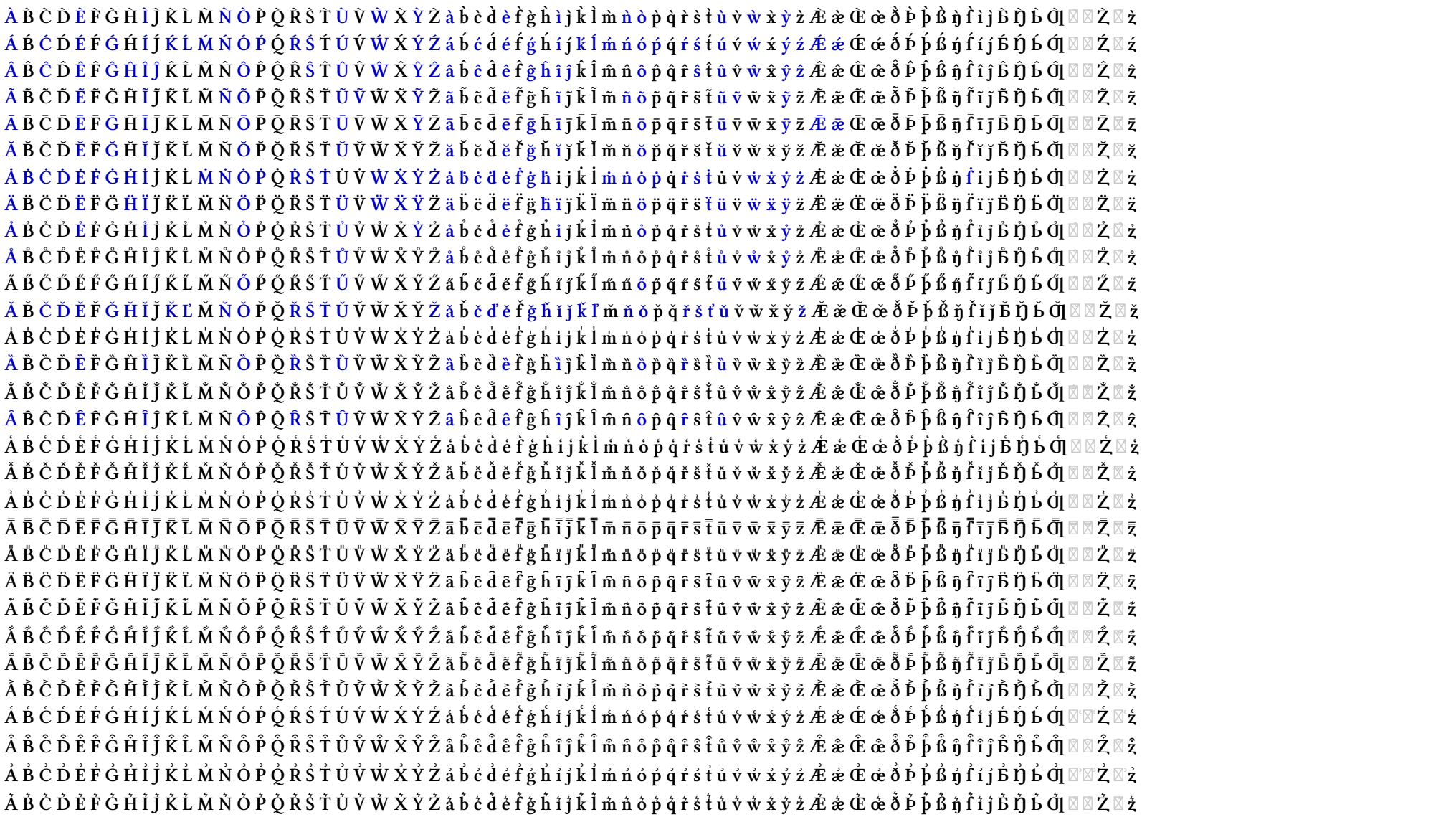
## Italic patched, Latin, Above



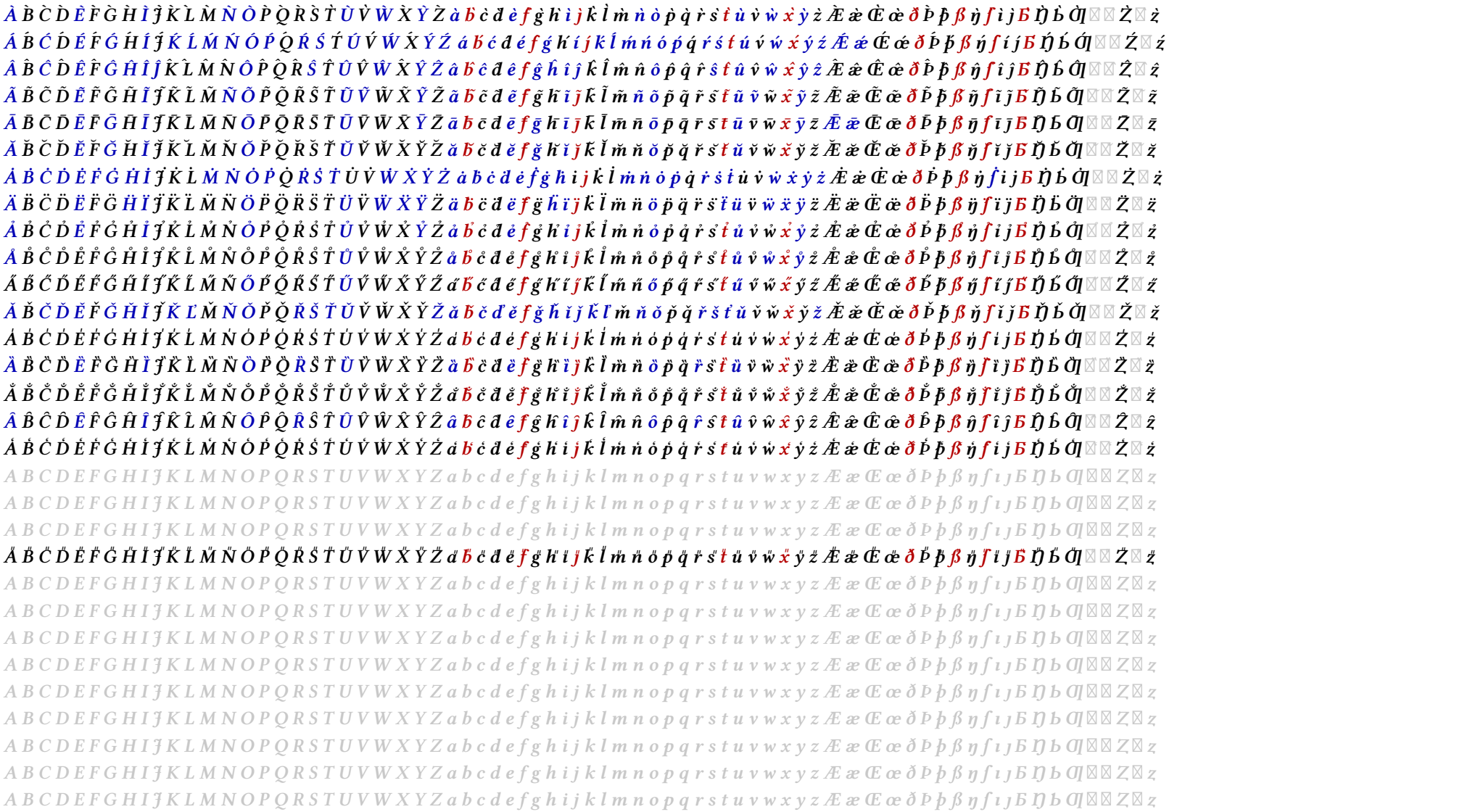
## Semibold, Latin, Above



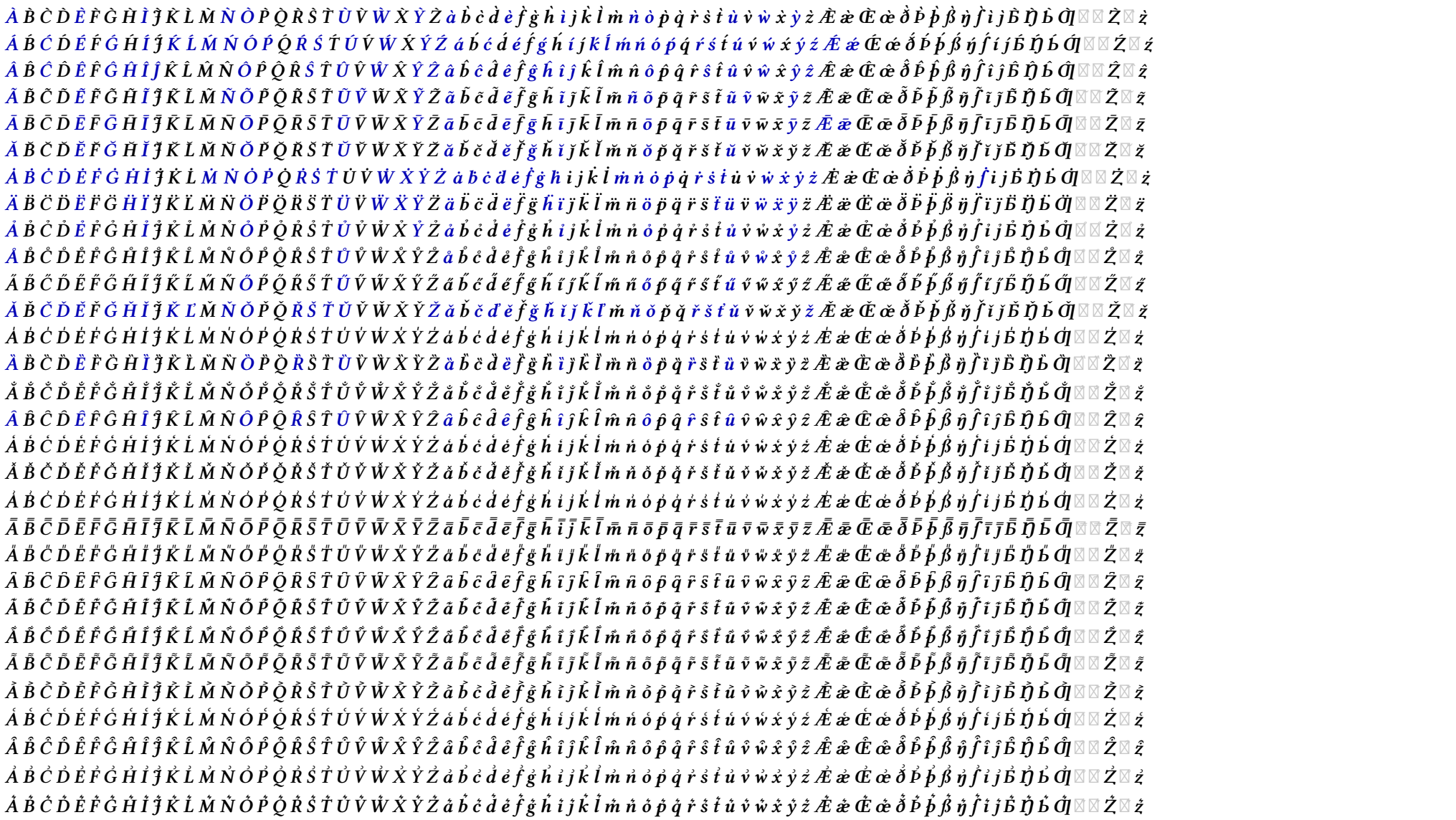
## Semibold patched, Latin, Above



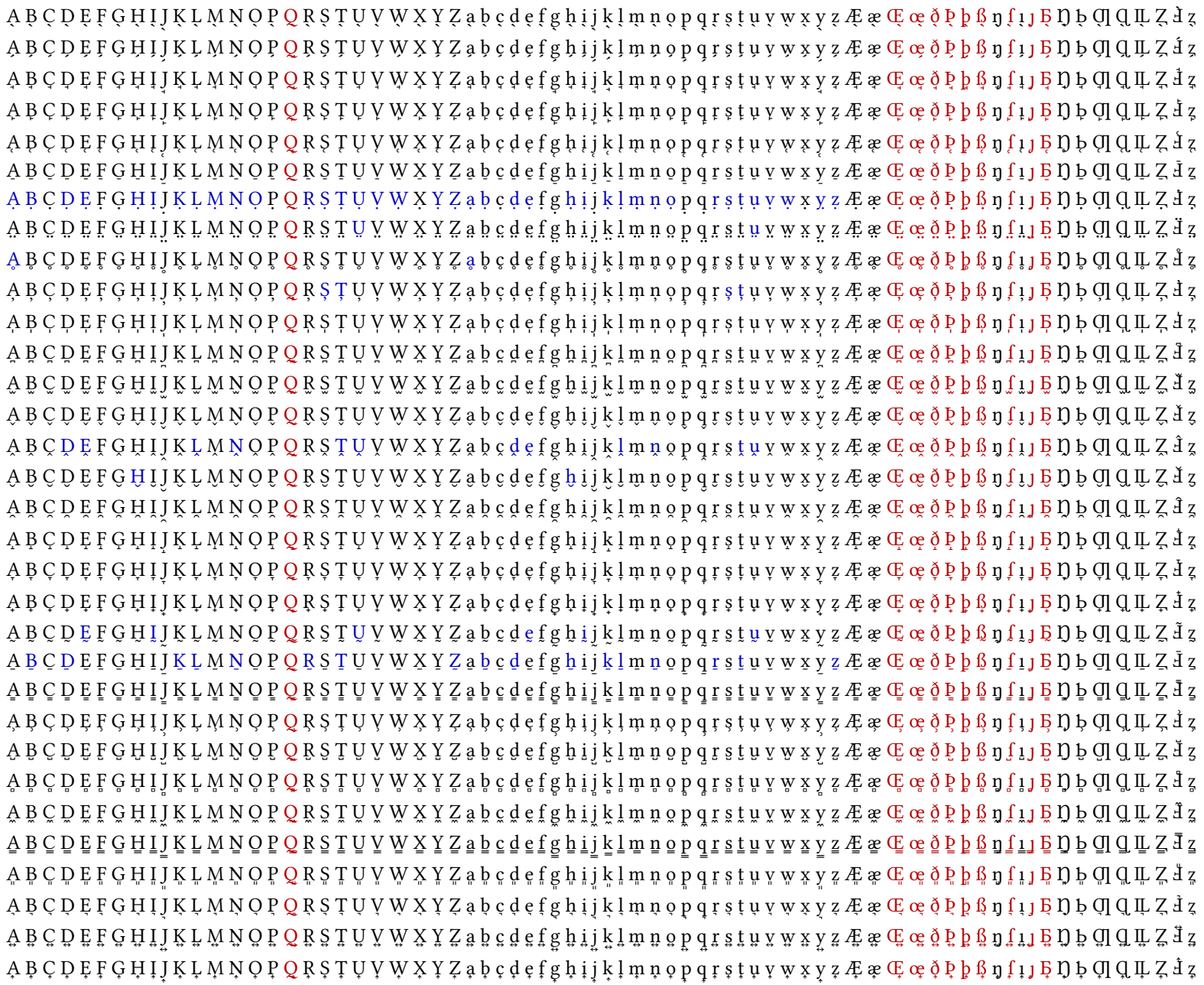
## Semibold italic, Latin, Above



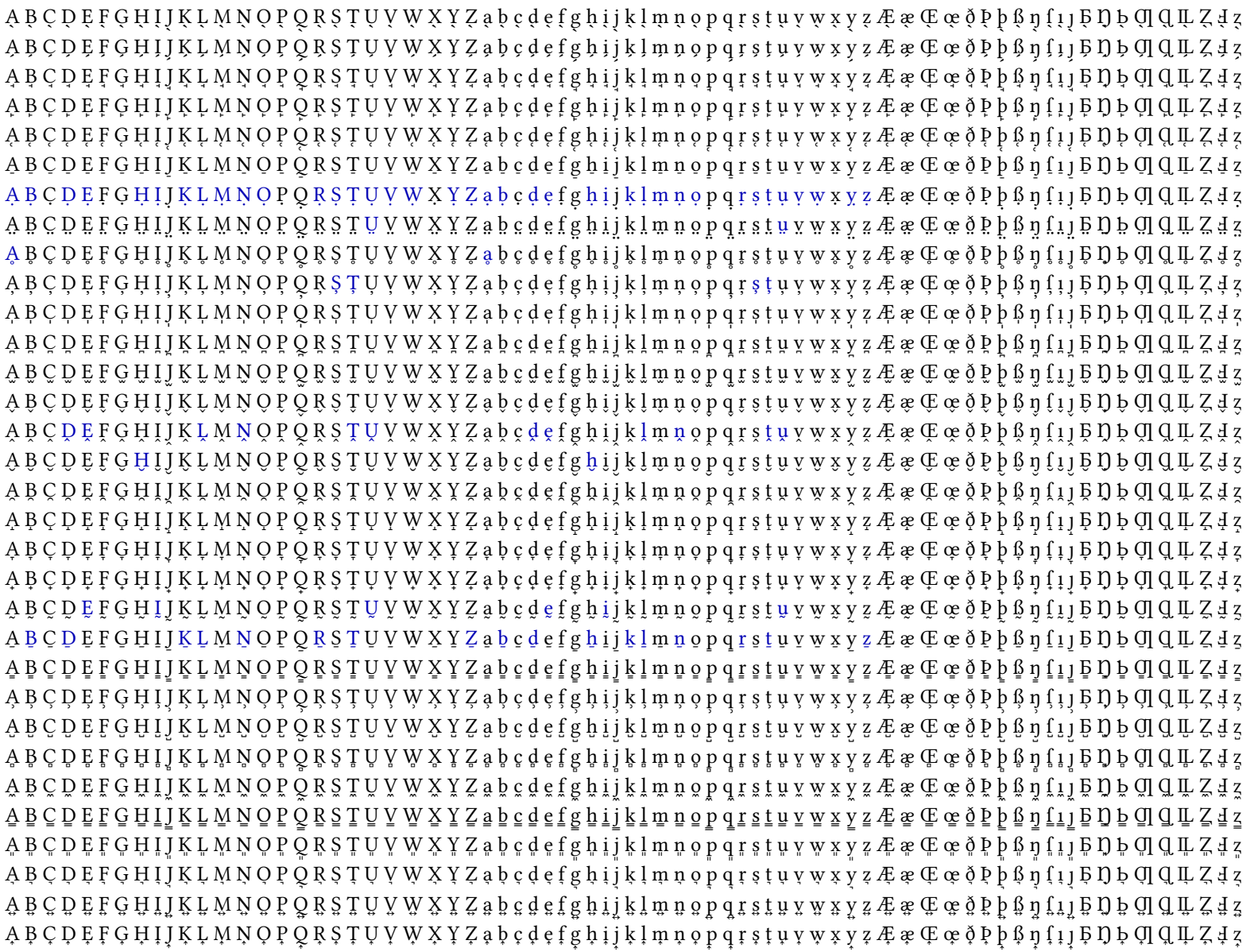
## Semibold italic patched, Latin, Above



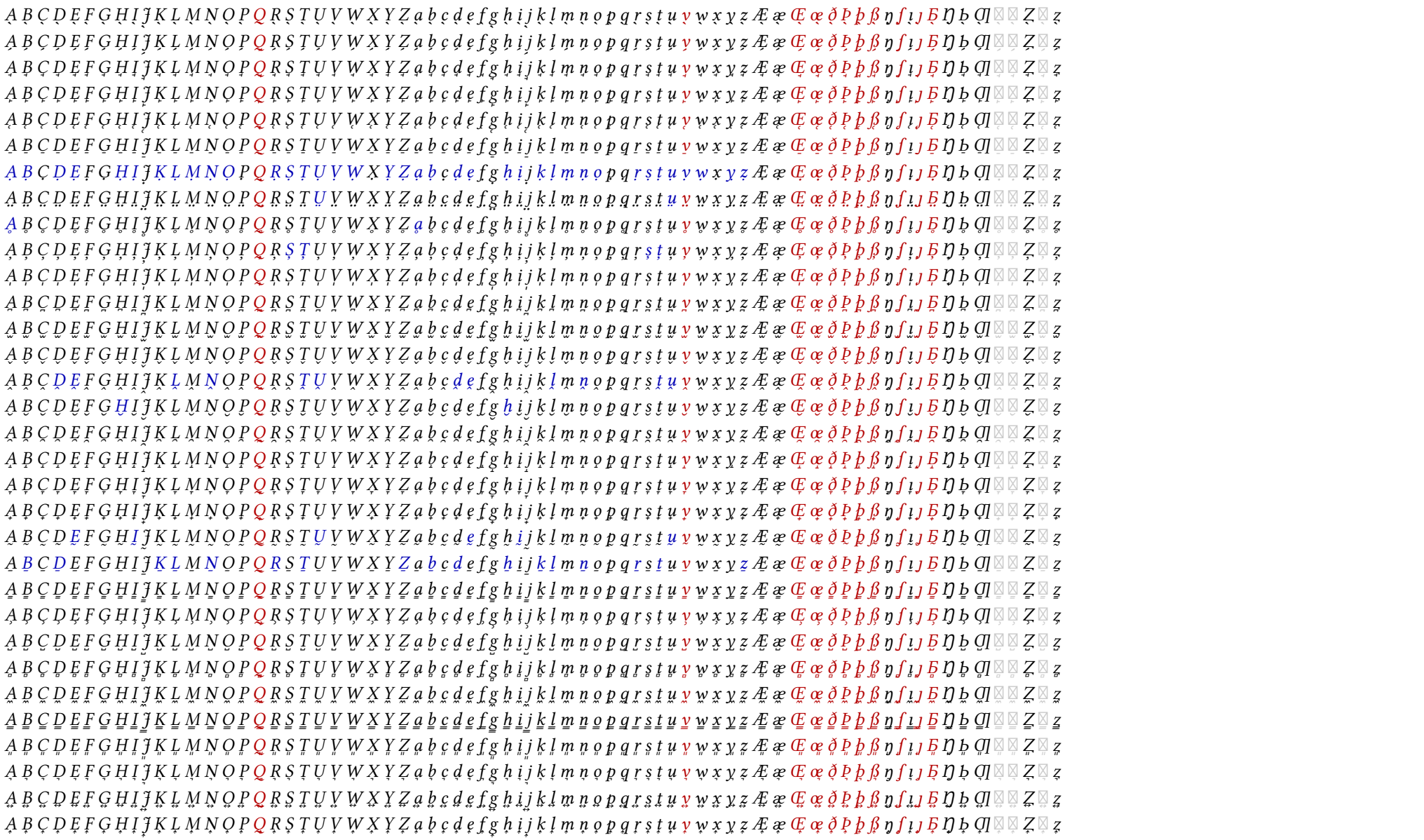
## Regular, Latin, Below



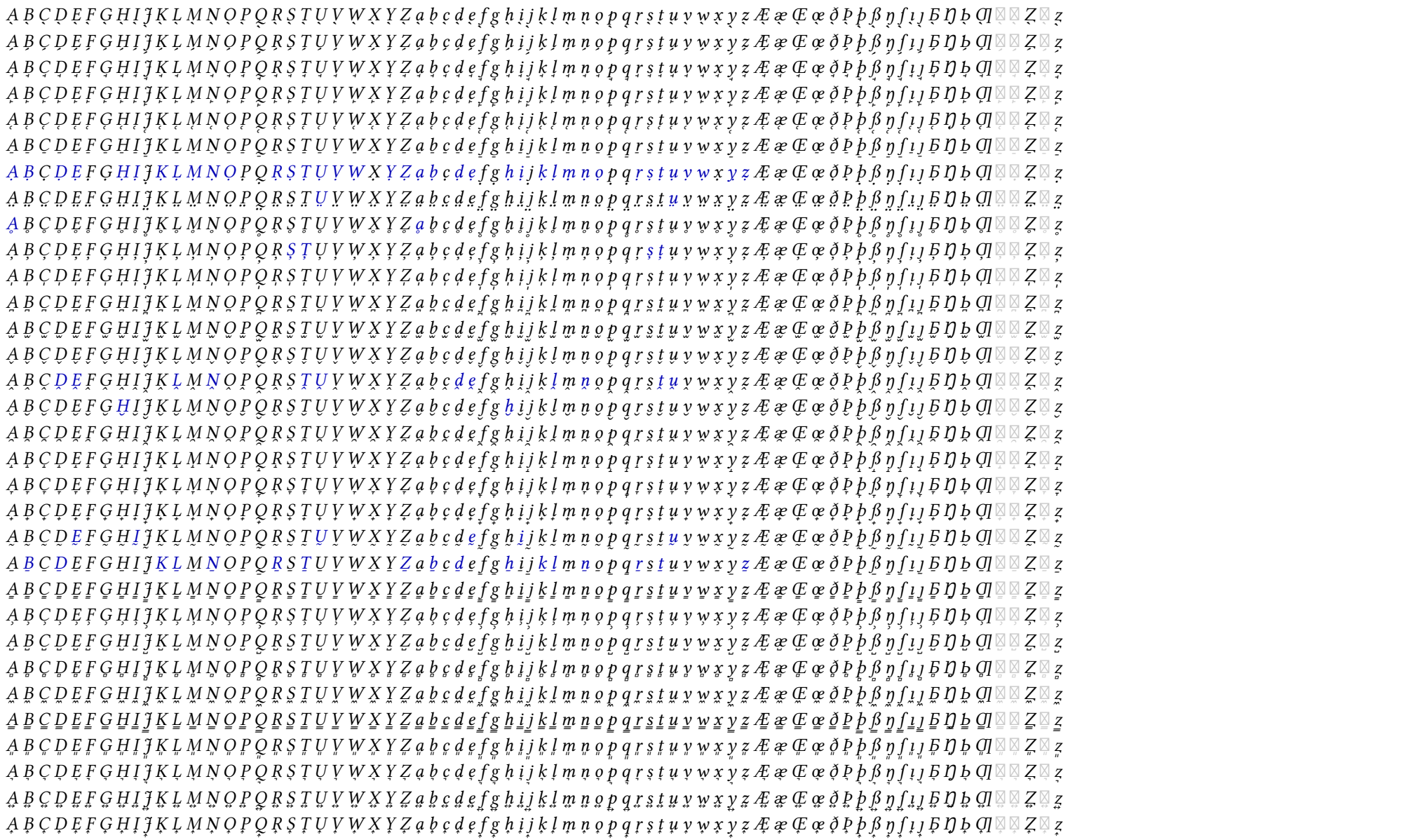
## Regular patched, Latin, Below



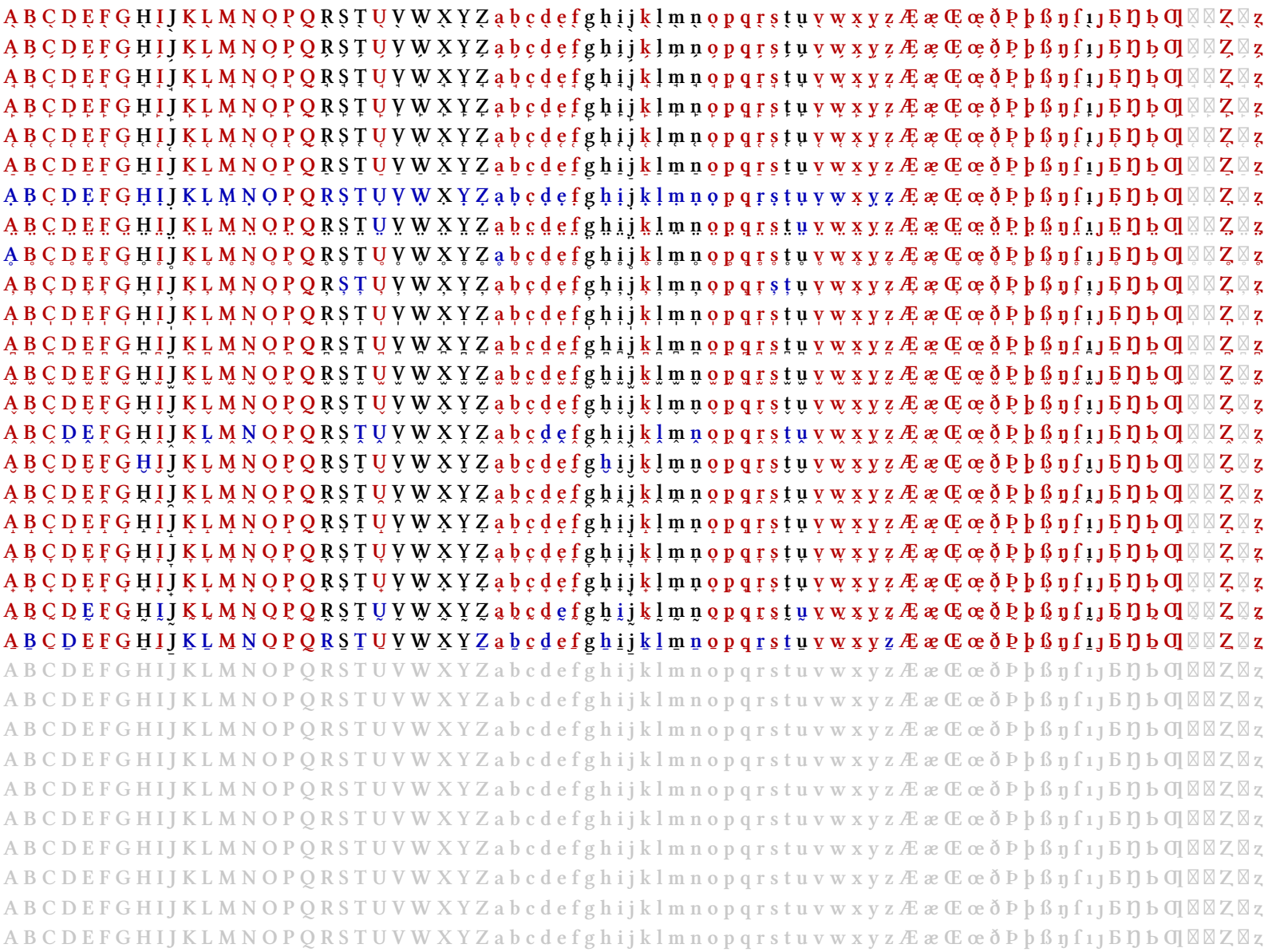
## Italic, Latin, Below



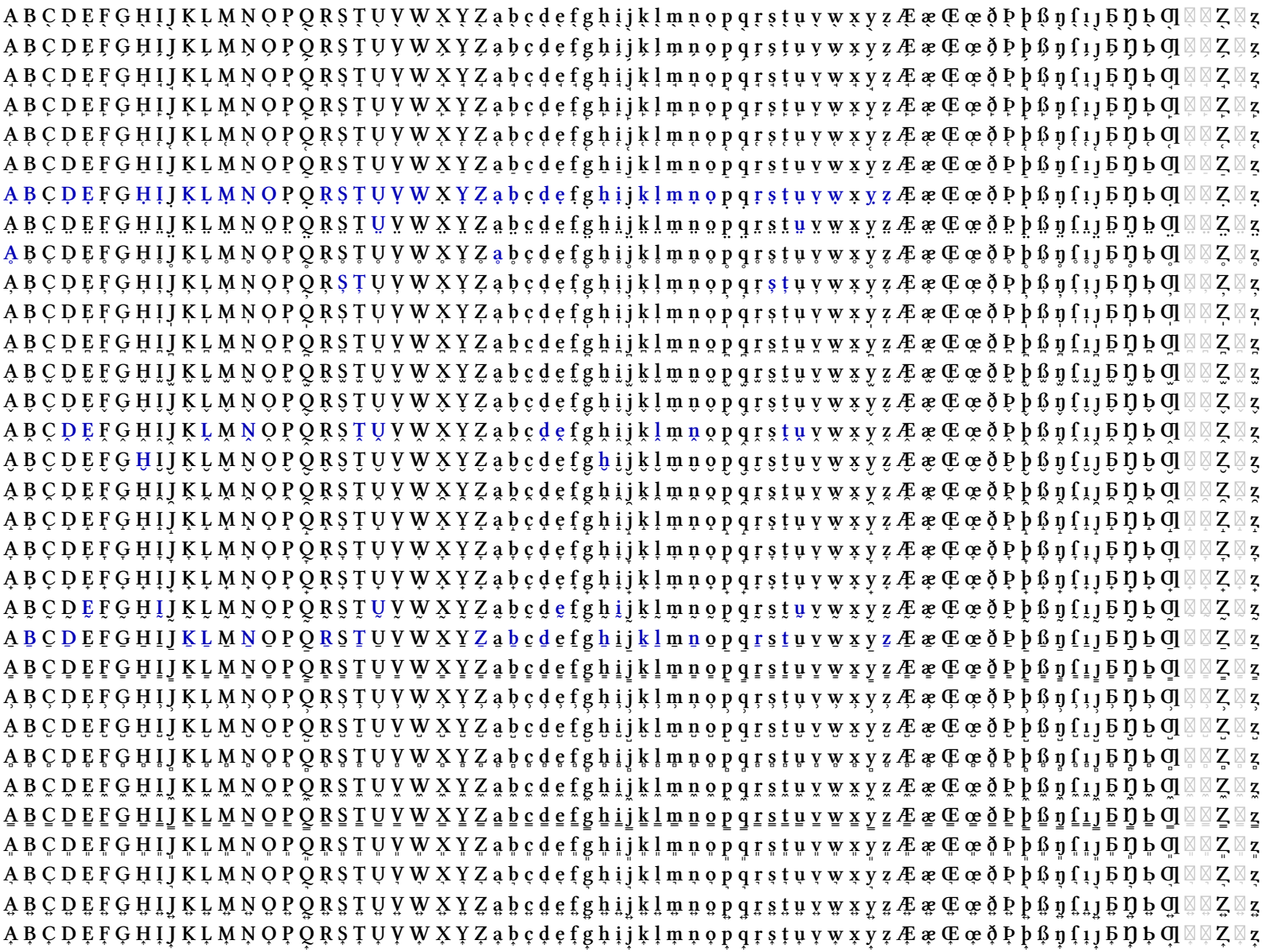
## Italic patched, Latin, Below



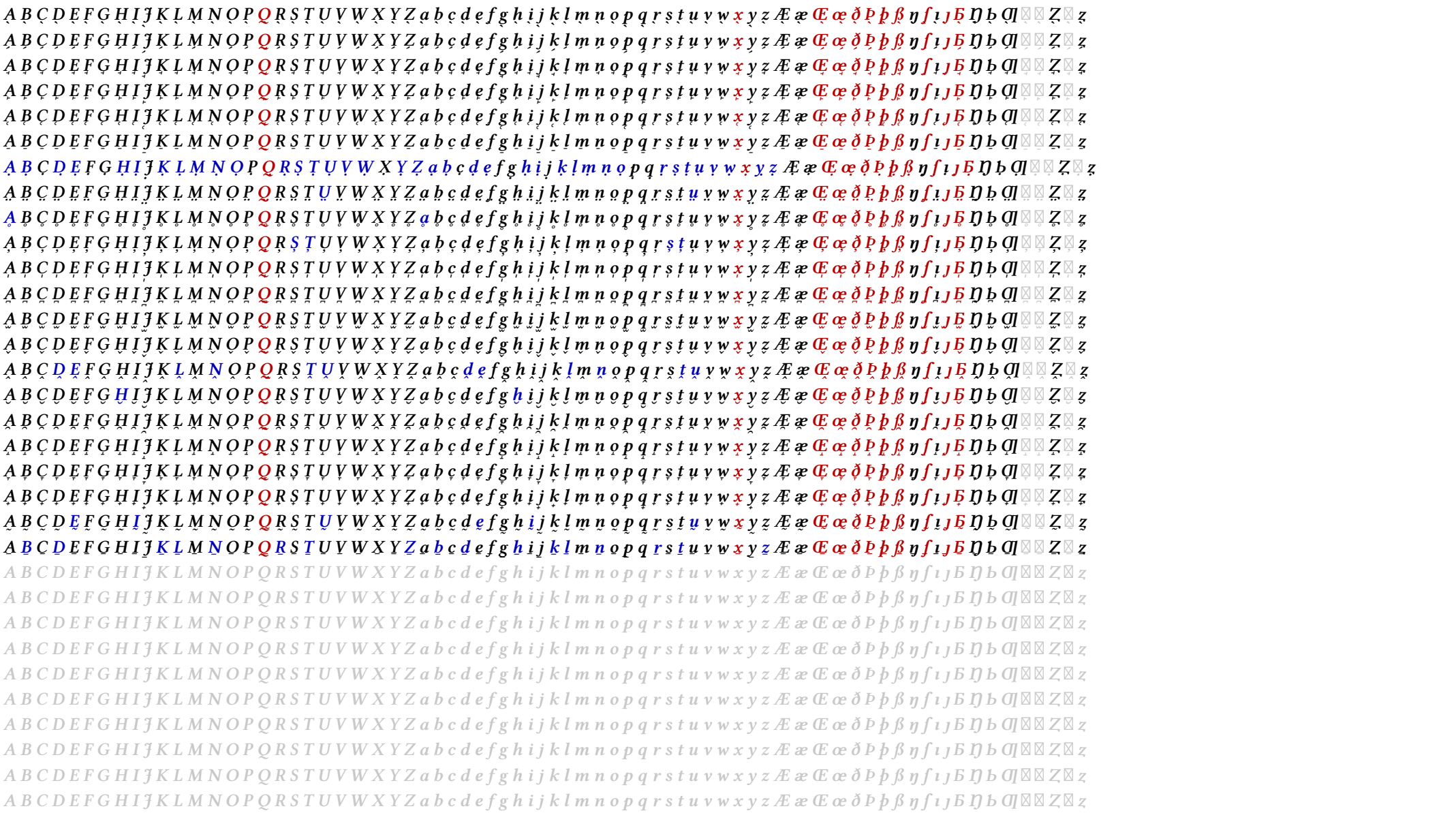
## Semibold, Latin, Below



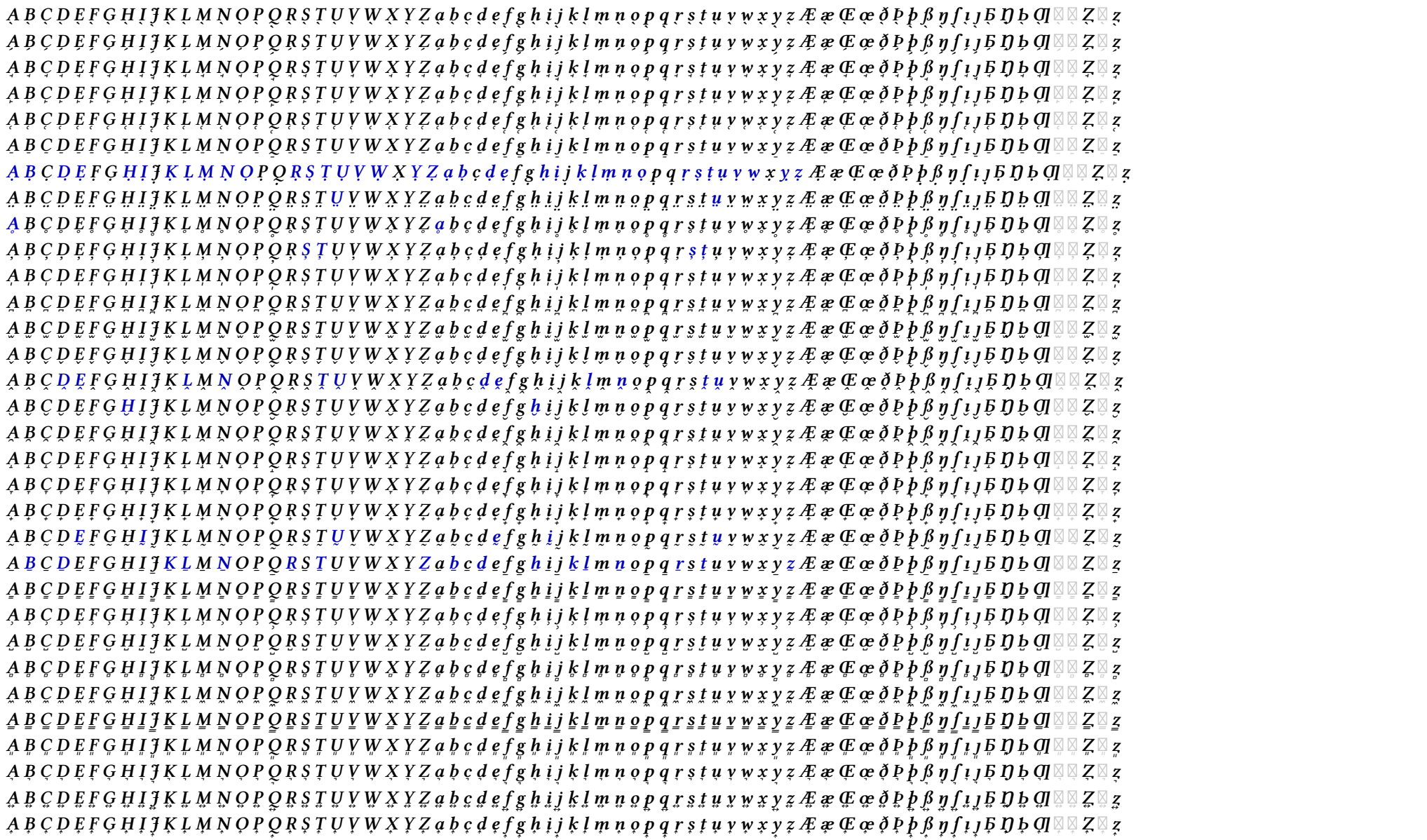
## Semibold patched, Latin, Below



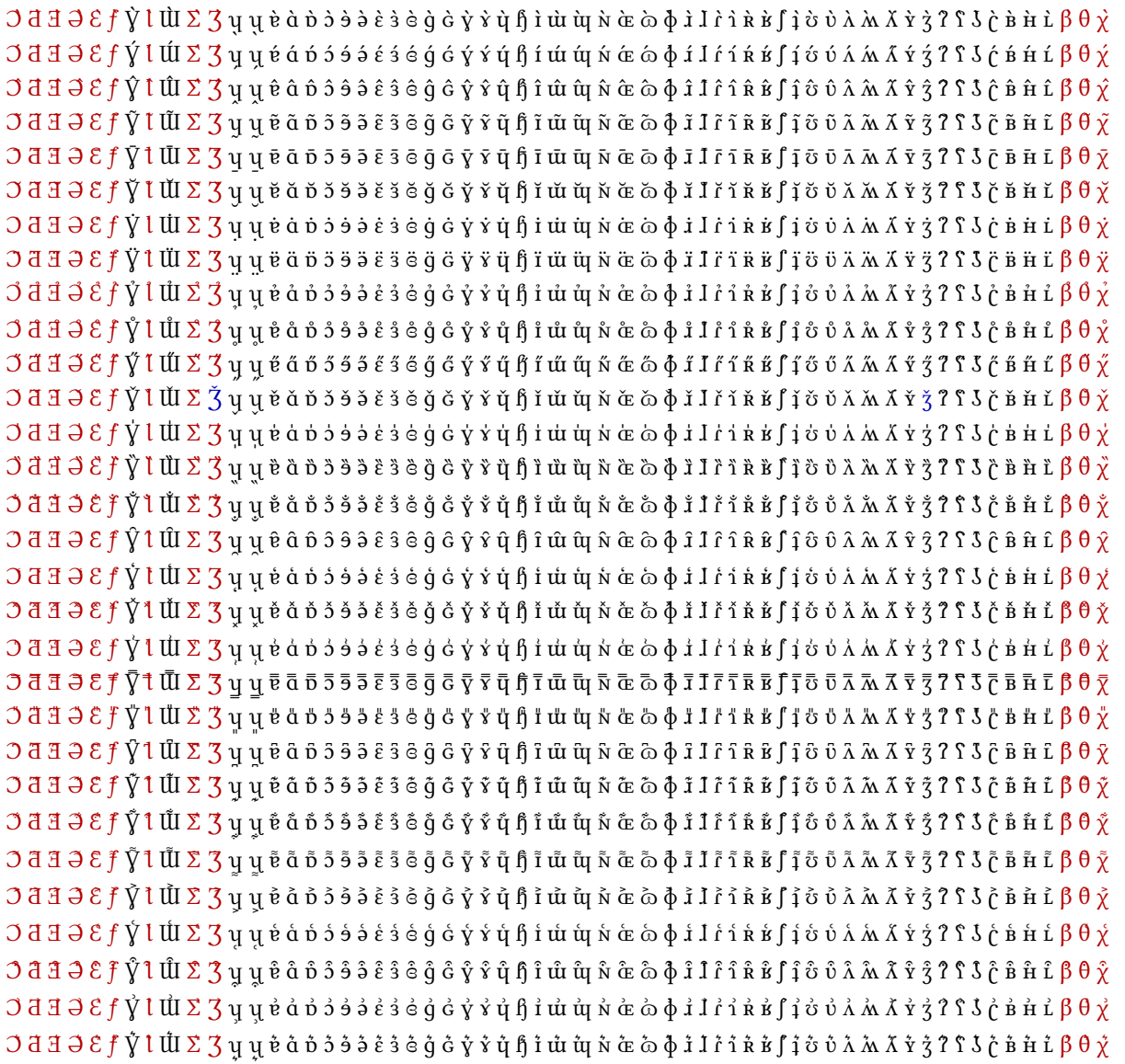
## Semibold italic, Latin, Below



## Semibold italic patched, Latin, Below



## Regular, IPA, Above



## Regular patched, IPA, Above

## Italic, IPA, Above



## Italic patched, IPA, Above



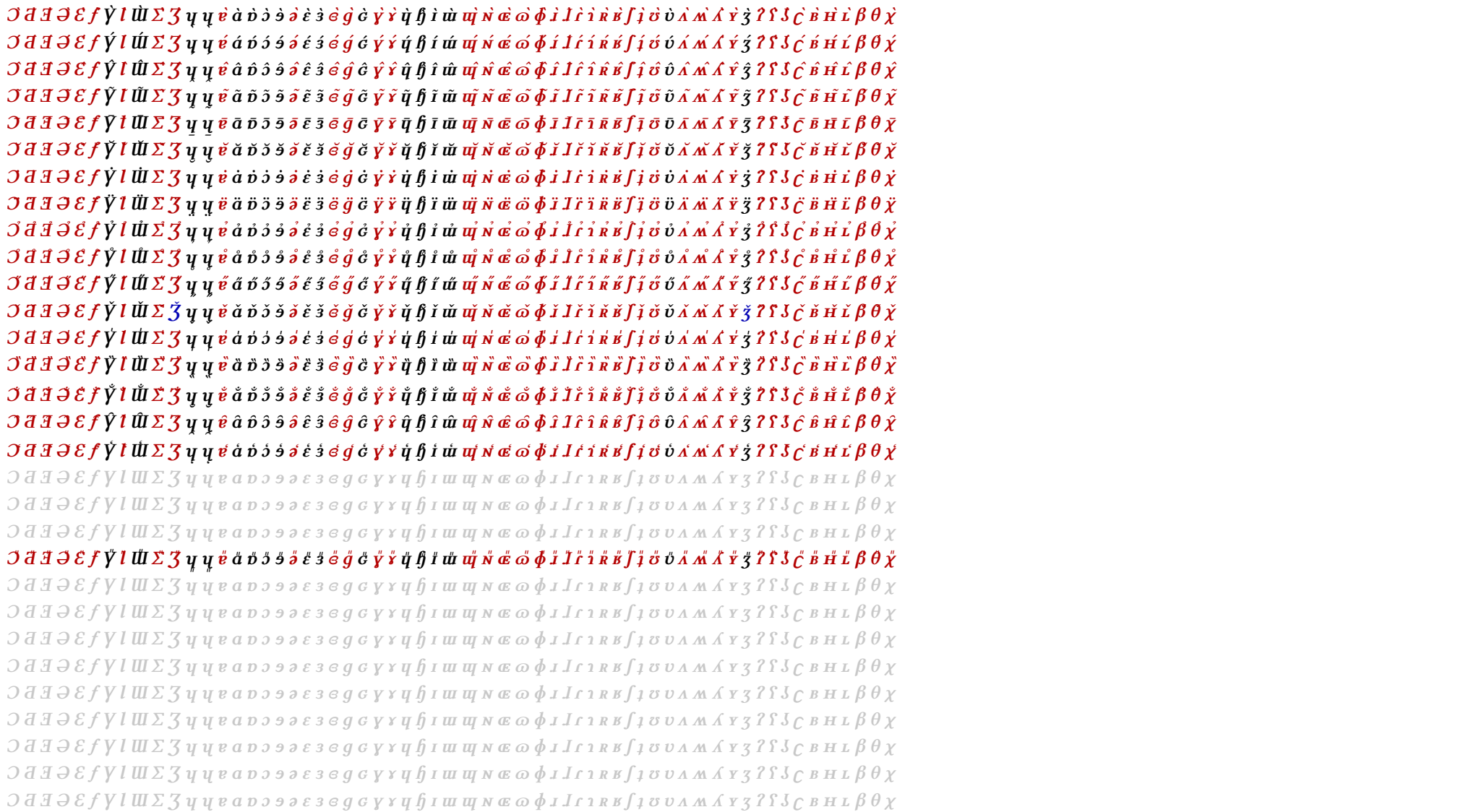
## Semibold, IPA, Above



## Semibold patched, IPA, Above

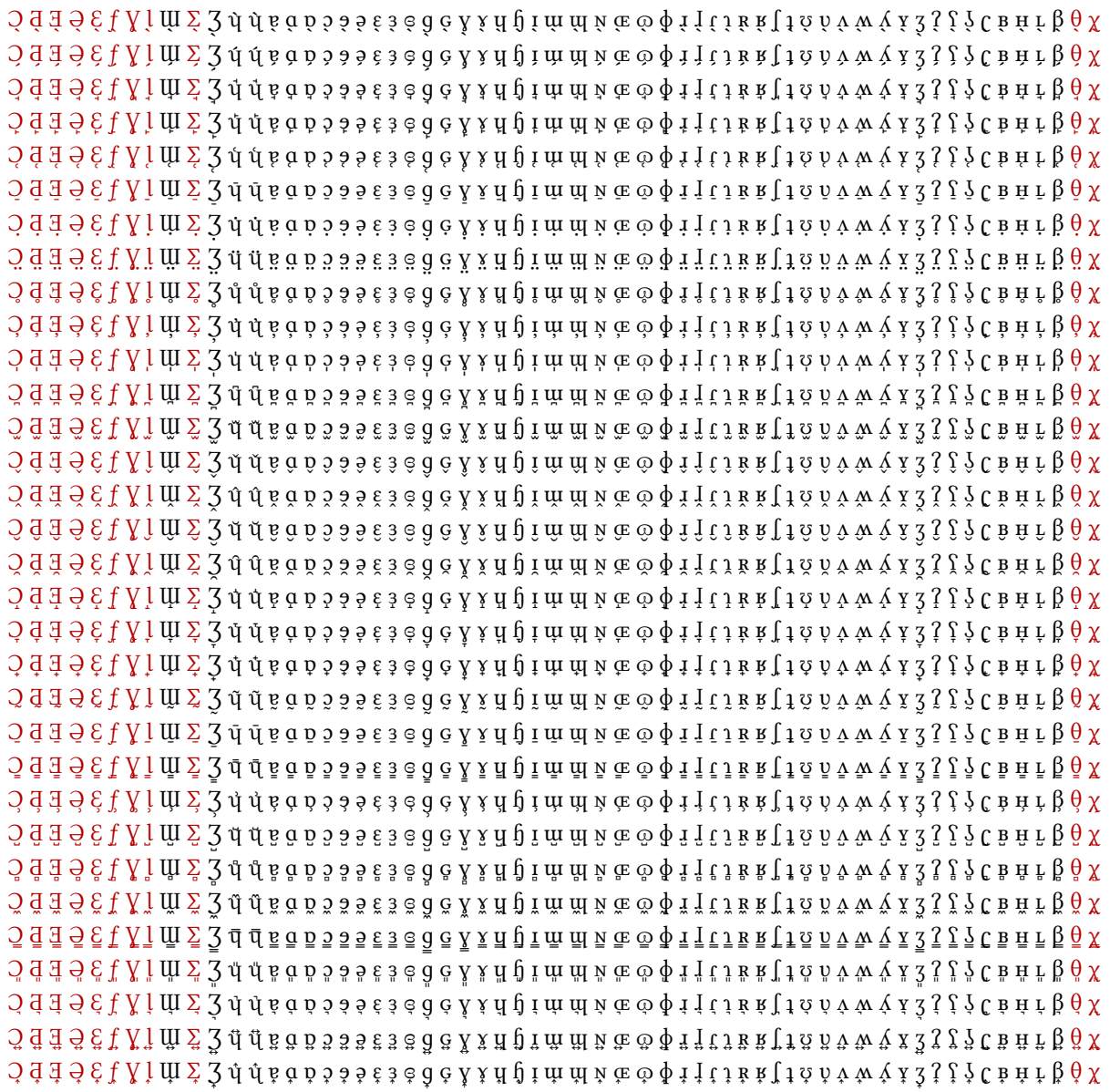


## Semibold italic, IPA, Above

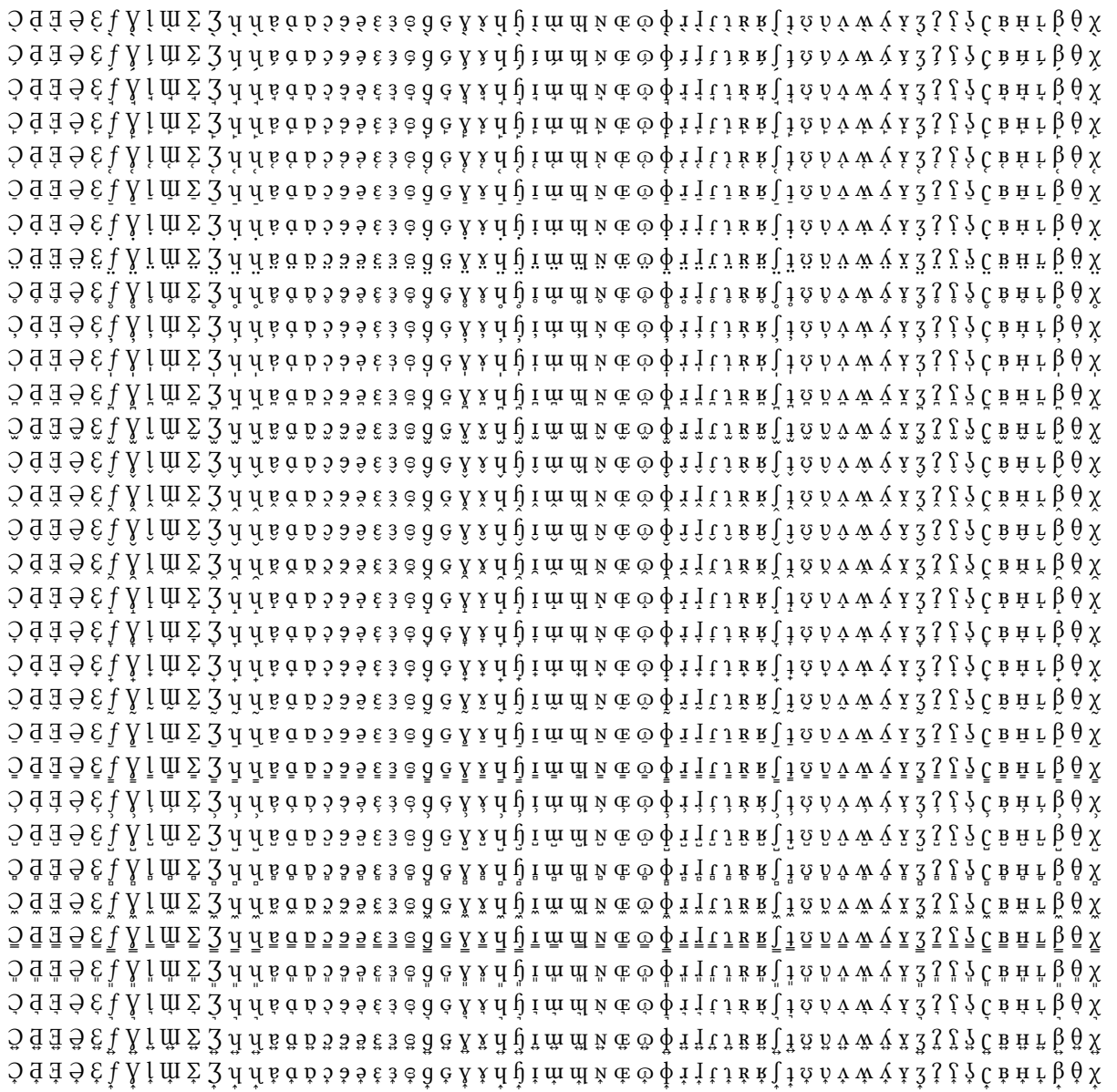


## Semibold italic patched, IPA, Above

## Regular, IPA, Below



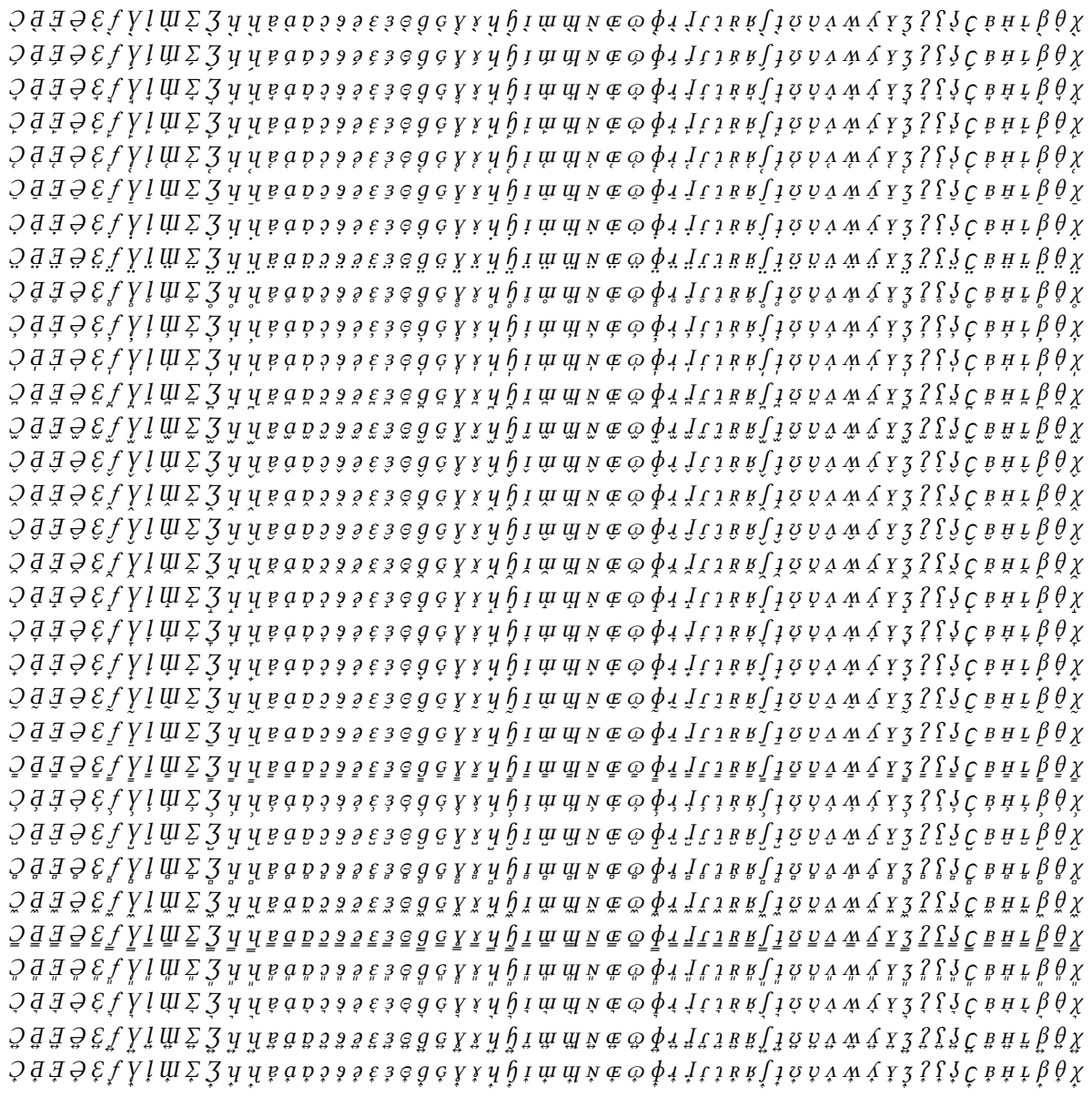
## Regular patched, IPA, Below



## Italic, IPA, Below

ɔ̗ ɑ̗ ɚ̗ ɘ̗ ɛ̗ ʃ̗ ʎ̗ ɭ̗ ɻ̗ ɹ̗ ɾ̗ ʃ̗ ʁ̗ ɾ̗ ɹ̗ ɻ̗ ɭ̗ ɽ̗ ɔ̗ ɓ̗ ɕ̗ ɛ̗ ɜ̗ ɞ̗ ɛ̗ ɜ̗ ɞ̗ ɔ̗ ɠ̗ ɦ̗ ɤ̗ ɥ̗ ɯ̗ ɱ̗ ɰ̗ ɲ̗ ɳ̗ ɵ̗ ɸ̗ ɷ̗ ɶ̗ ʏ̗ ʍ̗ ʌ̗ ɒ̗ ɐ̗ ɑ̗ ɻ̗ ɬ̗ ʒ̗ ɮ̗ ɺ̗ ɫ̗ ɴ̗ ɱ̗ ɽ̗ ʐ̗ ʑ̗ ɥ̗ ɧ̗ ɨ̗ ʝ̗ ɣ̗ χ̗ ʊ̗ ɯ̗ ʉ̗ ɤ̗ ɐ̗ ɑ̗ ɚ̗ ɘ̗ ɛ̗ ʃ̗ ʎ̗ ɭ̗ ɻ̗ ɹ̗ ɾ̗ ʃ̗ ʁ̗ ɾ̗ ɹ̗ ɻ̗ ɭ̗ ɽ̗ ɔ̗ ɓ̗ ɕ̗ ɛ̗ ɜ̗ ɞ̗ ɛ̗ ɜ̗ ɞ̗ ɔ̗ ɠ̗ ɦ̗ ɤ̗ ɥ̗ ɯ̗ ɱ̗ ɰ̗ ɲ̗ ɳ̗ ɵ̗ ɸ̗ ɷ̗ ɶ̗ ʏ̗ ʍ̗ ʌ̗ ɒ̗ ɐ̗ ɑ̗ ɻ̗ ɬ̗ ʒ̗ ɮ̗ ɺ̗ ɫ̗ ɴ̗ ɱ̗ ɽ̗ ʐ̗ ʑ̗ ɥ̗ ɧ̗ ɨ̗ ʝ̗ ɣ̗ χ̗ ʊ̗ ɯ̗ ʉ̗ ɤ̗

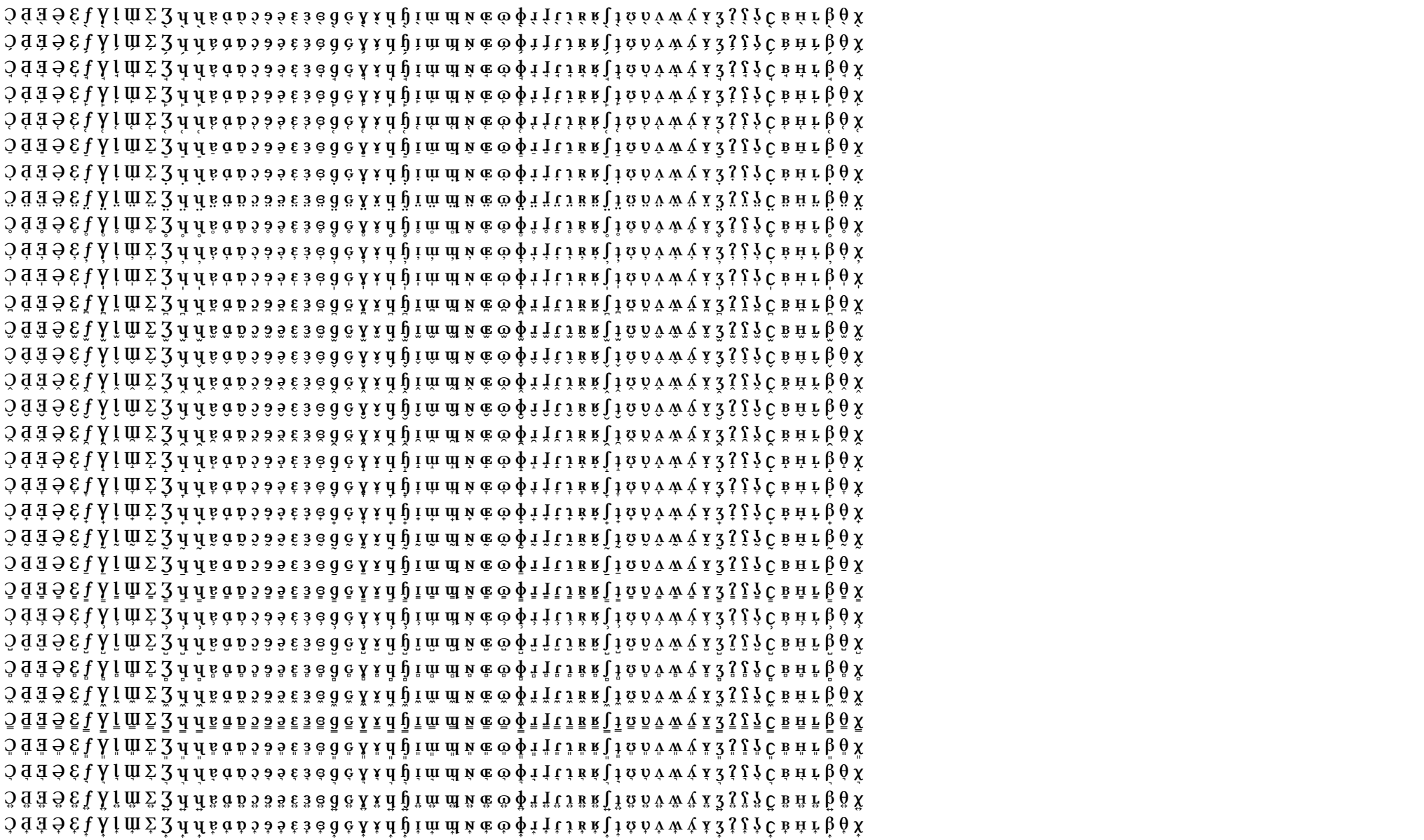
## Italic patched, IPA, Below



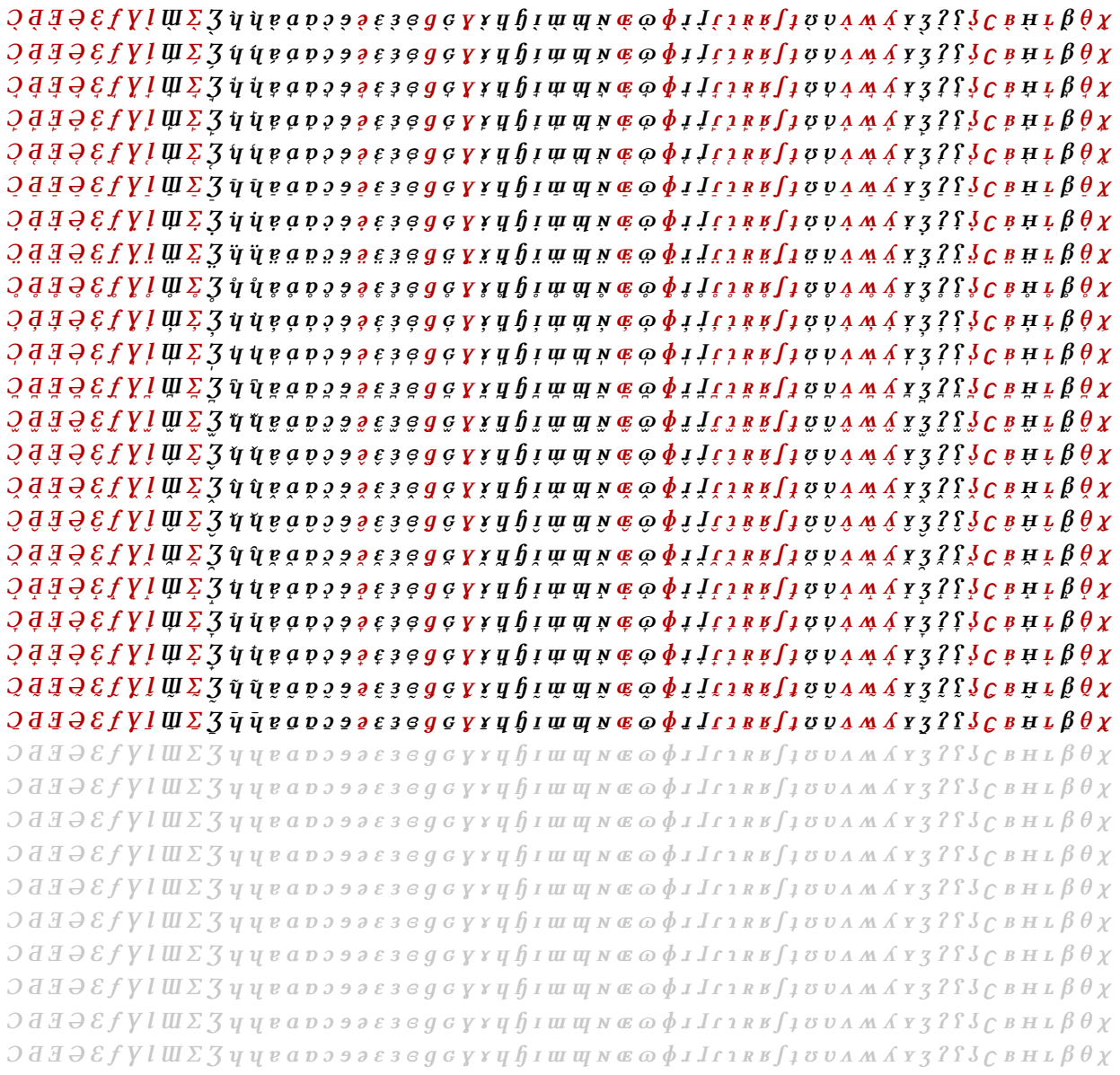
## Semibold, IPA, Below



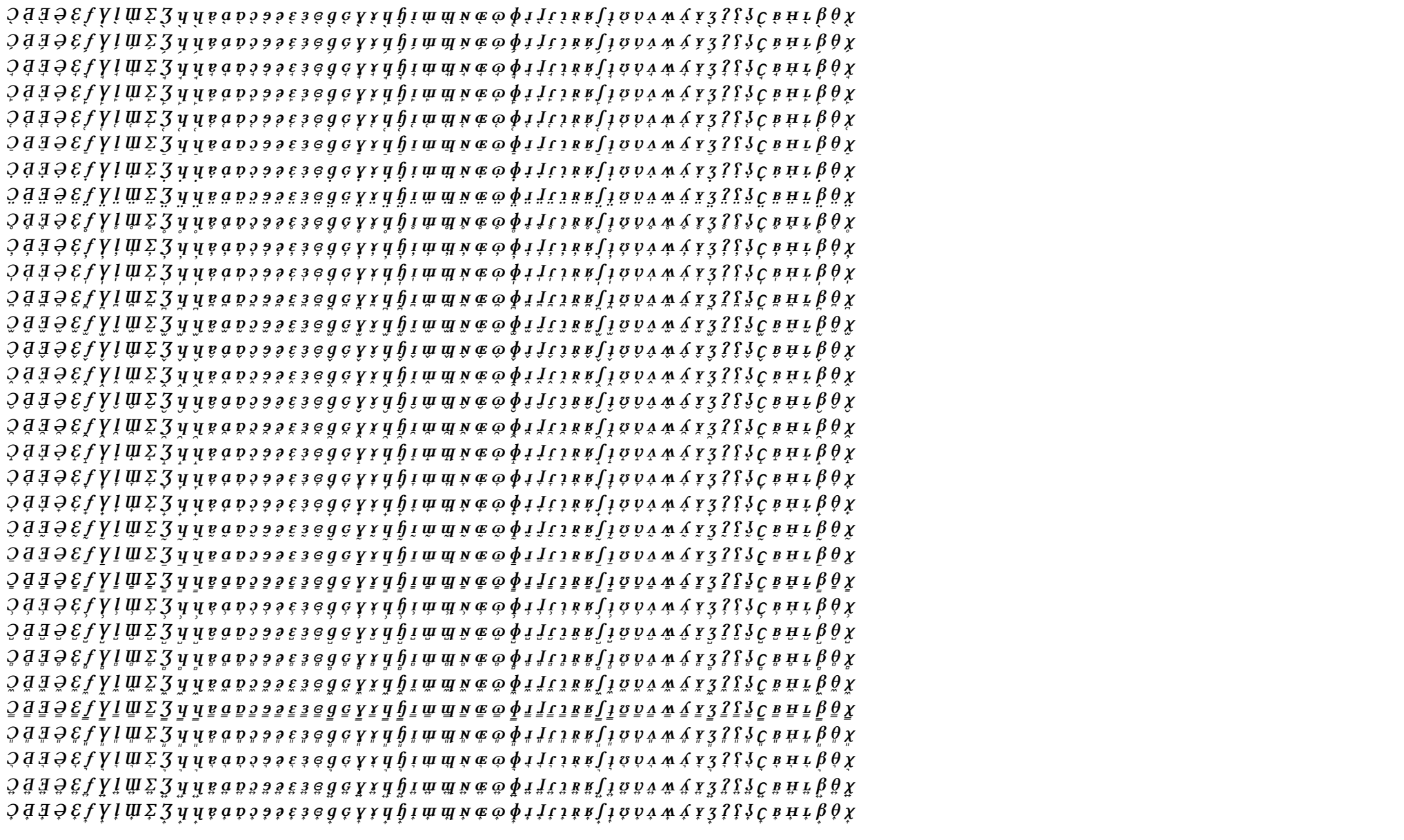
## Semibold patched, IPA, Below



## Semibold italic, IPA, Below



## Semibold italic patched, IPA, Below



## Capital alternates

A capital alternate (`grave.cap`, `acute.cap`, ...) is a form of an above-mark that sits lower over a capital, as the accents of the designer's precomposed capitals do. In the original fonts, `ccmp` substitutes only four of them, for the grave, acute, circumflex and breve, although the designer also drew the caron, double acute, double grave, inverted breve, dieresis and hook above (and, in regular and semibold, the dot above). The patched fonts substitute twelve, after every capital listed in `@capitals` of `data/fea/{style}/ccmp.fea` (stage 10): grave, acute, circumflex, tilde, breve, dot above, dieresis, hook above, double acute, caron, double grave and inverted breve. Stage 3 adds the two forms that were missing: `dotaccent.cap` in italic and semibold italic, the dot of the precomposed İ, and `tildecomb.cap` in every style (2026-11-08), the tilde of the precomposed tilded capitals (`tildecomb` itself, except in semibold italic, where the precomposed capitals use a taller tilde, that of Ã). The other above-marks have no capital form and attach with their own anchors. The ring above has none by decision: the precomposed Å sets the ring on the apex, and the combining ring keeps its clearance; nor do the vertical lines above U+030D and U+033E.

In stage 4 of the patched fonts (`font_patching_capmarks.py`), the mark anchor of each capital alternate is set from the designer's precomposed capitals A–Z. The mark's outline is found in each precomposed glyph that is a capital plus that mark, which gives how far the designer moved it there. The $Y$ coordinate is the capital anchor reference (850) less the median height by which the mark is raised in the precomposed glyphs. The $X$ coordinate of a symmetric mark is the centre of its outline on its own axis, $X = C + Y\tan\theta - K$. For the asymmetric grave, acute, double acute and double grave, it is the median position of the mark in the precomposed glyphs relative to the candidate above-anchor of the capital, $W/2 + 850\tan\theta - K$. $K$ is 150 in semibold italic and 0 elsewhere. The precomposed placement is reproduced exactly where a capital's above-anchor is on its candidate; where it is not, the composed mark is displaced by the same amount. Human-curated anchors (stage 7) overrule stage 4.

The table compares each composed capital (base, capital alternate, anchors) with the designer's precomposed glyph, over the capitals that have one. Vertically every capital alternate matches: the median difference is 0 in every style, and no single capital differs by more than 12, except Ỏ, whose precomposed hook above is 24 units higher than the composed one in all styles but semibold italic. The numbers are the median horizontal distance; $n$ is the number of precomposed capitals, and $^*$ marks a curated anchor.

| mark | $n$ | regular | italic | semibold | semibold italic |
|---|---|---|---|---|---|
| grave | 8 | 17 | $30^*$ | 4 | 10 |
| acute | 17 | 9 | $14^*$ | 16 | 28 |
| circumflex | 12–13 | 4 | $9^*$ | 14 | 16 |
| tilde | 7–8 | 5 | 10 | 10 | 25 |
| breve | 6 | 2 | $6^*$ | 10 | 10 |
| dot above | 20 | 9 | 13 | 10 | 15 |
| dieresis | 7 | 2 | 6 | 8 | 8 |
| hook above | 6 | 4 | 12 | 12 | 16 |
| double acute | 2 | 0 | $2^*$ | 18 | 6 |
| caron | 12–15 | 7 | $6^*$ | 16 | 11 |
| double grave | 6 | 12 | $4^*$ | 15 | 15 |
| inverted breve | 6 | 2 | $10^*$ | 12 | 9 |

Notes by style:

- Regular. All scripted (2026-11-08). The curated grave $(-175, 723)$ was removed in favour of stage 4's $(-162, 724)$, as in semibold italic: the median distance from the precomposed graves goes from 18 to 17 units, and the median offset from 18 units right to 6, since the graves vary letter by letter. The other curated anchors are reproduced by the script and are commented out in `fontanchors_human`.
- Italic. Eight curated anchors (2026-11-07) overrule stage 4: grave $-6$ (stage 4: 17), acute $-50$ ($-37$), circumflex 15 (24), breve $-14$ ($-1$), caron $-32$ ($-29$), double acute $-109$ ($-103$), double grave $-40$ ($-34$), inverted breve $-12$ ($-2$). Each is fitted so that the accent lands, as a median over the capitals, where the designer drew it in the precomposed glyphs, with the capitals' actual above-anchors (so the signed median is 0). The dieresis, dot above, hook above and tilde are scripted.
- Semibold. All scripted. The designer's capital above-anchors were near 805, with the capital alternates drawn to compensate; stage 5 moves the capitals to 850, the reference stage 4 computes from.

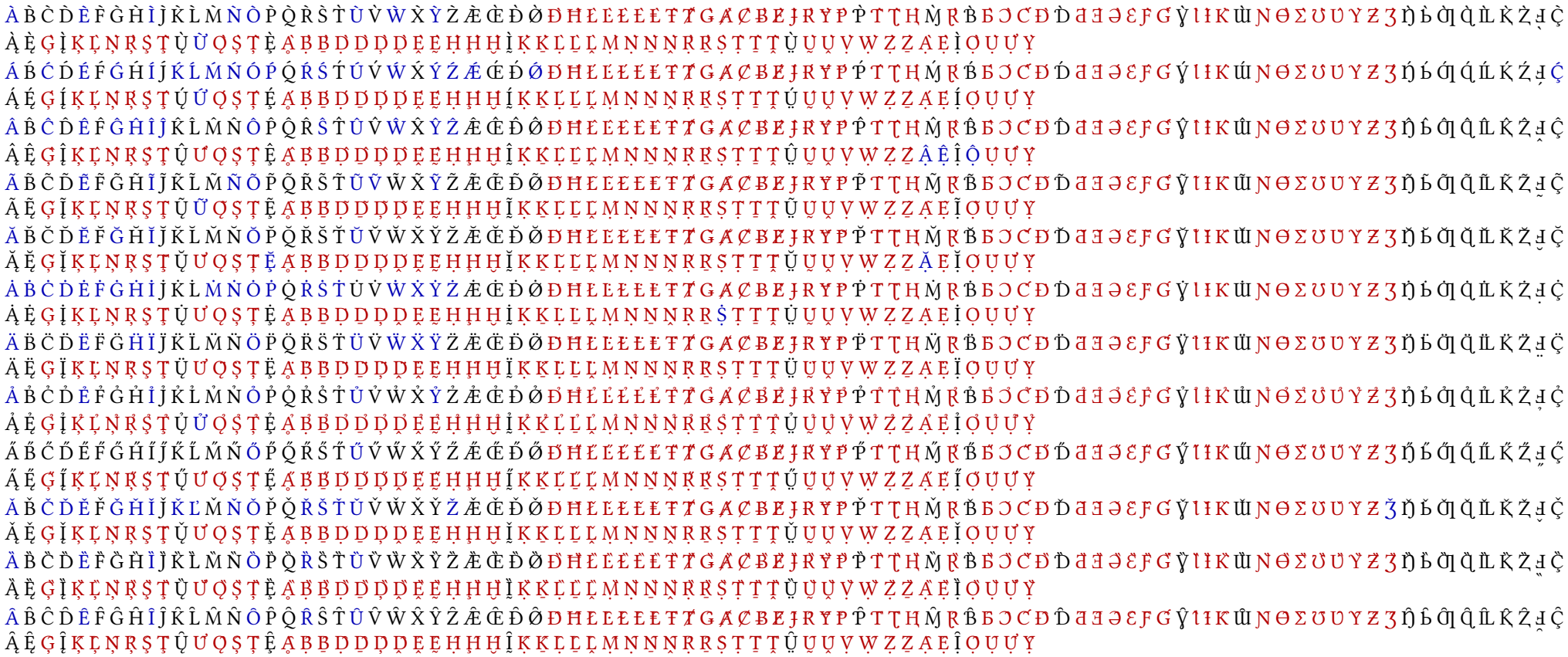
- Semibold italic. All scripted (2026-11-08). The acute, grave, double acute and double grave had been curated on 2026-11-06 at the mark-axis position, $X = C + Y \tan\theta + d - K$ with $d$ from regular ($-236$, $-194$, $-282$, $-214$, against stage 4's $-195$, $-168$, $-243$, $-167$). Those reproduced the precomposed capitals only on the designer's capital anchors, which are left of the candidates (median $k$ 173 against $K = 150$), because the designer drew the precomposed accents on those anchors. Stage 4 is preferred because the precomposed glyphs are used wherever they exist, so a composed capital without one should match them; and since H I J K L N O U V Y Æ were moved on 2026-11-07, it is also closer to them. The larger distances of the acute (28) and tilde (25) come from capitals whose above-anchors are off their candidates (A, B, D, Q, R, W, X; V, Y and E for the tilde), not from the mark anchors.

Notes by mark:

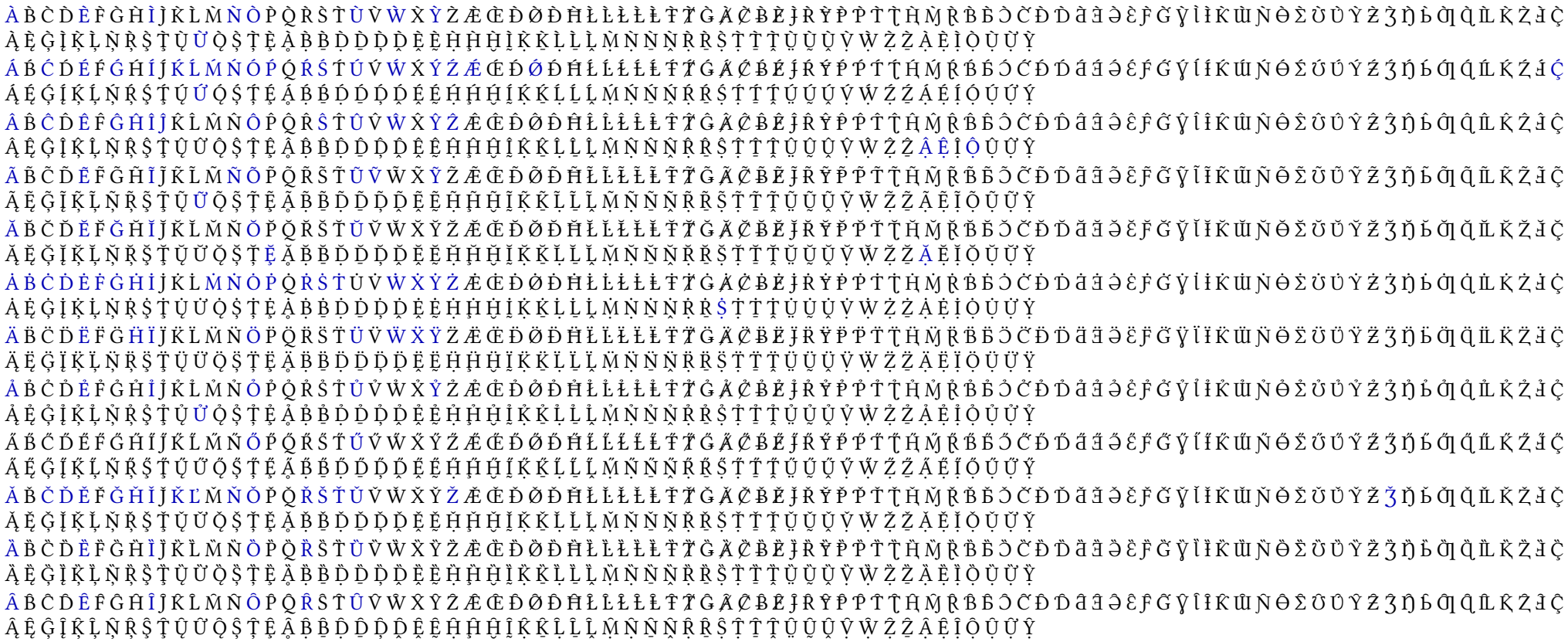
- The grave, acute and their doubles are placed by eye, letter by letter, in the precomposed glyphs, so the implied $X$ varies by up to 190 units over the bases (most for the acute, because of L), and one anchor is a compromise.
- Tilde. Unicode has no precomposed G̃, which Guarani writes, so G + U+0303 is composed. With the combining tilde it sat 25 to 47 units higher than the tilde of the precomposed Ñ, Ã, Õ beside it; `tildecomb.cap` matches their height exactly. Gentium Plus does the same with its `tildecomb.LP`.
- Dot above. `dotaccent.cap` is new in italic and semibold italic; in regular, stage 4 raises its anchor by 50 units to match the precomposed dotted capitals.
- Not used as evidence, because the mark outline differs in the precomposed glyph: Ÿ and Ẅ (dieresis), Ľ (comma-style caron), Ũ in semibold italic (tilde), and Ĵ in semibold italic.

The grids show these twelve marks on every Latin and IPA capital of `@capitals`: A–Z, Æ, Œ, Đ, Ø, Ɖ, Ħ, Ł, the IPA and African capitals, and the 59 precomposed capitals with a cedilla, ogonek, horn or below-mark that stage 9 gives an above-anchor (Ç Ą Ę Į Ų Ư Ḍ Ẹ Ọ …; added to `@capitals` 2026-11-08, so that, e.g., Lithuanian Ą́ and Yoruba Ẹ́ take the capital acute like Á and É). The Greek and Cyrillic capitals of `@capitals` are not shown: their combinations with these marks are all precomposed. Blue cells are precomposed glyphs, which the shaper substitutes for base + mark wherever the font has one, so they do not use the capital alternates. Black cells are composed: in the patched fonts, the base with the capital alternate; in the original fonts, the base with the alternate only for the grave, acute, circumflex and breve, and otherwise with the ordinary mark.

## Regular, Capitals, Marks with capital alternates



## Regular patched, Capitals, Marks with capital alternates



## Italic, Capitals, Marks with capital alternates

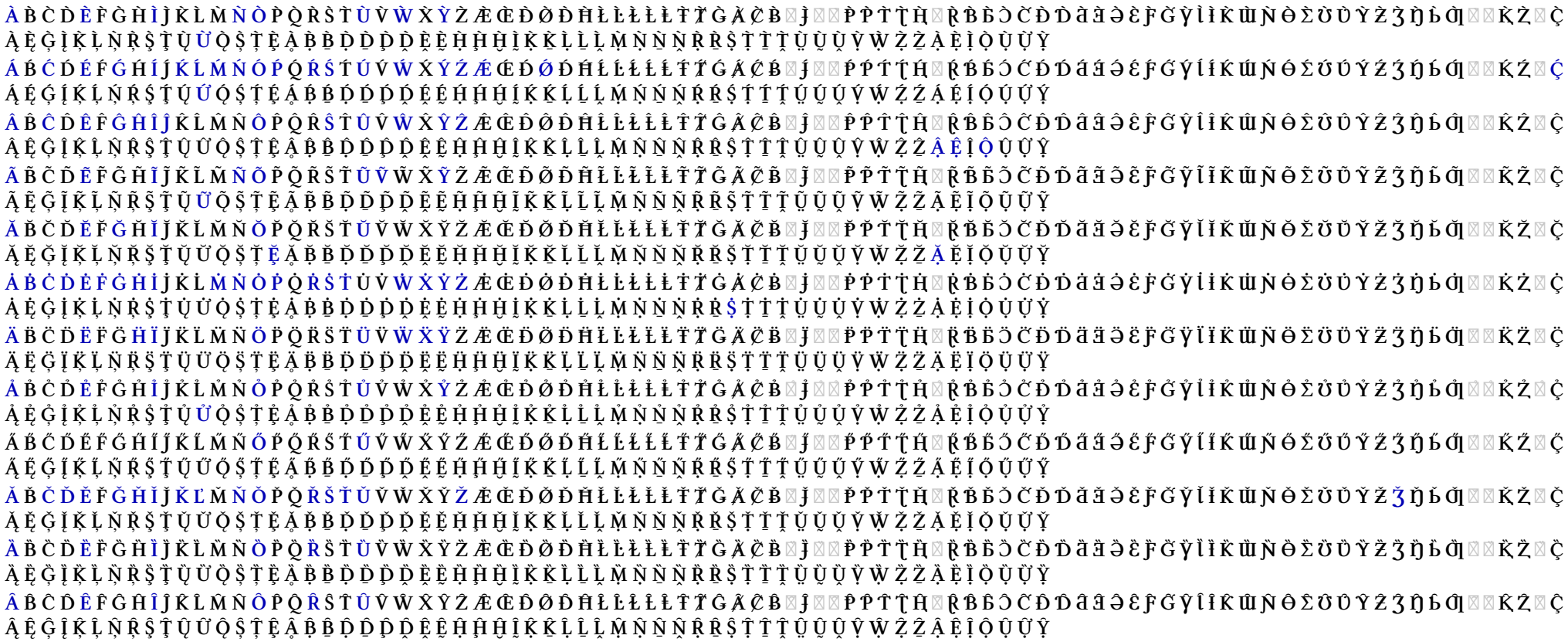


## Italic patched, Capitals, Marks with capital alternates

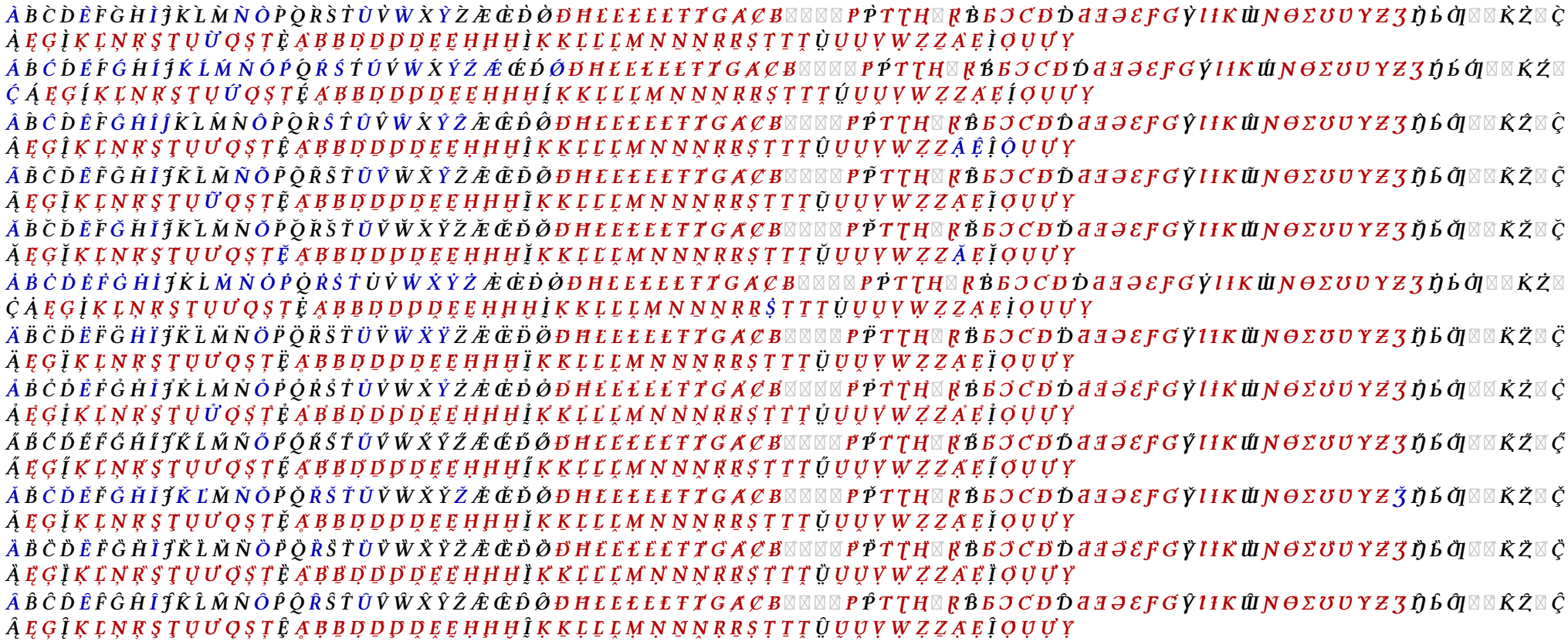
## Semibold, Capitals, Marks with capital alternates



## Semibold patched, Capitals, Marks with capital alternates



## Semibold italic, Capitals, Marks with capital alternates



## Semibold italic patched, Capitals, Marks with capital alternates

## Derived letters

Many Latin and IPA letters are a base letter with a bar, stroke, belt, hook, tail or curl: Đ is D with a bar, Ɗ is D with a hook, ɖ is d with a tail. Unlike the precomposed letters of stage 9, they have no Unicode decomposition; the base letter is assigned to each one in `data/derived_letters.py`. Stage 8 (`src/libertinus_analysis/font_patching_derivedanchors.py`, letter lists in `data/derived_letters.py`) gives each such letter its base letter's final anchor, moved by where the base letter sits inside it, on each side where the added part stays clear of the marks. The heuristic candidate of stage 6 cannot do this: a hook or bar moves the bounding box it is centred on (regular Ƒ above: candidate 183, F's anchor moved into Ƒ 276), and the designer often drew the derived letter in a shifted frame (regular Ɗ is D moved 80 units right).

Stage 8 measures how well the derived letter matches its base letter (the share of the base's outline that fits, in percent) and where the base sits in it (a shift, and a scale), in one of four ways:

- **Fit** (most letters). On each inherited side, the base letter's ink is fitted to the derived letter's with a shift and a scale of up to 15%, as in stage 9; the anchor moves with the fit, so a letter drawn a few percent wider keeps the anchor at the same place in the shape (italic Đ is D drawn 4% wider). At least 80% of the base must fit; otherwise the letter keeps its stage 6 candidate (or a curated anchor) on that side.
- **Stems** (`stem_match`: a hook, tail or curl at the bottom, and ɗ). The fit uses only the stems above the tail, at least 70%. The below anchor goes to the descender, as the stage 6 candidate does, where the tail lies under the mark; otherwise it stays at the base letter's $Y$ (Ɲ, Ⱨ).
- **Stem midline** (`stem_axis`: ɬ ɬ ɬ ɬ Ł ɪ ɟ, and T f Y 𝑇 𝑓). The anchor moves by the distance between the two stems, whatever the fit, for letters whose bar, belt, tilde or hook spoils the fit but not the stem.
- **Fixed** $Y$ (`derived_y`): ɓ ʝ ɠ ɾ ƥ ɋ f above, ɓ ɕ below take $X$ from the base letter and a $Y$ set by hand, usually because a hook rises to the ascender (ɠ ƥ ɋ f).

Where the added part reaches further than the base letter within 120 units of the mark's axis, $Y$ moves away by the difference (ɇ above $+31$ to $+39$, italic 𝑇 $+96$). A derived letter is curated (stage 7) only where stage 8 cannot inherit a good anchor: the match is too low, or the inherited anchor would be wrong; stage 8 runs after stage 7 and leaves those anchors alone. Where stage 8 cannot inherit and nothing is curated, the letter keeps stage 6's candidate.

| letters | inherit | note |
|---|---|---|
| Đ Ð Ɖ đ Ħ Ł ł Ø ø Θ ɵ Ɨ ᵿ ɬ ɬ ɬ ɬ Ł Ł Ŧ ŧ 𝑇 𝑡 G g Ꞡ ꞡ Ȼ ɇ ɕ | both | bar, stroke, belt or middle dot |
| Ƀ ƀ Ƶ ƶ Ɉ Ʀ ɾ Ɏ ɏ Ƥ Ƶ ƶ f f ƥ Ɬ ɬ ʔ ʕ | | |
| Ɓ Ɗ Ƈ Ɠ K ɗ ƈ ɠ Ƥ Y ƴ ƙ ƈ T ɓ ɖ ƈ̓ ᵾ | both | hook, topbar or ring clear of the marks |
| ħ ɦ | both (italics: below) | the bar or hook sits at the top of the italic ascender |
| ʒ ɚ ƥ ɋ f | below | the hook rises into the above mark (ƥ ɋ f above: fixed $Y$) |
| Ƒ Ɲ ɕ ɖ ɭ ɱ ɲ ɳ ɻ ɽ ʂ ʃ ʈ ʐ ʑ ʒ ʈ Ʈ ɖ ɭ ɳ ʈ Ʀ Ⱨ ɦ Ⱪ ɬ Ɱ y | both | tail at the bottom (stems) |
| ɨ ʝ | below | above: their dotless forms, from ɪ ɟ |
| ɮ ɻ U+E27F | above | below curated (ɮ) or the candidate |
| ɪ ɟ f | both | from ɪ, ɟ, and ʃ (Libertinus draws ƒ as ʃ with a bar) |
| ɓ ʝ ɠ ɾ | ɓ both, the others above | fixed $Y$, $X$ from b, j, script g, r |

Not derived, although the Unicode name says so: Ƀ ƀ (topbar), Ʊ ʊ ƒ ʒ ʂ Ŋ ʐ ɀ Ƴ ʋ ɾ, which match their base letters 5–50%, and Ƶ ƶ, whose base has no stems to match; they keep stage 6's candidate or a curated anchor. ħ Ħ take no marks in any orthography; ħ's copied anchors are deleted and both are in `BASE_NO_MARK`.

Fixes made on the way (2026-11-11): the copied x-height or baseline anchors of italic ɑ̈ ƙ, semibold italic ƙ and ʐ (three styles) are deleted, so that stage 6 fills them; the curated Ƴ of regular and semibold, at Y's own $X$, was 63–78 units left of Ƴ's stem and is retired; semibold italic ƴ, redrawn narrower than y, is curated at $(152, 645)$; italic ɾ is rebuilt in stage 2 (its SFD glyph referred to IJ and two acutes as well) and now inherits r's anchors. Inheritance carries the base letter's anchor, right or wrong: Ƈ and Ȼ follow C's below anchor, which the designer put right of centre. The stem-midline check (Table 28) confirms that every anchor of an l-like or i-like letter lies within 5 units of its own stem, except ɮ below, under the ezh by design.

In the tables, each column is a derived letter, ordered by base letter, with its base letter's codepoint below it; per style: the letter with circumflex and macron below, original and patched (`reg`, `patch`), and its base letter patched (`base`); then for the above anchor the shift and fit used (a~; `s` stems, `|` stem midline, italics if below the threshold), the designer's anchor (a), the candidate (a?), the inherited anchor (a<, bold where the patched font has it) and a curated anchor (a!); the same below (b).

What the tables decide, per letter and side: a bold a< is the anchor the patched font has, inherited in stage 8, replacing the designer's anchor and the candidate (216 anchors in regular, 186 in semibold, 167 in italic, 165 in semibold italic); an a! is curated because inheritance fails or would be wrong there (5 anchors in regular and semibold, 12 in each italic style, mostly tail letters whose stems do not match: italic ɖ ɭ ɽ ʐ ƶ, ʒ), and the a< beside it, if any, is not used; an a~ in italics means the match is below the threshold (3 in regular, 11 in semibold, 23 in italic, 21 in semibold italic: italic redrawings such as ɓ ƀ ɗ ƙ ɛ̀, and tails such as Ⱪ ⱪ Ɽ), and the letter keeps what it had before stage 8, the candidate (a?) or a designer anchor (a) that stage 1 kept; an empty a< means the side is not inherited, because the added part is in the mark's way (ǯ ɚ above, ħ ɦ above in the italics). The grids put the IPA marks on each base letter followed by its derived letters, patched fonts only, without `\cgj` (a cell shows the precomposed glyph in blue where one exists).

## Turned letters

Turned letters are drawn from upright letters too, rotated by 180°: Ⅲ ɐ ɒ ɥ ɯ ɰ ɹ ɾ ɻ ʁ ɬ ʌ ʍ ʎ ʖ ɥ ɥ ɟ in the base groups. Stage 8 does not derive their anchors: not even swapping the upright letter's above and below anchors makes sense, since the turned letter has its own reference lines (h's ascender is ɥ's descender), and the anchor's $Y$ and clearance come from those, not from the upright letter's. The designer's anchors on turned letters are bad: copied carelessly from other glyphs (a's (209, 645) and (196, −110) on ɰ ɹ ɾ ʁ ɬ ʌ ʍ ʎ ʖ in regular, and regular's numbers unslanted in the italics), or rotated along with the outline (ɥ ɥ ɟ, whose above anchor sat under the letter). So all of them are deleted in stage 1, except the few that happen to match the candidate within 4 units (regular and semibold italic ɥ above, semibold italic ɐ ɒ below), which stage 5 snaps to the reference line. The turned letters therefore get stage 6's candidate, or a curated anchor (stage 7) where the candidate is unsuitable.

- Where the letter is symmetric about its axis, both anchors are the candidate in every style: Ⅲ ɐ ɒ ɥ ɯ ʁ ɬ ʖ ɥ ɥ ɟ. ɯ's candidate lies on the central stem, within 2 to 7 units.
- Where a stem or a gap must be the centre and an arm pulls the candidate aside, the anchor is curated, in effect the upright letter's opposite anchor turned over: above, ɹ ɾ ɻ over the stem top, as r's below anchor sits on r's stem, and ʎ over the apex of the turned V, as y's below anchor; below, ʌ between the feet, ʍ under the middle foot (the candidate in semibold), and ɰ under the middle stem, at ɯ's candidate $Y$. Their other anchor is the candidate.
- ɰ is not derived from ɯ: it is narrower and its stems are spaced differently (outline match 21% in regular). ɻ is a derived letter of ɹ (a tail at the bottom) and inherits ɹ's above anchor in semibold and semibold italic; its above anchor is curated in regular and italic, and its below anchor follows stage 8's rule for a tail at the bottom (the candidate where the stems do not match, as in italic).

Table 21: Derived letters and inherited anchors 1/7

| hex | 023A | 0181 | 0243 | 0187 | 023B | 00D0 | 0110 | 0189 | 018A | 0246 | 0191 | 0193 | 01E4 | 0126 | 2C67 | 0197 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 0041 | 0042 | 0042 | 0043 | 0043 | 0044 | 0044 | 0044 | 0044 | 0045 | 0046 | 0047 | 0047 | 0048 | 0048 | 0049 |
| reg | A̋Ą | B̂Ḃ | B̄Ḃ | C̆Ç | C̓Ç | D̂Ḍ | D̄Ḍ | D̄Ḍ | D̂Ḍ | Ē̄Ę | F̂Ḟ | ĞĢ | Ğ̣Ģ | Ħ̄Ḥ | Ḥ̣Ḥ | Ī̄Į |
| patch | Ã̋Ą | B̂Ḃ | B̂Ḃ | C̆Ç | C̓Ç | D̂Ḍ | D̂Ḍ | D̂Ḍ | D̂Ḍ | Ē̄Ę | F̂Ḟ | ĞĢ | Ĝ̣Ģ | Ħ̂Ḥ | Ḥ̂Ḥ | Ī̂Į |
| base | ÂĄ | B̂Ḃ | B̂Ḃ | C̆Ç | Ĉ̣Ç | D̂Ḍ | D̂Ḍ | D̂Ḍ | D̂Ḍ | ÊĘ | F̂Ḟ | Ĝ̣Ģ | Ĝ̣Ģ | Ḥ̂Ḥ | Ḥ̂Ḥ | Ī̂Į |
| a | +0, 91% | +58, 98% | +0, 96% | +0, 100% | +0, 97% | +0, 96% | +0, 96% | +0, 96% | +79, 99% | +0, 81% | -8, 99% s | +0, 100% | +0, 95% | +0, 94% | -16, 100% s | +0, 94% |
| a | | 327, 805 | | | | 341, 850 | | | 306, 805 | | | | | | | |
| a? | 346, 850 | 303, 850 | 255, 850 | 369, 850 | 324, 850 | 333, 850 | 336, 850 | 333, 850 | 368, 850 | 250, 850 | 183, 850 | 372, 850 | 366, 850 | 362, 850 | 345, 850 | 148, 850 |
| a< | **361, 850** | **325, 850** | **267, 850** | **340, 850** | **340, 850** | **306, 850** | **306, 850** | **306, 850** | **385, 850** | **281, 850** | **276, 850** | **365, 850** | **365, 850** | **361, 850** | **345, 850** | **150, 850** |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 94% | +58, 96% | +0, 91% | +0, 100% | +0, 86% | +0, 100% | +0, 100% | +0, 100% | +79, 100% | *-2, 72%* | -8, 99% s | +0, 100% | +0, 92% | +0, 100% | -16, 100% s | +0, 100% |
| b | | 328, -108 | | | | | | | 369, -107 | | | | | | | |
| b? | 346, -218 | 303, -110 | 255, -110 | 369, -110 | 324, -218 | 333, -110 | 336, -110 | 333, -110 | 368, -110 | 250, -121 | 183, -257 | 372, -110 | 366, -110 | 362, -110 | 345, -226 | 148, -110 |
| b< | **339, -110** | **326, -110** | **268, -110** | **363, -110** | **363, -110** | **309, -110** | **309, -110** | **309, -110** | **388, -110** | 285, -110 | **202, -257** | **367, -110** | **367, -110** | **362, -110** | **346, -110** | **147, -110** |
| b! | | | | | | | | | | | | | | | | |
| it | *A̋Ą* | *B̂Ḃ* | *B̄Ḃ* | *C̆Ç* | *C̓Ç* | *D̂Ḍ* | *D̄Ḍ* | *D̄Ḍ* | *D̂Ḍ* | | *F̂Ḟ* | *ĞĢ* | *Ĝ̣Ģ* | *Ḥ̂Ḥ* | *Ḥ̣Ḥ* | *Ī̄Į* |
| patch | *Ã̋Ą* | *B̂Ḃ* | *B̂Ḃ* | *C̆Ç* | *C̓Ç* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | ⌧ | *F̂Ḟ* | *ĞĢ* | *Ĝ̣Ģ* | *Ĥ̂Ḥ* | *Ḥ̂Ḥ* | *Ī̂Į* |
| base | *ÂĄ* | *B̂Ḃ* | *B̂Ḃ* | *C̆Ç* | *Ĉ̣Ç* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *ÊĘ* | *F̂Ḟ* | *Ĝ̣Ģ* | *Ĝ̣Ģ* | *Ḥ̂Ḥ* | *Ḥ̂Ḥ* | *Ī̂Į* |
| a | +5, 93% | +58, 99% | +10, 84% | +32, 96% | +6, 96% | +0, 96% | +0, 96% | +27, 89% | +78, 98% | | +16, 100% s | +7, 99% | +7, 94% | +0, 94% | +10, 100% s | +0, 92% |
| a | | 327, 805 | | | | | | | 306, 805 | | | | | | | |
| a? | 528, 850 | 488, 850 | 477, 850 | 503, 850 | 503, 850 | 514, 850 | 514, 850 | 534, 850 | 553, 850 | | 419, 850 | 541, 850 | 524, 850 | 517, 850 | 528, 850 | 321, 850 |
| a< | **510, 850** | **505, 850** | **457, 850** | **532, 850** | **506, 850** | **471, 850** | **471, 850** | **498, 850** | **549, 850** | | **422, 850** | **520, 850** | **520, 850** | **504, 850** | **514, 850** | **316, 850** |
| a! | | | | | | | | | | | | | | | | |
| b | +10, 92% | +58, 100% | *+13, 74%* | +19, 94% | +13, 81% | +0, 100% | +0, 100% | +18, 98% | +78, 96% | | +16, 100% s | +13, 98% | +13, 90% | +0, 100% | +10, 100% s | +0, 100% |
| b | | 328, -108 | | | | | | | 369, -107 | | | | | | | |
| b? | 313, -164 | 284, -110 | 273, -110 | 299, -110 | 287, -165 | 310, -110 | 310, -110 | 330, -110 | 349, -110 | | 184, -257 | 337, -110 | 320, -110 | 313, -110 | 300, -223 | 117, -110 |
| b< | **318, -110** | **296, -110** | 251, -110 | **352, -110** | **346, -110** | **279, -110** | **279, -110** | **297, -110** | **357, -110** | | **113, -257** | **358, -110** | **358, -110** | **320, -110** | **331, -110** | **127, -110** |
| b! | | | | | | | | | | | | | | | | |
| sb | A̋Ą | B̂Ḃ | B̄Ḃ | C̆Ç | C̓Ç | D̂Ḍ | D̄Ḍ | D̄Ḍ | D̂Ḍ | | F̂Ḟ | ĞĢ | Ĝ̣Ģ | Ḥ̂Ḥ | Ḥ̣Ḥ | Ī̄Į |
| patch | Ã̋Ą | B̂Ḃ | B̄Ḃ | C̆Ç | C̓Ç | D̂Ḍ | D̂Ḍ | D̂Ḍ | D̂Ḍ | ⌧ | F̂Ḟ | ĞĢ | Ĝ̣Ģ | Ĥ̂Ḥ | Ḥ̂Ḥ | Ī̂Į |
| base | ÂĄ | B̂Ḃ | B̄Ḃ | C̆Ç | Ĉ̣Ç | D̂Ḍ | D̂Ḍ | D̂Ḍ | D̂Ḍ | ÊĘ | F̂Ḟ | Ĝ̣Ģ | Ĝ̣Ģ | Ḥ̂Ḥ | Ḥ̂Ḥ | Ī̂Į |
| a | +0, 92% | +55, 93% | +0, 95% | +0, 99% | +0, 94% | +0, 95% | +0, 95% | +0, 95% | +82, 97% | | *+18, 62% s* | +0, 100% | +0, 95% | +0, 92% | -17, 98% s | +0, 92% |
| a | | | | | | | | | | | | | | | | |
| a? | 350, 850 | 316, 850 | 294, 850 | 368, 850 | 332, 850 | 349, 850 | 349, 850 | 349, 850 | 381, 850 | | 236, 850 | 374, 850 | 368, 850 | 362, 850 | 345, 850 | 158, 850 |
| a< | **350, 850** | **338, 850** | **283, 850** | **332, 850** | **332, 850** | **352, 850** | **352, 850** | **352, 850** | **434, 850** | | | **385, 850** | **385, 850** | **362, 850** | **345, 850** | **160, 850** |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 94% | +58, 87% | +0, 89% | +0, 100% | +0, 87% | +0, 100% | +0, 100% | +0, 100% | +82, 98% | | *+18, 62% s* | +0, 100% | +0, 91% | +0, 100% | -17, 98% s | +0, 100% |
| b | | | | | | | | | | | | | | | | |
| b? | 350, -218 | 316, -110 | 294, -110 | 368, -110 | 332, -218 | 349, -110 | 349, -110 | 349, -110 | 381, -110 | | 236, -281 | 374, -110 | 368, -110 | 362, -110 | 345, -222 | 158, -110 |
| b< | **350, -110** | **358, -110** | **300, -110** | **332, -110** | **332, -110** | **352, -110** | **352, -110** | **352, -110** | **434, -110** | | **236, -281** | **368, -110** | **368, -110** | **362, -110** | **345, -110** | **160, -110** |
| b! | | | | | | | | | | | | | | | | |
| si | *A̋Ą* | *B̂Ḃ* | *B̄Ḃ* | *C̆Ç* | *C̓Ç* | *D̂Ḍ* | *D̄Ḍ* | *D̄Ḍ* | *D̂Ḍ* | | *F̂Ḟ* | *ĞĢ* | *Ĝ̣Ģ* | *Ḥ̂Ḥ* | *Ḥ̣Ḥ* | *Ī̄Į* |
| patch | *Ã̋Ą* | *B̂Ḃ* | *B̄Ḃ* | *C̆Ç* | *C̓Ç* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | ⌧ | *F̂Ḟ* | *ĞĢ* | *Ĝ̣Ģ* | *Ĥ̂Ḥ* | *Ḥ̂Ḥ* | *Ī̂Į* |
| base | *ÂĄ* | *B̂Ḃ* | *B̄Ḃ* | *C̆Ç* | *Ĉ̣Ç* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *D̂Ḍ* | *ÊĘ* | *F̂Ḟ* | *Ĝ̣Ģ* | *Ĝ̣Ģ* | *Ḥ̂Ḥ* | *Ḥ̂Ḥ* | *Ī̂Į* |
| a | +12, 93% | +67, 93% | +12, 95% | +25, 90% | +12, 94% | +0, 95% | +0, 96% | +38, 95% | +94, 95% | | *+15, 61% s* | +29, 85% | +12, 95% | +12, 92% | -44, 98% s | -17, 90% |
| a | | 327, 805 | | | | 341, 850 | | | 306, 805 | | | | | | | |
| a? | 382, 850 | 359, 850 | 319, 850 | 359, 850 | 357, 850 | 367, 850 | 367, 850 | 388, 850 | 424, 850 | | 288, 850 | 395, 850 | 388, 850 | 396, 850 | 370, 850 | 181, 850 |
| a< | **366, 850** | **334, 850** | **279, 850** | **365, 850** | **352, 850** | **306, 850** | **306, 850** | **344, 850** | **400, 850** | | | **394, 850** | **377, 850** | **408, 850** | **353, 850** | **177, 850** |
| a! | | | | | | | | | | | | | | | | |
| b | +12, 96% | +71, 87% | +12, 89% | +26, 92% | +12, 87% | +0, 100% | +0, 100% | +38, 96% | +94, 96% | | *+15, 61% s* | +31, 98% | +12, 91% | +12, 100% | -44, 98% s | -17, 96% |
| b | | 328, -108 | | | | | | | 369, -107 | | | | | | | |
| b? | 327, -218 | 326, -110 | 286, -110 | 326, -110 | 302, -218 | 334, -110 | 334, -110 | 355, -110 | 391, -110 | | 220, -281 | 362, -110 | 355, -110 | 363, -110 | 314, -222 | 148, -110 |
| b< | **320, -110** | **339, -110** | **280, -110** | **389, -110** | **375, -110** | **279, -110** | **279, -110** | **317, -110** | **373, -110** | | **220, -281** | **398, -110** | **379, -110** | **377, -110** | **322, -110** | **152, -110** |
| b! | | | | | | | | | | | | | | | | |

Table 22: Derived letters and inherited anchors 2/7

| hex | 0248 | 0198 | 2C69 | 013F | 0141 | 023D | 2C60 | 2C62 | 2C6E | 019D | 00D8 | 019F | 01A4 | 2C63 | 024C | 2C64 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 004A | 004B | 004B | 004C | 004C | 004C | 004C | 004C | 004D | 004E | 004F | 004F | 0050 | 0050 | 0052 | 0052 |
| reg | Ɉ̲Ɉ̲ | Ƙ̲Ƙ̲ | Ⱪ̲Ⱪ̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | M̲M̲ | N̲N̲ | Ø̲Ø̲ | Ɵ̲Ɵ̲ | P̲P̲ | P̲P̲ | Ɍ̲Ɍ̲ | Ɍ̲Ɍ̲ |
| patch | Ɉ̲Ɉ̲ | Ƙ̲Ƙ̲ | Ⱪ̲Ⱪ̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | M̲M̲ | N̲N̲ | Ø̲Ø̲ | Ɵ̲Ɵ̲ | P̲P̲ | P̲P̲ | Ɍ̲Ɍ̲ | Ɍ̲Ɍ̲ |
| base | Ɉ̲Ɉ̲ | Ƙ̲Ƙ̲ | Ⱪ̲Ⱪ̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | Ł̲Ł̲ | M̲M̲ | N̲N̲ | O̲O̲ | O̲O̲ | P̲P̲ | P̲P̲ | Ɍ̲Ɍ̲ | Ɍ̲Ɍ̲ |
| a | +50, 90% | *-6, 80%* | -8, 78% s | +0, 100% | +0, 87% | +0, 92% | +0, 85% | +0 \| | +0, 100% s | -1, 96% s | +0, 95% | +0, 92% | +122, 84% | +0, 95% | +0, 97% | +0, 100% s |
| a | | | 336, 798 | | | | | | 431, 850 | | 351, 850 | | 400, 850 | | | |
| a? | 186, 850 | 339, 850 | 326, 850 | 260, 850 | 256, 850 | 248, 850 | 260, 850 | 235, 850 | 399, 850 | 324, 850 | 352, 850 | 352, 850 | 318, 850 | 260, 850 | 301, 850 | 301, 850 |
| a< | **232, 850** | 330, 850 | **328, 850** | **151, 850** | **151, 850** | **151, 850** | **151, 850** | **151, 850** | **431, 850** | **345, 850** | **335, 850** | **335, 850** | **399, 850** | **277, 850** | **269, 850** | **269, 850** |
| a! | | | | | | | | | | | | | | | | |
| b | +50, 96% | +0, 100% | -8, 78% s | +0, 100% | +0, 100% | +0, 100% | +0, 100% | +0 \| | +0, 100% s | -1, 96% s | +0, 90% | +0, 100% | +123, 100% | +0, 100% | +0, 100% | +0, 100% s |
| b | | | 342, -109 | | | | | | 393, -110 | | 350, -110 | 350, -110 | 363, -110 | | | |
| b? | 186, -257 | 339, -110 | 326, -226 | 260, -110 | 256, -110 | 248, -110 | 260, -110 | 235, -110 | 399, -257 | 324, -272 | 352, -121 | 352, -110 | 318, -110 | 260, -110 | 301, -110 | 301, -257 |
| b< | **170, -257** | **342, -110** | **334, -110** | **262, -110** | **262, -110** | **262, -110** | **262, -110** | **262, -110** | **393, -257** | **347, -110** | **350, -110** | **350, -110** | **363, -110** | **240, -110** | **301, -110** | **301, -257** |
| b! | | | | | | | | | | | | | | | | |
| it | | *Ƙ̲Ƙ̲* | *Ⱪ̲Ⱪ̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | | *N̲N̲* | *Ø̲Ø̲* | *Ɵ̲Ɵ̲* | *P̲P̲* | *P̲P̲* | | *Ɍ̲Ɍ̲* |
| patch | ⌧ | *Ƙ̲Ƙ̲* | *Ⱪ̲Ⱪ̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | ⌧ | *N̲N̲* | *Ø̲Ø̲* | *Ɵ̲Ɵ̲* | *P̲P̲* | *P̲P̲* | ⌧ | *Ɍ̲Ɍ̲* |
| base | *Ɉ̲Ɉ̲* | *Ƙ̲Ƙ̲* | *Ⱪ̲Ⱪ̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *Ł̲Ł̲* | *M̲M̲* | *N̲N̲* | *O̲O̲* | *O̲O̲* | *P̲P̲* | *P̲P̲* | *Ɍ̲Ɍ̲* | *Ɍ̲Ɍ̲* |
| a | | *-10, 71%* | *+7, 52% s* | +0, 100% | +0, 87% | +0, 92% | +0, 85% | +1 \| | | +7, 99% s | +0, 96% | +7, 90% | +73, 95% | +0, 95% | | *+3, 56% s* |
| a | | | 336, 798 | | | | | | | | | | | | | |
| a? | | 499, 850 | 516, 850 | 440, 850 | 440, 850 | 440, 850 | 444, 850 | 444, 850 | | 528, 850 | 515, 850 | 515, 850 | 473, 850 | 430, 850 | | 465, 850 |
| a< | | 456, 850 | | **314, 850** | **314, 850** | **314, 850** | **314, 850** | **315, 850** | | **483, 850** | **493, 850** | **500, 850** | **499, 850** | **426, 850** | | |
| a! | | | | | | | | | | | | | | | | |
| b | | +8, 80% | *+7, 52% s* | +0, 100% | +0, 100% | +0, 100% | +0, 100% | +1 \| | | +7, 99% s | +0, 90% | +12, 98% | +71, 92% | +0, 100% | | *+3, 56% s* |
| b | | | 342, -109 | | | | | | | | 310, -113 | | | | | |
| b? | | 295, -110 | 288, -223 | 236, -110 | 236, -110 | 236, -110 | 240, -110 | 240, -110 | | 290, -272 | 302, -151 | 311, -110 | 269, -110 | 226, -110 | | 220, -305 |
| b< | | **300, -110** | | **252, -110** | **252, -110** | **252, -110** | **252, -110** | **253, -110** | | **337, -110** | **310, -110** | **322, -110** | **211, -110** | **140, -110** | | **220, -305** |
| b! | | | | | | | | | | | | | | | | |
| sb | **Ɉ̲Ɉ̲** | **Ƙ̲Ƙ̲** | **Ⱪ̲Ⱪ̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | | **N̲N̲** | **Ø̲Ø̲** | **Ɵ̲Ɵ̲** | **P̲P̲** | **P̲P̲** | | **Ɍ̲Ɍ̲** |
| patch | **Ɉ̲Ɉ̲** | **Ƙ̲Ƙ̲** | **Ⱪ̲Ⱪ̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | ⌧ | **N̲N̲** | **Ø̲Ø̲** | **Ɵ̲Ɵ̲** | **P̲P̲** | **P̲P̲** | ⌧ | **Ɍ̲Ɍ̲** |
| base | **Ɉ̲Ɉ̲** | **Ƙ̲Ƙ̲** | **Ⱪ̲Ⱪ̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **Ł̲Ł̲** | **M̲M̲** | **N̲N̲** | **O̲O̲** | **O̲O̲** | **P̲P̲** | **P̲P̲** | **Ɍ̲Ɍ̲** | **Ɍ̲Ɍ̲** |
| a | +0, 92% | *-6, 66%* | *+2, 12% s* | +0, 100% | +0, 85% | +0, 91% | +0, 85% | +0 \| | | *+8, 37% s* | +0, 92% | +0, 90% | +67, 96% | +0, 95% | | *-25, 68% s* |
| a | 203, 806 | | | | | | | | | | | | | | | |
| a? | 131, 850 | 338, 850 | 345, 850 | 267, 850 | 258, 850 | 263, 850 | 266, 850 | 238, 850 | | 336, 850 | 366, 850 | 366, 850 | 296, 850 | 276, 850 | | 292, 850 |
| a< | **203, 850** | 334, 850 | | **161, 850** | **161, 850** | **161, 850** | **161, 850** | **161, 850** | | | **363, 850** | **363, 850** | **345, 850** | **278, 850** | | |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 98% | +2, 96% | *+2, 12% s* | +0, 100% | +0, 100% | +0, 100% | +0, 100% | +0 \| | | *+8, 37% s* | +0, 87% | +0, 100% | +68, 96% | +0, 100% | | *-25, 68% s* |
| b | | | | | | | | | | | | | | | | |
| b? | 131, -267 | 338, -110 | 345, -221 | 267, -110 | 258, -110 | 263, -110 | 266, -110 | 238, -110 | | 336, -270 | 366, -153 | 366, -110 | 296, -110 | 276, -110 | | 292, -319 |
| b< | **131, -267** | **342, -110** | | **267, -110** | **267, -110** | **267, -110** | **267, -110** | **267, -110** | | **336, -270** | **366, -110** | **366, -110** | **346, -110** | **278, -110** | | **292, -319** |
| b! | | | | | | | | | | | | | | | | |
| si | | ***Ƙ̲Ƙ̲*** | ***Ⱪ̲Ⱪ̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | | ***N̲N̲*** | ***Ø̲Ø̲*** | ***Ɵ̲Ɵ̲*** | ***P̲P̲*** | ***P̲P̲*** | | ***Ɍ̲Ɍ̲*** |
| patch | ⌧ | ***Ƙ̲Ƙ̲*** | ***Ⱪ̲Ⱪ̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ⌧ | ***N̲N̲*** | ***Ø̲Ø̲*** | ***Ɵ̲Ɵ̲*** | ***P̲P̲*** | ***P̲P̲*** | ⌧ | ***Ɍ̲Ɍ̲*** |
| base | ***Ɉ̲Ɉ̲*** | ***Ƙ̲Ƙ̲*** | ***Ⱪ̲Ⱪ̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***Ł̲Ł̲*** | ***M̲M̲*** | ***N̲N̲*** | ***O̲O̲*** | ***O̲O̲*** | ***P̲P̲*** | ***P̲P̲*** | ***Ɍ̲Ɍ̲*** | ***Ɍ̲Ɍ̲*** |
| a | | *-36, 73%* | *-28, 11% s* | +12, 100% | +12, 81% | +12, 92% | +12, 85% | +14 \| | | -23, 99% s | +0, 93% | +1, 90% | +69, 96% | +12, 95% | | *-11, 69% s* |
| a | | | 336, 798 | | | | | | | | 351, 850 | | 400, 850 | | | |
| a? | | 353, 850 | 358, 850 | 310, 850 | 300, 850 | 300, 850 | 300, 850 | 300, 850 | | 383, 850 | 390, 850 | 387, 850 | 315, 850 | 296, 850 | | 307, 850 |
| a< | | 333, 850 | | **221, 850** | **221, 850** | **221, 850** | **221, 850** | **223, 850** | | **366, 850** | **390, 850** | **391, 850** | **385, 850** | **328, 850** | | |
| a! | | | | | | | | | | | | | | | | |
| b | | -28, 94% | *-28, 11% s* | +12, 100% | +12, 100% | +12, 100% | +12, 100% | +14 \| | | -23, 99% s | +0, 88% | +1, 100% | +70, 100% | +12, 100% | | *-11, 69% s* |
| b | | | 342, -109 | | | | | | | | 350, -110 | | 363, -110 | | | |
| b? | | 320, -110 | 302, -221 | 277, -110 | 267, -110 | 267, -110 | 267, -110 | 267, -110 | | 317, -270 | 349, -148 | 354, -110 | 282, -110 | 263, -110 | | 235, -302 |
| b< | | **314, -110** | | **272, -110** | **272, -110** | **272, -110** | **272, -110** | **274, -110** | | **288, -270** | **350, -110** | **351, -110** | **310, -110** | **252, -110** | | **235, -302** |
| b! | | | | | | | | | | | | | | | | |

Table 23: Derived letters and inherited anchors 3/7

| hex | 0166 | 01AC | 01AE | 023E | 01B3 | 024E | 01B5 | 1E9A | 2C65 | 0180 | 0183 | 0188 | 023C | 0255 | A793 | 0111 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 0054 | 0054 | 0054 | 0054 | 0059 | 0059 | 005A | 0061 | 0061 | 0062 | 0062 | 0063 | 0063 | 0063 | 0063 | 0064 |
| reg | Ŧ̲ | Ţ̲ | Ţ̲ | Ŧ̲ | Ẏ̲ | Ẏ̲ | Ẓ̲ | ả̲ | ⱥ̲ | ƀ̲ | ƀ̲ | c̲̓ | ȩ̲̓ | c̲̓ | ȩ̲̓ | đ̲ |
| patch | Ŧ̲ | Ţ̲ | Ţ̲ | Ŧ̲ | Ẏ̲ | Ẏ̲ | Ẓ̲ | ả̲ | ⱥ̲ | ƀ̲ | ƀ̲ | c̲̓ | ȩ̲̓ | c̲̓ | ȩ̲̓ | đ̲ |
| base | Ţ̲ | Ţ̲ | Ţ̲ | Ţ̲ | Ẏ̲ | Ẏ̲ | Ẓ̲ | ⱥ̲ | ⱥ̲ | ƀ̲ | ƀ̲ | c̲̓ | c̲̓ | c̲̓ | c̲̓ | đ̲ |
| a | +0, 96% | +22 \| | +0, 100% s | +0 \| | +65 \| | +0, 90% | +0, 94% | +0, 100% | +0, 84% | +0, 95% | -4, 93% | +1, 98% | +1, 85% | +1, 91% s | +85, 88% | +0, 95% |
| a | | | | | | | | | | | | | | 209, 645 | 327, 645 | |
| a? | 297, 850 | 308, 850 | 297, 850 | 297, 850 | 321, 850 | 288, 850 | 310, 850 | 272, 838 | 246, 690 | 231, 885 | 228, 885 | 264, 713 | 216, 668 | 222, 645 | 260, 645 | 270, 885 |
| a< | **307, 850** | **329, 850** | **307, 850** | **307, 850** | **360, 850** | **295, 850** | 320, 850 | **209, 645** | **209, 645** | **279, 885** | **275, 885** | **243, 669** | **243, 645** | **243, 645** | **327, 645** | **235, 885** |
| a! | | | | | | | 320, 850 | | | | | | | | | |
| b | +0, 100% | +22 \| | +0, 100% s | +0 \| | +65 \| | +0, 100% | +0, 100% | +0, 100% | +0, 96% | +0, 100% | -4, 100% | +0, 100% | +0, 85% | | +85, 88% | +0, 100% |
| b | | | | | | | | | | | 234, -109 | | | 212, -114 | 321, -110 | |
| b? | 297, -110 | 308, -110 | 297, -261 | 297, -165 | 321, -110 | 288, -110 | 310, -110 | 272, -110 | 246, -170 | 231, -110 | 228, -110 | 264, -110 | 216, -136 | 222, -214 | 260, -110 | 270, -110 |
| b< | **300, -110** | **322, -110** | **300, -261** | **300, -110** | **360, -110** | **295, -110** | 287, -110 | **196, -110** | **196, -162** | **234, -110** | **230, -110** | **218, -110** | **218, -110** | | **303, -110** | **235, -110** |
| b! | | | | | | | 287, -110 | | | | | | | | | |

| it | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | | *đ̲* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | ⌧ | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | ⌧ | *đ̲* |
| base | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ẏ̲* | *Ẏ̲* | *Ẓ̲* | *ⱥ̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *c̲̓* | *c̲̓* | *c̲̓* | *đ̲* |
| a | +0, 94% | +40 \| | +0, 100% s | -8 \| | +72 \| | | +0, 94% | +0, 100% | +1, 88% | *+4, 64%* | *-20, 64%* | *+18, 79%* | +22, 82% | +7, 83% s | | +0, 95% |
| a | | | | | | | | | | | | | | | | |
| a? | 453, 850 | 488, 850 | 453, 850 | 453, 850 | 501, 850 | | 470, 850 | 489, 843 | 385, 668 | 428, 890 | 413, 890 | 398, 713 | 352, 668 | 352, 645 | | 434, 890 |
| a< | **457, 850** | **497, 850** | **457, 850** | **469, 946** | **513, 850** | | 450, 850 | **341, 645** | **342, 645** | 432, 890 | 408, 890 | 361, 645 | **365, 645** | **350, 645** | | **434, 890** |
| a! | | | | | | | 489, 850 | | | | | | | | | |
| b | +0, 100% | +40 \| | +0, 100% s | -8 \| | +72 \| | | +0, 100% | +0, 100% | +0, 92% | *+15, 54%* | *-6, 54%* | *+5, 65%* | *+10, 44%* | | | +0, 100% |
| b | | | | | | | | | | | 230, -109 | | | 212, -114 | | |
| b? | 249, -110 | 284, -110 | 214, -274 | 237, -165 | 297, -110 | | 266, -110 | 286, -110 | 214, -136 | 216, -110 | 201, -110 | 223, -110 | 181, -137 | 171, -208 | | 221, -110 |
| b< | **277, -110** | **317, -110** | **242, -274** | **269, -110** | **342, -110** | | 267, -110 | **220, -110** | **220, -110** | 229, -110 | 208, -110 | 168, -110 | 173, -110 | | | **221, -110** |
| b! | | | | | | | 286, -110 | | | | | | | | | |

| sb | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | | *đ̲* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | ⌧ | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | ⌧ | *đ̲* |
| base | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ẏ̲* | *Ẏ̲* | *Ẓ̲* | *ⱥ̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *c̲̓* | *c̲̓* | *c̲̓* | *đ̲* |
| a | +0, 94% | +44 \| | +13, 79% s | +0 \| | +58 \| | | +0, 94% | +0, 100% | +0, 90% | +0, 95% | -14, 87% | +0, 95% | +0, 81% | +2, 75% s | | +0, 95% |
| a | | | | | | | | | | | | | | | | |
| a? | 299, 850 | 320, 850 | 306, 850 | 299, 850 | 333, 850 | | 309, 850 | 288, 840 | 255, 685 | 236, 885 | 223, 870 | 264, 708 | 200, 670 | 232, 645 | | 274, 885 |
| a< | **298, 850** | **342, 850** | **311, 850** | **298, 850** | **374, 850** | | 321, 850 | **219, 645** | **219, 645** | **236, 885** | **222, 885** | **224, 645** | **224, 645** | **226, 645** | | **274, 885** |
| a! | | | | | | | 316, 850 | | | | | | | | | |
| b | +0, 100% | +44 \| | +13, 79% s | +0 \| | +58 \| | | +0, 100% | +0, 100% | +0, 94% | +0, 100% | -14, 81% | +0, 100% | +0, 81% | | | +0, 100% |
| b | | | | | | | | | | | | | | | | |
| b? | 299, -110 | 320, -110 | 306, -281 | 299, -165 | 333, -110 | | 309, -110 | 288, -110 | 255, -170 | 236, -121 | 223, -124 | 264, -110 | 200, -129 | 232, -209 | | 274, -110 |
| b< | **301, -110** | **345, -110** | **314, -281** | **301, -110** | **374, -110** | | 308, -110 | **255, -110** | **255, -110** | **236, -121** | **222, -121** | **224, -110** | **224, -110** | | | **274, -110** |
| b! | | | | | | | 316, -110 | | | | | | | | | |

| si | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | | *đ̲* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | *Ŧ̲* | *Ţ̲* | *Ţ̲* | *Ŧ̲* | *Ẏ̲* | ⌧ | *Ẓ̲* | *ả̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *ȩ̲̓* | *c̲̓* | ⌧ | *đ̲* |
| base | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ţ̲* | *Ẏ̲* | *Ẏ̲* | *Ẓ̲* | *ⱥ̲* | *ⱥ̲* | *ƀ̲* | *ƀ̲* | *c̲̓* | *c̲̓* | *c̲̓* | *c̲̓* | *đ̲* |
| a | +12, 94% | +44 \| | +9, 79% s | +11 \| | +95 \| | | *+12, 62%* | +0, 100% | +11, 93% | +0, 94% | -16, 93% | +22, 93% | +13, 81% | *+1, 67% s* | | +0, 94% |
| a | | | | | | | | | | | | | | | | |
| a? | 325, 850 | 330, 850 | 314, 850 | 325, 850 | 368, 850 | | 332, 850 | 371, 844 | 234, 685 | 290, 890 | 269, 875 | 253, 708 | 204, 670 | 204, 645 | | 309, 890 |
| a< | **319, 850** | **351, 850** | **321, 850** | **318, 850** | **386, 850** | | 332, 850 | **251, 645** | **262, 645** | **290, 890** | **274, 890** | **270, 675** | **255, 645** | | | **309, 890** |
| a! | | | | | | | 312, 850 | | | | | | | | | |
| b | +12, 100% | +44 \| | +9, 79% s | +11 \| | +95 \| | | *+14, 65%* | +0, 100% | +12, 95% | +0, 100% | -16, 81% | *+19, 67%* | +12, 85% | | | +0, 100% |
| b | | | | | | | | | | | 234, -109 | | | 212, -114 | | |
| b? | 292, -110 | 297, -110 | 246, -281 | 281, -165 | 335, -110 | | 299, -110 | 339, -110 | 222, -170 | 248, -110 | 231, -110 | 249, -110 | 203, -129 | 192, -209 | | 268, -110 |
| b< | **312, -110** | **344, -110** | **280, -281** | **311, -110** | **372, -110** | | 301, -110 | **260, -110** | **270, -122** | **248, -110** | **232, -110** | 255, -110 | **248, -110** | | | **221, -110** |
| b! | | | | | | | 267, -110 | | | | | | | | | |

Table 24: Derived letters and inherited anchors 4/7

| hex | 018C | 0221 | 0256 | 0257 | 0247 | 01E5 | 0127 | 0266 | 2C68 | 0268 | 0249 | 0199 | 2C6A | 0140 | 0142 | 019A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 0064 | 0064 | 0064 | 0064 | 0065 | 0067 | 0068 | 0068 | 0068 | 0069 | 006A | 006B | 006B | 006C | 006C | 006C |
| reg | ƌ̈ | ɗ̈ | ɖ̈ | ƌ̈ | ɛ̈ | g̈ | ḧ | ɦ̈ | ⱨ̈ | ï | ǰ | k̈ | ⱪ̈ | ŧ̈ | ŧ̈ | ŧ̈ |
| patch | ƌ̂ | ɗ̂ | ɖ̂ | ƌ̂ | ɛ̂ | ĝ | ĥ | ɦ̂ | ⱨ̂ | î | ǰ | k̂ | ⱪ̂ | ŧ̂ | ŧ̂ | ŧ̂ |
| base | ɖ̂ | ɗ̂ | ɖ̂ | ɗ̂ | ɛ̂ | ĝ | ĥ | ɦ̂ | ⱨ̂ | î | ǰ | k̂ | ⱪ̂ | ł̂ | ł̂ | ł̂ |
| a | +5, 94% | -8, 91% s | -3, 95% s | -2, 94% s | +0, 85% | +0, 100% | +0, 95% | -10, 82% | -11, 96% s | +14, 86% | +0, 90% | +1, 85% | +2, 97% s | +0, 100% | +0, 89% | +0 \| |
| a | 214, 650 | | 218, 645 | 218, 645 | | | | 299, 565 | 299, 645 | 139, 645 | 140, 645 | | 275, 646 | | | |
| a? | 275, 885 | 326, 885 | 314, 885 | 322, 885 | 216, 668 | 256, 676 | 270, 885 | 262, 885 | 262, 885 | 150, 834 | 116, 834 | 262, 885 | 259, 885 | 163, 885 | 133, 885 | 132, 885 |
| a< | **240, 885** | **227, 885** | **232, 885** | **233, 885** | **243, 645** | **231, 645** | **282, 885** | **272, 885** | **271, 885** | 156, 645 | **152, 645** | **273, 885** | **274, 885** | **135, 885** | **135, 885** | **135, 885** |
| a! | | | | | | | | | | 156, 834 | | | | | | |
| b | +5, 100% | -8, 91% s | -3, 95% s | -2, 94% s | +0, 88% | +0, 89% | +0, 100% | -10, 96% | -11, 96% s | +14, 92% | +0, 94% | +1, 91% | +2, 97% s | +0, 100% | +0, 100% | +0 \| |
| b | 217, -107 | | 212, -114 | 212, -114 | | | | 240, -114 | 259, -111 | 134, -114 | | 263, -107 | 263, -107 | | | |
| b? | 275, -110 | 326, -210 | 314, -273 | 322, -110 | 216, -136 | 256, -319 | 270, -110 | 262, -110 | 262, -213 | 150, -110 | 116, -308 | 262, -110 | 259, -213 | 163, -110 | 133, -110 | 132, -110 |
| b< | **240, -110** | **227, -210** | **232, -273** | **233, -110** | **234, -110** | **229, -319** | **272, -110** | **262, -110** | **262, -213** | **156, -110** | **109, -319** | **262, -110** | **263, -213** | **135, -110** | **135, -110** | **135, -110** |
| b! | | | | | | | | | | | | | | | | |
| it | *ƌ̈* | *ɗ̈* | *ɖ̈* | *ƌ̈* | *ɛ̈* | *g̈* | *ḧ* | *ɦ̈* | *ⱨ̈* | *ï* | *ǰ* | *k̈* | *ⱪ̈* | *ŧ̈* | *ŧ̈* | *ŧ̈* |
| patch | *ƌ̂* | *ɗ̂* | *ɖ̂* | *ƌ̂* | *ɛ̂* | *ĝ* | *ĥ* | *ɦ̂* | *ⱨ̂* | *î* | *ǰ* | *k̂* | *ⱪ̂* | *ŧ̂* | *ŧ̂* | *ŧ̂* |
| base | *ɖ̂* | *ɗ̂* | *ɖ̂* | *ɗ̂* | *ɛ̂* | *ĝ* | *ĥ* | *ɦ̂* | *ⱨ̂* | *î* | *ǰ* | *k̂* | *ⱪ̂* | *ł̂* | *ł̂* | *ł̂* |
| a | *-7, 77%* | *+12, 4% s* | *+10, 4% s* | +0, 99% s | +0, 93% | +0, 100% | | | +12, 79% s | +14, 86% | +0, 86% | *+30, 48%* | *+24, 42% s* | +0, 100% | +0, 89% | +20 \| |
| a | 209, 650 | | 218, 645 | 218, 645 | 332, 645 | | | 299, 565 | 299, 645 | 267, 645 | 254, 645 | | 275, 646 | | | |
| a? | 423, 890 | 507, 890 | 465, 899 | 464, 890 | 348, 692 | 376, 645 | 449, 890 | 449, 890 | 448, 890 | 316, 810 | 315, 810 | 429, 890 | 453, 890 | 349, 890 | 322, 890 | 342, 890 |
| a< | 427, 890 | | | **434, 890** | **340, 684** | **360, 645** | | | **461, 890** | 289, 645 | **283, 645** | 462, 890 | | **312, 890** | **312, 890** | **332, 890** |
| a! | | | 440, 899 | | | | | | | 326, 810 | | | | | | |
| b | *-2, 51%* | *+12, 4% s* | *+10, 4% s* | +0, 99% s | +0, 91% | +0, 92% | +0, 100% | +0, 100% | +12, 79% s | +14, 97% | +0, 92% | *+14, 60%* | *+24, 42% s* | +0, 100% | +0, 100% | +20 \| |
| b | 212, -107 | | 212, -114 | 212, -114 | 158, -108 | | | 240, -114 | 259, -111 | 134, -114 | | 263, -107 | 263, -107 | | | |
| b? | 211, -110 | 274, -205 | 215, -273 | 251, -110 | 170, -143 | 171, -319 | 236, -110 | 236, -110 | 211, -223 | 121, -110 | 83, -281 | 216, -110 | 216, -223 | 136, -110 | 110, -110 | 130, -110 |
| b< | 219, -110 | **274, -205** | **215, -273** | **221, -110** | **153, -135** | **197, -319** | **229, -110** | **229, -110** | **216, -223** | **129, -110** | **0, -319** | 237, -110 | **216, -223** | **99, -110** | **99, -110** | **119, -110** |
| b! | | | | | | | | | | | | | | | | |
| sb | **ƌ̈** | **ɗ̈** | **ɖ̈** | **ƌ̈** | **ɛ̈** | **g̈** | **ḧ** | **ɦ̈** | **ⱨ̈** | **ï** | **ǰ** | **k̈** | **ⱪ̈** | **ŧ̈·** | **ŧ̈** | **ŧ̈** |
| patch | **ƌ̂** | **ɗ̂** | **ɖ̂** | **ƌ̂** | **ɛ̂** | **ĝ** | **ĥ** | **ɦ̂** | **ⱨ̂** | **î** | **ǰ** | **k̂** | **ⱪ̂** | **ŧ̂·** | **ŧ̂** | **ŧ̂** |
| base | **ɖ̂** | **ɗ̂** | **ɖ̂** | **ɗ̂** | **ɛ̂** | **ĝ** | **ĥ** | **ɦ̂** | **ⱨ̂** | **î** | **ǰ** | **k̂** | **ⱪ̂** | **ł̂** | **ł̂** | **ł̂** |
| a | -19, 87% | -2, 92% s | -2, 93% s | -4, 92% s | +0, 83% | +0, 100% | +0, 95% | -2, 80% | -15, 97% s | +0, 87% | +0, 90% | *-5, 66%* | *-2, 2% s* | +0, 100% | +0, 85% | +0 \| |
| a | | | | | | | | | | 151, 645 | | | | | | |
| a? | 261, 870 | 338, 885 | 322, 885 | 324, 885 | 232, 686 | 258, 676 | 275, 885 | 274, 885 | 264, 885 | 152, 818 | 112, 837 | 259, 885 | 273, 885 | 212, 885 | 142, 885 | 138, 885 |
| a< | **255, 885** | **272, 885** | **272, 885** | **270, 885** | **249, 678** | **231, 645** | **275, 885** | **273, 885** | **260, 885** | 156, 645 | **163, 645** | 257, 885 | | **150, 885** | **150, 885** | **150, 885** |
| a! | | | | | | | | | | 156, 837 | | | | | | |
| b | -20, 83% | -2, 92% s | -2, 93% s | -4, 92% s | -1, 81% | +0, 92% | +0, 100% | -3, 96% | -15, 97% s | +0, 92% | +0, 94% | -6, 91% | *-2, 2% s* | +0, 100% | +0, 100% | +0 \| |
| b | | | | | | | | | | 134, -114 | | | | | | |
| b? | 261, -110 | 338, -216 | 322, -271 | 324, -110 | 232, -142 | 258, -319 | 275, -110 | 274, -110 | 264, -222 | 152, -110 | 112, -319 | 259, -110 | 273, -222 | 212, -110 | 142, -110 | 138, -110 |
| b< | **254, -110** | **272, -216** | **272, -271** | **270, -110** | **231, -110** | **229, -319** | **277, -110** | **274, -110** | **262, -222** | **156, -110** | **109, -319** | **256, -110** | **273, -222** | **150, -110** | **150, -110** | **150, -110** |
| b! | | | | | | | | | | | | | | | | |
| si | ***ƌ̈*** | ***ɗ̈*** | ***ɖ̈*** | ***ƌ̈*** | ***ɛ̈*** | ***g̈*** | ***ḧ*** | ***ɦ̈*** | ***ⱨ̈*** | ***ï*** | ***ǰ*** | ***k̈*** | ***ⱪ̈*** | ***ŧ̈·*** | ***ŧ̈*** | ***ŧ̈*** |
| patch | ***ƌ̂*** | ***ɗ̂*** | ***ɖ̂*** | ***ƌ̂*** | ***ɛ̂*** | ***ĝ*** | ***ĥ*** | ***ɦ̂*** | ***ⱨ̂*** | ***î*** | ***ǰ*** | ***k̂*** | ***ⱪ̂*** | ***ŧ̂·*** | ***ŧ̂*** | ***ŧ̂*** |
| base | ***ɖ̂*** | ***ɗ̂*** | ***ɖ̂*** | ***ɗ̂*** | ***ɛ̂*** | ***ĝ*** | ***ĥ*** | ***ɦ̂*** | ***ⱨ̂*** | ***î*** | ***ǰ*** | ***k̂*** | ***ⱪ̂*** | ***ł̂*** | ***ł̂*** | ***ł̂*** |
| a | -15, 89% | -2, 74% s | -2, 81% s | -5, 82% s | +0, 98% | +4, 100% | | | -13, 71% s | +0, 86% | +0, 86% | *-5, 45%* | *-11, 60% s* | +12, 100% | +17, 85% | +18 \| |
| a | 214, 650 | | 218, 645 | 218, 645 | 243, 645 | | | 299, 565 | 299, 645 | 276, 645 | 290, 645 | | 275, 646 | | | |
| a? | 275, 875 | 362, 890 | 302, 890 | 318, 890 | 226, 684 | 220, 645 | 331, 890 | 302, 890 | 290, 890 | 180, 823 | 192, 823 | 283, 890 | 295, 890 | 234, 890 | 190, 890 | 174, 890 |
| a< | **294, 890** | **311, 890** | **311, 890** | **310, 890** | **249, 676** | **224, 645** | | | **315, 890** | 144, 645 | **171, 645** | 311, 890 | | **192, 890** | **197, 890** | **198, 890** |
| a! | | | | | | | | | | 180, 823 | | | | | | |
| b | *-13, 57%* | -2, 74% s | -2, 81% s | -5, 82% s | +0, 91% | +4, 92% | +0, 100% | *-4, 71%* | -13, 71% s | +2, 93% | +12, 92% | *-17, 44%* | *-11, 60% s* | +12, 100% | +17, 100% | +18 \| |
| b | 217, -107 | | 212, -114 | 212, -114 | 234, -110 | | | 240, -114 | 259, -111 | 136, -110 | 109, -319 | 263, -107 | 263, -107 | | | |
| b? | 237, -110 | 299, -216 | 227, -271 | 276, -110 | 221, -140 | 186, -319 | 290, -110 | 260, -110 | 225, -222 | 153, -110 | 127, -292 | 241, -110 | 230, -222 | 192, -110 | 148, -110 | 132, -110 |
| b< | 208, -110 | **200, -216** | **189, -271** | **220, -110** | **230, -132** | **203, -319** | **272, -110** | 268, -110 | **233, -222** | **155, -110** | **79, -319** | 258, -110 | **230, -222** | **150, -110** | **155, -110** | **156, -110** |
| b! | | | | | | | | | | | | | | | | |

Table 25: Derived letters and inherited anchors 5/7

| hex | 0234 | 026B | 026C | 026D | 026E | 2C61 | 0271 | 0235 | 0272 | 0273 | 00F8 | 0275 | 01A5 | 02A0 | 024D | 027D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 006C | 006C | 006C | 006C | 006C | 006C | 006D | 006E | 006E | 006E | 006F | 006F | 0070 | 0071 | 0072 | 0072 |
| reg | Ɫ̵ | ɬ̵ | ɮ̵ | ꞎ̵ | ꝁ̵ | ⱡ̵ | ɱ̵ | ɳ̵ | ɲ̵ | ŋ̵ | ø̵ | ɵ̵ | ƥ̵ | ʠ̵ | ɾ̵ | ɽ̵ |
| patch | Ɫ̵ | ɬ̵ | ɮ̵ | ꞎ̵ | ꝁ̵ | ⱡ̵ | ɱ̵ | ɳ̵ | ɲ̵ | ŋ̵ | ø̵ | ɵ̵ | ƥ̵ | ʠ̵ | ɾ̵ | ɽ̵ |
| base | Ɫ̵ | ɬ̵ | ɮ̵ | ꞎ̵ | ꝁ̵ | ⱡ̵ | ɱ̵ | ɳ̵ | ɲ̵ | ŋ̵ | o̵ | o̵ | ƥ̵ | q̵ | ɾ̵ | ɽ̵ |
| a | +0, 100% s | +12 \| | +90 \| | -3, 100% s | +0, 91% | +0 \| | -3, 97% s | -4, 99% s | -11, 96% s | -11, 96% s | +0, 91% | +0, 85% | | | +0, 93% | -16, 72% s |
| a | | 209, 645 | 209, 645 | 209, 645 | 209, 645 | | 383, 645 | 261, 646 | 264, 646 | 263, 645 | | 209, 645 | | 238, 645 | | 209, 645 |
| a? | 175, 885 | 144, 885 | 182, 885 | 182, 885 | 250, 885 | 132, 885 | 366, 645 | 333, 645 | 234, 645 | 296, 645 | 250, 668 | 252, 645 | 242, 885 | 320, 885 | 188, 645 | 186, 645 |
| a< | **135, 885** | **147, 885** | **225, 885** | **132, 885** | **135, 885** | **135, 885** | **389, 645** | **260, 645** | **253, 645** | **253, 645** | **238, 645** | **238, 645** | | | **191, 645** | **175, 645** |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 100% s | +12 \| | +90 \| | -3, 100% s | | +0 \| | -3, 97% s | -4, 99% s | -11, 96% s | -11, 96% s | +0, 90% | +0, 83% | +0, 100% | -5, 97% | +0, 96% | -16, 72% s |
| b | | 146, -114 | 218, -120 | 196, -110 | 196, -110 | | 383, -105 | 262, -111 | 265, -111 | 262, -112 | 243, -110 | 196, -110 | | 196, -110 | | 196, -110 |
| b? | 175, -216 | 144, -110 | 182, -110 | 182, -273 | 250, -319 | 132, -110 | 366, -306 | 333, -201 | 234, -306 | 296, -306 | 250, -136 | 252, -110 | 242, -319 | 320, -319 | 188, -110 | 186, -319 |
| b< | **135, -216** | **147, -110** | **225, -110** | **132, -273** | | **135, -110** | **398, -306** | **261, -201** | **254, -306** | **254, -306** | **243, -110** | **243, -110** | **120, -319** | **381, -319** | **140, -110** | **124, -319** |
| b! | | | | | 275, -319 | | | | | | | | | | | |
| it | *Ɫ̵* | *ɬ̵* | *ɮ̵* | *ꞎ̵* | *ꝁ̵* | *ⱡ̵* | *ɱ̵* | *ɳ̵* | *ɲ̵* | *ŋ̵* | *ø̵* | *ɵ̵* | *ƥ̵* | *ʠ̵* | *ɾ̵* | *ɽ̵* |
| patch | *Ɫ̵* | *ɬ̵* | *ɮ̵* | *ꞎ̵* | *ꝁ̵* | *ⱡ̵* | *ɱ̵* | *ɳ̵* | *ɲ̵* | *ŋ̵* | *ø̵* | *ɵ̵* | *ƥ̵* | *ʠ̵* | *ɾ̵* | *ɽ̵* |
| base | *Ɫ̵* | *ɬ̵* | *ɮ̵* | *ꞎ̵* | *ꝁ̵* | *ⱡ̵* | *ɱ̵* | *ɳ̵* | *ɲ̵* | *ŋ̵* | *o̵* | *o̵* | *ƥ̵* | *q̵* | *ɾ̵* | *ɽ̵* |
| a | +14, 91% s | +12 \| | +89 \| | +0, 91% s | +18, 89% | +0 \| | *-13, 70% s* | +0, 89% s | +18, 79% s | -1, 95% s | +0, 89% | +20, 84% | | | +0, 91% | *+4, 61% s* |
| a | | | | | | | 383, 645 | 261, 646 | 264, 646 | 263, 645 | | | | 238, 645 | 312, 646 | |
| a? | 389, 890 | 333, 890 | 368, 890 | 341, 890 | 447, 890 | 330, 890 | 515, 645 | 460, 645 | 396, 645 | 367, 645 | 370, 691 | 379, 645 | 429, 890 | 460, 871 | 515, 1008 | 320, 645 |
| a< | **326, 890** | **324, 890** | **401, 890** | 312, 890 | **330, 890** | **312, 890** | | **394, 645** | **414, 654** | **393, 645** | **359, 645** | **379, 645** | | | **312, 645** | |
| a! | | | | 312, 890 | | | | | | | | | | | | 359, 645 |
| b | +14, 91% s | +12 \| | +89 \| | +0, 91% s | | +0 \| | *-13, 70% s* | +0, 89% s | +18, 79% s | -1, 95% s | +0, 88% | +14, 83% | -17, 89% | *+4, 74%* | +0, 85% | *+4, 61% s* |
| b | | 146, -114 | 218, -120 | | | | 383, -105 | 262, -111 | 265, -111 | 262, -112 | 186, -107 | | | | 113, -110 | |
| b? | 155, -214 | 120, -110 | 156, -110 | 91, -287 | 192, -309 | 117, -110 | 310, -319 | 281, -197 | 191, -319 | 162, -319 | 190, -156 | 218, -110 | 172, -319 | 207, -319 | 246, -257 | 124, -279 |
| b< | **91, -214** | **111, -110** | **188, -110** | **61, -287** | | **99, -110** | **310, -319** | **217, -197** | **209, -319** | **190, -319** | **178, -148** | **200, -110** | **160, -319** | 304, -319 | **113, -110** | **124, -279** |
| b! | | | | | 207, -319 | | | | | | | | | | | |
| sb | **Ɫ̵** | **ɬ̵** | **ɮ̵** | **ꞎ̵** | **ꝁ̵** | **ⱡ̵** | **ɱ̵** | **ɳ̵** | **ɲ̵** | **ŋ̵** | **ø̵** | **ɵ̵** | **ƥ̵** | **ʠ̵** | **ɾ̵** | **ɽ̵** |
| patch | **Ɫ̵** | **ɬ̵** | **ɮ̵** | **ꞎ̵** | **ꝁ̵** | **ⱡ̵** | **ɱ̵** | **ɳ̵** | **ɲ̵** | **ŋ̵** | **ø̵** | **ɵ̵** | **ƥ̵** | **ʠ̵** | **ɾ̵** | **ɽ̵** |
| base | **Ɫ̵** | **ɬ̵** | **ɮ̵** | **ꞎ̵** | **ꝁ̵** | **ⱡ̵** | **ɱ̵** | **ɳ̵** | **ɲ̵** | **ŋ̵** | **o̵** | **o̵** | **ƥ̵** | **q̵** | **ɾ̵** | **ɽ̵** |
| a | -6, 94% s | +12 \| | +90 \| | +0, 100% s | +0, 89% | +0 \| | -10, 97% s | -12, 91% s | -13, 97% s | -15, 89% s | +1, 88% | +0, 89% | | | +0, 90% | -16, 92% s |
| a | | | | | | | | | | | | | | | | |
| a? | 212, 885 | 156, 885 | 190, 885 | 194, 885 | 274, 885 | 144, 885 | 363, 645 | 345, 645 | 246, 645 | 306, 645 | 260, 684 | 260, 645 | 236, 885 | 337, 885 | 210, 645 | 198, 645 |
| a< | **144, 885** | **162, 885** | **240, 885** | **150, 885** | **150, 885** | **150, 885** | **386, 645** | **278, 645** | **277, 645** | **275, 645** | **261, 645** | **260, 645** | | | **212, 645** | **196, 645** |
| a! | | | | | | | | | | | | | | | | |
| b | -6, 94% s | +12 \| | +90 \| | +0, 100% s | | +0 \| | -10, 97% s | -12, 91% s | -13, 97% s | -15, 89% s | +0, 85% | +0, 87% | -15, 93% | -4, 91% | +0, 100% | -16, 92% s |
| b | | 146, -114 | 218, -120 | | | | | | | | | 229, -114 | | | | |
| b? | 212, -223 | 156, -110 | 190, -110 | 194, -319 | 274, -319 | 144, -110 | 363, -319 | 345, -207 | 246, -319 | 306, -319 | 260, -142 | 260, -110 | 236, -319 | 337, -319 | 210, -110 | 198, -319 |
| b< | **144, -223** | **162, -110** | **240, -110** | **150, -319** | | **150, -110** | **386, -319** | **254, -207** | **253, -319** | **250, -319** | **260, -110** | **260, -110** | **239, -319** | **269, -319** | **152, -110** | **135, -319** |
| b! | | | | | 283, -319 | | | | | | | | | | | |
| si | ***Ɫ̵*** | ***ɬ̵*** | ***ɮ̵*** | ***ꞎ̵*** | ***ꝁ̵*** | ***ⱡ̵*** | ***ɱ̵*** | ***ɳ̵*** | ***ɲ̵*** | ***ŋ̵*** | ***ø̵*** | ***ɵ̵*** | ***ƥ̵*** | ***ʠ̵*** | ***ɾ̵*** | ***ɽ̵*** |
| patch | ***Ɫ̵*** | ***ɬ̵*** | ***ɮ̵*** | ***ꞎ̵*** | ***ꝁ̵*** | ***ⱡ̵*** | ***ɱ̵*** | ***ɳ̵*** | ***ɲ̵*** | ***ŋ̵*** | ***ø̵*** | ***ɵ̵*** | ***ƥ̵*** | ***ʠ̵*** | ***ɾ̵*** | ***ɽ̵*** |
| base | ***Ɫ̵*** | ***ɬ̵*** | ***ɮ̵*** | ***ꞎ̵*** | ***ꝁ̵*** | ***ⱡ̵*** | ***ɱ̵*** | ***ɳ̵*** | ***ɲ̵*** | ***ŋ̵*** | ***o̵*** | ***o̵*** | ***ƥ̵*** | ***q̵*** | ***ɾ̵*** | ***ɽ̵*** |
| a | +6, 89% s | +29 \| | +106 \| | +19, 93% s | +9, 87% | +11 \| | *-7, 66% s* | *-2, 67% s* | *-4, 69% s* | *-5, 67% s* | +0, 89% | +32, 82% | | | +0, 92% | *-10, 50% s* |
| a | | | | | | | 383, 645 | 261, 646 | 264, 646 | 263, 645 | | | | 238, 645 | 191, 646 | |
| a? | 233, 890 | 187, 890 | 223, 890 | 187, 890 | 301, 890 | 184, 890 | 373, 645 | 324, 645 | 254, 645 | 254, 645 | 235, 645 | 229, 645 | 271, 890 | 318, 890 | 203, 645 | 177, 645 |
| a< | **186, 890** | **209, 890** | **286, 890** | 205, 890 | **189, 890** | **191, 890** | | | | | **238, 645** | **270, 645** | | | **191, 645** | |
| a! | | | | 205, 890 | | | | | | | | | | | | 218, 645 |
| b | +6, 89% s | +29 \| | +106 \| | +19, 93% s | | +11 \| | *-7, 66% s* | *-2, 67% s* | *-4, 69% s* | *-5, 67% s* | +0, 89% | +25, 80% | -32, 88% | *-5, 65%* | +0, 85% | *-10, 50% s* |
| b | | 146, -114 | 218, -120 | | | | 383, -105 | 262, -111 | 265, -111 | 262, -112 | 243, -110 | | | | 135, -110 | |
| b? | 169, -223 | 145, -110 | 181, -110 | 103, -319 | 217, -319 | 142, -110 | 339, -319 | 313, -207 | 220, -319 | 220, -319 | 237, -138 | 238, -110 | 187, -319 | 234, -319 | 212, -110 | 143, -319 |
| b< | **121, -223** | **167, -110** | **244, -110** | **120, -319** | | **149, -110** | **339, -319** | **313, -207** | **220, -319** | **220, -319** | **243, -110** | **268, -110** | **147, -319** | 343, -319 | **135, -110** | **143, -319** |
| b! | | | | | 231, -319 | | | | | | | | | | | |

Table 26: Derived letters and inherited anchors 6/7

| hex | 0282 | 0167 | 01AB | 01AD | 0236 | 0288 | 2C66 | 0289 | AB52 | 01B4 | 024F | 1EFF | 01B6 | 0290 | 0291 | A765 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 0073 | 0074 | 0074 | 0074 | 0074 | 0074 | 0074 | 0075 | 0075 | 0079 | 0079 | 0079 | 007A | 007A | 007A | 00FE |
| reg | ʂ̂ʂ̲ | ŧ̂ŧ̲ | ʈ̂ʈ̲ | ɟ̂ɟ̲ | ȶ̂ȶ̲ | ʇ̂ʇ̲ | ɻ̂ɻ̲ | ʉ̂ʉ̲ | ᵾ̂ᵾ̲ | ʏ̂ʏ̲ | ɣ̂ɣ̲ | ỿ̂ỿ̲ | ƶ̂ƶ̲ | ʑ̂ʑ̲ | ʐ̂ʐ̲ | ƾ̂ƾ̲ |
| patch | ʂ̂ʂ̲ | ŧ̂ŧ̲ | ʈ̂ʈ̲ | ɟ̂ɟ̲ | ȶ̂ȶ̲ | ʇ̂ʇ̲ | ɻ̂ɻ̲ | ʉ̂ʉ̲ | ᵾ̂ᵾ̲ | ʏ̂ʏ̲ | ɣ̂ɣ̲ | ỿ̂ỿ̲ | ƶ̂ƶ̲ | ʑ̂ʑ̲ | ʐ̂ʐ̲ | ƾ̂ƾ̲ |
| base | ʂ̂ʂ̲ | ŧ̂ŧ̲ | ʈ̂ʈ̲ | ɟ̂ɟ̲ | ȶ̂ȶ̲ | ʇ̂ʇ̲ | ɻ̂ɻ̲ | ʉ̂ʉ̲ | ᵾ̂ᵾ̲ | ʏ̂ʏ̲ | ɣ̂ɣ̲ | ỿ̂ỿ̲ | ƶ̂ƶ̲ | ʑ̂ʑ̲ | ʐ̂ʐ̲ | ƾ̂ƾ̲ |
| a | +0, 100% s | +0, 91% | +0, 100% s | | -1, 94% s | -3, 86% s | +0, 89% | +0, 87% | *+119, 75%* | -5, 85% | +0, 90% | -4, 97% s | +0, 90% | +0, 100% s | -14, 79% s | +30, 93% |
| a | 209, 645 | | | | | 209, 645 | | 209, 645 | 376, 645 | 270, 644 | | 270, 644 | | 209, 645 | 209, 645 | |
| a? | 199, 645 | 160, 769 | 163, 769 | 186, 885 | 196, 770 | 158, 771 | 146, 769 | 271, 645 | 332, 645 | 316, 771 | 260, 645 | 256, 645 | 220, 645 | 262, 645 | 218, 662 | 240, 885 |
| a< | **206, 645** | **139, 769** | **139, 769** | | **138, 769** | **136, 769** | **139, 769** | **255, 645** | 374, 645 | **257, 645** | **262, 645** | **258, 645** | **220, 645** | **220, 645** | **206, 645** | **266, 885** |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 100% s | +0, 100% | +0, 100% s | +0 \| | -1, 94% s | -3, 86% s | +0, 83% | +0, 96% | +121, 98% | -5, 92% | +0, 97% | -4, 97% s | +0, 89% | +0, 100% s | -14, 79% s | +30, 100% |
| b | 196, -110 | | | 163, -107 | 163, -109 | 196, -110 | | 196, -110 | 360, -110 | | | | | 196, -110 | 196, -110 | |
| b? | 199, -261 | 160, -110 | 163, -230 | 186, -110 | 196, -201 | 158, -306 | 146, -170 | 271, -110 | 332, -110 | 316, -319 | 260, -319 | 256, -319 | 220, -110 | 262, -270 | 218, -207 | 240, -319 |
| b< | **196, -261** | **163, -110** | **163, -230** | **163, -110** | **162, -201** | **160, -306** | **163, -110** | **239, -110** | **360, -110** | **237, -319** | **242, -319** | **238, -319** | **207, -110** | **207, -270** | **192, -207** | **266, -319** |
| b! | | | | | | | | | | | | | | | | |

| it | *ʂ̂ʂ̲* | *ŧ̂ŧ̲* | *ʈ̂ʈ̲* | *ɟ̂ɟ̲* | *ȶ̂ȶ̲* | *ʇ̂ʇ̲* | *ɻ̂ɻ̲* | *ʉ̂ʉ̲* | | *ʏ̂ʏ̲* | | | *ƶ̂ƶ̲* | *ʑ̂ʑ̲* | *ʐ̂ʐ̲* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | *ʂ̂ʂ̲* | *ŧ̂ŧ̲* | *ʈ̂ʈ̲* | *ɟ̂ɟ̲* | *ȶ̂ȶ̲* | *ʇ̂ʇ̲* | *ɻ̂ɻ̲* | *ʉ̂ʉ̲* | ⌧ | *ʏ̂ʏ̲* | ⌧ | ⌧ | *ƶ̂ƶ̲* | *ʑ̂ʑ̲* | *ʐ̂ʐ̲* | ⌧ |
| base | *ʂ̂ʂ̲* | *ŧ̂ŧ̲* | *ʈ̂ʈ̲* | *ɟ̂ɟ̲* | *ȶ̂ȶ̲* | *ʇ̂ʇ̲* | *ɻ̂ɻ̲* | *ʉ̂ʉ̲* | *ᵾ̂ᵾ̲* | *ʏ̂ʏ̲* | *ɣ̂ɣ̲* | *ỿ̂ỿ̲* | *ƶ̂ƶ̲* | *ʑ̂ʑ̲* | *ʐ̂ʐ̲* | *ƾ̂ƾ̲* |
| a | *+11, 19% s* | +0, 90% | +3, 78% s | | +3, 80% s | +2, 84% s | +40, 90% | *-3, 55%* | | -4, 82% | | | +0, 89% | *-5, 22% s* | *-1, 15% s* | |
| a | | | | | | | | | | 270, 644 | | | | | | |
| a? | 314, 645 | 318, 775 | 316, 776 | 341, 890 | 352, 776 | 314, 776 | 364, 775 | 398, 645 | | 471, 754 | | | 343, 645 | 343, 645 | 352, 645 | |
| a< | | **318, 775** | **326, 798** | | **326, 798** | **325, 798** | **358, 775** | 383, 645 | | **363, 645** | | | **360, 645** | | | |
| a! | | | | | | | | 402, 645 | | | | | | 354, 645 | 354, 645 | |
| b | *+11, 19% s* | +0, 100% | +3, 78% s | +1 \| | +3, 80% s | +2, 84% s | *+40, 74%* | *-1, 46%* | | -4, 95% | | | +0, 90% | *-5, 22% s* | *-1, 15% s* | |
| b | | | | 163, -107 | 163, -109 | | | | | | | | | | | |
| b? | 121, -265 | 130, -110 | 102, -230 | 129, -110 | 144, -204 | 83, -311 | 170, -136 | 238, -110 | | 243, -319 | | | 182, -110 | 147, -276 | 171, -207 | |
| b< | **121, -265** | **133, -110** | **110, -230** | **134, -110** | **116, -204** | **92, -311** | 173, -110 | 225, -110 | | **236, -319** | | | **187, -110** | **147, -276** | **171, -207** | |
| b! | | | | | | | | 241, -110 | | | | | | | | |

| sb | **ʂ̂ʂ̲** | **ŧ̂ŧ̲** | **ʈ̂ʈ̲** | **ɟ̂ɟ̲** | **ȶ̂ȶ̲** | **ʇ̂ʇ̲** | **ɻ̂ɻ̲** | **ʉ̂ʉ̲** | | **ʏ̂ʏ̲** | | | **ƶ̂ƶ̲** | **ʑ̂ʑ̲** | **ʐ̂ʐ̲** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | **ʂ̂ʂ̲** | **ŧ̂ŧ̲** | **ʈ̂ʈ̲** | **ɟ̂ɟ̲** | **ȶ̂ȶ̲** | **ʇ̂ʇ̲** | **ɻ̂ɻ̲** | **ʉ̂ʉ̲** | ⌧ | **ʏ̂ʏ̲** | ⌧ | ⌧ | **ƶ̂ƶ̲** | **ʑ̂ʑ̲** | **ʐ̂ʐ̲** | ⌧ |
| base | **ʂ̂ʂ̲** | **ŧ̂ŧ̲** | **ʈ̂ʈ̲** | **ɟ̂ɟ̲** | **ȶ̂ȶ̲** | **ʇ̂ʇ̲** | **ɻ̂ɻ̲** | **ʉ̂ʉ̲** | **ᵾ̂ᵾ̲** | **ʏ̂ʏ̲** | **ɣ̂ɣ̲** | **ỿ̂ỿ̲** | **ƶ̂ƶ̲** | **ʑ̂ʑ̲** | **ʐ̂ʐ̲** | **ƾ̂ƾ̲** |
| a | +0, 100% s | +0, 91% | +0, 100% s | | -6, 88% s | -6, 94% s | +0, 91% | +0, 86% | | *+1, 52%* | | | +0, 90% | +0, 100% s | *-2, 65% s* | |
| a | | | | | | | | | | | | | | | | |
| a? | 201, 645 | 168, 782 | 168, 782 | 186, 885 | 206, 782 | 182, 782 | 146, 782 | 274, 645 | | 328, 768 | | | 222, 645 | 254, 645 | 230, 645 | |
| a< | **201, 645** | **168, 782** | **168, 782** | | **162, 782** | **162, 782** | **168, 782** | **264, 645** | | 265, 645 | | | **222, 645** | **222, 645** | | |
| a! | | | | | | | | | | | | | | | | |
| b | +0, 100% s | +0, 100% | +0, 100% s | -4 \| | -6, 88% s | -6, 94% s | -1, 89% | +0, 96% | | *+3, 36%* | | | +0, 89% | +0, 100% s | *-2, 65% s* | |
| b | | | | | | | | | | | | | | | | |
| b? | 201, -264 | 168, -110 | 168, -229 | 186, -110 | 206, -214 | 182, -319 | 146, -170 | 274, -110 | | 328, -319 | | | 222, -110 | 254, -276 | 230, -210 | |
| b< | **201, -264** | **163, -110** | **163, -229** | **159, -110** | **157, -214** | **157, -319** | **162, -110** | **248, -110** | | 267, -319 | | | **222, -110** | **222, -276** | **230, -210** | |
| b! | | | | | | | | | | | | | | | | |

| si | ***ʂ̂ʂ̲*** | ***ŧ̂ŧ̲*** | ***ʈ̂ʈ̲*** | ***ɟ̂ɟ̲*** | ***ȶ̂ȶ̲*** | ***ʇ̂ʇ̲*** | ***ɻ̂ɻ̲*** | ***ʉ̂ʉ̲*** | | ***ʏ̂ʏ̲*** | | | ***ƶ̂ƶ̲*** | ***ʑ̂ʑ̲*** | ***ʐ̂ʐ̲*** | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| patch | ***ʂ̂ʂ̲*** | ***ŧ̂ŧ̲*** | ***ʈ̂ʈ̲*** | ***ɟ̂ɟ̲*** | ***ȶ̂ȶ̲*** | ***ʇ̂ʇ̲*** | ***ɻ̂ɻ̲*** | ***ʉ̂ʉ̲*** | ⌧ | ***ʏ̂ʏ̲*** | ⌧ | ⌧ | ***ƶ̂ƶ̲*** | ***ʑ̂ʑ̲*** | ***ʐ̂ʐ̲*** | ⌧ |
| base | ***ʂ̂ʂ̲*** | ***ŧ̂ŧ̲*** | ***ʈ̂ʈ̲*** | ***ɟ̂ɟ̲*** | ***ȶ̂ȶ̲*** | ***ʇ̂ʇ̲*** | ***ɻ̂ɻ̲*** | ***ʉ̂ʉ̲*** | ***ᵾ̂ᵾ̲*** | ***ʏ̂ʏ̲*** | ***ɣ̂ɣ̲*** | ***ỿ̂ỿ̲*** | ***ƶ̂ƶ̲*** | ***ʑ̂ʑ̲*** | ***ʐ̂ʐ̲*** | ***ƾ̂ƾ̲*** |
| a | *+7, 39% s* | +0, 91% | *-22, 86% s* | | -9, 78% s | -9, 84% s | +12, 91% | *+15, 64%* | | *-91, 20%* | | | *-233, 10%* | *-4, 11% s* | *+0, 4% s* | |
| a | | | | | | | | | | 270, 644 | | | | | | |
| a? | 178, 645 | 175, 765 | 161, 787 | 183, 890 | 209, 787 | 172, 787 | 187, 765 | 255, 645 | | 317, 768 | | | 187, 645 | 202, 645 | 208, 645 | |
| a< | | **189, 765** | **171, 787** | | **184, 787** | **184, 787** | **201, 765** | 270, 645 | | 171, 645 | | | -13, 645 | | | |
| a! | | | | | | | | 275, 645 | | 152, 645 | | | | 244, 645 | 215, 645 | |
| b | *+7, 39% s* | +0, 100% | *-22, 86% s* | -23 \| | -9, 78% s | -9, 84% s | +10, 90% | *+16, 52%* | | *-37, 6%* | | | *+227, 11%* | *-4, 11% s* | *+0, 4% s* | |
| b | | | | 163, -107 | 163, -109 | | | | | | | | | | | |
| b? | 155, -265 | 159, -110 | 117, -228 | 142, -110 | 167, -214 | 109, -319 | 159, -170 | 264, -110 | | 262, -297 | | | 195, -110 | 176, -279 | 196, -210 | |
| b< | **155, -265** | **133, -110** | **87, -228** | **110, -110** | **103, -214** | **81, -319** | **137, -140** | 286, -110 | | 205, -319 | | | 434, -110 | **176, -279** | **196, -210** | |
| b! | | | | | | | | 284, -110 | | | | | | | | |

Table 27: Derived letters and inherited anchors 7/7

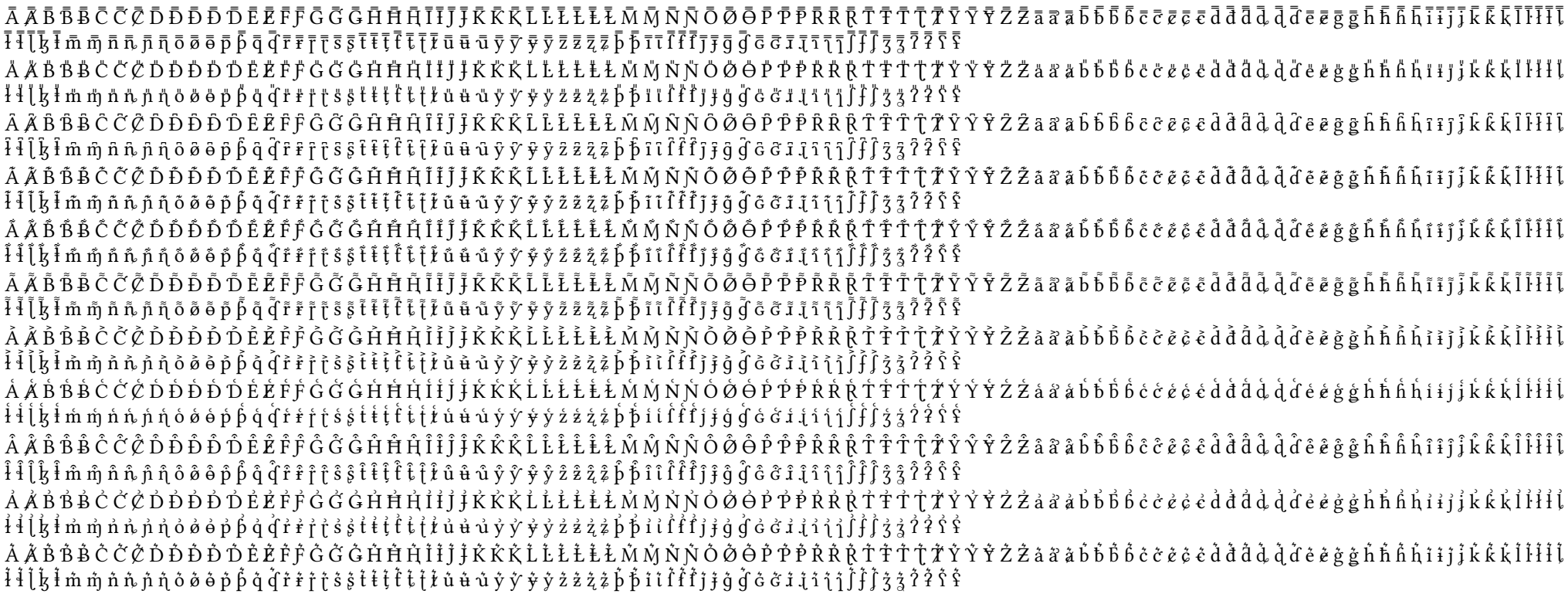
| hex | 0269 | 1E9C | 1E9D | 025F | 025A | 025D | 029B | 027B | 0285 | 0284 | 0286 | 0293 | 02A1 | 02A2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| base | 0131 | 017F | 017F | 0237 | 0259 | 025C | 0262 | 0279 | 027F | 0283 | 0283 | 0292 | 0294 | 0295 |
| reg | ɩ̰ | ſ̰ſ̰ | ſ̰ſ̰ | ȷ̰ɟ̰ | ə̰ɚ̰ | ɜ̰ɝ̰ | ɢ̰ʛ̰ | ɻ̰ɺ̰ | ʅ̰ | ʃ̰ʃ̰ | ʃ̰ʃ̰ | ʒ̰ʒ̰ | ʔ̰ʔ̰ | ʕ̰ʕ̰ |
| patch | ɩ̰ | ſ̰ſ̰ | ſ̰ſ̰ | ȷ̰ɟ̰ | ə̰ɚ̰ | ɜ̰ɝ̰ | ɢ̰ʛ̰ | ɻ̰ɺ̰ | ʅ̰ | ʃ̰ʃ̰ | ʃ̰ʃ̰ | ʒ̰ʒ̰ | ʔ̰ʔ̰ | ʕ̰ʕ̰ |
| base | ɩ̰ | ſ̰ſ̰ | ſ̰ſ̰ | ȷ̰ | ə̰ | ɜ̰ | ɢ̰ | ɻ̰ | ʅ̰ | ʃ̰ʃ̰ | ʃ̰ʃ̰ | ʒ̰ʒ̰ | ʔ̰ | ʕ̰ |
| a | -18 \| | +0, 92% | +0, 94% | +16 \| | | | +0, 100% | +20, 75% s | +0, 100% s | +0, 92% | +0, 100% s | *+4, 38% s* | +0, 92% | +0, 92% |
| a | 123, 645 | | | 169, 645 | 229, 645 | 209, 645 | 209, 645 | | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 | 209, 645 |
| a? | 170, 645 | 182, 885 | 190, 885 | 124, 645 | 306, 688 | 234, 795 | 294, 764 | 226, 645 | 170, 764 | 140, 885 | 150, 885 | 220, 645 | 211, 843 | 221, 843 |
| a< | **124, 645** | **190, 885** | **190, 885** | **168, 645** | | | **265, 663** | 247, 645 | **170, 645** | **140, 885** | **140, 885** | | **211, 850** | **221, 850** |
| a! | | | | | | | | 248, 645 | | | | 222, 645 | | |
| b | -18 \| | +0, 100% | +0, 100% | +16 \| | -13, 81% | +0, 100% | +0, 100% | +20, 75% s | | +0, 98% | +0, 100% s | *+4, 38% s* | +0, 92% | +0, 92% |
| b | 153, -114 | | | 196, -110 | 212, -114 | 212, -114 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 | 196, -110 |
| b? | 170, -110 | 182, -110 | 190, -110 | 124, -291 | 306, -110 | 234, -110 | 294, -110 | 226, -259 | 170, -299 | 140, -299 | 150, -310 | 220, -319 | 211, -110 | 221, -110 |
| b< | **124, -119** | **190, -110** | **190, -110** | **125, -319** | 199, -110 | **184, -110** | **248, -110** | **198, -259** | | **129, -319** | **129, -310** | **220, -319** | **202, -110** | **230, -110** |
| b! | | | | | 184, -110 | | | | | | | | | |
| it | *ɩ̰* | *ſ̰ſ̰* | *ſ̰ſ̰* | *ȷ̰ɟ̰* | *ə̰ɚ̰* | *ɜ̰ɝ̰* | *ɢ̰ʛ̰* | *ɻ̰ɺ̰* | *ʅ̰* | *ʃ̰ʃ̰* | *ʃ̰ʃ̰* | *ʒ̰ʒ̰* | *ʔ̰ʔ̰* | *ʕ̰ʕ̰* |
| patch | *ɩ̰* | *ſ̰ſ̰* | *ſ̰ſ̰* | *ȷ̰ɟ̰* | *ə̰ɚ̰* | *ɜ̰ɝ̰* | *ɢ̰ʛ̰* | *ɻ̰ɺ̰* | *ʅ̰* | *ʃ̰ʃ̰* | *ʃ̰ʃ̰* | *ʒ̰ʒ̰* | *ʔ̰ʔ̰* | *ʕ̰ʕ̰* |
| base | *ɩ̰* | *ſ̰ſ̰* | *ſ̰ſ̰* | *ȷ̰* | *ə̰* | *ɜ̰* | *ɢ̰* | *ɻ̰* | *ʅ̰* | *ʃ̰ʃ̰* | *ʃ̰ʃ̰* | *ʒ̰ʒ̰* | *ʔ̰* | *ʕ̰* |
| a | -19 \| | +0, 90% | +0, 94% | +43 \| | | | -9, 89% | *+111, 58% s* | +0, 100% s | +0, 92% | +0, 100% s | *+28, 26% s* | +0, 94% | +0, 94% |
| a | 123, 645 | | | | | | | | | | | | | |
| a? | 266, 645 | 339, 890 | 339, 890 | 305, 645 | 418, 687 | 413, 775 | 410, 796 | 352, 645 | 343, 900 | 343, 900 | 343, 900 | 355, 645 | 376, 864 | 378, 875 |
| a< | **256, 645** | **339, 890** | **339, 890** | **326, 645** | | | **379, 645** | | **307, 645** | **343, 900** | **343, 900** | | **376, 867** | **379, 878** |
| a! | | | | | | | | 389, 645 | | | | | | |
| b | -19 \| | +0, 100% | +0, 100% | +43 \| | *-10, 68%* | +0, 100% | -16, 89% | *+111, 58% s* | | +0, 90% | +0, 100% s | *+28, 26% s* | +0, 85% | +0, 84% |
| b | 153, -114 | | | | 212, -114 | 212, -114 | | | | | | | | |
| b? | 106, -110 | 82, -319 | 82, -319 | 100, -319 | 248, -110 | 225, -110 | 217, -110 | 160, -259 | 88, -299 | 88, -299 | 92, -283 | 133, -397 | 169, -110 | 169, -110 |
| b< | **96, -110** | **82, -319** | **82, -319** | **105, -320** | 159, -110 | **143, -110** | **232, -110** | **160, -259** | | **61, -319** | **69, -283** | 133, -397 | **202, -110** | **230, -110** |
| b! | | | | | 161, -110 | | | | | | | 133, -399 | | |
| sb | **ɩ̰** | | | **ȷ̰ɟ̰** | **ə̰ɚ̰** | **ɜ̰ɝ̰** | **ɢ̰ʛ̰** | **ɻ̰ɺ̰** | **ʅ̰** | **ʃ̰ʃ̰** | **ʃ̰ʃ̰** | **ʒ̰ʒ̰** | **ʔ̰ʔ̰** | **ʕ̰ʕ̰** |
| patch | **ɩ̰** | ⌧ | ⌧ | **ȷ̰ɟ̰** | **ə̰ɚ̰** | **ɜ̰ɝ̰** | **ɢ̰ʛ̰** | **ɻ̰ɺ̰** | **ʅ̰** | **ʃ̰ʃ̰** | **ʃ̰ʃ̰** | **ʒ̰ʒ̰** | **ʔ̰ʔ̰** | **ʕ̰ʕ̰** |
| base | **ɩ̰** | **ſ̰ſ̰** | **ſ̰ſ̰** | **ȷ̰** | **ə̰** | **ɜ̰** | **ɢ̰** | **ɻ̰** | **ʅ̰** | **ʃ̰ʃ̰** | **ʃ̰ʃ̰** | **ʒ̰ʒ̰** | **ʔ̰** | **ʕ̰** |
| a | -16 \| | | | +18 \| | | | +0, 100% | +55, 97% s | +0, 100% s | +0, 92% | -1, 100% s | *+9, 49% s* | +0, 94% | +0, 94% |
| a | | | | | | | | | | | | | | |
| a? | 164, 645 | | | 144, 645 | 319, 693 | 276, 757 | 290, 764 | 238, 645 | 152, 885 | 152, 885 | 162, 885 | 236, 645 | 206, 845 | 206, 845 |
| a< | **140, 645** | | | **181, 645** | | | **266, 645** | **260, 645** | **182, 645** | **152, 885** | **151, 885** | | **206, 850** | **206, 850** |
| a! | | | | | | | | | | | | 231, 645 | | |
| b | -16 \| | | | +18 \| | +6, 81% | +1, 100% | +0, 100% | +55, 97% s | | +0, 98% | -1, 100% s | *+9, 49% s* | +0, 96% | +0, 96% |
| b | | | | | | | | | | | | | | |
| b? | 164, -110 | | | 144, -319 | 319, -110 | 276, -110 | 290, -110 | 238, -257 | 152, -296 | 152, -296 | 162, -319 | 236, -413 | 206, -110 | 206, -110 |
| b< | **140, -120** | | | **96, -319** | 218, -110 | **205, -110** | **266, -110** | **196, -257** | | **141, -319** | **141, -319** | 236, -413 | **206, -110** | **206, -110** |
| b! | | | | | 199, -110 | | | | | | | 236, -415 | | |
| si | ***ɩ̰*** | | | ***ȷ̰ɟ̰*** | ***ə̰ɚ̰*** | ***ɜ̰ɝ̰*** | ***ɢ̰ʛ̰*** | ***ɻ̰ɺ̰*** | ***ʅ̰*** | ***ʃ̰ʃ̰*** | ***ʃ̰ʃ̰*** | ***ʒ̰ʒ̰*** | ***ʔ̰ʔ̰*** | ***ʕ̰ʕ̰*** |
| patch | ***ɩ̰*** | ⌧ | ⌧ | ***ȷ̰ɟ̰*** | ***ə̰ɚ̰*** | ***ɜ̰ɝ̰*** | ***ɢ̰ʛ̰*** | ***ɻ̰ɺ̰*** | ***ʅ̰*** | ***ʃ̰ʃ̰*** | ***ʃ̰ʃ̰*** | ***ʒ̰ʒ̰*** | ***ʔ̰ʔ̰*** | ***ʕ̰ʕ̰*** |
| base | ***ɩ̰*** | ***ſ̰ſ̰*** | ***ſ̰ſ̰*** | ***ȷ̰*** | ***ə̰*** | ***ɜ̰*** | ***ɢ̰*** | ***ɻ̰*** | ***ʅ̰*** | ***ʃ̰ʃ̰*** | ***ʃ̰ʃ̰*** | ***ʒ̰ʒ̰*** | ***ʔ̰*** | ***ʕ̰*** |
| a | -15 \| | | | -4 \| | | | +12, 100% | +35, 86% s | +0, 100% s | +0, 92% | +1, 100% s | *+3, 56% s* | +12, 94% | +12, 94% |
| a | 123, 645 | | | | | | | | | | | | | |
| a? | 138, 645 | | | 141, 645 | 299, 693 | 261, 762 | 278, 796 | 194, 645 | 195, 890 | 195, 890 | 189, 890 | 211, 645 | 227, 850 | 227, 850 |
| a< | **131, 654** | | | **167, 645** | | | **284, 681** | **246, 645** | **167, 645** | **195, 890** | **195, 890** | | **239, 850** | **239, 850** |
| a! | | | | | | | | | | | | 213, 645 | | |
| b | -15 \| | | | -4 \| | *-6, 60%* | +1, 100% | +12, 100% | +35, 86% s | | +0, 98% | +1, 100% s | *+3, 56% s* | +13, 96% | +12, 96% |
| b | 153, -114 | | | | 212, -114 | 212, -114 | | | | | | | | |
| b? | 147, -110 | | | 107, -319 | 298, -110 | 246, -110 | 256, -110 | 172, -257 | 116, -296 | 116, -296 | 105, -319 | 157, -413 | 194, -110 | 194, -110 |
| b< | **138, -110** | | | **85, -314** | 180, -110 | **194, -110** | **260, -110** | **185, -257** | | **88, -319** | **88, -319** | 157, -413 | **215, -110** | **242, -110** |
| b! | | | | | 188, -110 | | | | | | | 157, -415 | | |

Table 28: Stem-midline check: offset of the mark axis from the stem midline, patched fonts

| | l | Í | ļ | ĺ | ŀ | ł | ɬ | ɬ | ɬ | ɭ | ɮ | ɬ | i | ɪ | i | i.dl | ɩ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| hex | 006C | 013A | 013C | 013E | 0140 | 0142 | 019A | 026B | 026C | 026D | 026E | 2C61 | 0069 | 0131 | 0268 | dotless | 0269 |
| reg a | +0 | | +0 | | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 |
| b | +0 | +0 | | +0 | +0 | +0 | +0 | +0 | +0 | +0 | **+140** | +0 | +0 | +0 | +0 | +0 | +0 |
| it a | +2 | | +2 | | +2 | +2 | +2 | +2 | +2 | +2 | +0 | +2 | +2 | +0 | +2 | +0 | -1 |
| b | +1 | +1 | | +1 | +2 | +2 | +1 | +2 | +1 | +1 | **+134** | +1 | +0 | +0 | +0 | +0 | +0 |
| sb a | +0 | | +0 | | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 | +0 |
| b | +0 | +0 | | +0 | +0 | +0 | +0 | +0 | +0 | +0 | **+133** | +0 | +0 | +0 | +0 | +0 | +0 |
| si a | +3 | | +3 | | +5 | +4 | +3 | +3 | +3 | +3 | +0 | +3 | +2 | +0 | +0 | +0 | +0 |
| b | +2 | +2 | | +2 | +4 | +4 | +2 | +2 | +2 | +2 | **+126** | +3 | +3 | +1 | +3 | +1 | +1 |

**Regular patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above**





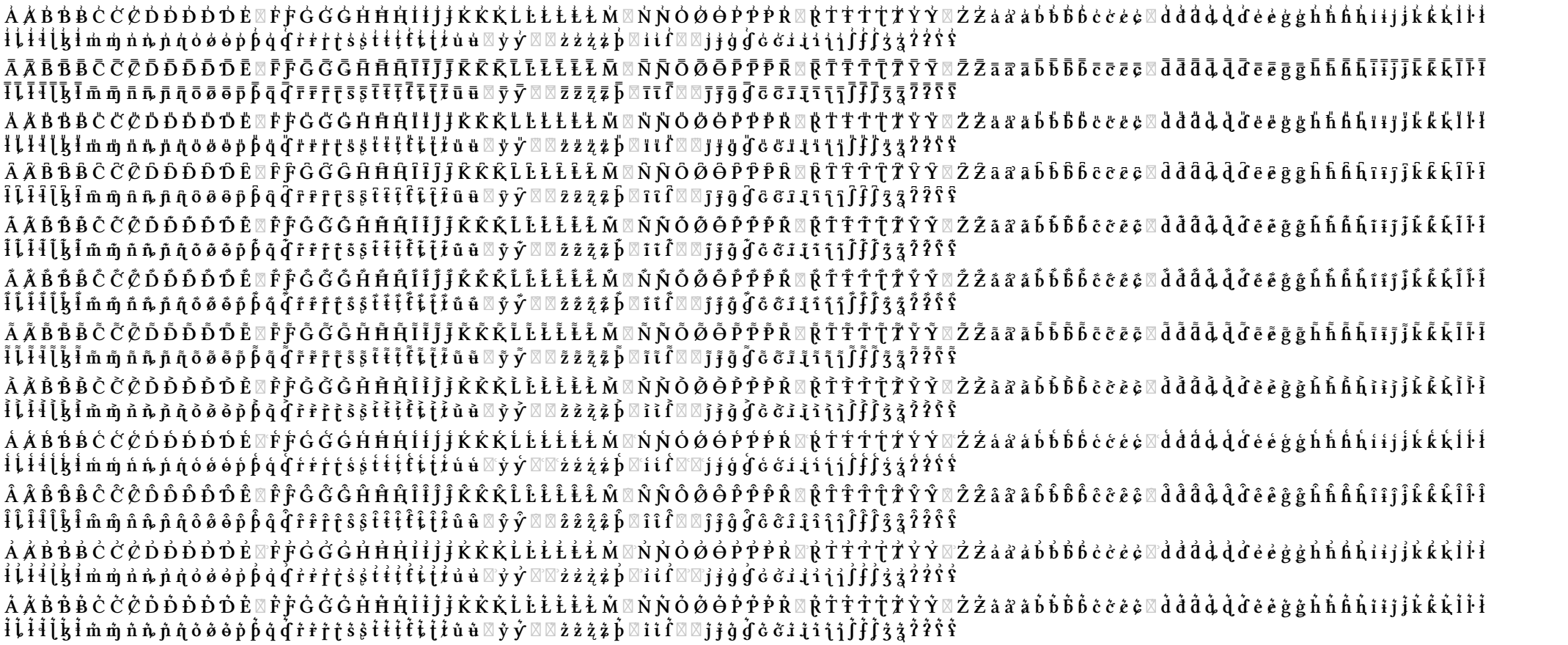
## Italic patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above



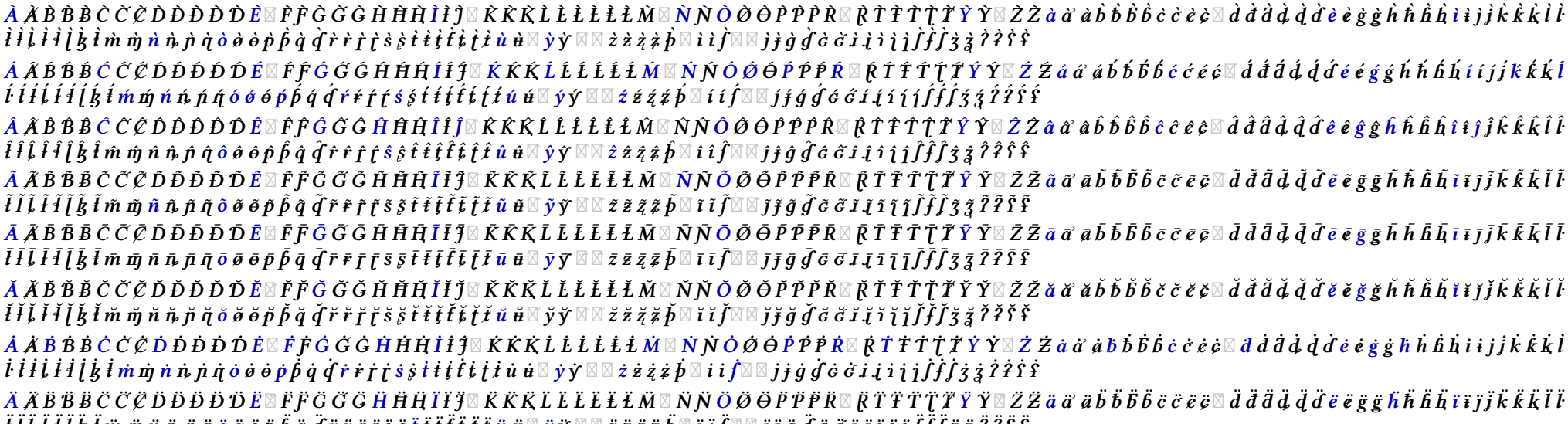
Á Ắ Ḃ Ḃ Ć Ḉ Ḋ Ḋ Ḍ Ḍ Ḍ É Ḟ Ḟ Ǵ Ḡ Ḡ Ḣ Ḧ Ḫ Ḯ Ḯ J́ Ḱ Ḱ Ḵ Ĺ Ḽ Ḷ Ḹ Ḿ Ṅ Ṅ Ó Ṓ Ṑ Ṕ Ṗ Ṕ Ŕ Ṛ Ṫ Ṫ Ṭ Ṯ Ṯ Ý Ẏ Ź Ẓ ả ắ ả ḃ ḃ ḃ ć ḉ é ḉ ḋ ḋ ḋ ḍ ḍ ḍ ė ḗ ḡ ḡ ḥ ḧ ḫ ḥ i ḭ j j́ ḱ ḱ ḵ l̀ l̀ l̀ ḷ ḷ́ ḷ̀ ḳ́ ḿ ṁ ṅ ṅ ṅ ṅ ó ṓ ṑ ṕ ṕ q́ q̀ ṛ ṛ́ ṛ̀ ṡ ṣ́ ṫ ṫ ṭ ṭ́ ṭ̀ ṭ ú ṵ ý ẏ ź ẓ ẓ́ ẓ̀ ṕ i i ̀ i ̂ i ̌ j j ̀ ǵ ḡ ć ć ̀ ḭ ḭ ́ i ̂ i ̌ i ̀ i ̏ i ̑ j ́ ǩ ̀ ḷ ̂ ḷ ̌ ḷ ̀ ḷ ̏ ḷ ̑ ṁ ̀ ṁ ̂ ṁ ̌ ṅ ́ ṅ ̀ ṅ ̂ ṅ ̌ ṅ ̏ ṅ ̑ ó ̀ ó ̂ ó ̌ ó ̏ ó ̑ ṕ ́ ṕ ̀ ṕ ̂ ṕ ̌ ṕ ̏ ṕ ̑ q́ ̀ q̀ ̀ ṛ ̀ ṛ ̂ ṛ ̌ ṛ ̏ ṛ ̑ ṛ ́ ṛ ̀ ṡ ̀ ṣ ́ ṡ ̀ ṫ ̀ ṫ ̂ ṫ ̌ ṫ ̏ ṫ ̑ ṭ ́ ṭ ̀ ṭ ̂ ṭ ̌ ṭ ̏ ṭ ̑ ṭ ́ ṭ ̀ ú ̀ ṵ ́ ṵ ̀ ý ̀ ẏ ́ ẏ ̀ ź ̀ ẓ ̀ ẓ ́ ẓ ̀ ṕ ̀ i ̀ i ̂ i ̌ i ̏ i ̑ j ́ ǵ ̀ ḡ ́ ć ̀ ć ́ ḭ ̀ ḭ ́ i ̂ i ̌

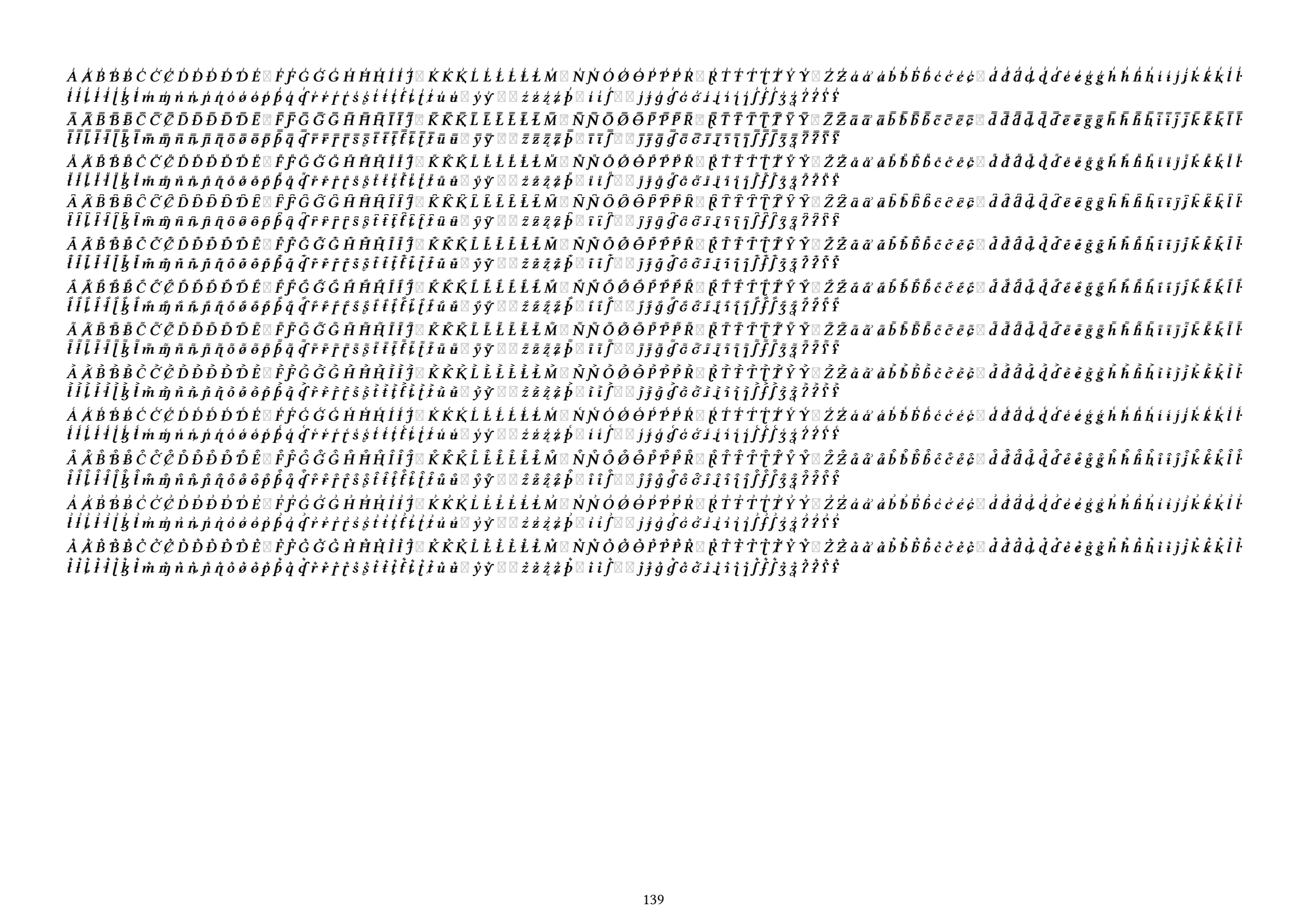
## Semibold patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above





**Semibold italic patched, Derived letters that may inherit an above anchor in stage 8, after their base letters, Above**



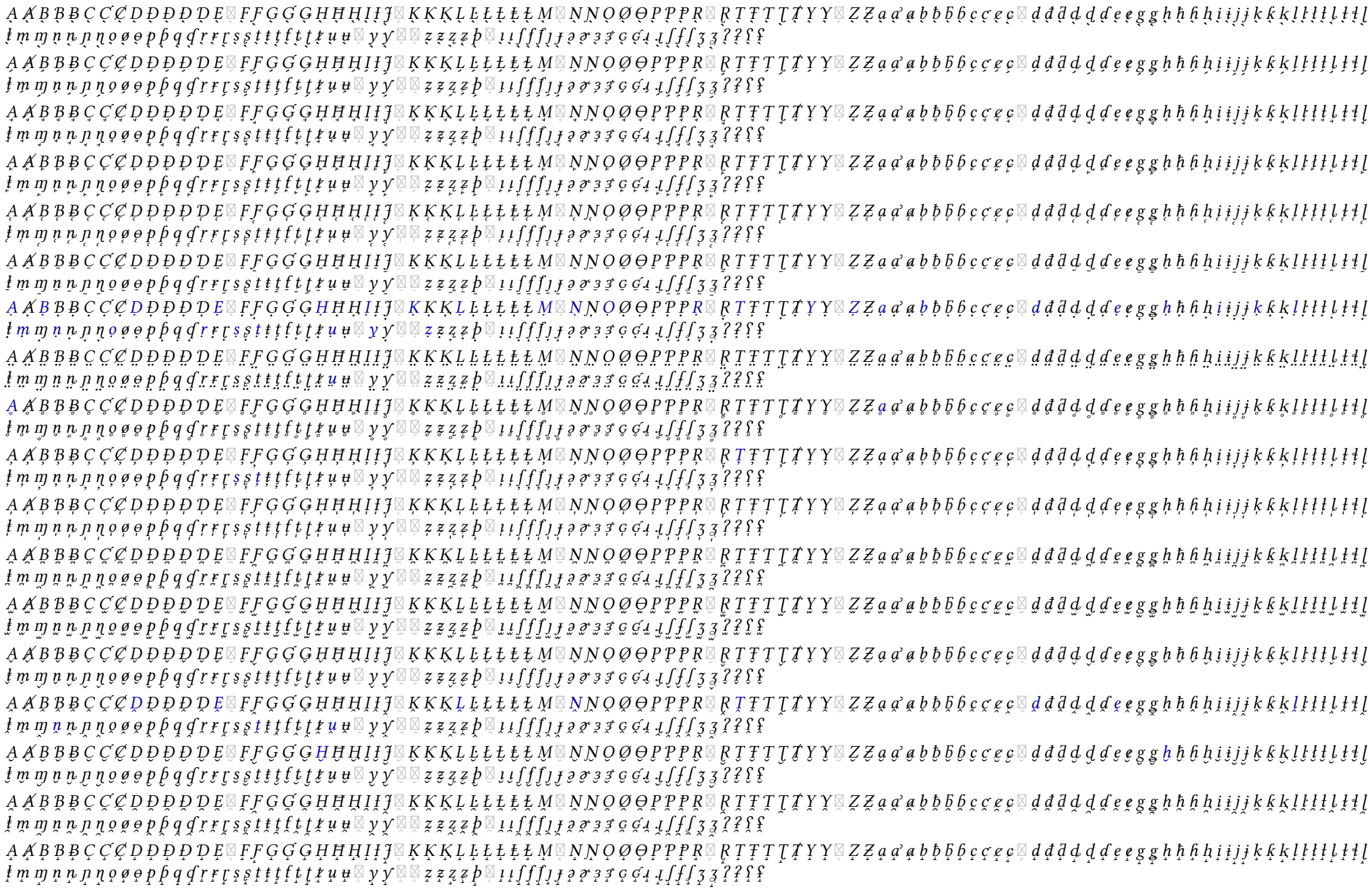


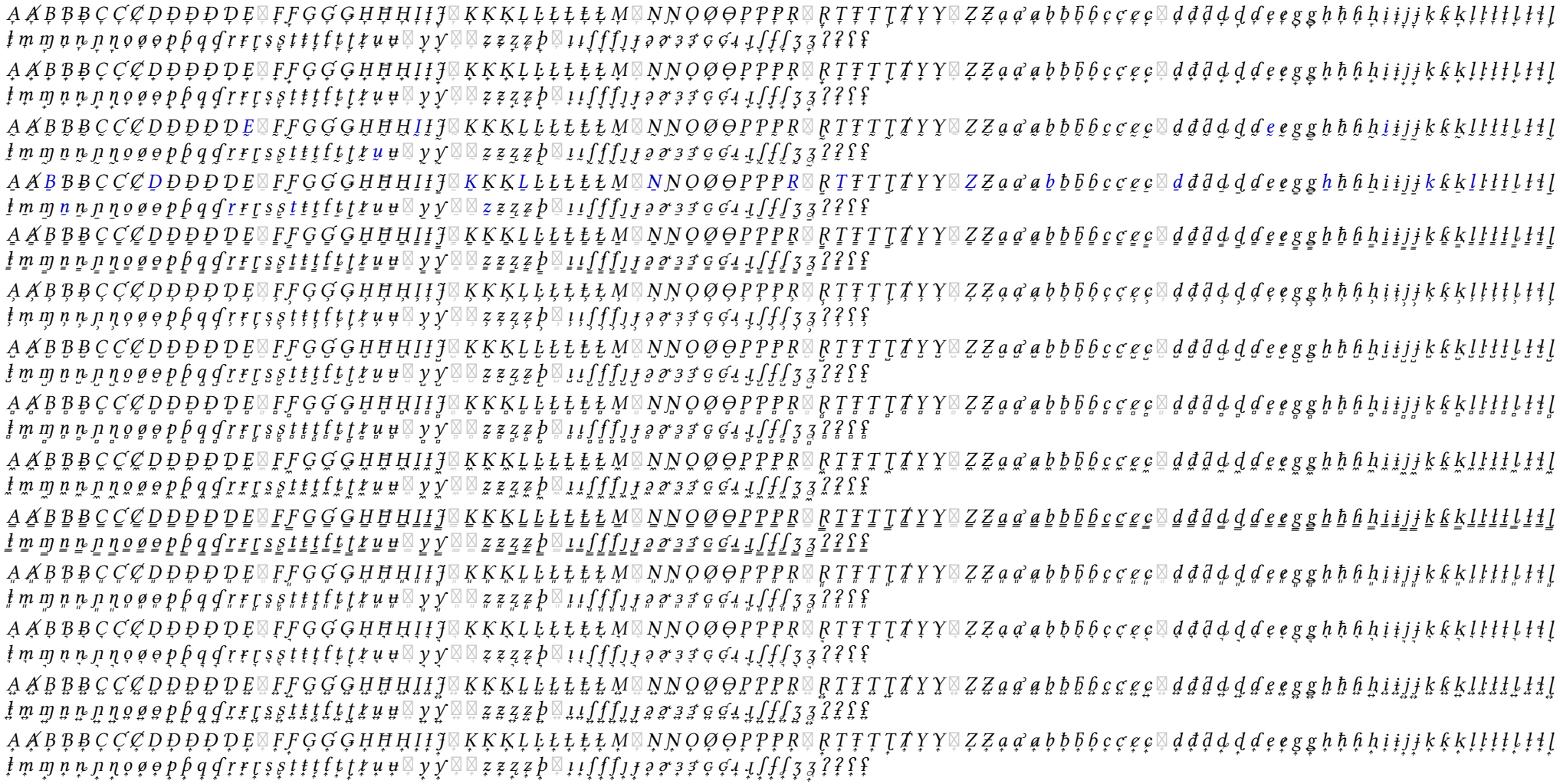
**Regular patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below**



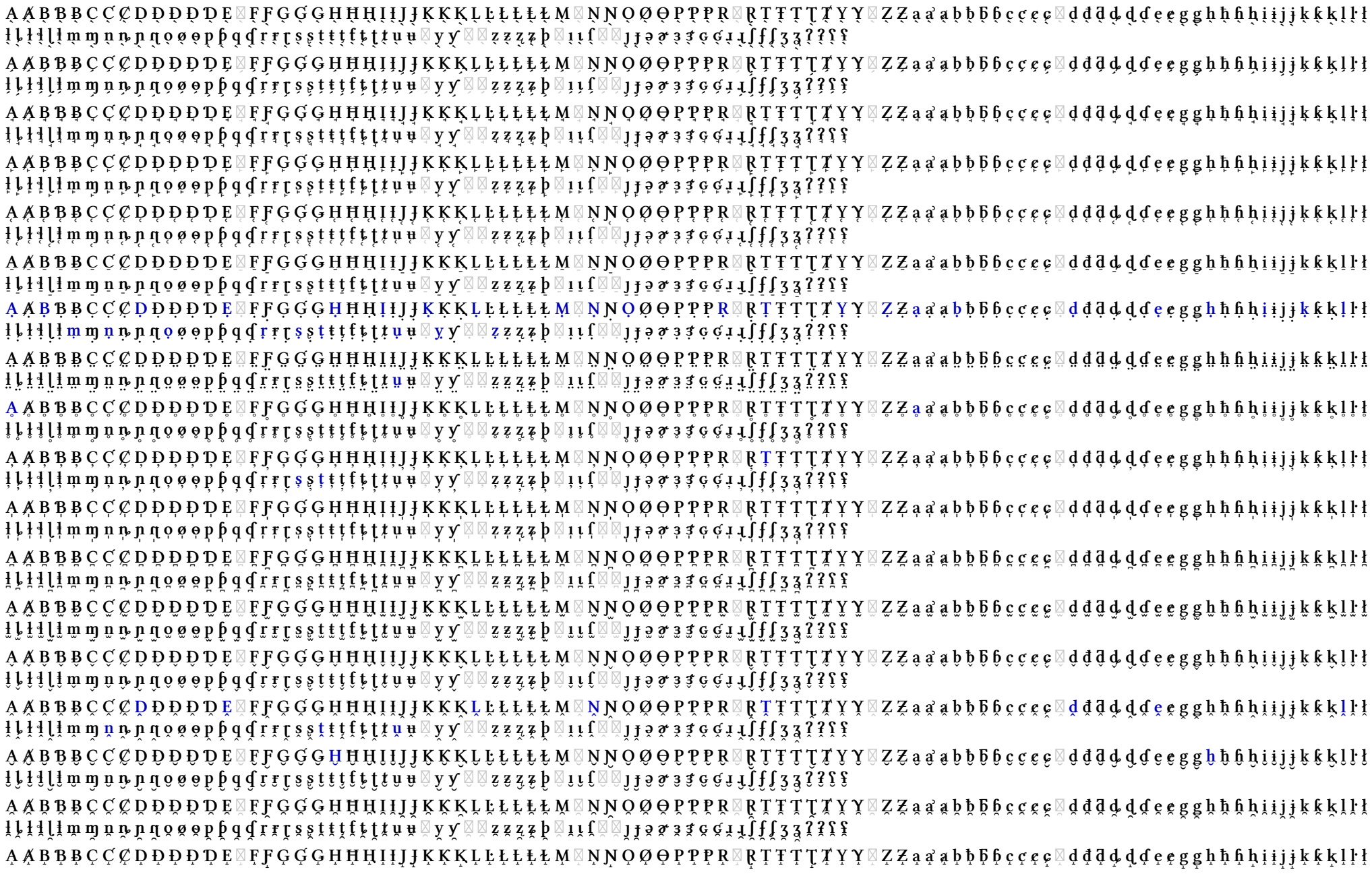


## Italic patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below

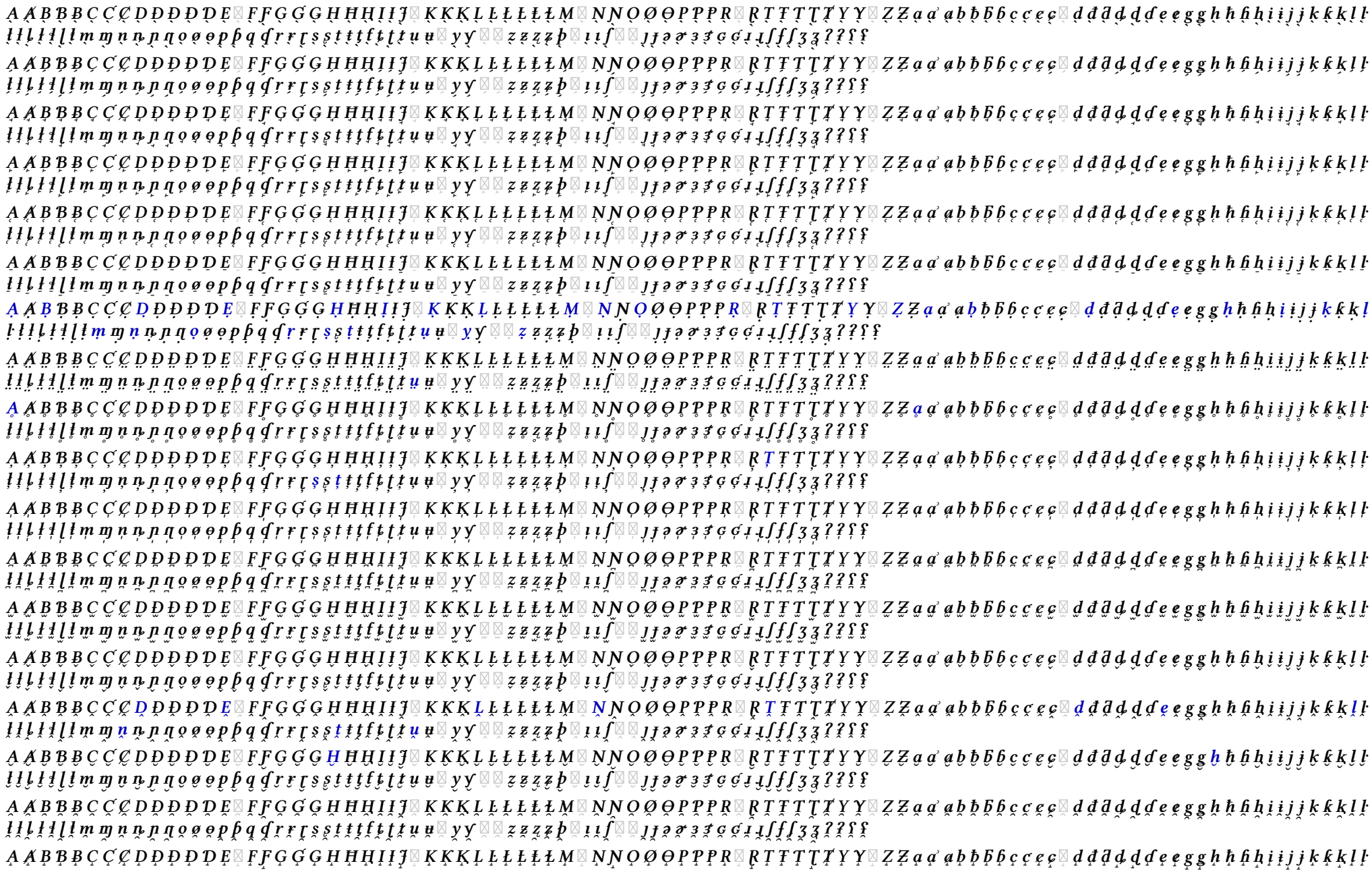


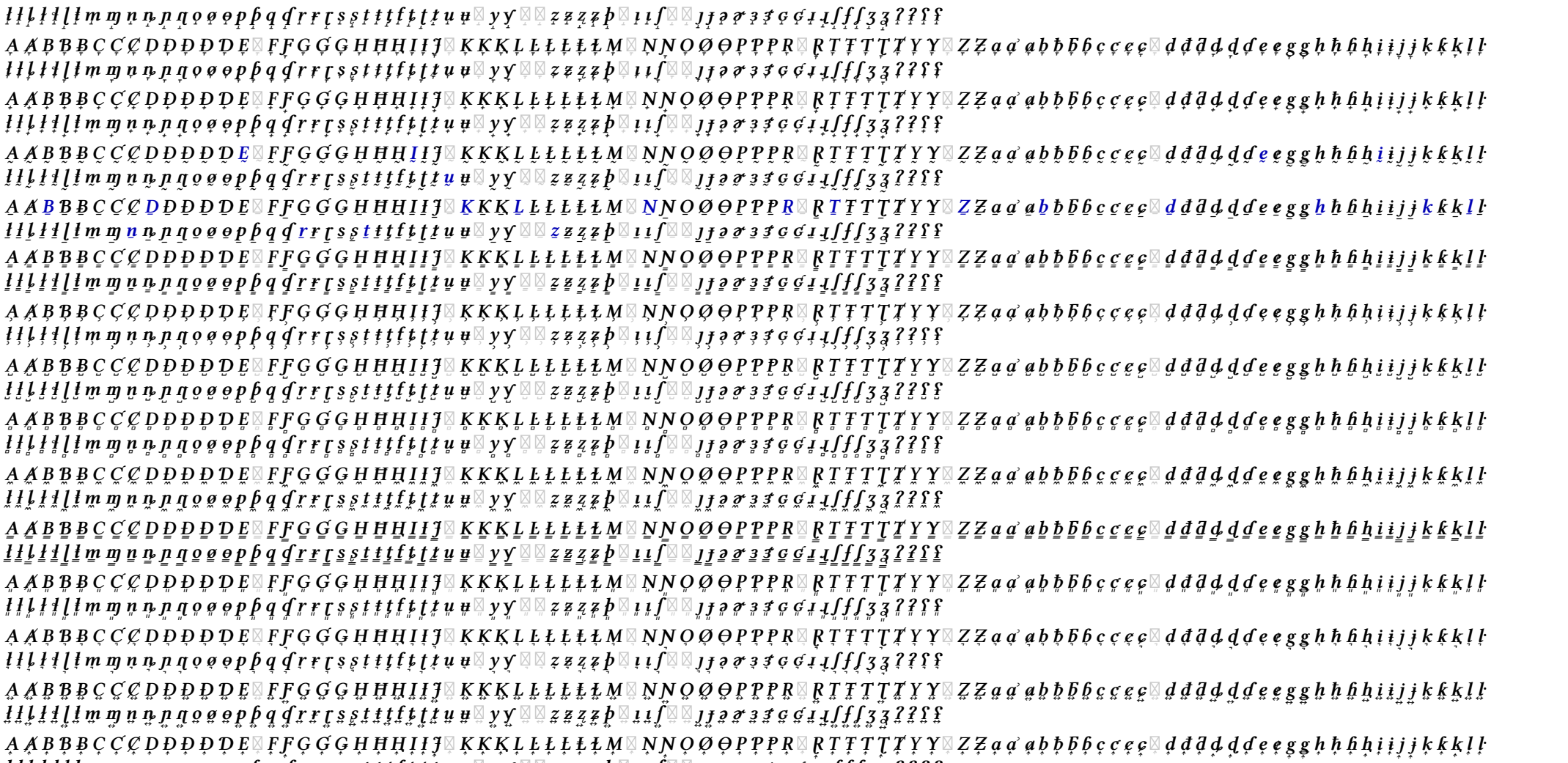


**Semibold patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below**



**Semibold italic patched, Derived letters that may inherit a below anchor in stage 8, after their base letters, Below**





## Precomposed characters

Stage 9 (`src/libertinus_analysis/font_patching_precomposed_anchors.py`, letter lists in `data/precomposed_letters.py`) gives a precomposed Latin letter the above- or below-anchor of its base letter where the other side is free: a letter with an above-mark (é, ñ) inherits a below-anchor, and a letter with a below-mark or attached below-mark (ẹ, ç, ą) inherits an above-anchor. (Letters with a bar or stroke, such as Ø, Ł, Đ, have no Unicode decomposition; they inherit in stage 8, previous section.) The stage also tests that the base letter's form is really there, since some precomposed letters were redrawn unlike their bases: it fits the base's ink on the inherited side to the precomposed glyph (a shift, and a scale of up to 15%), requires a fit of at least 80%, and moves the anchor by the fitted shift.

Precomposed letters inherit zealously. A designer anchor on a letter that can inherit is deleted in stage 1 (`"delete"` in `data/fontanchors_audit/overrides.py`) whenever it differs from the inherited anchor at all where the fit is 100%, or by more than 10 where the fit is lower; curated anchors that differed were commented out the same way. Most were copies of a base anchor that the patch has since changed, or copies that miss how the base sits in the precomposed glyph. Curated anchors (stage 7, `data/fontanchors_human/{style}.py`) are set on precomposed letters only where inheritance fails to find a good match. Every letter with a fit of 100% now has exactly the anchor it inherits (`run_precomposed_existing_report.py` checks this). The letters that do not simply inherit:

- ģ (U+0123) is formally g with a below-mark, a cedilla (U+0327), but is drawn with a comma above. It does inherit a below-anchor from g (fit 100% in every style), and gets no above-anchor: 30 IPA above-marks would collide with the comma.
- The horn letters Ơ ơ ư Ọ ọ ụ (U+01A0, U+01A1, U+01B0, U+1EE2, U+1EE3, U+1EF1) get no above-anchor in any style: 1–12 IPA above-marks per style would collide with the horn. Ơ ơ ư inherit a below-anchor (Ọ ọ ụ have a dot below), except semibold italic ư, whose below-anchor is curated (see the last item).
- **ą̄ ā** semibold ą (U+0105), above: the ogonek lowers the fit to 69%; curated at a's anchor plus the shift of ą's upper part (+12).
- *ĵ̲ ĵ̲* italic ĵ (U+0134), below: J moved 47 left with a shallower tail ($-166$, J $-200$), fit 65%; curated at the italic-model candidate from ĵ's own metrics, (130, $-251$).
- ***ĵ̲ ĵ̲*** semibold italic ĵ, below: a flat top serif and a deeper tail ($-196$, J $-164$), fit 64%; curated at the candidate from ĵ's own metrics, (125, $-281$).
- ***ư̲ ứ̲ ừ̲ ử̲ ữ̲ u̲*** semibold italic ư ứ ừ ử ữ (U+01B0, U+1EE9, U+1EEB, U+1EED, U+1EEF), below: the stems lie on u's but the bottom of the bowl is about 15 wider on each side, fit 75%; curated at u's anchor, (270, $-110$). The toned forms reference ư.

The grids show the IPA above-marks on the precomposed letters that inherit above and the below-marks on those that inherit below, in the patched fonts only. For completeness they include the letters whose anchor is curated instead of inherited (ą, ĵ, ư ứ ừ ử ữ). Samples and grid cells put `\cgj` between the letter and the mark, so that the mark is positioned by the letter's anchor and the shaper does not decompose and recompose the letter. The exception is a letter and mark that compose to one of the 24 precomposed letters with both an above- and a below-mark (Ç + acute = Ḉ, ạ + circumflex = ậ, Ȯ + dot below = Ọ̇): text gets the precomposed glyph, so the cell shows it, in blue, without `\cgj`. They are Ǫ ǫ Ḉ ḉ Ḝ ḝ Ḷ Ḯ Ḵ ḵ Ṣ ṣ Ậ ậ Ặ ặ Ệ ệ Ộ ộ, and, with the horn, Ợ ợ Ự ự.

## Regular patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ḝ ḝ Ģ Į Ķ ķ Ļ Į Ñ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ş ş Ţ ţ Ḝ ḝ Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḓ ḓ Ḝ ḝ Ḙ ḙ Ḫ ḫ Ḯ Ḳ ḳ Ḵ ḵ Ḻ Ḽ Ḻ Ḽ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṯ ṯ Ṳ ṳ Ṷ ṷ Ṷ ṷ Ṽ ṽ Ẇ ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ

Ç̄ ç̄ Ā ā Ē ē Ḡ ḡ Ī ī Ķ̄ ķ̄ Ļ̄ ļ̄ Ñ̄ ñ̄ Ō̄ ō̄ Ŗ̄ ŗ̄ Ś̄ ś̄ Ţ̄ ţ̄ Ū̄ ū̄ Ū̆ ū̆ Ō̆ ō̆ Ş̄ ş̄ Ţ̄ ţ̄ Ē̆ ē̆ Ā̆ ā̆ Ḇ̄ ḇ̄ B̄ b̄ Ḏ̄ ḏ̄ D̄ d̄ Ḏ̄ ḏ̄ Ḏ̄ ḏ̄ Ē̆ ē̆ Ę̄ ę̄ Ḥ̄ ḥ̄ Ḫ̄ ḫ̄ Ḫ̄ ḫ̄ Ī̆ Ķ̄ ķ̄ Ḵ̄ ḵ̄ Ļ̄ Ľ̄ Ī̆ Ľ̄ Ī̆ Ṁ̄ ṁ̄ Ñ̄ ñ̄ Ñ̄ ñ̄ Ñ̄ ñ̄ Ō̆ ō̆ Ŗ̄ ŗ̄ Ř̄ ř̄ Ś̄ ś̄ Ţ̄ ţ̄ T̄ t̄ Ţ̄ ţ̄ Ū̆ ū̆ Ū̄ ū̄ Ū̆ ū̆ V̄ v̄ Ẇ ẇ Ẓ̄ ẓ̄ Ẕ̄ ẕ̄ H̱ ẖ Ā̆ ā̆ Ḛ̄ ḛ̄ Ī̆ Ọ̄ ō̆ Ū̄ ū̄ Ū̆ Ȳ ȳ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

Ç̈ ç̈ Ä ä Ë ë Ḡ̈ ḡ̈ Ï ï Ḳ̈ ḳ̈ Ḷ̈ ḷ̈ Ñ̈ ñ̈ Ȫ ȫ Ṛ̈ ṛ̈ S̈ s̈ Ṯ̈ ṯ̈ Ǖ ǖ Ü̆ ü̆ Ö̆ ö̆ S̈ s̈ Ṯ̈ ṯ̈ Ë̆ ë̆ Ä̆ ä̆ Ḧ ḧ Ḯ Ọ̈ ö̇ Ṳ̈ ṳ̈ Ṻ ṻ Ÿ ÿ

## Italic patched, Precomposed letters that inherit an above anchor in stage 9, Above



## Semibold patched, Precomposed letters that inherit an above anchor in stage 9, Above

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ṭ ṭ Ẹ ẹ Ạ ạ Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḓ ḓ Ḍ ḍ Ḓ ḓ Ḕ ḕ Ḝ ḝ Ḥ ḥ Ḫ ḫ Ḥ ḫ Ḯ Ḳ ḳ Ḵ ḵ Ḷ Ḹ Ḽ Ḻ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṻ ṻ Ṷ ṷ Ṹ ṹ Ỿ ỿ Ẇ ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỵ ỵ

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ
Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Ļ Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ű Ǫ ǫ Ș ș Ț ț Ȩ ȩ Ḁ ą Ḇ ḇ Ḉ ḉ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḙ ḙ Ḫ ḫ Ḭ ḭ Ḵ ḵ Ḻ ḻ Ḽ ḽ Ḿ ḿ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṟ ṟ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ Ẇ
ẇ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỳ ỳ

**Semibold italic patched, Precomposed letters that inherit an above anchor in stage 9, Above**

Ç ç Ą ą Ę ę Ģ Į Ķ ķ Ļ Į Ņ ņ Ŗ ŗ Ş ş Ţ ţ Ų ų Ư Ǫ ǫ Ș ș Ț ț Ę ę Ą ą Ḅ ḅ Ḇ ḇ Ḍ ḍ Ḏ ḏ Ḑ ḑ Ḕ ḕ Ḗ ḗ Ḥ ḥ Ḫ ḫ Ḯ Ḳ ḳ Ḵ ḵ Ḷ Ḹ Ḻ Ḽ Ṃ ṃ Ṇ ṇ Ṉ ṉ Ṋ ṋ Ṛ ṛ Ṝ ṝ Ṣ ṣ Ṯ ṯ Ṱ ṱ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṽ ṽ
Ẁ ẁ Ẓ ẓ Ẕ ẕ ẖ Ạ ạ Ẹ ẹ Ị Ọ ọ Ụ ụ Ử Ỵ ỵ

## Regular patched, Precomposed letters that inherit a below anchor in stage 9, Below

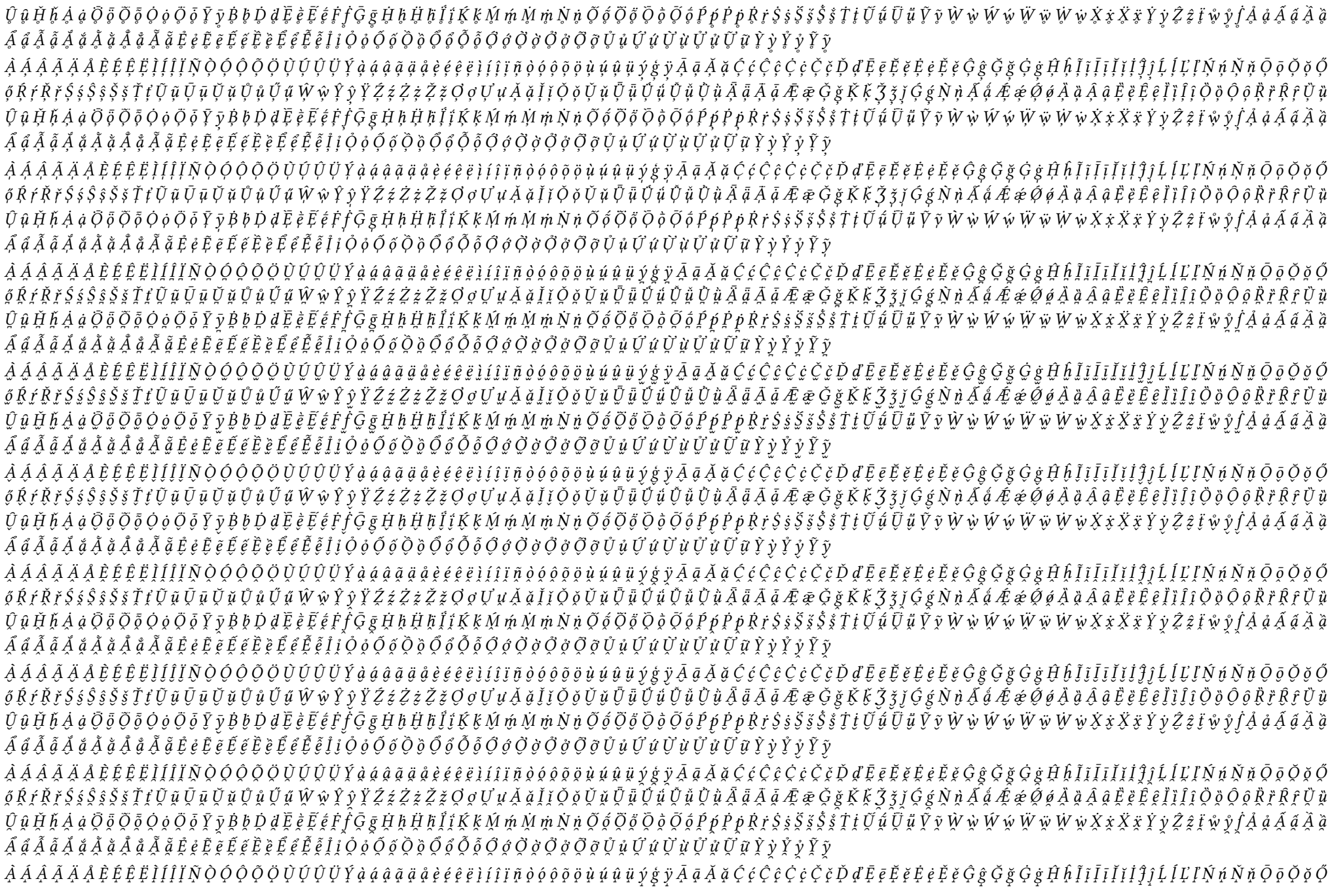


Ṛṛ Ṝṝ Ṟṟ Ṡṡ Ṣṣ Ṥṥ Ṧṧ Ṩṩ Ṫṫ Ṭṭ Ṯṯ Ṱṱ Ṳṳ Ṵṵ Ṷṷ Ṹṹ Ṻṻ Ṽṽ Ṿṿ Ẁẁ Ẃẃ Ẅẅ Ẇẇ Ẉẉ Ẋẋ Ẍẍ Ẏẏ Ỳỳ Ỵỵ Ỷỷ Ỹỹ Ỻỻ Ỽỽ Ỿỿ Ⱡⱡ Ɫɫ Ᵽᵽ Ɽɽ Ⱳⱳ Ⱨⱨ Ⱪⱪ Ⱬⱬ Ⱶⱶ ⱻⱻ Ⲃⲃ Ⲅⲅ Ⲇⲇ Ⲉⲉ Ⲋⲋ Ⲍⲍ Ⲏⲏ Ⲑⲑ Ⲓⲓ Ⲕⲕ Ⲗⲗ Ⲙⲙ Ⲛⲛ Ⲝⲝ Ⲟⲟ Ⲡⲡ Ⲣⲣ Ⲥⲥ Ⲧⲧ Ⲩⲩ Ⲫⲫ Ⲭⲭ Ⲯⲯ Ⲱⲱ Ⳁⳁ Ⳃⳃ Ⳅⳅ Ⳇⳇ Ⳉⳉ Ⳋⳋ Ⳍⳍ Ⳏⳏ Ⳑⳑ Ⳓⳓ Ⳕⳕ Ⳗⳗ Ⳙⳙ Ⳛⳛ Ⳝⳝ Ⳟⳟ Ⳡⳡ Ⳣⳣ Ⳬⳬ Ⳮⳮ Ⳳⳳ Ⴀⴀ Ⴁⴁ Ⴂⴂ Ⴃⴃ Ⴄⴄ Ⴅⴅ Ⴆⴆ Ⴇⴇ Ⴈⴈ Ⴉⴉ Ⴊⴊ Ⴋⴋ Ⴌⴌ Ⴍⴍ Ⴎⴎ Ⴏⴏ Ⴐⴐ Ⴑⴑ Ⴒⴒ Ⴓⴓ Ⴔⴔ Ⴕⴕ Ⴖⴖ Ⴗⴗ Ⴘⴘ Ⴙⴙ Ⴚⴚ Ⴛⴛ Ⴜⴜ Ⴝⴝ Ⴞⴞ Ⴟⴟ Ⴠⴠ Ⴡⴡ Ⴢⴢ Ⴣⴣ Ⴤⴤ Ⴥⴥ Ⴧⴧ Ⴭⴭ Ⴀ Ⴁ Ⴂ Ⴃ Ⴄ Ⴅ Ⴆ Ⴇ Ⴈ Ⴉ Ⴊ Ⴋ Ⴌ Ⴍ Ⴎ Ⴏ Ⴐ Ⴑ Ⴒ Ⴓ Ⴔ Ⴕ Ⴖ Ⴗ Ⴘ Ⴙ Ⴚ Ⴛ Ⴜ Ⴝ Ⴞ Ⴟ Ⴠ Ⴡ Ⴢ Ⴣ Ⴤ Ⴥ Ⴧ Ⴭ ⴀ ⴁ ⴂ ⴃ ⴄ ⴅ ⴆ ⴇ ⴈ ⴉ ⴊ ⴋ ⴌ ⴍ ⴎ ⴏ ⴐ ⴑ ⴒ ⴓ ⴔ ⴕ ⵡ ⵢ ⵣ ⵤ ⵥ ⵧ ⵯ ⶀ ⶁ ⶂ ⶃ ⶄ ⶅ ⶆ ⶇ ⶈ ⶉ ⶊ ⶋ ⶌ ⶍ ⶎ ⶏ ⶐ ⶑ ⶒ ⶓ ⶔ ⶕ ⶖ ⶗ ⶘ ⶙ ⶚ ⶛ ⶜ ⶝ ⶞ ⶟ ⶠ ⶡ ⶢ ⶣ ⶤ ⶥ ⶦ ⶧ ⶨ ⶩ ⶪ ⶫ ⶬ ⶭ ⶮ ⶯ ⶰ ⶱ ⶲ ⶳ ⶴ ⶵ ⶶ ⶷ ⶸ ⶹ ⶺ ⶻ ⶼ ⶽ ⶾ ⶿ ⷀ ⷁ ⷂ ⷃ ⷄ ⷅ ⷆ ⷇ ⷈ ⷉ ⷊ ⷋ ⷌ ⷍ ⷎ ⷏ ⷐ ⷑ ⷒ ⷓ ⷔ ⷕ ⷖ ⷗ ⷘ ⷙ ⷚ ⷛ ⷜ ⷝ ⷞ ⷟ ⷠ ⷡ ⷢ ⷣ ⷤ ⷥ ⷦ ⷧ ⷨ ⷩ ⷪ ⷫ ⷬ ⷭ ⷮ ⷯ ⷰ ⷱ ⷲ ⷳ ⷴ ⷵ ⷶ ⷷ ⷸ ⷹ ⷺ ⷻ ⷼ ⷽ ⷾ ⷿ

ǑǒǑǒǦǧǨǩǰǱǲǳǴǵǶǷǸǹǺǻǼǽǾǿȀȁȂȃȄȅȆȇȈȉȊȋȌȍȎȏȐȑȒȓȔȕȖȗȘșȚțȜȝȞȟȠȡȢȣȤȥȦȧȨȩȪȫȬȭȮȯȰȱȲȳȴȵȶȷȸȹȺȻȼȽȾȿɀɁɂɃɄɅɆɇɈɉɊɋɌɍɎɏɐɑɒɓɔɕɖɗɘəɚɛɜɝɞɟɠɡɢɣɤɥɦɧɨɩɪɫɬɭɮɯɰɱɲɳɴɵɶɷɸɹɺɻɼɽɾɿʀʁʂʃʄʅʆʇʈʉʊʋʌʍʎʏʐʑʒʓʔʕʖʗʘʙʚʛʜʝʞʟʠʡʢʣʤʥʦʧʨʩʪʫʬʭʮʯ
ΆΈΉΊΌΎΏΐΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩΪΫάέήίΰαβγδεζηθικλμνξοπρςστυφχψωϊϋόύώϐϑϒϓϔϕϖϗϘϙϚϛϜϝϞϟϠϡϢϣϤϥϦϧϨϩϪϫϬϭϮϯϰϱϲϳϴϵ϶ϷϸϹϺϻϼϽϾϿЀЁЂЃЄЅІЇЈЉЊЋЌЍЎЏАБВГДЕЖЗИЙКЛМНОПРСТУФХЦЧШЩЪЫЬЭЮЯабвгдежзийклмнопрстуфхцчшщъыьэюяѐёђѓєѕіїјљњћќѝўџѠѡѢѣѤѥѦѧѨѩѪѫѬѭѮѯѰѱѲѳѴѵѶѷѸѹѺѻѼѽѾѿҀҁ҂҃҄҅҆҇ҊҋҌҍҎҏҐґҒғҔҕҖҗҘҙҚқҜҝҞҟҠҡҢңҤҥҦҧҨҩҪҫҬҭҮүҰұҲҳҴҵҶҷҸҹҺһҼҽҾҿӀӁӂӃӄӅӆӇӈӉӊӋӌӍӎӏӐӑӒӓӔӕӖӗӘәӚӛӜӝӞӟӠӡӢӣӤӥӦӧӨөӪӫӬӭӮӯӰӱӲӳӴӵӶӷӸӹӺӻӼӽӾӿԀԁԂԃԄԅԆԇԈԉԊԋԌԍԎԏԐԑԒԓԔԕԖԗԘԙԚԛԜԝԞԟԠԡԢԣԤԥԦԧԨԩԪԫԬԭԮԯ

Ǒ ǒ Ő ő Ř ř Ř ř Ş ş Ŝ ŝ Š š Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ J̌ ǰ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ
Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ Ș ș Ț ț Ȟ ȟ Ȣ ȣ Ȥ ȥ Ȧ ȧ Ȩ ȩ Ȫ ȫ Ȭ ȭ Ȯ ȯ Ȱ ȱ Ȳ ȳ Ⱥ Ⱦ ⱥ ⱦ Ɂ ɂ Ƀ Ʉ Ɇ ɇ Ɉ ɉ Ɍ ɍ Ɏ ɏ ᾯ ᾼ ῌ ῼ ⁀ ⁔ ₮ ₱ ₲ ₴ ₵ ℀ ℁ ℂ ℃ ℄ ℅ ℆ ℇ ℈ ℉ ℊ ℋ ℌ ℍ ℎ ℏ ℐ ℑ ℒ ℓ ℔ ℕ № ℗ ℘ ℙ ℚ ℛ ℜ ℝ ℞ ℟ ℠ ℡ ™ ℣ ℤ ℥ Ω ℧ ℨ ℩ K Å ℬ ℭ ℮ ℯ ℰ ℱ Ⅎ ℳ ℴ ℵ ℶ ℷ ℸ ℹ ℺ ℻ ℼ ℽ ℾ ℿ ⅀ ⅁ ⅂ ⅃ ⅄ ⅅ ⅆ ⅇ ⅈ ⅉ ⅊ ⅋ ⅌ ⅍ ⅎ ⅏
Ⅰ Ⅱ Ⅲ Ⅳ Ⅴ Ⅵ Ⅶ Ⅷ Ⅸ Ⅹ Ⅺ Ⅻ Ⅼ Ⅽ Ⅾ Ⅿ ⅰ ⅱ ⅲ ⅳ ⅴ ⅵ ⅶ ⅷ ⅸ ⅹ ⅺ ⅻ ⅼ ⅽ ⅾ ⅿ ↀ ↁ ↂ Ↄ ↄ ↅ ↆ ↇ ↈ ↉ ↊ ↋ ↌ ↍ ↎ ↏ ← ↑ → ↓ ↔ ↕ ↖ ↗ ↘ ↙ ↚ ↛ ↜ ↝ ↞ ↟ ↠ ↡ ↢ ↣ ↤ ↥ ↦ ↧ ↨ ↩ ↪ ↫ ↬ ↭ ↮ ↯ ↰ ↱ ↲ ↳ ↴ ↵ ↶ ↷ ↸ ↹ ↺ ↻ ↼ ↽ ↾ ↿ ⇀ ⇁ ⇂ ⇃ ⇄ ⇅ ⇆ ⇇ ⇈ ⇉ ⇊ ⇋ ⇌ ⇍ ⇎ ⇏ ⇐ ⇑ ⇒ ⇓ ⇔ ⇕ ⇖ ⇗ ⇘ ⇙ ⇚ ⇛ ⇜ ⇝ ⇞ ⇟ ⇠ ⇡ ⇢ ⇣ ⇤ ⇥ ⇦ ⇧ ⇨ ⇩ ⇪ ⇫ ⇬ ⇭ ⇮ ⇯ ⇰ ⇱ ⇲ ⇳ ⇴ ⇵ ⇶ ⇷ ⇸ ⇹ ⇺ ⇻ ⇼ ⇽ ⇾ ⇿ ∀ ∁ ∂ ∃ ∄ ∅ ∆ ∇ ∈ ∉ ∊ ∋ ∌ ∍ ∎ ∏ ∐ ∑ − ∓ ∔ ∕ ∖ ∗ ∘ ∙ √ ∛ ∜ ∝ ∞ ∟ ∠ ∡ ∢ ∣ ∤ ∥ ∦ ∧ ∨ ∩ ∪ ∫ ∬ ∭ ∮ ∯ ∰ ∱ ∲ ∳ ∴ ∵ ∶ ∷ ∸ ∹ ∺ ∻ ∼ ∽ ∾ ∿ ≀ ≁ ≂ ≃ ≄ ≅ ≆ ≇ ≈ ≉ ≊ ≋ ≌ ≍ ≎ ≏ ≐ ≑ ≒ ≓ ≔ ≕ ≖ ≗ ≘ ≙ ≚ ≛ ≜ ≝ ≞ ≟ ≠ ≡ ≢ ≣ ≤ ≥ ≦ ≧ ≨ ≩ ≪ ≫ ≬ ≭ ≮ ≯ ≰ ≱ ≲ ≳ ≴ ≵ ≶ ≷ ≸ ≹ ≺ ≻ ≼ ≽ ≾ ≿ ⊀ ⊁ ⊂ ⊃ ⊄ ⊅ ⊆ ⊇ ⊈ ⊉ ⊊ ⊋ ⊌ ⊍ ⊎ ⊏ ⊐ ⊑ ⊒ ⊓ ⊔ ⊕ ⊖ ⊗ ⊘ ⊙ ⊚ ⊛ ⊜ ⊝ ⊞ ⊟ ⊠ ⊡ ⊢ ⊣ ⊤ ⊥ ⊦ ⊧ ⊨ ⊩ ⊪ ⊫ ⊬ ⊭ ⊮ ⊯ ⊰ ⊱ ⊲ ⊳ ⊴ ⊵ ⊶ ⊷ ⊸ ⊹ ⊺ ⊻ ⊼ ⊽ ⊾ ⊿ ⋀ ⋁ ⋂ ⋃ ⋄ ⋅ ⋆ ⋇ ⋈ ⋉ ⋊ ⋋ ⋌ ⋍ ⋎ ⋏ ⋐ ⋑ ⋒ ⋓ ⋔ ⋕ ⋖ ⋗ ⋘ ⋙ ⋚ ⋛ ⋜ ⋝ ⋞ ⋟ ⋠ ⋡ ⋢ ⋣ ⋤ ⋥ ⋦ ⋧ ⋨ ⋩ ⋪ ⋫ ⋬ ⋭ ⋮ ⋯ ⋰ ⋱ ⋲ ⋳ ⋴ ⋵ ⋶ ⋷ ⋸ ⋹ ⋺ ⋻ ⋼ ⋽ ⋾ ⋿ ⌀ ⌁ ⌂ ⌃ ⌄ ⌅ ⌆ ⌇ ⌈ ⌉ ⌊ ⌋ ⌌ ⌍ ⌎ ⌏ ⌐ ⌑ ⌒ ⌓ ⌔ ⌕ ⌖ ⌗ ⌘ ⌙ ⌚ ⌛ ⌜ ⌝ ⌞ ⌟ ⌠ ⌡ ⌢ ⌣ ⌤ ⌥ ⌦ ⌧ ⌨ 〈 〉 ⌫ ⌬ ⌭ ⌮ ⌯ ⌰ ⌱ ⌲ ⌳ ⌴ ⌵ ⌶ ⌷ ⌸ ⌹ ⌺ ⌻ ⌼ ⌽ ⌾ ⌿ ⍀ ⍁ ⍂ ⍃ ⍄ ⍅ ⍆ ⍇ ⍈ ⍉ ⍊ ⍋ ⍌ ⍍ ⍎ ⍏ ⍐ ⍑ ⍒ ⍓ ⍔ ⍕ ⍖ ⍗ ⍘ ⍙ ⍚ ⍛ ⍜ ⍝ ⍞ ⍟ ⍠ ⍡ ⍢ ⍣ ⍤ ⍥ ⍦ ⍧ ⍨ ⍩ ⍪ ⍫ ⍬ ⍭ ⍮ ⍯ ⍰ ⍱ ⍲ ⍳ ⍴ ⍵ ⍶ ⍷ ⍸ ⍹ ⍺ ⍻ ⍼ ⍽ ⍾ ⍿ ⎀ ⎁ ⎂ ⎃ ⎄ ⎅ ⎆ ⎇ ⎈ ⎉ ⎊ ⎋ ⎌ ⎍ ⎎ ⎏ ⎐ ⎑ ⎒ ⎓ ⎔ ⎕ ⎖ ⎗ ⎘ ⎙ ⎚ ⎛ ⎜ ⎝ ⎞ ⎟ ⎠ ⎡ ⎢ ⎣ ⎤ ⎥ ⎦ ⎧ ⎨ ⎩ ⎪ ⎫ ⎬ ⎭ ⎮ ⎯ ⎰ ⎱ ⎲ ⎳ ⎴ ⎵ ⎶ ⎷ ⎸ ⎹ ⎺ ⎻ ⎼ ⎽ ⎾ ⎿ ⏀ ⏁ ⏂ ⏃ ⏄ ⏅ ⏆ ⏇ ⏈ ⏉ ⏊ ⏋ ⏌ ⏍ ⏎ ⏏ ⏐ ⏑ ⏒ ⏓ ⏔ ⏕ ⏖ ⏗ ⏘ ⏙ ⏚ ⏛ ⏜ ⏝ ⏞ ⏟ ⏠ ⏡ ⏢ ⏣ ⏤ ⏥ ⏦ ⏧ ⏨ ⏩ ⏪ ⏫ ⏬ ⏭ ⏮ ⏯ ⏰ ⏱ ⏲ ⏳ ⏴ ⏵ ⏶ ⏷ ⏸ ⏹ ⏺ ⏻ ⏼ ⏽ ⏾ ⏿ ␀ ␁ ␂ ␃ ␄ ␅ ␆ ␇ ␈ ␉ ␊ ␋ ␌ ␍ ␎ ␏ ␐ ␑ ␒ ␓ ␔ ␕ ␖ ␗ ␘ ␙ ␚ ␛ ␜ ␝ ␞ ␟ ␠ ␡ ␢ ␣ ␤ ␥ ␦ ␧ ␨ ␩ ␪ ␫ ␬ ␭ ␮ ␯ ␰ ␱ ␲ ␳ ␴ ␵ ␶ ␷ ␸ ␹ ␺ ␻ ␼ ␽ ␾ ␿ ⑀ ⑁ ⑂ ⑃ ⑄ ⑅ ⑆ ⑇ ⑈ ⑉ ⑊ ⑋ ⑌ ⑍ ⑎ ⑏ ⑐ ⑑ ⑒ ⑓ ⑔ ⑕ ⑖ ⑗ ⑘ ⑙ ⑚ ⑛ ⑜ ⑝ ⑞ ⑟ ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑪ ⑫ ⑬ ⑭ ⑮ ⑯ ⑰ ⑱ ⑲ ⑳ ⑴ ⑵ ⑶ ⑷ ⑸ ⑹ ⑺ ⑻ ⑼ ⑽ ⑾ ⑿ ⒀ ⒁ ⒂ ⒃ ⒄ ⒅ ⒆ ⒇ ⒈ ⒉ ⒊ ⒋ ⒌ ⒍ ⒎ ⒏ ⒐ ⒑ ⒒ ⒓ ⒔ ⒕ ⒖ ⒗ ⒘ ⒙ ⒚ ⒛ ⒜ ⒝ ⒞ ⒟ ⒠ ⒡ ⒢ ⒣ ⒤ ⒥ ⒦ ⒧ ⒨ ⒩ ⒪ ⒫ ⒬ ⒭ ⒮ ⒯ ⒰ ⒱ ⒲ ⒳ ⒴ ⒵ Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ Ⓕ Ⓖ Ⓗ Ⓘ Ⓙ Ⓚ Ⓛ Ⓜ Ⓝ Ⓞ Ⓟ Ⓠ Ⓡ Ⓢ Ⓣ Ⓤ Ⓥ Ⓦ Ⓧ Ⓨ Ⓩ ⓐ ⓑ ⓒ ⓓ ⓔ ⓕ ⓖ ⓗ ⓘ ⓙ ⓚ ⓛ ⓜ ⓝ ⓞ ⓟ ⓠ ⓡ ⓢ ⓣ ⓤ ⓥ ⓦ ⓧ ⓨ ⓩ ⓪ ⓫ ⓬ ⓭ ⓮ ⓯ ⓰ ⓱ ⓲ ⓳ ⓴ ⓵ ⓶ ⓷ ⓸ ⓹ ⓺ ⓻ ⓼ ⓽ ⓾ ⓿ ─ ━ │ ┃ ┄ ┅ ┆ ┇ ┈ ┉ ┊ ┋ ┌ ┍ ┎ ┏ ┐ ┑ ┒ ┓ └ ┕ ┖ ┗ ┘ ┙ ┚ ┛ ├ ┝ ┞ ┟ ┠ ┡ ┢ ┣ ┤ ┥ ┦ ┧ ┨ ┩ ┪ ┫ ┬ ┭ ┮ ┯ ┰ ┱ ┲ ┳ ┴ ┵ ┶ ┷ ┸ ┹ ┺ ┻ ┼ ┽ ┾ ┿ ╀ ╁ ╂ ╃ ╄ ╅ ╆ ╇ ╈ ╉ ╊ ╋ ╌ ╍ ╎ ╏ ═ ║ ╒ ╓ ╔ ╕ ╖ ╗ ╘ ╙ ╚ ╛ ╜ ╝ ╞ ╟ ╠ ╡ ╢ ╣ ╤ ╥ ╦ ╧ ╨ ╩ ╪ ╫ ╬ ╭ ╮ ╯ ╰ ╱ ╲ ╳ ╴ ╵ ╶ ╷ ╸ ╹ ╺ ╻ ╼ ╽ ╾ ╿ ▀ ▁ ▂ ▃ ▄ ▅ ▆ ▇ █ ▉ ▊ ▋ ▌ ▍ ▎ ▏ ▐ ░ ▒ ▓ ▔ ▕ ▖ ▗ ▘ ▙ ▚ ▛ ▜ ▝ ▞ ▟ ■ □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩ ▪ ▫ ▬ ▭ ▮ ▯ ▰ ▱ ▲ △ ▴ ▵ ▶ ▷ ▸ ▹ ► ▻ ▼ ▽ ▾ ▿ ◀ ◁ ◂ ◃ ◄ ◅ ◆ ◇ ◈ ◉ ◊ ○ ◌ ◍ ◎ ● ◐ ◑ ◒ ◓ ◔ ◕ ◖ ◗ ◘ ◙ ◚ ◛ ◜ ◝ ◞ ◟ ◠ ◡ ◢ ◣ ◤ ◥ ◦ ◧ ◨ ◩ ◪ ◫ ◬ ◭ ◮ ◯ ◰ ◱ ◲ ◳ ◴ ◵ ◶ ◷ ◸ ◹ ◺ ◻ ◼ ◽ ◾ ◿ ☀ ☁ ☂ ☃ ☄ ★ ☆ ☇ ☈ ☉ ☊ ☋ ☌ ☍ ☎ ☏ ☐ ☑ ☒ ☓ ☔ ☕ ☖ ☗ ☘ ☙ ☚ ☛ ☜ ☝ ☞ ☟ ☠ ☡ ☢ ☣ ☤ ☥ ☦ ☧ ☨ ☩ ☪ ☫ ☬ ☭ ☮ ☯ ☰ ☱ ☲ ☳ ☴ ☵ ☶ ☷ ☸ ☹ ☺ ☻ ☼ ☽ ☾ ☿ ♀ ♁ ♂ ♃ ♄ ♅ ♆ ♇ ♈ ♉ ♊ ♋ ♌ ♍ ♎ ♏ ♐ ♑ ♒ ♓ ♔ ♕ ♖ ♗ ♘ ♙ ♚ ♛ ♜ ♝ ♞ ♟ ♠ ♡ ♢ ♣ ♤ ♥ ♦ ♧ ♨ ♩ ♪ ♫ ♬ ♭ ♮ ♯ ♰ ♱ ♲ ♳ ♴ ♵ ♶ ♷ ♸ ♹ ♺ ♻ ♼ ♽ ♾ ♿ ⚀ ⚁ ⚂ ⚃ ⚄ ⚅ ⚆ ⚇ ⚈ ⚉ ⚊ ⚋ ⚌ ⚍ ⚎ ⚏ ⚐ ⚑ ⚒ ⚓ ⚔ ⚕ ⚖ ⚗ ⚘ ⚙ ⚚ ⚛ ⚜ ⚝ ⚞ ⚟ ⚠ ⚡ ⚢ ⚣ ⚤ ⚥ ⚦ ⚧ ⚨ ⚩ ⚪ ⚫ ⚬ ⚭ ⚮ ⚯ ⚰ ⚱ ⚲ ⚳ ⚴ ⚵ ⚶ ⚷ ⚸ ⚹ ⚺ ⚻ ⚼ ⚽ ⚾ ⚿ ⛀ ⛁ ⛂ ⛃ ⛄ ⛅ ⛆ ⛇ ⛈ ⛉ ⛊ ⛋ ⛌ ⛍ ⛎ ⛏ ⛐ ⛑ ⛒ ⛓ ⛔ ⛕ ⛖ ⛗ ⛘ ⛙ ⛚ ⛛ ⛜ ⛝ ⛞ ⛟ ⛠ ⛡ ⛢ ⛣ ⛤ ⛥ ⛦ ⛧ ⛨ ⛩ ⛪ ⛫ ⛬ ⛭ ⛮ ⛯ ⛰ ⛱ ⛲ ⛳ ⛴ ⛵ ⛶ ⛷ ⛸ ⛹ ⛺ ⛻ ⛼ ⛽ ⛾ ⛿ ✀ ✁ ✂ ✃ ✄ ✅ ✆ ✇ ✈ ✉ ✊ ✋ ✌ ✍ ✎ ✏ ✐ ✑ ✒ ✓ ✔ ✕ ✖ ✗ ✘ ✙ ✚ ✛ ✜ ✝ ✞ ✟ ✠ ✡ ✢ ✣ ✤ ✥ ✦ ✧ ✨ ✩ ✪ ✫ ✬ ✭ ✮ ✯ ✰ ✱ ✲ ✳ ✴ ✵ ✶ ✷ ✸ ✹ ✺ ✻ ✼ ✽ ✾ ✿ ❀ ❁ ❂ ❃ ❄ ❅ ❆ ❇ ❈ ❉ ❊ ❋ ❌ ❍ ❎ ❏ ❐ ❑ ❒ ❓ ❔ ❕ ❖ ❗ ❘ ❙ ❚ ❛ ❜ ❝ ❞ ❟ ❠ ❡ ❢ ❣ ❤ ❥ ❦ ❧ ❨ ❩ ❪ ❫ ❬ ❭ ❮ ❯ ❰ ❱ ❲ ❳ ❴ ❵ ❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽ ❾ ❿ ➀ ➁ ➂ ➃ ➄ ➅ ➆ ➇ ➈ ➉ ➊ ➋ ➌ ➍ ➎ ➏ ➐ ➑ ➒ ➓ ➔ ➕ ➖ ➗ ➘ ➙ ➚ ➛ ➜ ➝ ➞ ➟ ➠ ➡ ➢ ➣ ➤ ➥ ➦ ➧ ➨ ➩ ➪ ➫ ➬ ➭ ➮ ➯ ➰ ➱ ➲ ➳ ➴ ➵ ➶ ➷ ➸ ➹ ➺ ➻ ➼ ➽ ➾ ⟀ ⟁ ⟂ ⟃ ⟄ ⟅ ⟆ ⟇ ⟈ ⟉ ⟊ ⟋ ⟌ ⟍ ⟎ ⟏ ⟐ ⟑ ⟒ ⟓ ⟔ ⟕ ⟖ ⟗ ⟘ ⟙ ⟚ ⟛ ⟜ ⟝ ⟞ ⟟ ⟠ ⟡ ⟢ ⟣ ⟤ ⟥ ⟦ ⟧ ⟨ ⟩ ⟪ ⟫ ⟬ ⟭ ⟮ ⟯ ⟰ ⟱ ⟲ ⟳ ⟴ ⟵ ⟶ ⟷ ⟸ ⟹ ⟺ ⟻ ⟼ ⟽ ⟾ ⟿ ⠀ ⠁ ⠂ ⠃ ⠄ ⠅ ⠆ ⠇ ⠈ ⠉ ⠊ ⠋ ⠌ ⠍ ⠎ ⠏ ⠐ ⠑ ⠒ ⠓ ⠔ ⠕ ⠖ ⠗ ⠘ ⠙ ⠚ ⠛ ⠜ ⠝ ⠞ ⠟ ⠠ ⠡ ⠢ ⠣ ⠤ ⠥ ⠦ ⠧ ⠨ ⠩ ⠪ ⠫ ⠬ ⠭ ⠮ ⠯ ⠰ ⠱ ⠲ ⠳ ⠴ ⠵ ⠶ ⠷ ⠸ ⠹ ⠺ ⠻ ⠼ ⠽ ⠾ ⠿ ⡀ ⡁ ⡂ ⡃ ⡄ ⡅ ⡆ ⡇ ⡈ ⡉ ⡊ ⡋ ⡌ ⡍ ⡎ ⡏ ⡐ ⡑ ⡒ ⡓ ⡔ ⡕ ⡖ ⡗ ⡘ ⡙ ⡚ ⡛ ⡜ ⡝ ⡞ ⡟ ⡠ ⡡ ⡢ ⡣ ⡤ ⡥ ⡦ ⡧ ⡨ ⡩ ⡪ ⡫ ⡬ ⡭ ⡮ ⡯ ⡰ ⡱ ⡲ ⡳ ⡴ ⡵ ⡶ ⡷ ⡸ ⡹ ⡺ ⡻ ⡼ ⡽ ⡾ ⡿ ⢀ ⢁ ⢂ ⢃ ⢄ ⢅ ⢆ ⢇ ⢈ ⢉ ⢊ ⢋ ⢌ ⢍ ⢎ ⢏ ⢐ ⢑ ⢒ ⢓ ⢔ ⢕ ⢖ ⢗ ⢘ ⢙ ⢚ ⢛ ⢜ ⢝ ⢞ ⢟ ⢠ ⢡ ⢢ ⢣ ⢤ ⢥ ⢦ ⢧ ⢨ ⢩ ⢪ ⢫ ⢬ ⢭ ⢮ ⢯ ⢰ ⢱ ⢲ ⢳ ⢴ ⢵ ⢶ ⢷ ⢸ ⢹ ⢺ ⢻ ⢼ ⢽ ⢾ ⢿ ⣀ ⣁ ⣂ ⣃ ⣄ ⣅ ⣆ ⣇ ⣈ ⣉ ⣊ ⣋ ⣌ ⣍ ⣎ ⣏ ⣐ ⣑ ⣒ ⣓ ⣔ ⣕ ⣖ ⣗ ⣘ ⣙ ⣚ ⣛ ⣜ ⣝ ⣞ ⣟ ⣠ ⣡ ⣢ ⣣ ⣤ ⣥ ⣦ ⣧ ⣨ ⣩ ⣪ ⣫ ⣬ ⣭ ⣮ ⣯ ⣰ ⣱ ⣲ ⣳ ⣴ ⣵ ⣶ ⣷ ⣸ ⣹ ⣺ ⣻ ⣼ ⣽ ⣾ ⣿ ⤀ ⤁ ⤂ ⤃ ⤄ ⤅ ⤆ ⤇ ⤈ ⤉ ⤊ ⤋ ⤌ ⤍ ⤎ ⤏ ⤐ ⤑ ⤒ ⤓ ⤔ ⤕ ⤖ ⤗ ⤘ ⤙ ⤚ ⤛ ⤜ ⤝ ⤞ ⤟ ⤠ ⤡ ⤢ ⤣ ⤤ ⤥ ⤦ ⤧ ⤨ ⤩ ⤪ ⤫ ⤬ ⤭ ⤮ ⤯ ⤰ ⤱ ⤲ ⤳ ⤴ ⤵ ⤶ ⤷ ⤸ ⤹ ⤺ ⤻ ⤼ ⤽ ⤾ ⤿ ⥀ ⥁ ⥂ ⥃ ⥄ ⥅ ⥆ ⥇ ⥈ ⥉ ⥊ ⥋ ⥌ ⥍ ⥎ ⥏ ⥐ ⥑ ⥒ ⥓ ⥔ ⥕ ⥖ ⥗ ⥘ ⥙ ⥚ ⥛ ⥜ ⥝ ⥞ ⥟ ⥠ ⥡ ⥢ ⥣ ⥤ ⥥ ⥦ ⥧ ⥨ ⥩ ⥪ ⥫ ⥬ ⥭ ⥮ ⥯ ⥰ ⥱ ⥲ ⥳ ⥴ ⥵ ⥶ ⥷ ⥸ ⥹ ⥺ ⥻ ⥼ ⥽ ⥾ ⥿ ⦀ ⦁ ⦂ ⦃ ⦄ ⦅ ⦆ ⦇ ⦈ ⦉ ⦊ ⦋ ⦌ ⦍ ⦎ ⦏ ⦐ ⦑ ⦒ ⦓ ⦔ ⦕ ⦖ ⦗ ⦘ ⦙ ⦚ ⦛ ⦜ ⦝ ⦞ ⦟ ⦠ ⦡ ⦢ ⦣ ⦤ ⦥ ⦦ ⦧ ⦨ ⦩ ⦪ ⦫ ⦬ ⦭ ⦮ ⦯ ⦰ ⦱ ⦲ ⦳ ⦴ ⦵ ⦶ ⦷ ⦸ ⦹ ⦺ ⦻ ⦼ ⦽ ⦾ ⦿ ⧀ ⧁ ⧂ ⧃ ⧄ ⧅ ⧆ ⧇ ⧈ ⧉ ⧊ ⧋ ⧌ ⧍ ⧎ ⧏ ⧐ ⧑ ⧒ ⧓ ⧔ ⧕ ⧖ ⧗ ⧘ ⧙ ⧚ ⧛ ⧜ ⧝ ⧞ ⧟ ⧠ ⧡ ⧢ ⧣ ⧤ ⧥ ⧦ ⧧ ⧨ ⧩ ⧪ ⧫ ⧬ ⧭ ⧮ ⧯ ⧰ ⧱ ⧲ ⧳ ⧴ ⧵ ⧶ ⧷ ⧸ ⧹ ⧺ ⧻ ⧼ ⧽ ⧾ ⧿ ⨀ ⨁ ⨂ ⨃ ⨄ ⨅ ⨆ ⨇ ⨈ ⨉ ⨊ ⨋ ⨌ ⨍ ⨎ ⨏ ⨐ ⨑ ⨒ ⨓ ⨔ ⨕ ⨖ ⨗ ⨘ ⨙ ⨚ ⨛ ⨜ ⨝ ⨞ ⨟ ⨠ ⨡ ⨢ ⨣ ⨤ ⨥ ⨦ ⨧ ⨨ ⨩ ⨪ ⨫ ⨬ ⨭ ⨮ ⨯ ⨰ ⨱ ⨲ ⨳ ⨴ ⨵ ⨶ ⨷ ⨸ ⨹ ⨺ ⨻ ⨼ ⨽ ⨾ ⨿ ⩀ ⩁ ⩂ ⩃ ⩄ ⩅ ⩆ ⩇ ⩈ ⩉ ⩊ ⩋ ⩌ ⩍ ⩎ ⩏ ⩐ ⩑ ⩒ ⩓ ⩔ ⩕ ⩖ ⩗ ⩘ ⩙ ⩚ ⩛ ⩜ ⩝ ⩞ ⩟ ⩠ ⩡ ⩢ ⩣ ⩤ ⩥ ⩦ ⩧ ⩨ ⩩ ⩪ ⩫ ⩬ ⩭ ⩮ ⩯ ⩰ ⩱ ⩲ ⩳ ⩴ ⩵ ⩶ ⩷ ⩸ ⩹ ⩺ ⩻ ⩼ ⩽ ⩾ ⩿ ⪀ ⪁ ⪂ ⪃ ⪄ ⪅ ⪆ ⪇ ⪈ ⪉ ⪊ ⪋ ⪌ ⪍ ⪎ ⪏ ⪐ ⪑ ⪒ ⪓ ⪔ ⪕ ⪖ ⪗ ⪘ ⪙ ⪚ ⪛ ⪜ ⪝ ⪞ ⪟ ⪠ ⪡ ⪢ ⪣ ⪤ ⪥ ⪦ ⪧ ⪨ ⪩ ⪪ ⪫ ⪬ ⪭ ⪮ ⪯ ⪰ ⪱ ⪲ ⪳ ⪴ ⪵ ⪶ ⪷ ⪸ ⪹ ⪺ ⪻ ⪼ ⪽ ⪾ ⪿ ⫀ ⫁ ⫂ ⫃ ⫄ ⫅ ⫆ ⫇ ⫈ ⫉ ⫊ ⫋ ⫌ ⫍ ⫎ ⫏ ⫐ ⫑ ⫒ ⫓ ⫔ ⫕ ⫖ ⫗ ⫘ ⫙ ⫚ ⫛ ⫝̸ ⫝ ⫞ ⫟ ⫠ ⫡ ⫢ ⫣ ⫤ ⫥ ⫦ ⫧ ⫨ ⫩ ⫪ ⫫ ⫬ ⫭ ⫮ ⫯ ⫰ ⫱ ⫲ ⫳ ⫴ ⫵ ⫶ ⫷ ⫸ ⫹ ⫺ ⫻ ⫼ ⫽ ⫾ ⫿ ⬀ ⬁ ⬂ ⬃ ⬄ ⬅ ⬆ ⬇ ⬈ ⬉ ⬊ ⬋ ⬌ ⬍ ⬎ ⬏ ⬐ ⬑ ⬒ ⬓ ⬔ ⬕ ⬖ ⬗ ⬘ ⬙ ⬚ ⬛ ⬜ ⬝ ⬞ ⬟ ⬠ ⬡ ⬢ ⬣ ⬤ ⬥ ⬦ ⬧ ⬨ ⬩ ⬪ ⬫ ⬬ ⬭ ⬮ ⬯ ⬰ ⬱ ⬲ ⬳ ⬴ ⬵ ⬶ ⬷ ⬸ ⬹ ⬺ ⬻ ⬼ ⬽ ⬾ ⬿ ⭀ ⭁ ⭂ ⭃ ⭄ ⭅ ⭆ ⭇ ⭈ ⭉ ⭊ ⭋ ⭌ ⭍ ⭎ ⭏ ⭐ ⭑ ⭒ ⭓ ⭔ ⭕ ⭖ ⭗ ⭘ ⭙ ⭚ ⭛ ⭜ ⭝ ⭞ ⭟ ⭠ ⭡ ⭢ ⭣ ⭤ ⭥ ⭦ ⭧ ⭨ ⭩ ⭪ ⭫ ⭬ ⭭ ⭮ ⭯ ⭰ ⭱ ⭲ ⭳ ⭶ ⭷ ⭸ ⭹ ⭺ ⭻ ⭼ ⭽ ⭾ ⭿ ⮀ ⮁ ⮂ ⮃ ⮄ ⮅ ⮆ ⮇ ⮈ ⮉ ⮊ ⮋ ⮌ ⮍ ⮎ ⮏ ⮐ ⮑ ⮒ ⮓ ⮔ ⮕ ⮖ ⮗ ⮘ ⮙ ⮚ ⮛ ⮜ ⮝ ⮞ ⮟ ⮠ ⮡ ⮢ ⮣ ⮤ ⮥ ⮦ ⮧ ⮨ ⮩ ⮪ ⮫ ⮬ ⮭ ⮮ ⮯ ⮰ ⮱ ⮲ ⮳ ⮴ ⮵ ⮶ ⮷ ⮸ ⮹ ⮺ ⮻ ⮼ ⮽ ⮾ ⮿ ⯀ ⯁ ⯂ ⯃ ⯄ ⯅ ⯆ ⯇ ⯈ ⯉ ⯊ ⯋ ⯌ ⯍ ⯎ ⯏ ⯐ ⯑ ⯒ ⯓ ⯔ ⯕ ⯖ ⯗ ⯘ ⯙ ⯚ ⯛ ⯜ ⯝ ⯞ ⯟ ⯠ ⯡ ⯢ ⯣ ⯤ ⯥ ⯦ ⯧ ⯨ ⯩ ⯪ ⯫ ⯬ ⯭ ⯮ ⯯ ⯰ ⯱ ⯲ ⯳ ⯴ ⯵ ⯶ ⯷ ⯸ ⯹ ⯺ ⯻ ⯼ ⯽ ⯾ ⯿ Ⰰ Ⰱ Ⰲ Ⰳ Ⰴ Ⰵ Ⰶ Ⰷ Ⰸ Ⰹ Ⰺ Ⰻ Ⰼ Ⰽ Ⰾ Ⰿ Ⱀ Ⱁ Ⱂ Ⱃ Ⱄ Ⱅ Ⱆ Ⱇ Ⱈ Ⱉ Ⱊ Ⱋ Ⱌ Ⱍ Ⱎ Ⱏ Ⱐ Ⱑ Ⱒ Ⱓ Ⱔ Ⱕ Ⱖ Ⱗ Ⱘ Ⱙ Ⱚ Ⱛ Ⱜ Ⱝ Ⱞ Ⱟ ⰰ ⰱ ⰲ ⰳ ⰴ ⰵ ⰶ ⰷ ⰸ ⰹ ⰺ ⰻ ⰼ ⰽ ⰾ ⰿ ⱀ ⱁ ⱂ ⱃ ⱄ ⱅ ⱆ ⱇ ⱈ ⱉ ⱊ ⱋ ⱌ ⱍ ⱎ ⱏ ⱐ ⱑ ⱒ ⱓ ⱔ ⱕ ⱖ ⱗ ⱘ ⱙ ⱚ ⱛ ⱜ ⱝ ⱞ ⱟ Ⱡ ⱡ Ɫ ɫ Ᵽ Ɽ ⱨ ⱪ ⱬ ⱳ ⱴ ⱶ ⱷ ⱸ ⱹ ⱺ ⱻ ⱼ ⱽ Ȿ Ɀ Ⲁ ⲁ Ⲃ ⲃ Ⲅ ⲅ Ⲇ ⲇ Ⲉ ⲉ Ⲋ ⲋ Ⲍ ⲍ Ⲏ ⲏ Ⲑ ⲑ Ⲓ ⲓ Ⲕ ⲕ Ⲗ ⲗ Ⲙ ⲙ Ⲛ ⲛ Ⲝ ⲝ Ⲟ ⲟ Ⲡ ⲡ Ⲣ ⲣ Ⲥ ⲥ Ⲧ ⲧ Ⲩ ⲩ Ⲫ ⲫ Ⲭ ⲭ Ⲯ ⲯ Ⲱ ⲱ Ⲳ ⲳ Ⲵ ⲵ Ⲷ ⲷ Ⲹ ⲹ Ⲻ ⲻ Ⲽ ⲽ Ⲿ ⲿ Ⳁ ⳁ Ⳃ ⳃ Ⳅ ⳅ Ⳇ ⳇ Ⳉ ⳉ Ⳋ ⳋ Ⳍ ⳍ Ⳏ ⳏ Ⳑ ⳑ Ⳓ ⳓ Ⳕ ⳕ Ⳗ ⳗ Ⳙ ⳙ Ⳛ ⳛ Ⳝ ⳝ Ⳟ ⳟ Ⳡ ⳡ Ⳣ ⳣ Ⳬ ⳬ Ⳮ ⳮ Ⳳ ⳳ ⳴ ⳵ ⳶ ⳷ ⳸ ⳹ ⳺ ⳻ ⳼ ⳽ ⳾ ⳿ ⴀ ⴁ ⴂ ⴃ ⴄ ⴅ ⴆ ⴇ ⴈ ⴉ ⴊ ⴋ ⴌ ⴍ ⴎ ⴏ ⴐ ⴑ ⴒ ⴓ ⴔ ⴕ ⴖ ⴗ ⴘ ⴙ ⴚ ⴛ ⴜ ⴝ ⴞ ⴟ ⴠ ⴡ ⴢ ⴣ ⴤ ⴥ ⴧ ⴭ ⴰ ⴱ ⴲ ⴳ ⴴ ⴵ ⴶ ⴷ ⴸ ⴹ ⴺ ⴻ ⴼ ⴽ ⵀ ⵁ ⵂ ⵃ ⵄ ⵅ ⵆ ⵇ ⵈ ⵉ ⵊ ⵋ ⵌ ⵍ ⵎ ⵏ ⵐ ⵑ ⵒ ⵓ ⵔ ⵕ ⵖ ⵗ ⵘ ⵙ ⵚ ⵛ ⵜ ⵝ ⵞ ⵟ ⵠ ⵡ ⵢ ⵣ ⵤ ⵥ ⵦ ⵧ ⵨ ⵩ ⵪ ⵫ ⵬ ⵭ ⵮ ⵯ ⵰ ⵱ ⵲ ⵳ ⵴ ⵵ ⵶ ⵷ ⵸ ⵹ ⵺ ⵻ ⵼ ⵽ ⵾ ⵿ ⶀ ⶁ ⶂ ⶃ ⶄ ⶅ ⶆ ⶇ ⶈ ⶉ ⶊ ⶋ ⶌ ⶍ ⶎ ⶏ ⶐ ⶑ ⶒ ⶓ ⶔ ⶕ ⶖ ⶠ ⶡ ⶢ ⶣ ⶤ ⶥ ⶦ ⶧ ⶨ ⶩ ⶪ ⶫ ⶬ ⶭ ⶮ ⶯ ⶰ ⶱ ⶲ ⶳ ⶴ ⶵ ⶶ ⶷ ⶸ ⶹ ⶺ ⶻ ⶼ ⶽ ⶾ ⶿ ⷀ ⷁ ⷂ ⷃ ⷄ ⷅ ⷆ ⷇ ⷈ ⷉ ⷊ ⷋ ⷌ ⷍ ⷎ ⷏ ⷐ ⷑ ⷒ ⷓ ⷔ ⷕ ⷖ ⷗ ⷘ ⷙ ⷚ ⷛ ⷜ ⷝ ⷞ ⷟ ⷠ ⷡ ⷢ ⷣ ⷤ ⷥ 

## Italic patched, Precomposed letters that inherit a below anchor in stage 9, Below



ő Ř ř Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẁ ẁ Ŵ ŵ Ẅ ẅ Ỳ ỳ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȅ ȅ Ȇ ȇ Ȉ ȉ Ȋ ȋ Ȍ ȍ Ȏ ȏ Ȑ ȑ Ȓ ȓ Ȕ ȕ Ȗ ȗ
Ụ ụ Ḫ ḫ Ḁ ḁ Ọ̄ ọ̄ Ō̩ ō̩ Ọ́ ọ́ Ö̩ ö̩ Ȳ ȳ Ḅ ḅ Ḍ ḍ Ē̩ ē̩ È̩ è̩ É̩ é̩ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṡ ṡ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ỹ ỹ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ẩ ẩ Ầ ầ Ấ ấ Ẫ ẫ Ẳ ẳ Ằ ằ Ắ ắ Ẵ ẵ Ẹ ẹ Ệ ệ Ễ ễ Ế ế Ề ề Ể ể Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỷ ỷ Ỷ ỷ Ỹ ỹ
Ậ ậ Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ä ä Å å Ẹ ẹ Ệ ệ Ễ ễ Ế ế Ề ề Ể ể Ễ ễ Ỉ ỉ Ị ị Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỷ ỷ Ỷ ỷ Ỹ ỹ
Ậ ậ Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ä ä Ọ̄ ọ̄ Ō̩ ō̩ Ọ́ ọ́ Ö̩ ö̩ Ȳ ȳ Ḅ ḅ Ḍ ḍ Ē̩ ē̩ È̩ è̩ É̩ é̩ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḭ ḭ Ḱ ḱ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṅ ṅ Ṇ ṇ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṍ ṍ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṡ ṡ Ṫ ṫ Ṳ ṳ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ỹ ỹ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẉ ẉ Ẋ ẋ Ẍ ẍ Ỳ ỳ Ẑ ẑ Ẓ ẓ Ẕ ẕ Ẩ ẩ Ầ ầ Ấ ấ Ẫ ẫ Ẳ ẳ Ằ ằ Ắ ắ Ẵ ẵ Ẹ ẹ Ệ ệ Ễ ễ Ế ế Ề ề Ể ể Ễ ễ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỷ ỷ Ỷ ỷ Ỹ ỹ
Ậ Ấ Ầ Ẩ Ẫ Ä Å Ẹ Ệ Ễ Ế Ề Ể Ễ Ỉ Ọ Ố Ồ Ổ Ỗ Ộ Ớ Ờ Ở Ỡ Ợ Ủ Ứ Ừ Ử Ữ Ự Ỷ Ỷ

## Semibold patched, Precomposed letters that inherit a below anchor in stage 9, Below

Ọ ọ Ỏ ỏ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ỏ ỏ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
Ỵ ỵ Ỷ ỷ Ỹ ỹ Ỻ ỻ Ỽ ỽ Ỿ ỿ Ἀ ἀ Ἁ ἁ Ἂ ἂ Ἃ ἃ Ἄ ἄ Ἅ ἅ Ἆ ἆ Ἇ ἇ Ἐ ἐ Ἑ ἑ Ἒ ἒ Ἓ ἓ Ἔ ἔ Ἕ ἕ Ἠ ἠ Ἡ ἡ Ἢ ἢ Ἣ ἣ Ἤ ἤ Ἥ ἥ Ἦ ἦ Ἧ ἧ Ἰ ἰ Ἱ ἱ Ἲ ἲ Ἳ ἳ Ἴ ἴ Ἵ ἵ Ἶ ἶ Ἷ ἷ Ὀ ὀ Ὁ ὁ Ὂ ὂ Ὃ ὃ Ὄ ὄ Ὅ ὅ Ὑ ὑ Ὓ ὓ Ὕ ὕ Ὗ ὗ Ὠ ὠ Ὡ ὡ Ὢ ὢ Ὣ ὣ Ὤ ὤ Ὥ ὥ Ὦ ὦ Ὧ ὧ Ὰ ὰ Ά ά Ὲ ὲ Έ έ Ὴ ὴ Ή ή Ὶ ὶ Ί ί Ὸ ὸ Ό ό Ὺ ὺ Ύ ύ Ὼ ὼ Ώ ώ ᾀ ᾁ ᾂ ᾃ ᾄ ᾅ ᾆ ᾇ ᾈ ᾉ ᾊ ᾋ ᾌ ᾍ ᾎ ᾏ ᾐ ᾑ ᾒ ᾓ ᾔ ᾕ ᾖ ᾗ ᾘ ᾙ ᾚ ᾛ ᾜ ᾝ ᾞ ᾟ ᾠ ᾡ ᾢ ᾣ ᾤ ᾥ ᾦ ᾧ ᾨ ᾩ ᾪ ᾫ ᾬ ᾭ ᾮ ᾯ ᾰ ᾱ ᾲ ᾳ ᾴ ᾶ ᾷ Ᾰ Ᾱ Ὰ Ά ᾼ ῁ ῂ ῃ ῄ ῆ ῇ Ὲ Έ Ὴ Ή ῌ ῍ ῎ ῏ ῐ ῑ ῒ ΐ ῖ ῗ Ῐ Ῑ Ὶ Ί ῝ ῞ ῟ ῠ ῡ ῢ ΰ ῤ ῥ Ὺ Ύ ῭ ΅ ` ῲ ῳ ῴ ῶ ῷ Ὸ Ό Ὼ Ώ ‼ ⁇ ⁈ ⁉ ₨ ℀ ℁ ℃ ℅ ℆ ℉ ℔ № ℗ ℠ ™ ℻ ⅀ ⅁ ⅂ ⅃ ⅄ ⅋ ⅌ ⅍ ⅎ ⅏ ↊ ↋ ∰ ∱ ∲ ∳ ⌀ ⌗ ⍰ ⍱ ⍲ ⍳ ⍴ ⎈ ⏨ ⏩ ⏪ ⏫ ⏬ ⏭ ⏮ ⏯ ⏰ ⏱ ⏲ ⏳ ⏴ ⏵ ⏶ ⏷ ⏸ ⏹ ⏺ ⏻ ⏼ ⏽ ⏾ ⏿ ␀ ␁ ␂ ␃ ␄ ␅ ␆ ␇ ␈ ␉ ␊ ␋ ␌ ␍ ␎ ␏ ␐ ␑ ␒ ␓ ␔ ␕ ␖ ␗ ␘ ␙ ␚ ␛ ␜ ␝ ␞ ␟ ␠ ␡ ␢ ␣ ␤ ␥ ␦ ␧ ␨ ␩ ␪ ␫ ␬ ␭ ␮ ␯ ␰ ␱ ␲ ␳ ␴ ␵ ␶ ␷ ␸ ␹ ␺ ␻ ␼ ␽ ␾ ␿ ⑀ ⑁ ⑂ ⑃ ⑄ ⑅ ⑆ ⑇ ⑈ ⑉ ⑊ ⑋ ⑌ ⑍ ⑎ ⑏ ⑐ ⑑ ⑒ ⑓ ⑔ ⑕ ⑖ ⑗ ⑘ ⑙ ⑚ ⑛ ⑜ ⑝ ⑞ ⑟ ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑪ ⑫ ⑬ ⑭ ⑮ ⑯ ⑰ ⑱ ⑲ ⑳ ⑴ ⑵ ⑶ ⑷ ⑸ ⑹ ⑺ ⑻ ⑼ ⑽ ⑾ ⑿ ⒀ ⒁ ⒂ ⒃ ⒄ ⒅ ⒆ ⒇ ⒈ ⒉ ⒊ ⒋ ⒌ ⒍ ⒎ ⒏ ⒐ ⒑ ⒒ ⒓ ⒔ ⒕ ⒖ ⒗ ⒘ ⒙ ⒚ ⒛ ⒜ ⒝ ⒞ ⒟ ⒠ ⒡ ⒢ ⒣ ⒤ ⒥ ⒦ ⒧ ⒨ ⒩ ⒪ ⒫ ⒬ ⒭ ⒮ ⒯ ⒰ ⒱ ⒲ ⒳ ⒴ ⒵ ⓐ ⓑ ⓒ ⓓ ⓔ ⓕ ⓖ ⓗ ⓘ ⓙ ⓚ ⓛ ⓜ ⓝ ⓞ ⓟ ⓠ ⓡ ⓢ ⓣ ⓤ ⓥ ⓦ ⓧ ⓨ ⓩ ⓪ ⓫ ⓬ ⓭ ⓮ ⓯ ⓰ ⓱ ⓲ ⓳ ⓴ ⓵ ⓶ ⓷ ⓸ ⓹ ⓺ ⓻ ⓼ ⓽ ⓾ ⓿ ─ ━ │ ┃ ┄ ┅ ┆ ┇ ┈ ┉ ┊ ┋ ┌ ┍ ┎ ┏ ┐ ┑ ┒ ┓ └ ┕ ┖ ┗ ┘ ┙ ┚ ┛ ├ ┝ ┞ ┟ ┠ ┡ ┢ ┣ ┤ ┥ ┦ ┧ ┨ ┩ ┪ ┫ ┬ ┭ ┮ ┯ ┰ ┱ ┲ ┳ ┴ ┵ ┶ ┷ ┸ ┹ ┺ ┻ ┼ ┽ ┾ ┿ ╀ ╁ ╂ ╃ ╄ ╅ ╆ ╇ ╈ ╉ ╊ ╋ ╌ ╍ ╎ ╏ ═ ║ ╒ ╓ ╔ ╕ ╖ ╗ ╘ ╙ ╚ ╛ ╜ ╝ ╞ ╟ ╠ ╡ ╢ ╣ ╤ ╥ ╦ ╧ ╨ ╩ ╪ ╫ ╬ ▀ ▁ ▂ ▃ ▄ ▅ ▆ ▇ ▉ ▊ ▋ ▌ ▍ ▎ ▏ ▐ ▔ ▕ ▖ ▗ ▘ ▙ ▚ ▛ ▜ ▝ ▞ ▟ ■ □ ▢ ▣ ▤ ▥ ▦ ▧ ▨ ▩ ▪ ▫ ▬ ▭ ▮ ▯ ▰ ▱ ▲ △ ▴ ▵ ▶ ▷ ▸ ▹ ► ▻ ▼ ▽ ▾ ▿ ◀ ◁ ◂ ◃ ◄ ◅ ◆ ◇ ◈ ◉ ◊ ○ ◌ ◍ ◎ ● ◐ ◑ ◒ ◓ ◔ ◕ ◖ ◗ ◘ ◙ ◚ ◛ ◜ ◝ ◞ ◟ ◠ ◡ ◢ ◣ ◤ ◥ ◦ ◧ ◨ ◩ ◪ ◫ ◬ ◭ ◮ ◯ ◰ ◱ ◲ ◳ ◴ ◵ ◶ ◷ ◸ ◹ ◺ ◻ ◼ ◽ ◾ ◿ ☀ ☁ ☂ ☃ ☄ ★ ☆ ☇ ☈ ☉ ☊ ☋ ☌ ☍ ☎ ☏ ☐ ☑ ☒ ☓ ☔ ☕ ☖ ☗ ☘ ☙ ☚ ☛ ☜ ☝ ☞ ☟ ☠ ☡ ☢ ☣ ☤ ☥ ☦ ☧ ☨ ☩ ☪ ☫ ☬ ☭ ☮ ☯ ☰ ☱ ☲ ☳ ☴ ☵ ☶ ☷ ☸ ☹ ☺ ☻ ☼ ☽ ☾ ☿ ♀ ♁ ♂ ♃ ♄ ♅ ♆ ♇ ♈ ♉ ♊ ♋ ♌ ♍ ♎ ♏ ♐ ♑ ♒ ♓ ♔ ♕ ♖ ♗ ♘ ♙ ♚ ♛ ♜ ♝ ♞ ♟ ♠ ♡ ♢ ♣ ♤ ♥ ♦ ♧ ♨ ♩ ♪ ♫ ♬ ♭ ♮ ♯ ♰ ♱ ♲ ♳ ♴ ♵ ♶ ♷ ♸ ♹ ♺ ♻ ♼ ♽ ♾ ♿ ⚀ ⚁ ⚂ ⚃ ⚄ ⚅ ⚆ ⚇ ⚈ ⚉ ⚊ ⚋ ⚌ ⚍ ⚎ ⚏ ⚐ ⚑ ⚒ ⚓ ⚔ ⚕ ⚖ ⚗ ⚘ ⚙ ⚚ ⚛ ⚜ ⚝ ⚞ ⚟ ⚠ ⚡ ⚢ ⚣ ⚤ ⚥ ⚦ ⚧ ⚨ ⚩ ⚪ ⚫ ⚬ ⚭ ⚮ ⚯ ⚰ ⚱ ⚲ ⚳ ⚴ ⚵ ⚶ ⚷ ⚸ ⚹ ⚺ ⚻ ⚼ ⚽ ⚾ ⚿ ⛀ ⛁ ⛂ ⛃ ⛄ ⛅ ⛆ ⛇ ⛈ ⛉ ⛊ ⛋ ⛌ ⛍ ⛎ ⛏ ⛐ ⛑ ⛒ ⛓ ⛔ ⛕ ⛖ ⛗ ⛘ ⛙ ⛚ ⛛ ⛜ ⛝ ⛞ ⛟ ⛠ ⛡ ⛢ ⛣ ⛤ ⛥ ⛦ ⛧ ⛨ ⛩ ⛪ ⛫ ⛬ ⛭ ⛮ ⛯ ⛰ ⛱ ⛲ ⛳ ⛴ ⛵ ⛶ ⛷ ⛸ ⛹ ⛺ ⛻ ⛼ ⛽ ⛾ ⛿ ✀ ✁ ✂ ✃ ✄ ✅ ✆ ✇ ✈ ✉ ✊ ✋ ✌ ✍ ✎ ✏ ✐ ✑ ✒ ✓ ✔ ✕ ✖ ✗ ✘ ✙ ✚ ✛ ✜ ✝ ✞ ✟ ✠ ✡ ✢ ✣ ✤ ✥ ✦ ✧ ✨ ✩ ✪ ✫ ✬ ✭ ✮ ✯ ✰ ✱ ✲ ✳ ✴ ✵ ✶ ✷ ✸ ✹ ✺ ✻ ✼ ✽ ✾ ✿ ❀ ❁ ❂ ❃ ❄ ❅ ❆ ❇ ❈ ❉ ❊ ❋ ❌ ❍ ❎ ❏ ❐ ❑ ❒ ❓ ❔ ❕ ❖ ❗ ❘ ❙ ❚ ❛ ❜ ❝ ❞ ❟ ❠ ❡ ❢ ❣ ❤ ❥ ❦ ❧ ❨ ❩ ❪ ❫ ❬ ❭ ❮ ❯ ❰ ❱ ❲ ❳ ❴ ❵ ❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽ ❾ ❿ ➀ ➁ ➂ ➃ ➄ ➅ ➆ ➇ ➈ ➉ ➊ ➋ ➌ ➍ ➎ ➏ ➐ ➑ ➒ ➓ ➔ ➕ ➖ ➗ ➘ ➙ ➚ ➛ ➜ ➝ ➞ ➟ ➠ ➡ ➢ ➣ ➤ ➥ ➦ ➧ ➨ ➩ ➪ ➫ ➬ ➭ ➮ ➯ ➰ ➱ ➲ ➳ ➴ ➵ ➶ ➷ ➸ ➹ ➺ ➻ ➼ ➽ ➾ ⟀ ⟁ ⟂ ⟃ ⟄ ⟅ ⟆ ⟇ ⟈ ⟉ ⟊ ⟋ ⟌ ⟍ ⟎ ⟏ ⟐ ⟑ ⟒ ⟓ ⟔ ⟕ ⟖ ⟗ ⟘ ⟙ ⟚ ⟛ ⟜ ⟝ ⟞ ⟟ ⟠ ⟡ ⟢ ⟣ ⟤ ⟥ ⟦ ⟧ ⟨ ⟩ ⟪ ⟫ ⟬ ⟭ ⟮ ⟯ ⟰ ⟱ ⟲ ⟳ ⟴ ⟵ ⟶ ⟷ ⟸ ⟹ ⟺ ⟻ ⟼ ⟽ ⟾ ⟿ ⠀ ⠁ ⠂ ⠃ ⠄ ⠅ ⠆ ⠇ ⠈ ⠉ ⠊ ⠋ ⠌ ⠍ ⠎ ⠏ ⠐ ⠑ ⠒ ⠓ ⠔ ⠕ ⠖ ⠗ ⠘ ⠙ ⠚ ⠛ ⠜ ⠝ ⠞ ⠟ ⠠ ⠡ ⠢ ⠣ ⠤ ⠥ ⠦ ⠧ ⠨ ⠩ ⠪ ⠫ ⠬ ⠭ ⠮ ⠯ ⠰ ⠱ ⠲ ⠳ ⠴ ⠵ ⠶ ⠷ ⠸ ⠹ ⠺ ⠻ ⠼ ⠽ ⠾ ⠿ ⡀ ⡁ ⡂ ⡃ ⡄ ⡅ ⡆ ⡇ ⡈ ⡉ ⡊ ⡋ ⡌ ⡍ ⡎ ⡏ ⡐ ⡑ ⡒ ⡓ ⡔ ⡕ ⡖ ⡗ ⡘ ⡙ ⡚ ⡛ ⡜ ⡝ ⡞ ⡟ ⡠ ⡡ ⡢ ⡣ ⡤ ⡥ ⡦ ⡧ ⡨ ⡩ ⡪ ⡫ ⡬ ⡭ ⡮ ⡯ ⡰ ⡱ ⡲ ⡳ ⡴ ⡵ ⡶ ⡷ ⡸ ⡹ ⡺ ⡻ ⡼ ⡽ ⡾ ⡿ ⢀ ⢁ ⢂ ⢃ ⢄ ⢅ ⢆ ⢇ ⢈ ⢉ ⢊ ⢋ ⢌ ⢍ ⢎ ⢏ ⢐ ⢑ ⢒ ⢓ ⢔ ⢕ ⢖ ⢗ ⢘ ⢙ ⢚ ⢛ ⢜ ⢝ ⢞ ⢟ ⢠ ⢡ ⢢ ⢣ ⢤ ⢥ ⢦ ⢧ ⢨ ⢩ ⢪ ⢫ ⢬ ⢭ ⢮ ⢯ ⢰ ⢱ ⢲ ⢳ ⢴ ⢵ ⢶ ⢷ ⢸ ⢹ ⢺ ⢻ ⢼ ⢽ ⢾ ⢿ ⣀ ⣁ ⣂ ⣃ ⣄ ⣅ ⣆ ⣇ ⣈ ⣉ ⣊ ⣋ ⣌ ⣍ ⣎ ⣏ ⣐ ⣑ ⣒ ⣓ ⣔ ⣕ ⣖ ⣗ ⣘ ⣙ ⣚ ⣛ ⣜ ⣝ ⣞ ⣟ ⣠ ⣡ ⣢ ⣣ ⣤ ⣥ ⣦ ⣧ ⣨ ⣩ ⣪ ⣫ ⣬ ⣭ ⣮ ⣯ ⣰ ⣱ ⣲ ⣳ ⣴ ⣵ ⣶ ⣷ ⣸ ⣹ ⣺ ⣻ ⣼ ⣽ ⣾ ⣿ ⤀ ⤁ ⤂ ⤃ ⤄ ⤅ ⤆ ⤇ ⤈ ⤉ ⤊ ⤋ ⤌ ⤍ ⤎ ⤏ ⤐ ⤑ ⤒ ⤓ ⤔ ⤕ ⤖ ⤗ ⤘ ⤙ ⤚ ⤛ ⤜ ⤝ ⤞ ⤟ ⤠ ⤡ ⤢ ⤣ ⤤ ⤥ ⤦ ⤧ ⤨ ⤩ ⤪ ⤫ ⤬ ⤭ ⤮ ⤯ ⤰ ⤱ ⤲ ⤳ ⤴ ⤵ ⤶ ⤷ ⤸ ⤹ ⤺ ⤻ ⤼ ⤽ ⤾ ⤿ ⥀ ⥁ ⥂ ⥃ ⥄ ⥅ ⥆ ⥇ ⥈ ⥉ ⥊ ⥋ ⥌ ⥍ ⥎ ⥏ ⥐ ⥑ ⥒ ⥓ ⥔ ⥕ ⥖ ⥗ ⥘ ⥙ ⥚ ⥛ ⥜ ⥝ ⥞ ⥟ ⥠ ⥡ ⥢ ⥣ ⥤ ⥥ ⥦ ⥧ ⥨ ⥩ ⥪ ⥫ ⥬ ⥭ ⥮ ⥯ ⥰ ⥱ ⥲ ⥳ ⥴ ⥵ ⥶ ⥷ ⥸ ⥹ ⥺ ⥻ ⥼ ⥽ ⥾ ⥿ ⦀ ⦁ ⦂ ⦃ ⦄ ⦅ ⦆ ⦇ ⦈ ⦉ ⦊ ⦋ ⦌ ⦍ ⦎ ⦏ ⦐ ⦑ ⦒ ⦓ ⦔ ⦕ ⦖ ⦗ ⦘ ⦙ ⦚ ⦛ ⦜ ⦝ ⦞ ⦟ ⦠ ⦡ ⦢ ⦣ ⦤ ⦥ ⦦ ⦧ ⦨ ⦩ ⦪ ⦫ ⦬ ⦭ ⦮ ⦯ ⦰ ⦱ ⦲ ⦳ ⦴ ⦵ ⦶ ⦷ ⦸ ⦹ ⦺ ⦻ ⦼ ⦽ ⦾ ⦿ ⧀ ⧁ ⧂ ⧃ ⧄ ⧅ ⧆ ⧇ ⧈ ⧉ ⧊ ⧋ ⧌ ⧍ ⧎ ⧏ ⧐ ⧑ ⧒ ⧓ ⧔ ⧕ ⧖ ⧗ ⧘ ⧙ ⧚ ⧛ ⧜ ⧝ ⧞ ⧟ ⧠ ⧡ ⧢ ⧣ ⧤ ⧥ ⧦ ⧧ ⧨ ⧩ ⧪ ⧫ ⧬ ⧭ ⧮ ⧯ ⧰ ⧱ ⧲ ⧳ ⧴ ⧵ ⧶ ⧷ ⧸ ⧹ ⧺ ⧻ ⧼ ⧽ ⧾ ⧿ ⨀ ⨁ ⨂ ⨃ ⨄ ⨅ ⨆ ⨇ ⨈ ⨉ ⨊ ⨋ ⨌ ⨍ ⨎ ⨏ ⨐ ⨑ ⨒ ⨓ ⨔ ⨕ ⨖ ⨗ ⨘ ⨙ ⨚ ⨛ ⨜ ⨝ ⨞ ⨟ ⨠ ⨡ ⨢ ⨣ ⨤ ⨥ ⨦ ⨧ ⨨ ⨩ ⨪ ⨫ ⨬ ⨭ ⨮ ⨯ ⨰ ⨱ ⨲ ⨳ ⨴ ⨵ ⨶ ⨷ ⨸ ⨹ ⨺ ⨻ ⨼ ⨽ ⨾ ⨿ ⩀ ⩁ ⩂ ⩃ ⩄ ⩅ ⩆ ⩇ ⩈ ⩉ ⩊ ⩋ ⩌ ⩍ ⩎ ⩏ ⩐ ⩑ ⩒ ⩓ ⩔ ⩕ ⩖ ⩗ ⩘ ⩙ ⩚ ⩛ ⩜ ⩝ ⩞ ⩟ ⩠ ⩡ ⩢ ⩣ ⩤ ⩥ ⩦ ⩧ ⩨ ⩩ ⩪ ⩫ ⩬ ⩭ ⩮ ⩯ ⩰ ⩱ ⩲ ⩳ ⩴ ⩵ ⩶ ⩷ ⩸ ⩹ ⩺ ⩻ ⩼ ⩽ ⩾ ⩿ ⪀ ⪁ ⪂ ⪃ ⪄ ⪅ ⪆ ⪇ ⪈ ⪉ ⪊ ⪋ ⪌ ⪍ ⪎ ⪏ ⪐ ⪑ ⪒ ⪓ ⪔ ⪕ ⪖ ⪗ ⪘ ⪙ ⪚ ⪛ ⪜ ⪝ ⪞ ⪟ ⪠ ⪡ ⪢ ⪣ ⪤ ⪥ ⪦ ⪧ ⪨ ⪩ ⪪ ⪫ ⪬ ⪭ ⪮ ⪯ ⪰ ⪱ ⪲ ⪳ ⪴ ⪵ ⪶ ⪷ ⪸ ⪹ ⪺ ⪻ ⪼ ⪽ ⪾ ⪿ ⫀ ⫁ ⫂ ⫃ ⫄ ⫅ ⫆ ⫇ ⫈ ⫉ ⫊ ⫋ ⫌ ⫍ ⫎ ⫏ ⫐ ⫑ ⫒ ⫓ ⫔ ⫕ ⫖ ⫗ ⫘ ⫙ ⫚ ⫛ ⫝̸ ⫝ ⫞ ⫟ ⫠ ⫡ ⫢ ⫣ ⫤ ⫥ ⫦ ⫧ ⫨ ⫩ ⫪ ⫫ ⫬ ⫭ ⫮ ⫯ ⫰ ⫱ ⫲ ⫳ ⫴ ⫵ ⫶ ⫷ ⫸ ⫹ ⫺ ⫻ ⫼ ⫽ ⫾ ⫿ ⬀ ⬁ ⬂ ⬃ ⬄ ⬅ ⬆ ⬇ ⬈ ⬉ ⬊ ⬋ ⬌ ⬍ ⬎ ⬏ ⬐ ⬑ ⬒ ⬓ ⬔ ⬕ ⬖ ⬗ ⬘ ⬙ ⬚ ⬛ ⬜ ⬝ ⬞ ⬟ ⬠ ⬡ ⬢ ⬣ ⬤ ⬥ ⬦ ⬧ ⬨ ⬩ ⬪ ⬫ ⬬ ⬭ ⬮ ⬯ ⬰ ⬱ ⬲ ⬳ ⬴ ⬵ ⬶ ⬷ ⬸ ⬹ ⬺ ⬻ ⬼ ⬽ ⬾ ⬿ ⭀ ⭁ ⭂ ⭃ ⭄ ⭅ ⭆ ⭇ ⭈ ⭉ ⭊ ⭋ ⭌ ⭍ ⭎ ⭏ ⭐ ⭑ ⭒ ⭓ ⭔ ⭕ ⭖ ⭗ ⭘ ⭙ ⭚ ⭛ ⭜ ⭝ ⭞ ⭟ ⭠ ⭡ ⭢ ⭣ ⭤ ⭥ ⭦ ⭧ ⭨ ⭩ ⭪ ⭫ ⭬ ⭭ ⭮ ⭯ ⭰ ⭱ ⭲ ⭳ ⭶ ⭷ ⭸ ⭹ ⭺ ⭻ ⭼ ⭽ ⭾ ⭿ ⮀ ⮁ ⮂ ⮃ ⮄ ⮅ ⮆ ⮇ ⮈ ⮉ ⮊ ⮋ ⮌ ⮍ ⮎ ⮏ ⮐ ⮑ ⮒ ⮓ ⮔ ⮕ ⮖ ⮗ ⮘ ⮙ ⮚ ⮛ ⮜ ⮝ ⮞ ⮟ ⮠ ⮡ ⮢ ⮣ ⮤ ⮥ ⮦ ⮧ ⮨ ⮩ ⮪ ⮫ ⮬ ⮭ ⮮ ⮯ ⮰ ⮱ ⮲ ⮳ ⮴ ⮵ ⮶ ⮷ ⮸ ⮹ ⮺ ⮻ ⮼ ⮽ ⮾ ⮿ ⯀ ⯁ ⯂ ⯃ ⯄ ⯅ ⯆ ⯇ ⯈ ⯉ ⯊ ⯋ ⯌ ⯍ ⯎ ⯏ ⯐ ⯑ ⯒ ⯓ ⯔ ⯕ ⯖ ⯗ ⯘ ⯙ ⯚ ⯛ ⯜ ⯝ ⯞ ⯟ ⯠ ⯡ ⯢ ⯣ ⯤ ⯥ ⯦ ⯧ ⯨ ⯩ ⯪ ⯫ ⯬ ⯭ ⯮ ⯯ ⯰ ⯱ ⯲ ⯳ ⯴ ⯵ ⯶ ⯷ ⯸ ⯹ ⯺ ⯻ ⯼ ⯽ ⯾ ⯿ ⸀ ⸁ ⸂ ⸃ ⸄ ⸅ ⸆ ⸇ ⸈ ⸉ ⸊ ⸋ ⸌ ⸍ ⸎ ⸏ ⸐ ⸑ ⸒ ⸓ ⸔ ⸕ ⸖ ⸗ ⸘ ⸙ ⸚ ⸛ ⸜ ⸝ ⸞ ⸟ ⸠ ⸡ ⸢ ⸣ ⸤ ⸥ ⸦ ⸧ ⸨ ⸩ ⸪ ⸫ ⸬ ⸭ ⸮ ⸯ ⸰ ⸱ ⸲ ⸳ ⸴ ⸵ ⸶ ⸷ ⸸ ⸹ ⸺ ⸻ ⸼ ⸽ ⸾ ⸿ ⹀ ⹁ ⹂ ⹃ ⹄ ⹅ ⹆ ⹇ ⹈ ⹉ ⹊ ⹋ ⹌ ⹍ ⹎ ⹏ ⹐ ⹑ ⹒ ⹓ ⹔ ⹕ ⹖ ⹗ ⹘ ⹙ ⹚ ⹛ ⹜ ⹝ ⹞ ⹟ ⹠ ⹡ ⹢ ⹣ ⹤ ⹥ ⹦ ⹧ ⹨ ⹩ ⹪ ⹫ ⹬ ⹭ ⹮ ⹯ ⹰ ⹱ ⹲ ⹳ ⹴ ⹵ ⹶ ⹷ ⹸ ⹹ ⹺ ⹻ ⹼ ⹽ ⹾ ⹿ ⺀ ⺁ ⺂ ⺃ ⺄ ⺅ ⺆ ⺇ ⺈ ⺉ ⺊ ⺋ ⺌ ⺍ ⺎ ⺏ ⺐ ⺑ ⺒ ⺓ ⺔ ⺕ ⺖ ⺗ ⺘ ⺙ ⺛ ⺜ ⺝ ⺞ ⺟ ⺠ ⺡ ⺢ ⺣ ⺤ ⺥ ⺦ ⺧ ⺨ ⺩ ⺪ ⺫ ⺬ ⺭ ⺮ ⺯ ⺰ ⺱ ⺲ ⺳ ⺴ ⺵ ⺶ ⺷ ⺸ ⺹ ⺺ ⺻ ⺼ ⺽ ⺾ ⺿ ⻀ ⻁ ⻂ ⻃ ⻄ ⻅ ⻆ ⻇ ⻈ ⻉ ⻊ ⻋ ⻌ ⻍ ⻎ ⻏ ⻐ ⻑ ⻒ ⻓ ⻔ ⻕ ⻖ ⻗ ⻘ ⻙ ⻚ ⻛ ⻜ ⻝ ⻞ ⻟ ⻠ ⻡ ⻢ ⻣ ⻤ ⻥ ⻦ ⻧ ⻨ ⻩ ⻪ ⻫ ⻬ ⻭ ⻮ ⻯ ⻰ ⻱ ⻲ ⻳ ⼀ ⼁ ⼂ ⼃ ⼄ ⼅ ⼆ ⼇ ⼈ ⼉ ⼊ ⼋ ⼌ ⼍ ⼎ ⼏ ⼐ ⼑ ⼒ ⼓ ⼔ ⼕ ⼖ ⼗ ⼘ ⼙ ⼚ ⼛ ⼜ ⼝ ⼞ ⼟ ⼠ ⼡ ⼢ ⼣ ⼤ ⼥ ⼦ ⼧ ⼨ ⼩ ⼪ ⼫ ⼬ ⼭ ⼮ ⼯ ⼰ ⼱ ⼲ ⼳ ⼴ ⼵ ⼶ ⼷ ⼸ ⼹ ⼺ ⼻ ⼼ ⼽ ⼾ ⼿ ⽀ ⽁ ⽂ ⽃ ⽄ ⽅ ⽆ ⽇ ⽈ ⽉ ⽊ ⽋ ⽌ ⽍ ⽎ ⽏ ⽐ ⽑ ⽒ ⽓ ⽔ ⽕ ⽖ ⽗ ⽘ ⽙ ⽚ ⽛ ⽜ ⽝ ⽞ ⽟ ⽠ ⽡ ⽢ ⽣ ⽤ ⽥ ⽦ ⽧ ⽨ ⽩ ⽪ ⽫ ⽬ ⽭ ⽮ ⽯ ⽰ ⽱ ⽲ ⽳ ⽴ ⽵ ⽶ ⽷ ⽸ ⽹ ⽺ ⽻ ⽼ ⽽ ⽾ ⽿ ⾀ ⾁ ⾂ ⾃ ⾄ ⾅ ⾆ ⾇ ⾈ ⾉ ⾊ ⾋ ⾌ ⾍ ⾎ ⾏ ⾐ ⾑ ⾒ ⾓ ⾔ ⾕ ⾖ ⾗ ⾘ ⾙ ⾚ ⾛ ⾜ ⾝ ⾞ ⾟ ⾠ ⾡ ⾢ ⾣ ⾤ ⾥ ⾦ ⾧ ⾨ ⾩ ⾪ ⾫ ⾬ ⾭ ⾮ ⾯ ⾰ ⾱ ⾲ ⾳ ⾴ ⾵ ⾶ ⾷ ⾸ ⾹ ⾺ ⾻ ⾼ ⾽ ⾾ ⾿ ⿀ ⿁ ⿂ ⿃ ⿄ ⿅ ⿆ ⿇ ⿈ ⿉ ⿊ ⿋ ⿌ ⿍ ⿎ ⿏ ⿐ ⿑ ⿒ ⿓ ⿔ ⿕ 〄 〒 〓 〠 〶 〷 〾 〿 ぀ ぁ あ ぃ い ぅ う ぇ え ぉ お か が き ぎ く ぐ け げ こ ご さ ざ し じ す ず せ ぜ そ ぞ た だ ち ぢ っ つ づ て で と ど な に ぬ ね の は ば ぱ ひ び ぴ ふ ぶ ぷ へ べ ぺ ほ ぼ ぽ ま み む め も ゃ や ゅ ゆ ょ よ ら り る れ ろ ゎ わ ゐ ゑ を ん ゔ ゕ ゖ ゗ ゘ ゙ ゚ ゛ ゜ ゝ ゞ ゟ ゠ ァ ア ィ イ ゥ ウ ェ エ ォ オ カ ガ キ ギ ク グ ケ ゲ コ ゴ サ ザ シ ジ ス ズ セ ゼ ソ ゾ タ ダ チ ヂ ッ ツ ヅ テ デ ト ド ナ ニ ヌ ネ ノ ハ バ パ ヒ ビ ピ フ ブ プ ヘ ベ ペ ホ ボ ポ マ ミ ム メ モ ャ ヤ ュ ユ ョ ヨ ラ リ ル レ ロ ヮ ワ ヰ ヱ ヲ ン ヴ ヵ ヶ ヷ ヸ ヹ ヺ ・ ー ヽ ヾ ヿ ㄅ ㄆ ㄇ ㄈ ㄉ ㄊ ㄋ ㄌ ㄍ ㄎ ㄏ ㄐ ㄑ ㄒ ㄓ ㄔ ㄕ ㄖ ㄗ ㄘ ㄙ ㄚ ㄛ ㄜ ㄝ ㄞ ㄟ ㄠ ㄡ ㄢ ㄣ ㄤ ㄥ ㄦ ㄧ ㄨ ㄩ ㄪ ㄫ ㄬ ㄭ ㄱ ㄲ ㄳ ㄴ ㄵ ㄶ ㄷ ㄸ ㄹ ㄺ ㄻ ㄼ ㄽ ㄾ ㄿ ㅀ ㅁ ㅂ ㅃ ㅄ ㅅ ㅆ ㅇ ㅈ ㅉ ㅊ ㅋ ㅌ ㅍ ㅎ ㅏ ㅐ ㅑ ㅒ ㅓ ㅔ ㅕ ㅖ ㅗ ㅘ ㅙ ㅚ ㅛ ㅜ ㅝ ㅞ ㅟ ㅠ ㅡ ㅢ ㅣ ㅤ ㅥ ㅦ ㅧ ㅨ ㅩ ㅪ ㅫ ㅬ ㅭ ㅮ ㅯ ㅰ ㅱ ㅲ ㅳ ㅴ ㅵ ㅶ ㅷ ㅸ ㅹ ㅺ ㅻ ㅼ ㅽ ㅾ ㅿ ㆀ ㆁ ㆂ ㆃ ㆄ ㆅ ㆆ ㆇ ㆈ ㆉ ㆊ ㆋ ㆌ ㆍ ㆎ ㆏ ㆐ ㆑ ㆒ ㆓ ㆔ ㆕ ㆖ ㆗ ㆘ ㆙ ㆚ ㆛ ㆜ ㆝ ㆞ ㆟ ㆠ ㆡ ㆢ ㆣ ㆤ ㆥ ㆦ ㆧ ㆨ ㆩ ㆪ ㆫ ㆬ ㆭ ㆮ ㆯ ㆰ ㆱ ㆲ ㆳ ㆴ ㆵ ㆶ ㆷ ㆸ ㆹ ㆺ ㇀ ㇁ ㇂ ㇃ ㇄ ㇅ ㇆ ㇇ ㇈ ㇉ ㇊ ㇋ ㇌ ㇍ ㇎ ㇏ ㇐ ㇑ ㇒ ㇓ ㇔ ㇕ ㇖ ㇗ ㇘ ㇙ ㇚ ㇛ ㇜ ㇝ ㇞ ㇟ ㇠ ㇡ ㇢ ㇣ ㇤ ㇥ ㇦ ㇧ ㇨ ㇩ ㇪ ㇫ ㇬ ㇭ ㇮ ㇯ ㇰ ㇱ ㇲ ㇳ ㇴ ㇵ ㇶ ㇷ ㇸ ㇹ ㇺ ㇻ ㇼ ㇽ ㇾ ㇿ ㈀ ㈁ ㈂ ㈃ ㈄ ㈅ ㈆ ㈇ ㈈ ㈉ ㈊ ㈋ ㈌ ㈍ ㈎ ㈏ ㈐ ㈑ ㈒ ㈓ ㈔ ㈕ ㈖ ㈗ ㈘ ㈙ ㈚ ㈛ ㈜ ㈝ ㈞ ㈠ ㈡ ㈢ ㈣ ㈤ ㈥ ㈦ ㈧ ㈨ ㈩ ㈪ ㈫ ㈬ ㈭ ㈮ ㈯ ㈰ ㈱ ㈲ ㈳ ㈴ ㈵ ㈶ ㈷ ㈸ ㈹ ㈺ ㈻ ㈼ ㈽ ㈾ ㈿ ㉀ ㉁ ㉂ ㉃ ㉄ ㉅ ㉆ ㉇ ㉐ ㉑ ㉒ ㉓ ㉔ ㉕ ㉖ ㉗ ㉘ ㉙ ㉚ ㉛ ㉜ ㉝ ㉞ ㉟ ㉠ ㉡ ㉢ ㉣ ㉤ ㉥ ㉦ ㉧ ㉨ ㉩ ㉪ ㉫ ㉬ ㉭ ㉮ ㉯ ㉰ ㉱ ㉲ ㉳ ㉴ ㉵ ㉶ ㉷ ㉸ ㉹ ㉺ ㉻ ㉼ ㉽ ㉾ ㊀ ㊁ ㊂ ㊃ ㊄ ㊅ ㊆ ㊇ ㊈ ㊉ ㊊ ㊋ ㊌ ㊍ ㊎ ㊏ ㊐ ㊑ ㊒ ㊓ ㊔ ㊕ ㊖ ㊗ ㊘ ㊙ ㊚ ㊛ ㊜ ㊝ ㊞ ㊟ ㊠ ㊡ ㊢ ㊣ ㊤ ㊥ ㊦ ㊧ ㊨ ㊩ ㊪ ㊫ ㊬ ㊭ ㊮ ㊯ ㊰ ㊱ ㊲ ㊳ ㊴ ㊵ ㊶ ㊷ ㊸ ㊹ ㊺ ㊻ ㊼ ㊽ ㊾ ㊿ ㋀ ㋁ ㋂ ㋃ ㋄ ㋅ ㋆ ㋇ ㋈ ㋉ ㋊ ㋋ ㋌ ㋍ ㋎ ㋏ ㋐ ㋑ ㋒ ㋓ ㋔ ㋕ ㋖ ㋗ ㋘ ㋙ ㋚ 

Ō ō Ŏ ŏ Ő ő Ř ř Ř ř Ŝ ŝ Ŝ ŝ Š š Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â È è Ê ê Ì ì Î î
Ò ò Ô ô Ř ř Ř ř Ù ù Û û Ḧ ḧ Ạ ạ Ọ ọ Ő ő Ọ ọ Ỏ ỏ Ỳ ỳ Ḅ ḅ Ḍ ḍ Ẹ ẹ Ẽ ẽ Ḝ ḝ Ḟ ḟ Ḡ ḡ Ḥ ḥ Ḧ ḧ Ḫ ḫ Ḯ ḯ Ḱ ḱ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṃ ṃ Ṇ ṇ Ṍ ṍ Ṏ ṏ Ṑ ṑ Ṓ ṓ Ṕ ṕ Ṗ ṗ Ṛ ṛ Ṣ ṣ Ṣ ṣ Ṩ ṩ Ṭ ṭ Ṳ ṳ Ṷ ṷ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẅ ẅ Ẋ ẋ
Ẍ ẍ Ỳ ỳ Ẓ ẓ Ỻ ỻ Ẃ ẃ Ỷ ỷ Ỻ ỻ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ầ ầ Ể ể Ễ ễ Ệ ệ Ễ ễ Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỳ ỳ Ỷ ỷ Ỹ ỹ
Ạ Ấ Ậ Ã Ä Ả È É Ẹ Ẽ Ë Ì Í Ị Ĩ Ï Ñ Ò Ó Ọ Õ Ö Ù Ú Ụ Ũ Ý à á ạ ã ä ả è é ẹ ẽ ë ì í ị ĩ ï ñ ò ó ọ õ ö ù ú ụ ũ ý ģ ÿ Ā ā Ă ă Ç ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ħ ħ Ĩ ĩ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ļ ļ Ń ń Ň ň
Ō ō Ŏ ŏ Ő ő Ř ŕ

Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ų ų Ẁ ẁ Ẃ ẃ Ẅ ẅ Ỳ ỳ Ý ý Ÿ ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ À à Â â È è Ê ê Ì ì Î î
Ò ò Ô ô Ř ř Ŗ ŗ Ù ù Û û Ḧ ḧ Ạ ạ Ọ ọ Ộ ộ Ờ ờ Ỡ ỡ Ỳ ỳ Ḅ ḅ Ḍ ḍ Ẹ ẹ Ḥ ḥ Ḭ ḭ Ḳ ḳ Ḿ ḿ Ṁ ṁ Ṇ ṇ Ọ ọ Ṓ ṓ Ṑ ṑ Ṛ ṛ Ṣ ṣ Ṩ ṩ Ṯ ṯ Ṳ ṳ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ Ẉ ẉ Ẋ ẋ
Ẍ ẍ Ỳ ỳ Ẓ ẓ Ẕ ẕ W̊ ẘ Ỷ ỷ Ỻ ỻ Ả ả Ấ ấ Ầ ầ Ẩ ẩ Ẫ ẫ Ậ ậ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẻ ẻ Ẽ ẽ Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ị ị Ỏ ỏ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ủ ủ Ứ ứ Ừ ừ Ử ử Ữ ữ Ự ự Ỹ ỹ Ỷ ỷ Ỹ ỹ
À Á Â Ã Ä Å È É Ê Ë Ì Í Î Ï Ñ Ò Ó Ô Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǧ ÿ Ā ā Ă ă Ç ç Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ě ě Ġ ġ Ğ ğ Ġ ġ Ḧ ḧ Ǐ ǐ Ī ī Ĭ ĭ İ Ĵ ĵ Ĺ ĺ Ľ ľ Ń ń Ň ň
Ō ō Ŏ ŏ Ő ő Ŕ ŕ Ř ř Ś ś Ŝ ŝ Ş ş Š š Ţ ţ Ť ť Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ

## Semibold italic patched, Precomposed letters that inherit a below anchor in stage 9, Below



Ê ê Ì ì Î î Ò ò Ô ô Ř ř Ř ř Ù ù Û û Ḧ ḧ Ȧ ȧ Ǫ ǫ Ǭ ǭ Ȯ ȯ Ȱ ȱ Ỳ ỳ Ƀ ƀ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḫ ḫ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṩ ṩ Ṩ ṩ Ṩ ṩ Ṫ ṫ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ
Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẗ ẘ ẙ Ả ả Ấ ấ Ậ ậ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỵ ỵ Ỷ ỷ Ỹ ỹ
Ạ Ạ Ậ Ậ Ặ Ặ Ẹ Ẹ Ệ Ệ Ị Ị Ị Ị Ï Ï Ñ Ọ Ọ Ộ Ộ Ỏ Ỏ Ù Ù Ú Ú Û Û Ü Ü Ý ý á á â â ã ã ä ä å å è è é é ê ê ë ë ì ì í í î î ï ï ñ ò ò ó ó ô ô õ õ ö ö ù ù ú ú û û ü ü ý ý ÿ ÿ Ā ā Ă ă Ć ć Ĉ ĉ Ċ ċ Č č Ď ď Ē ē Ĕ ĕ Ė ė Ę ę Ě ě Ĝ ĝ Ğ ğ Ġ ġ Ģ ģ Ĥ ĥ Ĩ ĩ Ī ī Ĭ ĭ İ İ Ĵ ĵ Ĺ ĺ Ļ ļ Ľ ľ Ń ń
Ň ň Ŏ ŏ Ő ő Ř ř Ř ř Ś ś Ş ş Š š Ţ ţ Ũ ũ Ū ū Ŭ ŭ Ů ů Ű ű Ẁ ẁ Ŷ ŷ Ÿ ÿ Ź ź Ż ż Ž ž Ơ ơ Ư ư Ǎ ǎ Ǐ ǐ Ǒ ǒ Ǔ ǔ Ǖ ǖ Ǘ ǘ Ǚ ǚ Ǜ ǜ Ǟ ǟ Ǡ ǡ Ǣ ǣ Ǧ ǧ Ǩ ǩ Ǯ ǯ Ǵ ǵ Ǹ ǹ Ǻ ǻ Ǽ ǽ Ǿ ǿ Ȁ ȁ Ȃ ȃ Ȇ ȇ
Ê ê Ì ì Î î Ò ò Ô ô Ř ř Ř ř Ù ù Û û Ḧ ḧ Ȧ ȧ Ǫ ǫ Ǭ ǭ Ȯ ȯ Ȱ ȱ Ỳ ỳ Ƀ ƀ Ḓ ḓ Ḕ ḕ Ḗ ḗ Ḟ ḟ Ḡ ḡ Ḫ ḫ Ḧ ḧ Ḯ ḯ Ḱ ḱ Ḿ ḿ Ṁ ṁ Ṅ ṅ Ṓ ṓ Ṑ ṑ Ṏ ṏ Ṕ ṕ Ṗ ṗ Ṙ ṙ Ṩ ṩ Ṩ ṩ Ṩ ṩ Ṫ ṫ Ṷ ṷ Ṹ ṹ Ṻ ṻ Ṽ ṽ Ẁ ẁ Ẃ ẃ Ẅ ẅ
Ẇ ẇ Ẋ ẋ Ẍ ẍ Ẏ ẏ Ẑ ẑ ẗ ẘ ẙ Ả ả Ấ ấ Ậ ậ Ẩ ẩ Ẫ ẫ Ắ ắ Ằ ằ Ẳ ẳ Ẵ ẵ Ặ ặ Ẹ ẹ Ẽ ẽ Ế ế Ề ề Ể ể Ễ ễ Ệ ệ Ỉ ỉ Ọ ọ Ố ố Ồ ồ Ổ ổ Ỗ ỗ Ộ ộ Ớ ớ Ờ ờ Ở ở Ỡ ỡ Ợ ợ Ụ ụ Ứ ứ Ừ ừ Ử ử Ữ ữ Ỵ ỵ Ỷ ỷ Ỹ ỹ
Ạ Ạ Ậ Ã Ä Ả Ẹ Ẹ Ệ Ë Ì Í Î Ï Ñ Ọ Ọ Ộ Õ Ö Ù Ú Û Ü Ý à á â ã ä å è é ê ë ì í î ï ñ ò ó ô õ ö ù ú û ü ý ǵ